

Conditional good minimal models for compact Kähler fourfolds

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Abstract

Assuming orbifold Iitaka subadditivity and abundance for nef fourfold adjoints of nonnegative Kodaira dimension, we prove the existence of good minimal models for globally strongly $\mathbb{Q}$ -factorial compact Kähler klt fourfold pairs with effective rational boundary and analytically pseudo-effective actual $\mathbb{Q}$ -Cartier adjoint. The ordinary fourfold minimal model program supplies the nef endpoint. We prove nonvanishing by fibration arguments and, in algebraic dimension zero, by singular metrics, holomorphic foliations, and extension from a reduced boundary. The projective abundance argument used in the proof is included in full.

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Part I

Compact Kähler fourfolds

1 Introduction

The good minimal model problem asks whether a pseudo-effective canonical adjoint can be made nef and generated by its sections on a birational model. It joins two different questions: construction of a minimal model and abundance of its nef adjoint. In the compact Kähler category, the existence of the first pluricanonical section is itself a substantial part of the problem. Positivity is analytic, and even the passage from a cohomological identity to an identity of holomorphic line bundles must be kept explicit.

The projective minimal model program provides the birational framework [56, 9]. In dimension three the Kähler theory has developed through nonvanishing, minimal models and abundance [27, 47, 16, 17, 41]. Work on fourfolds and on pseudo-effective adjoints supplies further analytic and birational tools [19, 46]. Our concern is the precise additional nonvanishing argument needed when a pseudo-effective fourfold MMP and abundance after nonvanishing are available.

Section 2 records three inputs: full orbifold Iitaka subadditivity in Fujiki class $\mathcal{C}$, the specified pseudo-effective klt fourfold MMP, and abundance for nef klt fourfold adjoints of nonnegative Kodaira dimension. The MMP input is supplied by [62, Theorem 6.24 and Proposition 3.4], independently of subadditivity, as detailed after Assumption 2.3. Assumption 2.3 records the precise program used below; the remaining conditional premises are orbifold Iitaka subadditivity and effective fourfold abundance. We prove the good-minimal-model assertion in the following category.

Theorem 1.1 (Conditional good minimal models). *Assume Assumptions 2.2, 2.3, and 2.4. Let X be a normal connected compact Kähler fourfold, and let B be an effective rational Weil divisor. Suppose that (X, B) is klt, that its actual adjoint $D = K_X + B$ is $\mathbb{Q}$ -Cartier and analytically pseudo-effective, and that X is globally strongly $\mathbb{Q}$ -factorial in the sense of Definition 2.1.*

Then there are a normal connected globally strongly $\mathbb{Q}$ -factorial compact Kähler fourfold Y and a bimeromorphic map $\phi : X \dashrightarrow Y$ such that:

- (i) ϕ extracts no prime divisor, and $B_Y = \phi_* B$;
- (ii) (Y, B_Y) is klt and its actual adjoint $D_Y = K_Y + B_Y$ is $\mathbb{Q}$ -Cartier and analytically nef;
- (iii) $a(E; Y, B_Y) \geq a(E; X, B)$ for every prime divisor over the models, with strict inequality for each prime on X contracted by ϕ ;
- (iv) for some $m > 0$, the Cartier line bundle $\mathcal{O}_Y(mD_Y)$ is globally generated.

The exponent may depend on the pair. Zero boundary, smooth fourfolds and all numerical dimensions are included. Global strong $\mathbb{Q}$ -factoriality concerns all rank-one reflexive sheaves on the whole space; it is not an assumption of local analytic $\mathbb{Q}$ -factoriality on every open subset. The conclusion concerns the actual holomorphic adjoint line on the specified endpoint.

Assumption 2.3 already produces a nef minimal model with properties (i)–(iii). The new conclusion needed before Assumption 2.4 can be applied is the following.

Theorem 1.2 (Nonvanishing on the nef endpoint). *Under Assumptions 2.2, 2.3, and 2.4, let (Y, Δ) be an ordinary klt pair on a normal connected globally strongly $\mathbb{Q}$ -factorial compact Kähler fourfold. If Δ is effective and rational and the actual $\mathbb{Q}$ -Cartier adjoint $J = K_Y + \Delta$ is analytically nef, then $\kappa(Y, J) \geq 0$.*

1.1 Structure of the proof

The projective case uses conditional projective log abundance. We include its complete proof in Appendices A–I; Lemma 3.1 derives its precise logarithmic-Iitaka premise from Assumption 2.2. Thus the projective branch introduces no further conditional premise. For a nonprojective endpoint, rational quotients, the Albanese map and lower-dimensional nonvanishing reduce the problem to canonical nonvanishing on a smooth non-uniruled fourfold, principally with irregularity zero.

When algebraic dimension is positive, we first construct an actual canonical pullback model over a projective base. The construction uses sufficiently large ample twists, whose sections are already known, and steps of a single empty-boundary program. Relative rationality allows the twist to be chosen separately at each stage without changing that program. The base has dimension at most three. For a surface base, fiber powers convert subadditivity into the intersection inequality needed for nonvanishing. For a threefold base, the genus-one Hodge line and two modular forms give an explicit crepant klt adjoint on the base. Their common divisorial order is determined by one integrability calculation, including exceptional base valuations.

The algebraic-dimension-zero argument has four parts.

1. A nef ordinary klt adjoint has a metric of minimal singularities with zero Lelong numbers. The proof combines a volume-normalized capacity estimate, a differentiated Monge–Ampère equation and a Bochner transport estimate. The comparison at the end controls a concentrating residual measure on the same sublevel set.
2. If the divisorial locus is empty and no signed canonical frame exists, holomorphic forms produce a transverse spherical structure. A fixed point on the full compact convex set of positive currents and the compactification of its holonomy lead to a contradiction with algebraic dimension zero.
3. Relative analytic constructions produce nef reduced-boundary models while retaining actual line identities and the global reflexive-sheaf condition. The required dlt special termination is proved in the dimension order used by the construction.
4. Adjunction and gluing give a section on the *whole* reduced floor. In the nonprojective torsion case, the ambient Kähler class controls the pluricanonical scalars around gluing cycles. This extra structure is what makes the cycle argument finite.

Finally, finite divisorial support and hard Lefschetz with multiplier ideals extend a high power of the floor section if ambient nonvanishing were to fail. Two choices of reduced boundary, according to whether a signed meromorphic pluricanonical tensor exists, finish canonical nonvanishing. The result then returns to the original nef endpoint of the stipulated MMP.

Several of these constructions have a scope beyond their immediate use here: the metric lemma is dimension independent; the relative descent argument treats global reflexive sheaves directly; the genus-one calculation determines actual crepant orders; and the floor argument isolates the role of a single ambient Kähler class in nonprojective gluing. The long projective appendices are organized separately so that the nonprojective argument can be read with their precise theorem statement, while every new argument used by that theorem remains available in the same manuscript.

Figure 1 records these uses and the return to the original nef endpoint. In the projective appendices, Assumption B.1 is the lower-dimensional induction hypothesis, discharged in Appendix I.

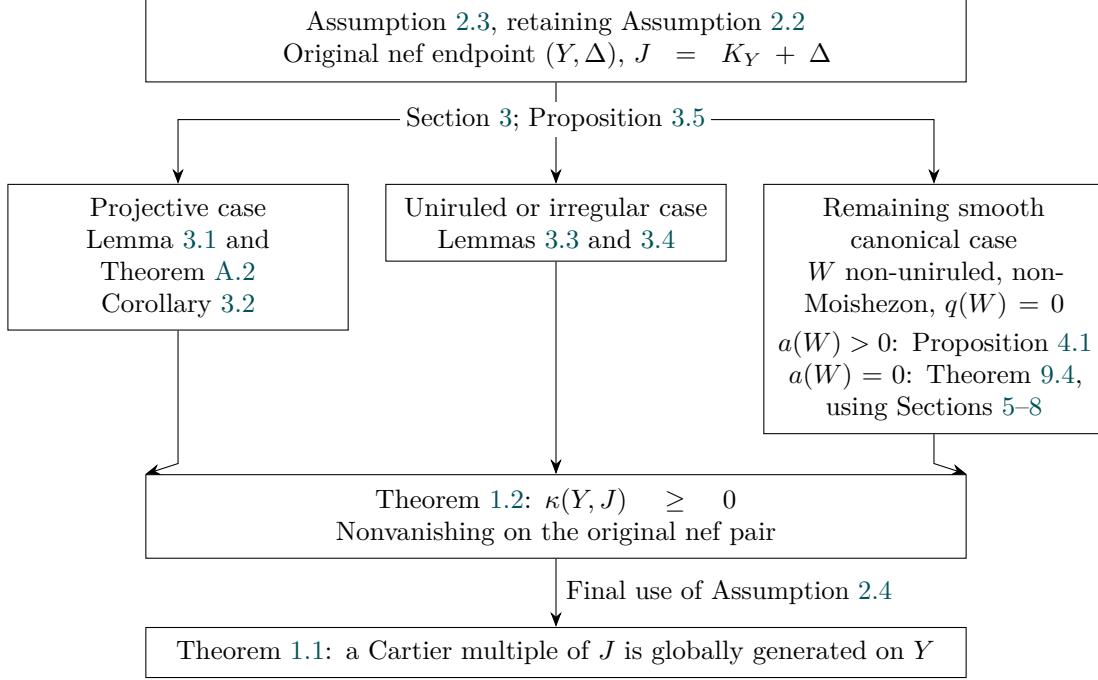


Figure 1: Reading map under Assumptions 2.2–2.4. Lemma 3.1 derives the sole premise of Theorem A.2 from Assumption 2.2, so the projective appendices add no independent conditional input. The groupings indicate uses in this proof, not the full scope of the component lemmas. Assumption 2.4 also enters intermediate constructions. All routes return nonvanishing to the original nef pair before its final application on that endpoint.

2 Conventions and the three inputs

All spaces are over $\mathbb{C}$, unless a statement in the projective appendices explicitly specifies another field of characteristic zero. A compact Kähler space means a complex analytic space with a Kähler form given by local smooth strictly plurisubharmonic potentials on local embeddings. Our spaces are normal and connected, hence irreducible. Smooth compact Kähler resolutions and resolutions of meromorphic maps are obtained by projective modifications. We use the usual resolution and Kähler modification theorems [45, 74].

2.1 Actual adjoints and positivity

The canonical object on a normal space X is its reflexive canonical sheaf ω_X . For an effective rational Weil divisor B , the notation $D = K_X + B$ is an actual rational holomorphic line bundle when, for some positive integer r clearing the coefficients of B ,

$$L_r = (\omega_X^{[r]} \otimes \mathcal{O}_X(rB))^{**}$$

is invertible. Its positive tensor powers define $\mathcal{O}_X(krD)$. Here $\omega_X^{[r]} = (\omega_X^{\otimes r})^{**}$, and divisorial sheaves and their products are interpreted reflexively. An identity $D_1 \sim_{\mathbb{Q}} D_2$ means an isomorphism of actual holomorphic line bundles after a common positive integral multiple. Numerical equivalence alone is never used to infer this identity.

Canonical divisor calculations can be performed with compatible local canonical divisors on a common smooth model $p : W \rightarrow X$. We use log discrepancies

$$a(E; X, B) = 1 + \text{coeff}_E(K_W - p^*(K_X + B)). \quad (1)$$

The pair is klt if these numbers are positive for every prime divisor over X , including primes on X . In particular the boundary coefficients are less than one. We do not assume at the outset that a positive canonical power has a nonzero meromorphic section.

The adjoint D is *analytically pseudo-effective* if $c_1(L_r)/r$ is represented by a closed positive $(1, 1)$ -current with local potentials. It is *analytically nef* if this class is in the closure of the Kähler cone in real Bott–Chern cohomology. Equivalently, for a fixed Kähler form ω and every $\varepsilon > 0$, the line bundle has a smooth Hermitian metric whose curvature, divided by r , is bounded below by $-\varepsilon\omega$. On singular spaces the forms and potentials are understood through local embeddings. Nonnegativity on compact curves is a consequence of nefness, not its definition.

The Iitaka dimension $\kappa(X, D)$ is $-\infty$ if all positive Cartier multiples have zero sections. Otherwise it is the maximum dimension of the images of their complete linear systems. A rational line bundle is *semiample* if some positive Cartier multiple is globally generated. In particular, a semiample rational line of Iitaka dimension zero is torsion as an actual rational line bundle. We write $a(X)$ for algebraic dimension, and $q(M) = h^1(M, \mathcal{O}_M)$ on smooth compact Kähler models.

Definition 2.1 (Global strong $\mathbb{Q}$ -factoriality). A normal compact space X is globally strongly $\mathbb{Q}$ -factorial if every coherent rank-one reflexive sheaf $\mathcal{F}$ on the whole X has an invertible positive reflexive power $\mathcal{F}^{[m]} = (\mathcal{F}^{\otimes m})^{**}$.

Definition 2.1 is a condition on global sheaves. It does not assert the same condition on every analytic open subset. Smooth spaces satisfy it. The constructions below retain this global condition when it is required; local analytic factoriality is not inserted as an intermediate assumption.

2.2 Orbifold subadditivity

We give the model convention in the first hypothesis because the distinction between the invariant base and the base on an arbitrary model is used in the proof.

For a smooth compact space Z with an effective rational SNC boundary Γ having coefficients in $[0, 1]$, set

$$m_\Gamma(E) = \begin{cases} (1 - \text{coeff}_E \Gamma)^{-1}, & \text{coeff}_E \Gamma < 1, \\ \infty, & \text{coeff}_E \Gamma = 1 \end{cases}$$

for every prime divisor E . Thus the multiplicity off the boundary is one; finite multiplicities need not be integers. If $g : (Z, \Gamma) \rightarrow S$ is a surjective morphism with connected fibers and smooth base, put

$$\begin{aligned} m(g, \Gamma; P) &= \min_{E: g(E)=P} (\text{ord}_E(g^*P) m_\Gamma(E)), \\ B(g, \Gamma) &= \sum_P \left(1 - \frac{1}{m(g, \Gamma; P)} \right) P. \end{aligned} \quad (2)$$

Only divisors dominating P enter the minimum; the convention is $1/\infty = 0$. The sum has finite support. This is the inf-multiplicity convention.

An elementary equivalence of smooth source/base fibrations is a commuting bimeromorphic diagram

$$\begin{array}{ccc} (Z', \Gamma') & \xrightarrow{p} & (Z, \Gamma) \\ g' \downarrow & & \downarrow g \\ S' & \xrightarrow{q} & S, \end{array}$$

where the maps are proper holomorphic modifications, $p_*\Gamma' = \Gamma$, and p is an orbifold morphism: for every prime D on Z and every prime E on Z' with $t = \text{ord}_E(p^*D) > 0$,

$$t m_{\Gamma'}(E) \geq m_{\Gamma}(D). \quad (3)$$

Infinite multiplicities are compared in the usual order, with $\infty \geq \infty$. All boundaries in these diagrams are rational SNC boundaries in $[0, 1]$. Equivalence is generated by these diagrams with arrows allowed in either direction, not by arbitrary boundary changes on a fixed space. Define

$$\kappa(g \mid \Gamma) = \inf_{g' \sim g} \kappa(S', K_{S'} + B(g', \Gamma')). \quad (4)$$

For an initially normal possibly singular base, first resolve that base and the main fiber product. On the resulting smooth source use the strict transform of the original boundary plus the full reduced exceptional divisor, and resolve the total support to SNC. These modifications are isomorphisms over very general base points. The invariant is independent of these choices.

A smooth-base fibration is *neat* if there is a proper bimeromorphic orbifold morphism of its source pair to a smooth pair, with the boundary pushing forward, which contracts every source prime mapping to codimension at least two in the given base. The hypothesis below includes existence of suitable neat models and the formula

$$\kappa(g \mid \Gamma) = \kappa(S, K_S + B(g, \Gamma)) \quad \text{on a neat model.} \quad (5)$$

Assumption 2.2 (Full orbifold Iitaka subadditivity). Let X be a smooth compact connected manifold in Fujiki class $\mathcal{C}$, let Δ be an effective rational SNC boundary with coefficients in $[0, 1]$, and let $f : X \rightarrow Y$ be a surjective holomorphic map with connected fibers onto a normal compact irreducible complex space. In arbitrary dimensions,

$$\kappa(X, K_X + \Delta) \geq \kappa(F, K_F + \Delta|_F) + \kappa(f \mid \Delta), \quad (6)$$

where F is a very general smooth fiber with SNC boundary restriction, and the invariant base is defined by (2)–(5).

Here very general means outside a countable union of proper closed analytic subsets, as well as the critical values and unsuitable boundary-stratum loci. The zero boundary is allowed, a point has Kodaira dimension zero, and a sum with $-\infty$ is $-\infty$. Neither source nor base need be projective. There is no abundance, positivity or good-model premise in Assumption 2.2. The displayed statement is the orbifold subadditivity theorem of [63, Theorem 1.1]. For later use, on a smooth base the invariant is at least $\kappa(Y, K_Y)$, by effectivity of the orbifold boundaries and smooth birational invariance.

2.3 The pseudo-effective fourfold program

The closed analytic cone $\overline{\text{NA}}(X)$ consists of positive closed bidimension- $(1, 1)$ currents, modulo their pairings with all real Bott–Chern $(1, 1)$ -classes. We use its negative extremal rays in the following hypothesis.

Assumption 2.3 (Pseudo-effective fourfold MMP). Assume Assumption 2.2. Start from an ordinary klt pair (X, B) , where X is a normal irreducible globally strongly $\mathbb{Q}$ -factorial compact Kähler fourfold, B is effective and rational, and the actual adjoint $D = K_X + B$ is $\mathbb{Q}$ -Cartier and analytically pseudo-effective. No initial modification is inserted. The following program exists and terminates.

At every non-nef stage (X_i, B_i) there is a nonzero D_i -negative extremal ray of $\overline{\text{NA}}(X_i)$, and every such ray R may be chosen. It has a nef supporting class α with

$$\overline{\text{NA}}(X_i) \cap \alpha^\perp = R, \quad \alpha - c_1(D_i) \text{ Kähler.}$$

For every chosen ray there is a projective surjective bimeromorphic contraction with connected fibers $f_i : X_i \rightarrow Z_i$, where Z_i is normal compact Kähler, $\rho_{\text{BC}}(X_i/Z_i) = 1$, and $-D_i$ is relatively ample. For some Kähler class ω_i on Z_i ,

$$\overline{\text{NA}}(X_i) \cap (f_i^* \omega_i)^\perp = R.$$

A divisorial contraction gives the next pair by pushforward. For a small contraction the canonical log-canonical-positive flip is the relative analytic Proj of

$$\bigoplus_{m \geq 0} (f_i)_* \mathcal{O}_{X_i}(mrD_i)$$

for a sufficiently divisible positive Cartier index r . This algebra is locally finitely generated. Its Proj is normal and its map to Z_i is projective, small and has connected fibers. The boundary is strictly transformed; the new actual adjoint is relatively ample, and its divisible Cartier multiple is the tautological line bundle. Passing to a Veronese does not change the model.

Each new pair is klt, compact Kähler and globally strongly $\mathbb{Q}$ -factorial, with $\mathbb{Q}$ -Cartier analytically pseudo-effective actual adjoint. Every nonterminal finite prefix can be extended. Every program so obtained has finitely many steps, irrespective of the choices of negative rays. It ends at an analytically nef pair. The composite map extracts no prime and satisfies the minimal-model discrepancy inequalities, weakly for every prime over the models and strictly for every original prime contracted.

Assumption 2.3 follows from [62, Theorem 6.24; Lemma 3.2; Propositions 3.4 and 6.23] in the intrinsic canonical-sheaf formulation and the pseudo-effective case. Global strong $\mathbb{Q}$ -factoriality supplies the global Weil $\mathbb{Q}$ -factoriality and the canonical rational line bundle required there. The program starts on the given pair, allows every negative extremal ray, and preserves the strong factoriality condition. The cited ordinary-step proposition identifies its flips with the canonical relative Proj models, and the finite-program theorem gives termination for every choice of rays and the stated discrepancy inequalities. This input is independent of Assumption 2.2. The assumption gives neither a first plurisecion nor semiampleness. It includes no non-pseudo-effective program and no dlt or generalized-pair extension. Fiber-type contractions are not stopping outcomes in this program. The Cartier index may change at every stage; no positive-side relative Bott–Chern dimension or identification of absolute Bott–Chern spaces across a flip is assumed. Trivial flops, inserted blowups and arbitrary birational walks are not steps of the program. These distinctions matter when auxiliary boundaries are used below.

2.4 Abundance after nonvanishing

Assumption 2.4 (Effective fourfold abundance). Let X be a normal connected compact Kähler fourfold and Δ an effective rational boundary such that (X, Δ) is klt and the actual adjoint

$J = K_X + \Delta$ is $\mathbb{Q}$ -Cartier. If J is analytically nef and $\kappa(X, J) \geq 0$, then a positive Cartier multiple of J is globally generated.

Assumption 2.4 is the abundance-after-nonvanishing theorem of [61, Theorem 1.1]. It needs no strong factoriality hypothesis and includes the actual torsion conclusion when $\kappa = 0$. Its nonvanishing premise is explicit. No auxiliary statement from its proof is granted separately.

The proof uses these three inputs. Apart from them, established literature is invoked at its stated scope. The separate projective abundance theorem used in the Moishezon branch is proved in Appendices A–I; Lemma 3.1 establishes its sole subadditivity premise from Assumption 2.2.

3 Reductions to canonical nonvanishing

We first discharge the hypothesis of the projective theorem proved in the appendices. We then reduce the remaining problem to canonical nonvanishing on a smooth compact Kähler fourfold of irregularity zero. Throughout, a fibration means a surjective holomorphic map with connected fibers. Smooth source and target spaces do not mean that the map is everywhere a submersion.

3.1 The projective case

Lemma 3.1 (The logarithmic premise). *Assumption 2.2 implies the following statement. Let $f : M \rightarrow S$ be a fibration between smooth connected complex projective varieties, and let D_M, D_S be reduced effective simple normal crossing divisors, possibly zero, such that*

$$\text{Supp}(f^*D_S) \subseteq \text{Supp}(D_M). \quad (7)$$

For a very general smooth fiber F , with $D_F = D_M|_F$, one has

$$\kappa(M, K_M + D_M) \geq \kappa(F, K_F + D_F) + \kappa(S, K_S + D_S). \quad (8)$$

The usual convention for a $-\infty$ summand applies.

Proof. It suffices to prove

$$\kappa(f|D_M) \geq \kappa(S, K_S + D_S), \quad (9)$$

because Assumption 2.2 can then be applied to the original pair (M, D_M) . We construct a neat model in the precise equivalence class occurring in that assumption.

Flatten f by a projective modification of its base and resolve that base, obtaining $q : S' \rightarrow S$ with S' smooth projective. The main strict transform before resolving the source can be taken equidimensional over S' . Indeed, start with the flat strict transform provided by flattening and then base-change to the smooth resolved base. Flatness persists; the unchanged smooth connected generic fiber is irreducible, and flatness rules out vertical components. The main space consequently has pure fibers of dimension $\dim M - \dim S$. Resolve it and its boundary to obtain a commuting diagram

$$\begin{array}{ccc} M' & \xrightarrow{p} & M \\ f' \downarrow & & \downarrow f \\ S' & \xrightarrow{q} & S. \end{array}$$

Here p is a projective birational morphism, M' is smooth projective, and f' has connected fibers: its general fiber is connected, so Stein factorization over the normal base S' has trivial finite factor. Set

$D_{M'}$ equal to the strict transform of D_M plus the full reduced p -exceptional divisor, and arrange that its support is simple normal crossing.

This is an allowed orbifold modification of the source pair. It pushes $D_{M'}$ to D_M . If a prime upstairs has positive pullback order over a coefficient-one prime of D_M , it is either that prime's strict transform or a p -exceptional prime; in both cases its coefficient upstairs is one. Thus the infinite-multiplicity condition in the definition of an orbifold morphism is satisfied.

Moreover, p witnesses neatness. Let $E \subset M'$ be a divisor whose image in S' has codimension $c \geq 2$. In the equidimensional main space its image has dimension at most

$$(\dim S - c) + (\dim M - \dim S) = \dim M - c.$$

Its image in M therefore has codimension at least two, so E is p -exceptional. Subsequent resolutions preserve this argument.

Let $B_{f'}$ be the orbifold base divisor on S' . Every component of a pullback over a prime of $(q^{-1} \text{Supp } D_S)_{\text{red}}$ belongs to $D_{M'}$: this follows from (7) if it is not exceptional over M , and from the definition of $D_{M'}$ otherwise. The multiplicity assigned to each such source component is infinite. Consequently

$$B_{f'} \geq (q^{-1} \text{Supp } D_S)_{\text{red}}. \quad (10)$$

There is also an effective q -exceptional divisor E_S such that

$$K_{S'} + (q^{-1} \text{Supp } D_S)_{\text{red}} = q^*(K_S + D_S) + E_S \quad (11)$$

for compatible canonical divisors. Above the boundary, this is the nonnegativity of log discrepancies of the simple normal crossing pair (S, D_S) ; away from the boundary, it is the nonnegativity of the ordinary discrepancies of the smooth space S . Equivalently, a blowup of a smooth center of codimension c contained in s boundary components contributes $c - s$ if the center lies in the boundary, and $c - 1$ otherwise, both nonnegative.

The neat-model formula in Assumption 2.2, followed by (10) and (11), now gives

$$\kappa(f \mid D_M) = \kappa(S', K_{S'} + B_{f'}) \geq \kappa(S, K_S + D_S).$$

This proves (9). In particular, the infimum in the definition of the invariant has been computed on an allowed neat model, rather than replaced by the divisor of an arbitrary base model. $\square$

Corollary 3.2 (Projective abundance in the present setting). *Under Assumption 2.2, a nef $\mathbb{Q}$ -Cartier adjoint of a normal projective log canonical pair over $\mathbb{C}$, with effective rational boundary, is semiample.*

Proof. Lemma 3.1 is exactly the logarithmic-Iitaka hypothesis of Theorem A.2. Apply that theorem, whose full proof is included in the appendices. $\square$

In particular, the Moishezon case of Theorem 1.2 is settled. A compact Kähler Moishezon space with rational singularities is projective by Namikawa's criterion [60]. Analytic klt singularities are rational; this follows as well from relative vanishing on a projective log resolution [31]. Thus a Moishezon klt endpoint (Y, Δ) of Assumption 2.3 is projective. Its rational analytic boundary and adjoint are algebraic by Chow's theorem and GAGA, and analytic nefness implies nonnegative degree on every curve. Corollary 3.2 applies to its actual adjoint line.

3.2 Lower-dimensional inputs and restriction to fibers

We use smooth compact Kähler resolutions of spaces, pairs, and graphs. The resolution and Kähler-modification theorems ensure that these can be obtained by projective modifications and remain Kähler [45, 74]. A space in Fujiki class $\mathcal{C}$ has a smooth compact Kähler model. The quantities $\kappa(K_W)$, $q(W) = h^0(W, \Omega_W^1)$, and $a(W)$ for smooth compact Kähler W are bimeromorphic invariants.

We recall precisely the lower-dimensional nonvanishing facts used below. A smooth non-uniruled compact Kähler manifold of dimension at most three has nonnegative canonical Kodaira dimension. In nonprojective dimension three this is [47, Corollary 1.4]. More explicitly, take a terminal minimal model by Höring–Petersen and apply canonical nonvanishing [27, Theorem 0.3] to its nef canonical class. Terminal discrepancy comparison pulls its plurisections back to the smooth model. In the projective case, the klt minimal model program and log abundance in dimensions at most three imply nonvanishing for every effective rational klt pair with pseudo-effective adjoint [53, 51, 52]. Finally, Ou’s Theorem 1.1 identifies non-uniruledness of a smooth compact Kähler manifold with analytic pseudo-effectivity of its canonical class [64]. These inputs have no four-dimensional nonvanishing conclusion.

We will repeatedly use the following elementary parameter observation. Let $f : Z \rightarrow S$ be a proper fibration between smooth compact complex manifolds, and let a pseudo-effective rational adjoint on Z have a closed positive representative with local potentials. Its restriction to a smooth fiber is defined and positive for almost every parameter: local plurisubharmonic potentials restrict unless they are identically minus infinity, and Fubini excludes the latter event for almost every parameter. This full-measure set meets the complement of any countable union of proper analytic subsets. If the lower-dimensional results give nonvanishing on those fibers, a fixed multiple has sections generically. Indeed, on a connected smooth parameter open set the loci

$$\{s : h^0(Z_s, mL|_{Z_s}) \geq 1\},$$

for divisible positive integers m , are analytic jumping loci by proper semicontinuity. If every one were proper, their union would have measure zero. Some such locus is therefore the whole open set. Generic base change gives a direct image of positive rank. The same reasoning allows all required very-general smoothness and boundary conditions to be imposed simultaneously.

3.3 The uniruled and irregular cases

Lemma 3.3 (The uniruled reduction). *Assume Assumption 2.2. Let (Y, Δ) be a non-Moishezon compact Kähler klt fourfold with effective rational boundary and pseudo-effective $\mathbb{Q}$ -Cartier adjoint $J = K_Y + \Delta$. If a smooth compact Kähler resolution of Y is uniruled, then $\kappa(Y, J) \geq 0$.*

Proof. Take the almost-holomorphic rational-chain quotient of a smooth resolution W of Y . Campana’s rational quotient theorem in class $\mathcal{C}$ gives a quotient with base in that class and very general fibers rationally connected after resolution [14, Theorem 2.6]. Resolve the quotient and the pair simultaneously to a fibration

$$f : W' \longrightarrow T, \quad p : W' \longrightarrow Y,$$

with W', T smooth compact Kähler. The very general fibers are projective: rational-chain-connected manifolds in class $\mathcal{C}$ are Moishezon by Campana’s algebraic connectedness criterion [13], and a smooth Kähler Moishezon manifold is projective. They are then rationally connected.

The quotient base is not uniruled; see also [64, Lemma 8.10]. For completeness, suppose otherwise and choose a covering rational curve through a very general point of T . Pull the fibration back to its

normalization $\mathbb{P}^1$ and resolve the main component, obtaining a smooth compact Kähler space Z over $\mathbb{P}^1$ with rationally connected very general fiber. Such a fiber has no holomorphic one- or two-forms. A global two-form on Z restricts to zero on those fibers; the relative differential sequence over the smooth locus then places it in the base-one-form times relative-one-form piece. Its restriction there is again zero, so the two-form vanishes on a dense open and hence everywhere. Thus $H^0(Z, \Omega_Z^2) = 0$. Rational approximation of a Kähler class, followed by Kodaira's criterion, makes Z projective. The Graber–Harris–Starr theorem supplies a section of $Z \rightarrow \mathbb{P}^1$ [39].

The chosen base point can be taken so that its original quotient fiber is not contained in any of the countably many analytic exceptional sets in the very-general quotient property. This follows from the proper fiber-dimension theorem on the resolved quotient. Consequently the rational curve is contained in none of the corresponding bad parameter sets. Choose two distinct good parameters on it. In each smooth projective rationally connected fiber, rational chains join a very general point to the point of the section. The section joins these two chains. Their projection to W is a rational chain joining points of different very general rational-quotient fibers, where the original almost-holomorphic quotient is defined. This contradicts the quotient property. Hence T is not uniruled. Its dimension is positive, since a point quotient would make W Moishezon, and is at most three; the fiber dimension is likewise between one and three.

On the simultaneous log resolution write

$$K_{W'} + \Gamma \sim_{\mathbb{Q}} p^*J + E, \quad 0 \leq \Gamma < 1, \quad E \geq 0, \quad (12)$$

where Γ is simple normal crossing and E is p -exceptional. Concretely, take the nonnegative part of the crepant log pullback boundary; its negative part becomes E . The adjoint in (12) is pseudo-effective. At a very general smooth fiber F we can impose the klt simple normal crossing restriction and also restrict its positive current, by the parameter observation above. Projective nonvanishing in dimension at most three gives $\kappa(F, K_F + \Gamma|_F) \geq 0$. The non-uniruled base similarly satisfies $\kappa(T, K_T) \geq 0$.

For a smooth base, the invariant orbifold-base dimension is at least its ordinary canonical Kodaira dimension: on every equivalent smooth model the orbifold divisor is effective, and the smooth canonical Kodaira dimension is birationally invariant. Assumption 2.2 therefore gives $\kappa(W', K_{W'} + \Gamma) \geq 0$. A resulting section pushes through (12) to a section of a positive multiple of J . The exceptional pole allowance disappears away from a codimension-two subset of the normal target, and reflexive extension fills that subset. $\square$

Lemma 3.4 (Irregular canonical nonvanishing). *Assume Assumption 2.2. If W is a smooth non-uniruled compact Kähler fourfold with $q(W) > 0$, then $\kappa(W, K_W) \geq 0$.*

Proof. Resolve the Stein-factor base of the Albanese map and its main pullback. This gives a fibration on smooth compact Kähler models, with a positive-dimensional smooth base T mapping generically with full rank to the Albanese torus. A suitable wedge of the invariant one-forms on that torus pulls back to a nonzero section of K_T . The very general fiber has dimension at most three; if its dimension is zero, it is a point. The pseudo-effective canonical class of the total space restricts to the canonical class of almost every smooth fiber. Lower-dimensional canonical nonvanishing and the parameter observation supply $\kappa(K_F) \geq 0$ on very general fibers. Assumption 2.2, with zero boundary, now gives canonical nonvanishing on the resolved total space and hence on W . $\square$

Proposition 3.5 (Canonical reduction). *Under the three assumptions of the main theorem, it is enough for Theorem 1.2 to prove canonical nonvanishing for every smooth non-uniruled non-Moishezon compact Kähler fourfold W with $q(W) = 0$.*

Proof. Let (Y, Δ) be the nef klt endpoint in Theorem 1.2, and put $J = K_Y + \Delta$. The Moishezon case was settled by Corollary 3.2. Let W be a smooth compact Kähler resolution in the remaining case. If W is uniruled, Lemma 3.3 applies. Otherwise Ou’s theorem makes K_W pseudo-effective. A section of a divisible canonical power on W pushes to the corresponding reflexive canonical power on Y , and multiplication by the effective boundary section, at a common multiple, gives a section of J . Thus canonical nonvanishing on W suffices. Lemma 3.4 handles $q(W) > 0$, leaving exactly the stated case.

Once nonvanishing on this original endpoint is established, Assumption 2.4 makes its adjoint semiample. Its other good-minimal-model and discrepancy properties have already been supplied by Assumption 2.3. None of the auxiliary models constructed later needs to replace this endpoint. $\square$

4 Positive algebraic dimension

Proposition 4.1. *Assume Assumptions 2.2, 2.3, and 2.4. Let W be a smooth non-uniruled compact Kähler fourfold with $q(W) = 0$ and $0 < a(W) < 4$. Then $\kappa(W, K_W) \geq 0$.*

The first step constructs an actual canonical pullback over a projective base. We then prove nonvanishing of that base line in each of its three possible dimensions.

4.1 A contraction index bound

Lemma 4.2 (A local rationality bound). *Let (Z, Δ) be an ordinary rational klt pair on a normal n -dimensional compact Kähler space, with actual $\mathbb{Q}$ -Cartier adjoint $J = K_Z + \Delta$. Let $\pi : Z \rightarrow T$ be a projective bimeromorphic morphism with connected fibers to a normal compact Kähler space. Assume that $-J$ is relatively ample and that all contracted curves have class in one J -negative ray R of $\overline{\text{NA}}(Z)$. Let $k > 0$ be an integer such that kJ is Cartier, and let L be a line bundle with $L \cdot R > 0$. For any nontrivial fiber F of π ,*

$$r_F := \sup\{u : L + uJ \text{ has nonnegative degree on all curves of } F\} \geq \frac{1}{k(n+1)}. \quad (13)$$

No local analytic $\mathbb{Q}$ -factoriality is required. In particular, this applies to the contractions in Assumption 2.3, with $n = 4$.

Proof. Every curve C in F has nonzero class, since a Kähler class has positive degree on it, and its class belongs to R . Therefore

$$r_F = \frac{L \cdot C}{-J \cdot C} > 0$$

is independent of C . The restriction of L to the projective fiber is numerically a positive multiple of the relatively ample rational line $-J$. It is thus ample on F , and the openness of fiberwise ampleness makes L relatively ample after shrinking around the image point of F .

Work over a small Stein neighborhood of that point, with compactum equal to the point. At r_F , the restriction of $L + r_F J$ is numerically trivial, hence nef but not ample on the nontrivial fiber. Fujino’s relative rationality theorem applies to this finite positive threshold: in lowest terms its denominator is at most $k(\dim F + 1)$ [32, Theorem 4.3.1]. The non-lc-locus condition is empty for a klt pair, and the singleton compactum satisfies the required local condition on the Stein base. The relative numerical space over it is finite dimensional, as is also seen by restriction to the projective fiber. The theorem consequently gives $r_F \geq 1/(k(\dim F + 1))$, which implies (13).

The line-bundle formulation is expressly permitted by [32, Remark 4.3.4]; the nearby ampleness assertion is [32, Lemma 2.2.4]. Canonical divisors in that theorem may be handled locally, in its formal

canonical-class convention. Alternatively, over the Stein neighborhood the proper direct image of a rank-one canonical sheaf is coherent of rank one, since the contraction is bimeromorphic. Cartan's theorem supplies a generically nonzero section, hence a meromorphic canonical representative there. This requires no global meromorphic canonical frame on the compact source and no local $\mathbb{Q}$ -factoriality. The Cartier index used in the bound belongs to this stage alone. $\square$

4.2 An actual canonical pullback

Put $d = a(W)$. Resolve a map defined by d algebraically independent meromorphic functions on W . Stein factorization has a projective base, since its finite map has projective image. Resolving that base and the main graph, we may replace W by a smooth compact Kähler model on which there is a fibration

$$b : W \longrightarrow S_0, \quad S_0 \text{ smooth connected projective, } \dim S_0 = d.$$

The generic fibers remain connected through these modifications; Stein factorization over the normal resolved base gives connected fibers everywhere. Fix a very ample line bundle H on S_0 .

Since K_W is pseudo-effective and $4 - d \leq 3$, the restriction and parameter argument in Section 3 gives $\text{rk } b_* \mathcal{O}_W(\ell K_W) > 0$ for some positive integer ℓ . Coherence and ample twisting on the projective base imply that $b_* \mathcal{O}_W(\ell K_W) \otimes H^{\ell N}$ has a nonzero global section for every sufficiently large integer N . Thus there is N_0 such that

$$\kappa(K_W + Nb^*H) \geq 0 \quad \text{for every integer } N \geq N_0. \quad (14)$$

Proposition 4.3 (The pullback model). *There are a canonical globally strongly $\mathbb{Q}$ -factorial compact Kähler fourfold V , a normal projective variety T of dimension d , a fibration $g : V \rightarrow T$, and an actual rational line bundle $A_T \in \text{Pic}(T) \otimes \mathbb{Q}$, such that*

$$K_V \sim_{\mathbb{Q}} g^* A_T. \quad (15)$$

The space V is obtained from the current smooth W by a finite empty-boundary K -program allowed by Assumption 2.3, and K_V remains pseudo-effective.

Proof. At a stage V_i of the empty-boundary program that still maps to S_0 , write $b_i : V_i \rightarrow S_0$ and $H_i = b_i^* H$. All these stages are canonical. Indeed, a common projective resolution of a negative canonical step gives the comparison

$$p^* K_{\text{before}} \sim_{\mathbb{Q}} q^* K_{\text{after}} + G, \quad G \geq 0, \quad (16)$$

with G exceptional over the after model. The comparison follows from the projective analytic negativity lemma over the contraction base: there is no extraction, and canonical negativity supplies the relative sign. For flips both sides and their common resolution are projective over that base. Compatible local canonical comparisons glue to the intrinsic discrepancy divisor in (16). Discrepancies therefore do not decrease. Starting from smooth W , this preserves canonicity. The remaining category and pseudo-effectivity properties are part of Assumption 2.3.

Choose a Cartier index k_i for K_{V_i} , and choose an integer

$$N \geq N_0, \quad N > 5k_i. \quad (17)$$

This choice is made anew at the current stage. The actual line $K_{V_i} + NH_i$ has an effective rational klt boundary representative: take a general divisor in a sufficiently high free multiple of NH_i and divide by that multiple. On a fixed log resolution the system has no fixed exceptional component,

and Bertini makes the chosen member transverse to the exceptional strata; its small coefficient preserves klt. Its adjoint is pseudo-effective because K_{V_i} is pseudo-effective and H_i is semipositive.

If this adjoint is not analytically nef, Assumption 2.3 supplies an extremal ray on which $K_{V_i} + NH_i$ is negative. Since H_i is the pullback of a semipositive form, the ray is also K_{V_i} -negative. Take its contraction and its negative step for the empty-boundary program. In fact

$$H_i \cdot R_i = 0. \quad (18)$$

Otherwise Lemma 4.2, with $L = H_i$ and $J = K_{V_i}$, gives

$$\frac{1}{5k_i} \leq \frac{H_i \cdot C}{-K_{V_i} \cdot C} < \frac{1}{N}$$

on any curve of a nontrivial contraction fiber, contrary to (17).

Every connected projective fiber of the contraction maps to a point under b_i . If its image had positive dimension, some curve in that fiber would map nontrivially to the projective base and have positive H_i -degree. Proper descent over the normal contraction target therefore factors b_i through it. In a flip, composition with the positive-side morphism gives b_{i+1} . Thus the actual pullback line H_i , not merely its numerical class, continues to be the pullback of the fixed H .

Repeat this procedure as long as the selected twist is not nef. Every step actually taken is a step of the single empty-boundary K -program starting from W . The changing integers N select rays; they do not change that program's boundary. If the procedure were infinite, it would contradict the arbitrary termination assertion in Assumption 2.3. It therefore reaches a stage V_i at which its selected $K_{V_i} + NH_i$ is nef. In particular, if the canonical program reaches a nef canonical class, the selected semipositive twist is nef there as well.

For this final value of N , (14) already gave a section on the original W . Clear the finitely many Cartier indices of the chosen program. Since no step extracts a prime and all maps agree to S_0 , that section pushes through every step as a section of the corresponding actual twisted canonical power. Reflexive extension across codimension two on each normal target justifies the pushforward. Hence $\kappa(K_{V_i} + NH_i) \geq 0$ before abundance is invoked. Assumption 2.4, applied to the klt representative above, makes this actual line semiample.

Set $V = V_i$, choose a free integral multiple $m(K_V + NH_i)$, and let $h : V \rightarrow \mathbb{P}^r$ be its morphism. Take the Stein factorization $g : V \rightarrow T$ of (b_i, h) . The variety T is normal and finite over the projective image, hence projective. Let $a : T \rightarrow S_0$ and $c : T \rightarrow \mathbb{P}^r$ be the induced maps. We have $\dim T \geq d$ because a is surjective, and $\dim T \leq a(V) = d$ because T is projective. Define

$$A_T = \frac{1}{m} c^* \mathcal{O}_{\mathbb{P}^r}(1) - Na^*H \quad \text{in } \text{Pic}(T) \otimes \mathbb{Q}. \quad (19)$$

The defining evaluation isomorphism for h and the actual identity $H_i = g^*a^*H$ prove (15). $\square$

Choose a smooth projective resolution $\mu : S \rightarrow T$, and resolve the main graph to obtain a smooth compact Kähler M with $p : M \rightarrow V$ and a fibration $f : M \rightarrow S$. Put $A = \mu^*A_T$. Canonicity and the fixed actual isomorphism in (15) give

$$K_M \sim_{\mathbb{Q}} f^*A + R, \quad R \geq 0 \text{ and } R \text{ is } p\text{-exceptional}. \quad (20)$$

This is an identity of rational line bundles; all further resolutions use its canonical transform. Pullback of holomorphic one-forms gives $q(S) \leq q(M) = q(W) = 0$.

The rational line A is pseudo-effective. To see this with the analytic definition, pull a positive representative Θ of $c_1(K_V)$, with local potentials, to M . The map p is dominant, so its

plurisubharmonic potentials do not become identically minus infinity. If ω is a Kähler form on M , then

$$f_*(p^*\Theta \wedge \omega^{4-d})$$

is a closed positive $(1, 1)$ -current. The projection formula puts its class in $v c_1(A)$, where the degree-zero closed current $f_*(\omega^{4-d}) = v$ is the positive constant volume of a smooth general fiber. Division by v proves the assertion. Since S is projective, this also gives numerical divisor pseudo-effectivity [12].

It remains to prove $\kappa(S, A) \geq 0$. A section then pulls back through (20), multiplied by the effective exceptional section, to a canonical plurisecion on M , and hence on W . If $d = 1$, the smooth projective curve S has $q(S) = 0$, so it is $\mathbb{P}^1$; a pseudo-effective rational line on it has nonnegative degree and has a section at a divisible multiple. The two remaining dimensions require more information from the fibration.

4.3 A surface base and fiber powers

Assume $d = 2$. Fix a sufficiently divisible positive integer m_0 for (20). On very general smooth fibers one has

$$h^0(F, mK_F) = 1 \quad (m > 0, m_0 \mid m). \quad (21)$$

At least one section is supplied by the effective divisor $R|_F$. If for a fixed divisible m the generic dimension were at least two, generic base change and ample twisting of $f_*\mathcal{O}_M(mK_M)$ would give two sections whose ratio is nonconstant on a general fiber. Together with meromorphic functions from the two-dimensional projective base this would give algebraic dimension at least three, contrary to $a(M) = 2$. The analytic jumping loci for the countably many m 's can be excluded simultaneously, proving (21).

Lemma 4.4 (A base intersection inequality). *There are nonnegative rational numbers c_P , indexed by the finitely many μ -exceptional curves on S , such that*

$$\left(A - K_S + \sum_P c_P P \right) \cdot H' \geq 0 \quad (22)$$

for every ample divisor H' on S .

Proof. Choose a dense open set of S over which f is smooth and the generator supplied by (20) is not identically zero on any fiber. This requires deleting a proper analytic subset: a horizontal divisor cannot contain a whole fiber over a divisor in the base without being vertical, by the fiber-dimension theorem; the remaining degeneracies are proper analytic images or jumping loci. Include all μ -exceptional curves among the finitely many complementary curves.

For each such curve P , fix a component D of f^*P dominating P , write

$$e = \text{ord}_D(f^*P), \quad h = \text{coeff}_D R,$$

and make these choices before choosing any ample test curve or fiber power. If P is not exceptional over T , take D with $h = 0$. Indeed, over a general point of its image divisor in the normal T , a local equation pulls back under the holomorphic g to a divisor on V dominating that base divisor. The strict transform of one such component has zero coefficient in the p -exceptional R . For an exceptional P , set

$$c_P = h/e. \quad (23)$$

Replace H' by a sufficiently large very ample multiple and choose a smooth member C of genus at least one. It can be chosen transverse to all complementary curves at their general points, avoiding their finitely many excluded points and any isolated bad base points. Moreover, $M_C = f^{-1}(C)$ is smooth by Bertini for the pulled-back free system, and connected because f has connected fibers. To retain (21), choose C through one parameter where all those countably many equalities hold. A high enough linear system through that point is free away from it; on its fiber f is a submersion, so the same Bertini conclusion holds there. Each jumping locus then cuts a proper analytic subset of C . Thus very general fibers of $f_C : M_C \rightarrow C$ satisfy all the equalities in (21).

By adjunction,

$$K_{M_C/C} \sim_{\mathbb{Q}} f_C^*((A - K_S)|_C) + R_C, \quad R_C = R|_{M_C} \geq 0. \quad (24)$$

At each $z \in C \cap P$, a general point of the chosen component D over z admits coordinates (x, y_1, y_2) on M_C and a parameter t on C in which

$$t = x^e, \quad R_C = h\{x = 0\} \quad (25)$$

in that chart. A unit in the pullback equation is absorbed into x . Such charts exist over general points of P : the map $D \rightarrow P$ is generically submersive, and its intersections with the other divisorial supports have smaller generic fiber dimension. Properness and the fiber-dimension theorem allow the exceptional base points to be discarded before C is chosen. These choices are independent of the next integer r .

For $r \geq 1$, take the main component of the r -fold fiber product of M_C over C , and a smooth compact Kähler resolution Z_r of it. Over the good open set this fiber product is smooth with connected fibers F^r , hence has a unique main component. It is an analytic subspace of a product of compact Kähler spaces; the required Kähler resolution and connected-fiber morphism to C follow as above. The general fiber has a canonical section by (21), and $\kappa(C) \geq 0$. Assumption 2.2, with zero boundary and in dimension $2r + 1$, gives a nonzero section of mK_{Z_r} for some $m > 0$. By taking a power, make m divisible by m_0 and all indices in use. It may depend on r and C .

Over good parameters, the product formula and (21) make the fiberwise pluriform space one-dimensional. The chosen section is therefore the product of the r relative generators in (24), times a base pluriform and a scalar base coefficient. This coefficient is a section on the good open of the line

$$\mathcal{O}_C(m(K_C + r(A - K_S)|_C)). \quad (26)$$

Indeed, its quotient by the displayed product is constant on very general fibers; local submersion sections and analytic continuation give meromorphic descent to that open. Evaluation at points where the product does not vanish identically on the fiber makes the descended coefficient holomorphic there.

We compute its order at a missing point z . In (25), a frame of $\mathcal{O}_C(m(A - K_S)|_C)$ maps, up to a unit, to

$$x^{mh} \left(\frac{dx \wedge dy_1 \wedge dy_2}{dt} \right)^m.$$

On a normalization branch of the local r -fold product we have $x_1 = x$, $x_j = \zeta_j x$, where $\zeta_j^e = 1$. Multiplying the r relative factors by one base factor $(dt)^m$, and using $dt = ex^{e-1}dx$, gives order

$$m(rh - (r - 1)(e - 1)) \quad (27)$$

along $x = 0$. This smooth normalization branch belongs to the main component, being a closure of points with $t \neq 0$. The global pluriform on Z_r has no pole at its generic divisor, by birational

comparison. Its scalar coefficient u , pulled back by $t = x^e$, therefore extends meromorphically, with

$$e \operatorname{ord}_z(u) + m(rh - (r-1)(e-1)) \geq 0.$$

In particular,

$$\operatorname{ord}_z(u) \geq -\frac{mrh}{e}. \quad (28)$$

There is no pole allowance at a nonexceptional base curve, where $h = 0$. The exceptional allowances are exactly mrc_P , with the fixed c_P in (23).

The nonzero meromorphic coefficient in (26), with these pole bounds, implies

$$\deg K_C + r \left(A - K_S + \sum_P c_P P \right) \cdot C \geq 0.$$

Divide by r and let $r \rightarrow \infty$, keeping C fixed. Since C is a positive multiple of H' , this proves (22). $\square$

Take the surface Zariski decomposition [4]

$$A = P_0 + N_0, \quad (29)$$

where P_0 is nef, $N_0 \geq 0$ has negative-definite support if nonzero, and P_0 is orthogonal to every component of N_0 . The negative-part linear equations give rational coefficients, so $P_0 = A - N_0$ remains an actual rational line. For a μ -exceptional curve P , $A \cdot P = 0$. If P belongs to $\operatorname{Supp} N_0$, orthogonality gives $P_0 \cdot P = 0$; otherwise both $P_0 \cdot P$ and $N_0 \cdot P$ are nonnegative, and their sum is zero. Hence every such P is orthogonal to P_0 .

If $P_0 \equiv 0$, the equality $q(S) = 0$ makes a divisible multiple of P_0 torsion, so P_0 is zero in $\operatorname{Pic}(S) \otimes \mathbb{Q}$. If $P_0^2 > 0$, the nef line P_0 is big. Either case supplies sections of a positive multiple of A . In the remaining case, $P_0 \not\equiv 0$ and $P_0^2 = 0$. Test (22) on ample classes approaching P_0 . The exceptional terms and $A \cdot P_0$ vanish, giving $K_S \cdot P_0 \leq 0$. For large divisible ℓ ,

$$\chi(S, \ell P_0) = \chi(\mathcal{O}_S) - \frac{\ell}{2} K_S \cdot P_0 > 0, \quad (30)$$

since $\chi(\mathcal{O}_S) = 1 + h^{2,0}(S) > 0$. Serre duality gives $H^2(S, \ell P_0) = 0$ for large ℓ : the divisor $K_S - \ell P_0$ has negative degree against a fixed ample divisor. Thus $h^0(S, \ell P_0) > 0$; multiplication by the section of ℓN_0 proves $\kappa(S, A) \geq 0$.

4.4 A threefold base and genus-one orders

Assume $d = 3$. The general fibers of $g : V \rightarrow T$ are smooth connected genus-one curves. Indeed, the singular locus of normal V has dimension at most two, so misses a general fiber; generic smoothness then applies, and (15) makes its canonical line torsion. Also all components of R in (20) are vertical over S : their images on V have dimension at most two.

Choose one integer $m > 0$, divisible by 12, that clears the Cartier data and the actual isomorphism in (15). It clears (20) on every smooth model subsequently used. In fact, the pullback of a local frame of the Cartier line $\mathcal{O}_V(mK_V)$, regarded on the regular locus as an m -pluriform, has integral orders on every resolution. Those orders are precisely $m \operatorname{coeff}_E R$. There is thus no need to choose a new index after a base blowup.

4.4.1 The actual Hodge line and its growth

Shrink to a dense open $U \subset T_{\text{reg}}$, also viewed on S , on which the family is smooth and agrees with its resolution. Let $\mathcal{H}$ be its line of holomorphic fiberwise one-forms. It is a holomorphic line by Grauert base change; periods against local integral cycles are holomorphic. Give it the Hodge norm, so that $\|\alpha\|^2$ is, up to a fixed convention, the fiber integral of $\sqrt{-1}\alpha \wedge \bar{\alpha}$.

Evaluation $f^*\mathcal{H} \rightarrow \omega_{M/U}$ is an isomorphism: a nonzero holomorphic one-form on a smooth genus-one curve has no zero. Its m -th power, together with the fixed actual pullback identity, gives an isomorphism of pullbacks of lines on U . Applying f_* and $f_*\mathcal{O}_M = \mathcal{O}_U$ yields the actual identification

$$\mathcal{H}^{\otimes m} \simeq \mathcal{L}_m|_U, \quad \mathcal{L}_m = \mathcal{O}_S(m(A - K_S)). \quad (31)$$

It is compatible on all birational base models over their common open. No section of the genus-one fibration is needed: its first integral cohomology, periods, and invariant differentials are defined without an origin. In particular, (31) leaves no unspecified flat line factor.

Let E_4 and Δ_{mod} be the classical modular forms of weights four and twelve for $\text{SL}_2(\mathbb{Z})$ [68, Chapter VII]. A local symplectic period basis gives a parameter τ in the upper half-plane and a normalized Hodge frame with periods $1, \tau$. The weight transformation laws compensate the frame transformation, so these forms define sections of $\mathcal{H}^4$ and $\mathcal{H}^{12}$. Define sections of $\mathcal{L}_m|_U$ by

$$e_1 = E_4^{m/4}, \quad e_2 = \Delta_{\text{mod}}^{m/12}, \quad (32)$$

where the tensor powers of the normalized frame are understood.

Near a general point of a complementary prime divisor on any smooth base model, let $t = 0$ be its equation. There are positive constants c, C, b , locally uniform in the remaining coordinates, such that

$$c \leq (\|e_1\| + \|e_2\|)^{2/m} \leq C(1 + |\log |t||)^b \quad (t \neq 0). \quad (33)$$

Here is the full growth argument. In the standard fundamental domain the normalized frame has squared norm proportional to $\text{Im } \tau$. Both scalar modular forms are bounded there, Δ_{mod} has no zero in the upper half-plane, and $E_4 \rightarrow 1$ at the cusp. On the compact part their joint norm has a positive minimum; at the cusp the E_4 term supplies the same positive lower bound. This also covers the case where e_1 is identically zero on the given family.

For the upper bound, restrict to punctured disks transverse to the prime, with the other parameters in a small compact set and with $t \neq 0$ inside U . Lift the period map to universal covers. Schwarz–Pick makes it distance-decreasing for the hyperbolic metrics. The radial hyperbolic distance in a punctured disk, from a fixed radius to $|t|$, is $O(1) + O(\log |\log |t||)$. Starting period representatives at that radius can be chosen in a bounded part of the fundamental domain, uniformly in angle and the other parameters, by compactness inside the smooth locus. Since $\log \text{Im } \tau$ changes by at most hyperbolic distance, both $\text{Im } \tau$ and its inverse are bounded by powers of $1 + |\log |t||$ along the lifted radial paths. Returning to the fundamental domain preserves such a bound, because for $\gamma \in \text{SL}_2(\mathbb{Z})$,

$$\text{Im}(\gamma\tau) \leq \max\{\text{Im } \tau, (\text{Im } \tau)^{-1}\}.$$

The bounded scalar modular forms and the Hodge-frame norm now prove the upper inequality in (33).

4.4.2 The integration threshold and meromorphic extension

Fix a prime $P \subset S$, a local parameter t at its general point, and a frame s of $\mathcal{L}_m$. On a simultaneous log resolution of R and f^*P , recompute R by the canonical comparison in (20). Define

$$t_P = \min_{E \rightarrow P} \frac{1 + \text{coeff}_E R}{\text{ord}_E(f^*P)}. \quad (34)$$

The minimum is taken over components above the general point of P . It is positive because $R \geq 0$.

Lemma 4.5 (The divisorial order identity). *The sections e_1, e_2 extend meromorphically to S . If*

$$\ell_P = \min_i \text{ord}_P(e_i/s),$$

omitting an identically zero section, then

$$\frac{\ell_P}{m} = 1 - t_P. \quad (35)$$

The same identity holds on every subsequent smooth base model, using its actual line $\mathcal{O}(m(A - K_S))$ and the fixed integer m .

Proof. First identify the weighted integrability threshold of the frame:

$$t_P = \sup \left\{ u \in \mathbb{R} : |t|^{-2u} \|s\|^{2/m} \text{ is locally integrable near general } P \right\}. \quad (36)$$

Multiply s by a local m -canonical base frame. Under (20), the resulting total-space pluriform has divisor mR . On a smooth good fiber it is the m -th tensor of a one-form, up to a scalar, so integrating its $2/m$ -density along the fiber gives $\|s\|^{2/m}$ times the smooth base coordinate density. Fubini and change of variables therefore identify the weighted base integral with the total-space integral. In simple normal crossing coordinates the latter has, at a component E , a factor

$$|x_E|^{2(h_E - ue_E)}, \quad h_E = \text{coeff}_E R, \quad e_E = \text{ord}_E(f^*P).$$

Such a factor is integrable exactly when $h_E - ue_E > -1$. Choose the point of P generally to exclude components not dominating it, and use properness for a finite covering of its inverse image. The conditions are exactly those in (34), proving (36).

We next establish a polynomial lower bound on the Hodge norm of s , without assuming meromorphic extension of the modular coefficients. Choose one component E above general P , of multiplicity e and coefficient h in R . At a general point of E , the map has local coordinates

$$(t, z_1, z_2) = (x^e, z_1, z_2)$$

on the base, with one further fiber coordinate y on the source. The map $E \rightarrow P$ is submersive there, units have been absorbed into x , and no other divisorial support meets the chart. The total pluriform associated with s is a unit times $x^{mh}(dx \wedge dz_1 \wedge dz_2 \wedge dy)^m$. After division by the base pluriform, its relative coefficient is a unit times $x^{m(h-e+1)}$. Integrating on a fixed smaller disk in the y -coordinate gives

$$\|s\|^{2/m} \geq c |t|^{2(h-e+1)/e}. \quad (37)$$

These charts are available above every point of P outside a proper analytic subset. Indeed, the non-submersive locus and intersections with the removed supports are proper subsets of E ; those that dominate P have strictly smaller generic fiber dimension. Properness and the fiber-dimension

theorem show that a general fiber of $E \rightarrow P$ is not exhausted by them. We also omit the singular and intersection loci of complementary base divisors.

On the punctured chart, write $e_i = a_i s$. Combining (33) with (37) gives an upper bound for $|a_i|$ by a fixed power of $|t|^{-1}$, after absorbing a logarithmic power into an arbitrarily small additional power. Thus a_i has at most a pole across general P . The integer pole allowance can be fixed for that prime because the exponent in (37) depends on the one fixed component E . There are finitely many complementary prime divisors. Twist $\mathcal{L}_m$ by their bounded pole allowances; both sections are then holomorphic away from an analytic set of codimension at least two. Hartogs extension on the smooth S gives global meromorphic sections of the original $\mathcal{L}_m$.

At a general point of P , the two meromorphic scalar coefficients satisfy

$$|a_1| + |a_2| \asymp |t|^{\ell_P}.$$

Equation (33) therefore gives

$$c|t|^{-2\ell_P/m} \leq \|s\|^{2/m} \leq C|t|^{-2\ell_P/m}(1 + |\log |t||)^b. \quad (38)$$

The supremum of the exponents u for which the weighted expression is integrable is consequently $1 - \ell_P/m$: it is integrable for strictly smaller u , and the lower bound makes it nonintegrable for strictly larger u . Its possible behavior at the endpoint does not change the supremum. Comparison with (36) proves (35). At a good divisor the identity is immediate.

On a further smooth base modification the same argument uses its canonical line and a simultaneous resolution of the total space. The integer m still clears the actual identity, as explained above. The sections agree on the common open by (31), and hence over the meromorphic field. Thus the calculation applies with the same m on every model needed below. $\square$

For a prime P that is not exceptional over T , one also has

$$t_P \leq 1. \quad (39)$$

Indeed, as in the surface argument, a component of the inverse image of its image divisor on T is a divisor on V dominating it. Its strict transform has coefficient zero in R and positive integral pullback multiplicity. It is one of the components tested in (34). No such upper bound is required for a base-exceptional prime.

4.4.3 A klt adjoint on the original base

Resolve the meromorphic pencil $[e_1 : e_2]$ by projective blowups of S , and then resolve the divisorial supports, including the exceptional locus over T . A constant pencil is allowed. If $\rho : S' \rightarrow S$ is such a smooth modification, its actual line is

$$\mathcal{L}'_m = \rho^* \mathcal{L}_m \otimes \mathcal{O}_{S'}(-mK_{S'/S}). \quad (40)$$

Consequently both section divisors transform by their pullbacks minus the same relative canonical Jacobian term. Resolving the pencil and subtracting its full signed fixed divisor therefore leaves a free moving system; subsequent blowups pull back that free system and preserve its freeness.

Rename this smooth model S . Its fixed divisor is $\sum_P \ell_P P$, with simple normal crossing support. A general complex linear combination e of e_1, e_2 has

$$\frac{1}{m} \operatorname{div}_S(e) = \sum_P (1 - t_P) P + \frac{1}{m} Q =: B_S, \quad (41)$$

where Q is a general member of the free moving system, possibly zero. Bertini makes it smooth, reduced, and transverse to all the chosen strata, with no fixed component. Every coefficient of this simple normal crossing divisor is strictly less than one: the fixed coefficients are $1 - t_P < 1$, and the moving coefficient is $1/m < 1$. The fixed coefficients over exceptional primes may be negative.

Regard e downstairs as a rational section of the rank-one reflexive difference $mA_T - mK_T$, and define

$$\Xi = \frac{1}{m} \operatorname{div}_T(e). \quad (42)$$

Equation (39) shows that $\Xi \geq 0$. There is an actual rational-linear identity

$$K_T + \Xi \sim_{\mathbb{Q}} A_T. \quad (43)$$

To verify both this identity and crepancy, choose over the rational function field a frame a of mA_T and a rational canonical form ω , and write $e = u a / \omega^m$. With canonical divisor chosen by ω ,

$$m(K_T + \Xi) = \operatorname{div}(u) + \operatorname{div}(a).$$

The right side is Cartier. Pull it back to S and subtract $m \operatorname{div}_S(\omega)$; the result is precisely $\operatorname{div}_S(e)$, with the canonical Jacobian term in (40). Thus

$$K_S + B_S = \mu^*(K_T + \Xi) \quad (44)$$

for compatible rational divisor representatives. This proves that (T, Ξ) is klt: B_S is its simple normal crossing crepant boundary with every coefficient less than one. Only the sum $K_T + \Xi$ has been proved $\mathbb{Q}$ -Cartier; a separate $\mathbb{Q}$ -Gorenstein hypothesis on T was not used.

Finally put $\Gamma_S = (B_S)_+$. It is an effective rational klt simple normal crossing boundary, and

$$K_S + \Gamma_S \sim_{\mathbb{Q}} A + E_S, \quad E_S \geq 0 \text{ and } E_S \text{ is } \mu\text{-exceptional.}$$

The exceptionality follows from $\Xi \geq 0$: negative coefficients of the crepant boundary occur only over T 's codimension-two locus. This adjoint is numerically pseudo-effective on the smooth projective threefold S . Projective klt threefold nonvanishing, in the scope recalled in Section 3, gives a section of a positive multiple of $K_S + \Gamma_S$. Push it to T . The exceptional pole allowance disappears in codimension one, and normality extends the section of the actual line mA_T across codimension two, after taking a common multiple. Hence $\kappa(T, A_T) \geq 0$, as required.

Proof of Proposition 4.1. Proposition 4.3 gives (20) with A pseudo-effective and $q(S) = 0$. The curve, surface, and threefold arguments above each give $\kappa(S, A) \geq 0$ in the corresponding dimension. Pull a divisible section back to M and multiply by the section of mR in (20). This gives a nonzero canonical plurisecion on M , and smooth birational invariance gives $\kappa(W, K_W) \geq 0$. $\square$

5 Minimal singularities of nef klt adjoints

We prove the metric statement needed in the remaining nonprojective case. The proof does not require sections of a positive twist. We use $\operatorname{dd}^c = \sqrt{-1} \partial \bar{\partial}$, and identify the class of a line bundle with its curvature class; thus the usual factor 2π in its first Chern class is understood. Weights on a rational line bundle mean weights obtained by taking a root of a metric on a fixed Cartier power. All the line identifications below are actual rational line-bundle identifications.

Lemma 5.1 (Minimal metric of a nef klt adjoint). *Let $p: M \rightarrow Z$ be a compact Kähler log resolution of a normal compact Kähler klt pair (Z, Δ) , where $\Delta \geq 0$ is rational and $K_Z + \Delta$ is a nef rational line bundle. Then a semipositive metric with minimal singularities on*

$$L = p^*(K_Z + \Delta)$$

has zero Lelong numbers at every point of M .

Proof. We may suppose that M is connected. The assertion is immediate in dimension zero, so write $n = \dim M > 0$.

The proof is by contradiction from a positive Lelong number. We construct the normalized Monge–Ampère family (54) and transport its tail to obtain (86). Keeping the potentially concentrating residual measure, we compare on the same sublevel set to obtain (89). We then take $k \rightarrow \infty$ at fixed t ; the upper Lelong estimate (90) contradicts the lower bound (93) as $t \rightarrow 0$.

Resolution data and regularized measures

The log pullback gives

$$L = K_M + \sum_i g_i D_i, \quad g_i < 1, \quad g_i < 0 \implies D_i \text{ is } p\text{-exceptional}, \quad (45)$$

with simple normal crossings support. Choose a smooth weight ℓ on L , pulled back from downstairs, and write θ for its curvature. Fix Kähler forms ω on M and ω_Z on Z , and a smooth probability volume dV on M . On a singular space, smooth forms and weights are understood through smooth local potentials.

For $0 < t \leq 1$, put

$$\beta_t^0 = \theta + tp^*\omega_Z. \quad (46)$$

Analytic nefness gives a smooth function $q_t \leq 0$ such that $\beta_t^0 + \text{dd}^c q_t$ is semipositive and positive definite on a dense open set. Indeed one can spend only half the available $t\omega_Z$ in the nef approximation downstairs. Consequently

$$\int_M (\beta_t^0)^n > 0, \quad \int_{D_i} (\beta_t^0)^{n-1} = 0 \quad \text{if } g_i < 0. \quad (47)$$

The second equality follows from the dimension of the image of an exceptional divisor and the fact that β_t^0 is pulled back.

Nefness and weak compactness give a positive current in $[\theta]$. Let

$$\Phi = V_\theta := (\sup\{v \in \text{PSH}(M, \theta) : v \leq 0\})^*. \quad (48)$$

The weight $\ell + \Phi$ has minimal singularities. Suppose, for a contradiction, that its Lelong number is positive at a point. In local coordinates x centered at that point, choose $\lambda > 0$ such that

$$\Phi(x) \leq \lambda \log |x|^2 + O(1). \quad (49)$$

In a local frame for $\mathcal{O}(D_i)$, write s_i for its canonical section and $d_i = \log |s_i|^2$. Set $b = \sum_i g_i d_i$. The actual identification (45) defines a global volume μ_b : locally its density is $e^{\ell-b}$, with the coordinate volume associated with the canonical frame. Choose smooth weights d_i^0 on $\mathcal{O}(D_i)$ and set

$$d_{i,\varepsilon} = \log(|s_i|^2 + \varepsilon e^{d_i^0}), \quad \mu_\varepsilon = e^{\ell - \sum_i g_i d_{i,\varepsilon}}, \quad 0 < \varepsilon \leq 1. \quad (50)$$

These local expressions respect the line transitions. The measures μ_ε are smooth and positive, and their densities converge almost everywhere to that of μ_b . The simple normal crossings description and $g_i < 1$ give constants $p > 1$ and C such that

$$\left\| \frac{\mu_\varepsilon}{dV} \right\|_{L^p(dV)} \leq C, \quad \int_M \left(\frac{\mu_b}{dV} \right)^{-s/(1-s)} dV < \infty \quad (51)$$

for some fixed $0 < s < 1$. To obtain the second assertion, take s small enough for the finitely many negative coefficients in (45). Fix

$$C_0 > \frac{n}{s\lambda}. \quad (52)$$

Let $t \downarrow 0$ along a sequence, and for each t take $k = 1, 2, \dots$. We shall choose

$$0 < \delta = \delta_{t,k} \leq \min(t, 1/k), \quad \beta = \beta_t^0 + \delta\omega, \quad V = \int_M \beta^n.$$

The class of β is Kähler, although the chosen smooth representative β need not be positive. After choosing δ , choose a smooth β -psh function $\psi = \psi_{t,k}$ such that

$$\begin{aligned} q_t - k - 1 &\leq \psi \leq 1, \\ \|\psi - \max(\Phi, q_t - k)\|_{L^1(dV)} &\leq \min(t, 1/k). \end{aligned} \quad (53)$$

The maximum is bounded and β_t^0 -psh. Regularization with arbitrarily small curvature loss gives smooth approximants because the bounded potential has no positive Lelong numbers [22]. A smooth convex regularized maximum with $q_t - k$, and the upper bound from Hartogs' lemma, give the stated pointwise bounds. Finally choose $0 < \varepsilon = \varepsilon_{t,k} \leq \min(t, 1/k)$. Additional smallness requirements on δ and ε will be imposed below, in this order. None of these data depends on the parameter c .

For $0 \leq c \leq C_0$, solve

$$T = \beta + \text{dd}^c u > 0, \quad \rho = \frac{T^n}{V} = e^{(1+c)u - c\psi} \mu_\varepsilon = e^{u + cH} \mu_\varepsilon, \quad H = u - \psi. \quad (54)$$

Here and below ρ denotes a probability measure. The Aubin–Yau theorem applies in the Kähler class [2, 76]. More precisely, the positive coefficient $1 + c$ is the negative- λ case of [3, Theorem 7.14]; it gives uniqueness and an invertible linearized operator. Thus u depends smoothly on c .

A capacity estimate independent of the class volume

We first establish estimates uniform in $t, k, \delta, \varepsilon, c$ whenever the displayed conditions hold. All the representatives β have a common upper bound by a fixed multiple of ω . We use the following standard uniform integrability consequence of Skoda's theorem and semicontinuity of complex singularity exponents: for fixed B , there are $a_B, C_B > 0$ such that

$$\sup_M v = 0, \quad \text{dd}^c v \geq -B\omega \implies \int_M e^{-a_B v} dV \leq C_B. \quad (55)$$

One obtains uniformity by compactness of normalized quasi-psh functions, local Skoda integrability at each limit, and Demailly–Kollár semicontinuity [70, 26].

Put $E = V_\beta$, with the same envelope convention as in (48). For each of our Kähler classes, E is bounded and $\sup E = 0$. Define the normalized capacity by

$$\text{cap}_\beta(A) = \frac{1}{V} \sup_{\substack{v \in \text{PSH}(M, \beta) \\ E-1 \leq v \leq E}} \int_A (\beta + \text{dd}^c v)^n. \quad (56)$$

We suppress the subscript when there is no ambiguity. In particular, $0 \leq \text{cap}(A) \leq 1$.

There are constants $a, C > 0$, independent of the class volume V , such that

$$\mu_\varepsilon(A) \leq C \exp(-a \text{cap}(A)^{-1/n}) \quad (57)$$

for every Borel set A , with the usual value zero on the right when the capacity is zero. Here is the normalization argument. For a compact nonpluripolar set A , let F_A be the upper regularization of the global β -psh extremal with obstacle zero on A , and put $m = \sup_M F_A$. The compact Kähler extremal theory gives a bounded function, at most zero on A outside a pluripolar set, maximal outside A , with Monge–Ampère mass V carried by A [40, Sections 5–7].

These statements also hold for our possibly nonpositive smooth representative. Indeed the defining family has a bounded competitor. If its suprema were unbounded, sup-normalization, quasi-psh compactness and a rapidly convergent weighted sum would produce a quasi-psh pole on A , contradicting nonpluripolarity. The obstacle constraint survives upper regularization outside a pluripolar set. A quasi-everywhere constraint gives the same envelope: mix a competitor with arbitrarily small weight on a nonpositive β -psh function having a pole on the exceptional pluripolar set, whose existence is the global pluripolarity theorem in a Kähler class. Finally local balayage off A proves maximality there. This explains why no normalization $E = 0$ is being assumed.

By its defining property and sup-normalization,

$$E \leq F_A \leq E + m.$$

If $b_A = \max(m, 1)$, then

$$E + \frac{F_A - E}{b_A} - 1$$

is admissible in (56). Positivity of mixed products and the mass support of F_A give

$$\text{cap}(A) \geq \frac{1}{b_A^n V} \int_A (\beta + \text{dd}^c F_A)^n = b_A^{-n}. \quad (58)$$

In particular $b_A \geq \text{cap}(A)^{-1/n}$. The volume has canceled exactly. The function $F_A - m$ satisfies (55), and is at most $-m$ on A almost everywhere. Hence $dV(A) \leq C e^{-am}$. If the capacity is less than one, (58) implies $m \geq \text{cap}(A)^{-1/n}$; capacity one is absorbed in the constant. Hölder and (51) prove (57). Pluripolar compact sets have zero volume, and inner approximation proves the Borel-set assertion.

Initial comparison and a uniformly small shell

First,

$$\sup_M u \leq C. \quad (59)$$

Otherwise normalize u by its supremum and take an almost-everywhere convergent subsequence using quasi-psh compactness. The limits are finite almost everywhere. The measures μ_ε also subconverge almost everywhere to a density positive almost everywhere, including when $\varepsilon \rightarrow 0$. Since $\psi \leq 1$, Fatou's lemma applied to (54) would make its total mass tend to infinity. This contradicts $\int_M \rho = 1$.

Fix

$$\frac{C_0}{1 + C_0} < r < 1, \quad v_* = r\psi + (1 - r)E.$$

For a capacity test v and $0 < \tau \leq 1 - r$, put

$$v_\tau = r\psi + (1 - r - \tau)E + \tau v, \quad A_l = \{u < v_* - l\}.$$

Since $v_* - \tau \leq v_\tau \leq v_*$, the set $B = \{u < v_\tau - l\}$ satisfies $A_{l+\tau} \subset B \subset A_l$. On B , for $l \geq 0$,

$$\begin{aligned} (1 + c)u - c\psi &\leq ((1 + c)r - c)\psi + (1 + c)(1 - r)E - (1 + c)l \\ &\leq C - l. \end{aligned}$$

We used $(1 + c)r - c > 0$, $\psi \leq 1$, and $E \leq 0$. The Bedford–Taylor comparison principle on this same set B and mixed-product positivity therefore give

$$\tau^n \text{cap}(A_{l+\tau}) \leq Ce^{-l} \mu_\varepsilon(A_l). \quad (60)$$

For completeness, write $g(l) = \text{cap}(A_l)^{1/n}$. A fixed $\tau > 0$ in (60) first makes $g(l)$ uniformly small for large l . Then (57) gives, after enlarging a uniform constant,

$$\tau g(l + \tau) \leq C_1 g(l)^2.$$

Choose an initial l so that $2C_1 g(l) \leq 1 - r$, and successively take $\tau = 2C_1 g(l)$. Each step halves g , and the sum of the increments is bounded by a geometric series. Monotonicity gives zero capacity above the limiting level. The resulting almost everywhere inequality extends to the quasi-psh representatives. Thus

$$u \geq r\psi + (1 - r)E - C. \quad (61)$$

Since $E \geq \psi - 1$, and ψ has a uniform lower L^1 bound by (53), we obtain

$$H \geq -C, \quad -C \leq \sup_M u \leq C. \quad (62)$$

In particular, u and H have two-sided pointwise bounds depending on t, k , but independent of the extra smallness of δ, ε , and independent of the permissible choice of ψ .

In the rest of the proof, $o(1)$ means a quantity tending to zero as $t \rightarrow 0$, uniformly in k and c , after the choices specified below. We can choose one fixed large R for which

$$dV\{H > R - 1\} = o(1). \quad (63)$$

To see the uniformity, consider any sequence with $t \rightarrow 0$, allowing both k and c to vary. A subsequence of u converges in L^1 and almost everywhere to a θ -psh function bounded above by $\Phi + C$. Moreover,

$$\|(\Phi - \psi)_+\|_{L^1(dV)} \leq t.$$

The asserted tail estimate follows by taking $R > C + 2$. Absolute continuity for μ_b , the uniform L^p bound for μ_ε , and (54) now imply that the shell

$$\mathcal{S} = \{R \leq H \leq R + 1\} \quad (64)$$

has $o(1)$ mass for $\mu_b, \mu_\varepsilon, \rho$. At $c = 0$, the whole tail $\{H > R\}$ has ρ -mass $o(1)$. At a general c , only the bounded-shell assertion has so far been proved.

Differentiating the Monge–Ampère equation

Set

$$h = -\partial_c u, \quad \Delta_T = \text{tr}_T \text{dd}^c.$$

Differentiating both sides of (54) gives

$$\partial_c \rho = (H - (1+c)h)\rho = -\Delta_T h \rho, \quad \Delta_T h = (1+c)h - H. \quad (65)$$

Also

$$\Delta_T H = n - \text{tr}_T(\beta + \text{dd}^c \psi) \leq n.$$

At a minimum of h , the last equation in (65) and $H \geq -C$ give $h \geq -C$. If $F = (1+c)h - H$, then $\Delta_T F \geq (1+c)F - n$; the maximum principle gives

$$h \geq -C, \quad (1+c)h \leq H + \frac{n}{1+c}. \quad (66)$$

Consequently

$$|\partial_c \rho| \leq C(1 + H_+)\rho. \quad (67)$$

For every fixed $a > 0$, integration by parts on the compact manifold gives

$$a \int_M e^{-ah} |\partial h|_T^2 \rho = \int_M e^{-ah} \Delta_T h \rho \leq n \int_M e^{-ah} \rho \leq C_a. \quad (68)$$

Choose a fixed C_* so large that $h + C_* > 0$, and put

$$G = (h + C_*) \frac{nT^{n-1}}{V}. \quad \text{dd}^c G = -\partial_c \rho. \quad (69)$$

The last identity is spatial: at a fixed value of c , T is closed, so no parameter derivative of T^{n-1} enters it.

The order of the smallness choices

Set

$$M_i = \max(0, \lceil -g_i \rceil), \quad N = \sum_i M_i D_i, \quad d_{N,\varepsilon} = \sum_i M_i d_{i,\varepsilon},$$

and let s_N be the canonical section of N . We impose

$$\begin{aligned} \sup_M (h + C_*) \int_M |\text{tr}_T(\text{dd}^c d_{N,\varepsilon})| \rho &= o(1), \\ \int_M (1 - |s_N|_{d_{N,\varepsilon}}^2) |\partial_c \rho| &= o(1). \end{aligned} \quad (70)$$

We verify that these choices respect the order already prescribed.

For fixed t, k , Equations (53), (59), and (66) give a bound $B_{t,k}$ for $\sup(h + C_*)$ before choosing δ , ψ , or ε . The signed trace integral is

$$\int_M \text{tr}_T(\text{dd}^c d_{N,\varepsilon}) \rho = \frac{n(2\pi)c_1(N)[\beta]^{n-1}}{V}. \quad (71)$$

At $\delta = 0$, its numerator is zero by (47), and its denominator is positive for this fixed t . Choose δ , still at most $\min(t, 1/k)$, so that $B_{t,k}$ times the modulus of (71) is at most t . Now choose ψ as in (53).

It remains to control an absolute trace, not just its signed integral. The regularization satisfies

$$\mathrm{dd}^c d_{i,\varepsilon} \geq -C \frac{\varepsilon}{|s_i|^2 e^{-d_i^0} + \varepsilon} \omega. \quad (72)$$

For fixed t, k, δ , the measures $T^{n-1} \wedge \omega$ are controlled uniformly in ε, c by a multiple of ordinary ω -capacity. Here is a direct way to make the dependence explicit. We have $\beta \leq A\omega$ and $|u| \leq B$, for constants fixed at this stage. Take $L_0 \geq \max(2A, 2B, 1)$, enlarging it if necessary, and set $v = (u - B)/L_0$. Then $-1 \leq v \leq 0$, $\omega + \mathrm{dd}^c v \geq \omega/2$, and $T \leq L_0(\omega + \mathrm{dd}^c v)$. Thus

$$T^{n-1} \wedge \omega \leq 2L_0^{n-1}(\omega + \mathrm{dd}^c v)^n.$$

The capacity of tubes shrinking to a divisor tends to zero. The factor on the right of (72) is uniformly bounded and tends uniformly to zero outside any fixed such tube. Consequently the negative part of the trace integral tends to zero uniformly in c as $\varepsilon \rightarrow 0$, at this fixed t, k, δ . The elementary inequality

$$\int |f| \rho \leq \left| \int f \rho \right| + 2 \int f_- \rho$$

then proves the first assertion of (70), with an error $O(t)$, after decreasing ε .

For the second assertion, observe that $0 \leq |s_N|_{d_{N,\varepsilon}}^2 \leq 1$, with convergence to one almost everywhere. At this stage $|\partial_c \rho| \leq C_{t,k} \mu_\varepsilon$. Uniform integrability from (51) gives the assertion, again with an error at most t after decreasing ε . We have therefore chosen δ , then ψ , then ε , for every t, k ; all errors are $O(t)$, and all choices are independent of c .

A Bochner estimate with a residual term

Choose a smooth nondecreasing cutoff χ with

$$\chi(H) = 0 \text{ for } H \leq R, \quad \chi(H) = 1 \text{ for } H \geq R + 1.$$

Choose a smooth convex function f on $\mathbb{R}$ and a constant d such that

$$C_0 + 2 \leq f' \leq d - 1, \quad f'' > 0 \text{ on } [-R - 1, -R].$$

Define

$$W_0 = e^{-f(-H) - cH}. \quad (73)$$

It is bounded above on $\mathcal{S}$. Since $H \geq -C$, the lower bound for f' and (66) give, with fixed positive constants,

$$W_0 \geq C^{-1} e^{2H}, \quad W_0 \geq C^{-1} (1 + H_+^2), \quad W_0 \geq C^{-1} (h + C_*), \quad W_0 \geq C^{-1} e^h. \quad (74)$$

On $M^\circ = M \setminus \bigcup_i D_i$, use the following weight on $-K_M + N$:

$$P = -\ell - u + b + \sum_i M_i d_i + f(-H). \quad (75)$$

The divisor weights are pluriharmonic on this open set. Therefore

$$\begin{aligned} \mathrm{dd}^c P &= -T + (\beta - \theta) - f'(-H) \mathrm{dd}^c H + f''(-H) \sqrt{-1} \partial H \wedge \bar{\partial} H \\ &\geq -dT + f''(-H) \sqrt{-1} \partial H \wedge \bar{\partial} H. \end{aligned} \quad (76)$$

We used $\beta - \theta = tp^*\omega_Z + \delta\omega \geq 0$ and $\text{dd}^c H = T - (\beta + \text{dd}^c \psi) \leq T$.

We claim that there is an N -valued section $w = s_N \chi(H) - U$ on M° such that

$$\int_{M^\circ} \left(|U|_{d_{N,\varepsilon}}^2 + |\bar{\partial} w|_{d_{N,\varepsilon},T}^2 \right) W_0 \rho = o(1). \quad (77)$$

The residual derivative in this assertion is essential: the curvature lower bound in (76) is not semipositive.

Identify N -valued sections with top-degree forms valued in $-K_M + N$. Choose a complete Kähler form Ω on M° , using logarithmic cusp terms along the simple normal crossings divisor, and use the complete metrics $T_\kappa = T + \kappa\Omega$, $\kappa > 0$. Let

$$g = \bar{\partial}(s_N \chi(H)) = s_N \chi'(H) \bar{\partial} H.$$

This is a closed square-integrable $(n, 1)$ -form for fixed parameters. For a $\bar{\partial}$ -closed $(n, 1)$ -form ζ in the domain of $\bar{\partial}^*$, complete-metric Bochner–Kodaira and (76) yield

$$|\langle g, \zeta \rangle|^2 \leq C \left(\int_{\mathcal{S}} e^{u-f(-H)} \mu_b \right) (\|\bar{\partial}^* \zeta\|^2 + d\|\zeta\|^2). \quad (78)$$

For clarity, on $(n, 1)$ -forms the curvature contribution of $-dT$ is bounded below by $-d\|\zeta\|^2$, because $T \leq T_\kappa$. The positive rank-one curvature term controls contraction with $\bar{\partial} H$. Cauchy–Schwarz in that direction gives (78), since $(\chi')^2/f''(-H)$ is bounded on the shell. The squared top-form density of s_N in the weight P is exactly $e^{u-f(-H)} \mu_b$. This density, and hence the estimate’s constant, is independent of the complete base metric. Approximation on a complete Kähler manifold justifies the indicated weak tests; see [23, Sections 4–5].

The integral in (78) is $o(1)$ by the shell estimate. Apply the Riesz representation theorem to the functional defined on pairs

$$(\bar{\partial}^* \zeta, \sqrt{d} \zeta) \mapsto \langle g, \zeta \rangle.$$

After extending this bounded functional, and projecting its second component to the closed subspace $\ker \bar{\partial}$, we obtain

$$\bar{\partial} U + \sqrt{d} V_1 = g, \quad \bar{\partial} V_1 = 0, \quad \|U\|^2 + \|V_1\|^2 = o(1). \quad (79)$$

Although the initial tests were closed, this is an equation against all appropriate tests. Indeed, orthogonal projection of a test $\zeta \in \text{Dom } \bar{\partial}^*$ to $\ker \bar{\partial}$ preserves its adjoint domain and adjoint value: the range of $\bar{\partial}$ from $(n, 0)$ -forms is contained in that kernel. It also preserves its pairings with the closed forms g, V_1 .

Let $\kappa \downarrow 0$. The norms of top-degree $(n, 0)$ -forms are unchanged by the base metric, while the $(n, 1)$ -norms increase to the norm for T . For a countable decreasing sequence of κ ’s, weak compactness in each preceding norm and a diagonal subsequence give limits satisfying (79) for T . Monotone convergence of the norms preserves its bound.

Finally, the ratio of the P -density for sections to the density $|\cdot|_{d_{N,\varepsilon}}^2 W_0 \rho$ is

$$\prod_i \left(\frac{|s_i|^2 + \varepsilon e^{d_i^0}}{|s_i|^2} \right)^{g_i + M_i} \geq 1. \quad (80)$$

The same comparison applies to derivative norms measured with T . Since $\bar{\partial} w = \sqrt{d} V_1$, this proves (77).

For fixed parameters, its smooth positive weights imply ordinary L^2 bounds on w and its displayed derivative up to the divisor. To extend the derivative distributionally, use simple normal

crossings cutoffs whose derivatives are $O(\tau^{-1})$ on tubes of volume $O(\tau^2)$. Their derivative L^2 norms are bounded. The boundary error is bounded by this fixed bound times the L^2 norm of w on the shrinking tubes, and tends to zero by absolute continuity. Thus $\bar{\partial}w$ extends as its distributional derivative across the divisor. Ellipticity on sections gives

$$w \in W^{1,2}(M, N). \quad (81)$$

The nonholomorphic logarithm and one-sided transport

Write $d_N = d_{N,\varepsilon}$ for this subsection and put $z = 1 + |w|_{d_N}^2$. For a smooth section, set

$$A_w = \frac{\langle \bar{\partial}w, w \rangle_{d_N}}{z},$$

where the Hermitian pairing is linear in the first variable. In a holomorphic frame normal for the line metric at the point, direct differentiation gives

$$\begin{aligned} \mathrm{dd}^c \log z &= \sqrt{-1} \partial A_w - \sqrt{-1} \bar{\partial} A_w - \frac{|w|_{d_N}^2}{z} \mathrm{dd}^c d_N \\ &\quad + \frac{\sqrt{-1}}{z^2} (\partial w \wedge \bar{\partial} \bar{w} - \partial \bar{w} \wedge \bar{\partial} w). \end{aligned} \quad (82)$$

The last line is evaluated in that normal frame, or equivalently using the Chern derivatives. Its first summand is nonnegative; its negative summand is controlled by $|\bar{\partial}w|^2$. Thus no estimate for the positive ∂w term is required.

Integrate (82) against the positive form G in (69). By (74) and (77), the negative-gradient contribution is at most

$$C \int_M (h + C_*) |\bar{\partial}w|_{d_N, T}^2 \rho = o(1).$$

The curvature cost is bounded by the first error in (70). For the two exact terms, integration by parts differentiates only h , since T is closed. Using $|w|/(1 + |w|^2) \leq 1/2$, their absolute value is at most

$$C \left(\int_M |\bar{\partial}w|_{d_N, T}^2 W_0 \rho \int_M |\partial h|_T^2 W_0^{-1} \rho \right)^{1/2} = o(1). \quad (83)$$

The second integral is bounded by (68) and $W_0^{-1} \leq C e^{-h}$. Approximation by smooth sections in $W^{1,2}$ justifies these calculations for (81): the logarithm has bounded first and second derivatives as a function of the section, and the derivative products converge in L^1 for fixed smooth data. We conclude

$$\int_M \mathrm{dd}^c \log z \wedge G \geq -o(1), \quad \int_M \log z \partial_c \rho \leq o(1). \quad (84)$$

The second assertion follows from (69) by integration by parts.

We can replace $\log z$ in the last integral by $\log(1 + \chi(H)^2)$. First the globally Lipschitz radial function $\xi \mapsto \log(1 + |\xi|^2)$, followed by Cauchy–Schwarz, gives

$$\begin{aligned} &\int_M \left| \log(1 + |w|_{d_N}^2) - \log(1 + |s_N \chi(H)|_{d_N}^2) \right| |\partial_c \rho| \\ &\leq C \int_M |U|_{d_N} (1 + H_+) \rho = o(1). \end{aligned}$$

Here $W_0 \geq C^{-1}(1 + H_+^2)$ pays for the second Cauchy–Schwarz factor. Next,

$$\left| \log(1 + |s_N|_{d_N}^2 \chi^2) - \log(1 + \chi^2) \right| \leq 1 - |s_N|_{d_N}^2,$$

so the second error in (70) pays for this replacement.

Set

$$\eta(H) = \frac{\log(1 + \chi(H)^2)}{\log 2}.$$

The preceding estimate, $\partial_c H = -h \leq C$, and the shell support of the bounded nonnegative function η' give

$$\partial_c \int_M \eta(H) \rho = \int_M \eta(H) \partial_c \rho + \int_M \eta'(H)(-h) \rho \leq o(1). \quad (85)$$

The section w need not be chosen smoothly in c : its estimate was used separately at each c , whereas the left side here depends only on the smooth family (54). At $c = 0$ the integral is $o(1)$ by the initial tail estimate. Integrating (85) up to C_0 proves, uniformly in k ,

$$\int_{\{H > R+1\}} \rho = o(1) \quad (c = C_0). \quad (86)$$

Canceling the residual measure on the comparison set

Work now at $c = C_0$. Take a smooth cutoff $\xi(H)$ which is zero for $H \leq R + 1$, one for $H \geq R + 2$, and lies between zero and one. The smooth positive measure $\xi(H)\rho + t dV$ has mass $a_{t,k} > 0$, with $a_{t,k} \leq m_t \rightarrow 0$ uniformly in k , by (86). Solve the prescribed-volume equation [76]

$$\frac{(\beta + dd^c v)^n}{V} = \frac{\xi(H)\rho + t dV}{a_{t,k}}, \quad \sup_M v = 0.$$

The complementary part of ρ has $H \leq R + 2$. Therefore

$$\rho \leq C_R \mu_\varepsilon + m_t \frac{(\beta + dd^c v)^n}{V}. \quad (87)$$

Choose $r_t \rightarrow 0$ with $r_t^n \geq m_t$. We may assume $r_t \leq 1/2$ for all the remaining t 's.

We prove a uniform estimate

$$u \geq r_t v + (1 - r_t)E - C. \quad (88)$$

Put $v_* = r_t v + (1 - r_t)E$, and for an arbitrary capacity test φ and $0 < \tau \leq 1 - r_t$, put

$$v_\tau = r_t v + (1 - r_t - \tau)E + \tau \varphi, \quad A_l = \{u < v_* - l\}, \quad B = \{u < v_\tau - l\}.$$

Again $A_{l+\tau} \subset B \subset A_l$. On precisely this same set B , comparison and positivity give

$$\begin{aligned} r_t^n \int_B \frac{(\beta + dd^c v)^n}{V} + \tau^n \int_B \frac{(\beta + dd^c \varphi)^n}{V} \\ \leq \int_B \frac{(\beta + dd^c v_\tau)^n}{V} \leq \int_B \rho \\ \leq C_R \mu_\varepsilon(B) + m_t \int_B \frac{(\beta + dd^c v)^n}{V}. \end{aligned}$$

Since $r_t^n \geq m_t$, the residual measure cancels on B . Taking the supremum over φ yields

$$\tau^n \text{cap}(A_{l+\tau}) \leq C_R \mu_\varepsilon(A_l). \quad (89)$$

There is no comparison of residual masses on different sets.

To start the iteration uniformly, note that $v_* \leq 0$, so $A_l \subset \{u < -l\}$. Equations (55) and (62), followed by Hölder with (51), make $\mu_\varepsilon(A_l)$ uniformly small as $l \rightarrow \infty$. A fixed $\tau = 1/4$ in (89) therefore makes its capacity uniformly small. Combining with (57), the same quadratic iteration used after (60) applies; the allowed upper bound $1 - r_t$ is at least $1/2$. It proves (88) with a constant independent of t, k .

The fixed- t limit and Lelong contradiction

Fix one sufficiently small $t > 0$, and let $k \rightarrow \infty$. Quasi-psh compactness, (62), and $\sup v = 0$ give a subsequence converging in L^1 and almost everywhere to β_t^0 -psh functions u_t, v_t . Since $E \geq q_t$, Equation (88) gives

$$u_t \geq r_t v_t - C_t, \quad \nu(u_t, x) \leq r_t \nu(v_t, x) \leq C r_t. \quad (90)$$

The last constant is independent of t : the normalized potentials v_t have a common lower curvature bound, so their Lelong numbers are uniformly bounded, for example by (55). The additive constant C_t is allowed to depend on t and does not affect Lelong numbers.

At fixed t , Equation (53) gives $\psi_{t,k} \rightarrow \Phi$ in L^1 . Choose the subsequence also to converge almost everywhere. Since $\varepsilon_{t,k} \rightarrow 0$, Fatou's lemma in (54) at $c = C_0$ gives

$$\int_M e^{(1+C_0)u_t - C_0\Phi} \mu_b \leq 1. \quad (91)$$

Writing $f_b = \mu_b/dV$, Hölder and the second moment in (51) yield

$$\int_M e^{s(1+C_0)u_t - sC_0\Phi} dV \leq \left(\int_M e^{(1+C_0)u_t - C_0\Phi} f_b dV \right)^s \left(\int_M f_b^{-s/(1-s)} dV \right)^{1-s} < \infty.$$

By (49), this implies near the chosen point

$$\int e^{au_t(x)} |x|^{-2b_0} dV(x) < \infty, \quad a = s(1+C_0), \quad b_0 = sC_0\lambda > n. \quad (92)$$

Add a local smooth potential for β_t^0 to make u_t plurisubharmonic. Its exponential is subharmonic, and the smooth addition changes the following estimates only by bounded factors. For $x \neq 0$ sufficiently close to zero, the submean inequality on $B(x, |x|/2)$ gives

$$\begin{aligned} e^{au_t(x)} &\leq C_t |x|^{-2n} \int_{B(x, |x|/2)} e^{au_t(y)} dV(y) \\ &\leq C'_t |x|^{2b_0 - 2n}, \end{aligned}$$

where the second inequality uses (92) and $|y| \asymp |x|$ on that ball. Hence

$$u_t(x) \leq \frac{sC_0\lambda - n}{s(1+C_0)} \log |x|^2 + O_t(1). \quad (93)$$

The coefficient is positive and independent of t , contradicting (90) as $t \rightarrow 0$. Thus Φ has zero Lelong numbers everywhere. Any two metrics with minimal singularities differ by bounded weights, so the conclusion holds for every such metric on L . $\square$

6 The empty divisorial locus

The obstruction considered in this section is the absence of a meromorphic pluricanonical section, including one with poles. We keep this distinction throughout: a rational line bundle on a nonalgebraic compact complex space need not have a global meromorphic frame.

Proposition 6.1. *Assume Assumption 2.2. Let Y be a normal compact Kähler fourfold with klt singularities, and suppose that the actual rational canonical line bundle K_Y is analytically nef. Suppose that Y contains no prime divisors. Let $p: M \rightarrow Y$ be a smooth compact Kähler log resolution. If $a(M) = 0$ and K_M is analytically pseudo-effective, then some positive tensor power of K_M has a nonzero meromorphic section.*

We prove the proposition by contradiction. Until the end of the section, assume its hypotheses and, in addition, assume that

$$H^0(M, \mathcal{M}_M \otimes \mathcal{O}_M(mK_M)) = 0 \quad \text{for every integer } m > 0, \quad (94)$$

where $\mathcal{M}_M$ is the sheaf of meromorphic functions. Put

$$L = p^*K_Y, \quad K_M = L + C, \quad \alpha = 2\pi c_1(L). \quad (95)$$

Here C is the rational exceptional discrepancy divisor, and the middle identity is an identity of actual rational line bundles, using the canonical comparison on the isomorphism locus of p . No sign is imposed on C . Every prime divisor on M is p -exceptional, because Y has none. We use the weight convention $\text{dd}^c = \sqrt{-1}\partial\bar{\partial}$, so a metric weight on L has curvature class α .

6.1 Virtual vanishings and holomorphic forms

The starting Euler-characteristic/Hodge-theoretic route to holomorphic forms was informed by the related forms strategy of Vikash [75]. That work considers smooth minimal fourfolds without effective divisors or surfaces. The present normal klt setting has different hypotheses, and the required vanishings and forms construction are proved below.

Lemma 6.2. *Let $f: N \rightarrow M$ be a proper surjective generically finite morphism, where N is a connected smooth compact Kähler manifold. Then*

$$a(N) = q(N) = 0, \quad K_N \text{ is pseudo-effective.} \quad (96)$$

and no positive tensor power of K_N has a nonzero meromorphic section. These conclusions also hold after any further smooth compact Kähler modification or finite cover followed by resolution.

Proof. First consider meromorphic functions and tensors. A proper generically finite map of normal complex spaces factors, by Stein factorization, as a proper modification followed by a finite map. Meromorphic functions on a modification descend to the normal target, and finite maps have local meromorphic traces and norms. These constructions use local meromorphic function fields and therefore do not require a global meromorphic frame.

If u is a meromorphic function on N , its characteristic polynomial over M , computed on the finite part of the factorization, has meromorphic coefficients on M . These coefficients are constant because $a(M) = 0$. Thus u satisfies a polynomial with constant coefficients. Connectedness then makes u constant, and $a(N) = 0$.

Suppose that s is a nonzero meromorphic section of mK_N . Over a coordinate neighborhood $U \subset M$, choose a nowhere vanishing local frame τ of mK_M . The differential pullback $f^*\tau$ is a holomorphic pluricanonical tensor, generically nonzero, so $s/(f^*\tau)$ is a meromorphic function over $f^{-1}U$; zeros of the Jacobian merely contribute a meromorphic divisor to this quotient. If d is the generic degree of f , its norm is the coefficient of a meromorphic section of mdK_M : replacing τ by $g\tau$ divides the norm by g^d . The local sections consequently glue and are nonzero. This contradicts (94). The ordinary Jacobian formula

$$K_N = f^*K_M + R_f, \quad R_f \geq 0,$$

proves pseudo-effectivity of K_N .

It remains to prove $q(N) = 0$. If $q(N) > 0$, the Albanese map has a positive-dimensional image. Let $h: N \rightarrow B$ be its Stein factorization onto its normal image. The space B is finite over

a subvariety of a torus. On a smooth compact Kähler model B' of B , the pullbacks of suitable ambient one-forms have a nonzero wedge of degree $\dim B$. Hence $\kappa(B') \geq 0$. On every neat smooth model used to compute the invariant base in Assumption 2.2, the orbifold boundary is effective; birational invariance of ordinary Kodaira dimension therefore gives

$$\kappa(h \mid 0) \geq \kappa(B') \geq 0.$$

This comparison concerns the invariant base term, not a boundary on an arbitrarily chosen initial model.

Choose a positive singular metric on K_N . Its local potentials restrict to almost every smooth fiber of h , by local integrability and Fubini's theorem. We may choose such a fiber outside the countably many proper analytic subsets excluded by the very-general-fiber convention in Assumption 2.2. If F is that fiber, then K_F is pseudo-effective and $\dim F \leq 3$. Ordinary canonical nonvanishing for smooth compact Kähler manifolds of dimension at most three gives $\kappa(F) \geq 0$. In dimension three this follows from the Kähler threefold minimal model theorem and abundance for nef normal compact Kähler threefold pairs; see [47, 21]. In dimensions at most two it is the classical curve and surface statement. Applying Assumption 2.2 to $(N, 0)$ and h gives

$$\kappa(N) \geq \kappa(F) + \kappa(h \mid 0) \geq 0,$$

contrary to the absence of a meromorphic pluricanonical section on N . The same proof applies to any further morphism of the stated kind. $\square$

In particular, $q(M) = 0$. Moreover,

$$\alpha \neq 0. \tag{97}$$

Indeed, if $c_1(L) = 0$ in real cohomology, a multiple of L has torsion integral first Chern class. After a further multiple that class vanishes. The exponential sequence and $H^1(M, \mathcal{O}_M) = 0$ make that line bundle trivial. Equation (95) would then give a meromorphic section of a positive multiple of K_M .

Lemma 6.3. *For every holomorphic vector bundle $\mathcal{V}$ on M ,*

$$H^0(M, \mathcal{V} \otimes \mathcal{O}_M(mL)) = 0$$

for every sufficiently large positive integer m divisible by the index of L . Furthermore,

$$\chi(M, \mathcal{O}_M) = 0, \quad h^{2,0}(M) \geq 1, \quad h^{3,0}(M) \geq 2. \tag{98}$$

Proof. Choose an integer $r > 0$ such that $A = rL$ is a line bundle. A section of $\mathcal{V} \otimes A^k$ is a morphism $A^{-k} \rightarrow \mathcal{V}$. Among all nonzero sections with $k > 0$, choose a tuple $s_1, \dots, s_t$, of twists $k_1, \dots, k_t$, which is generically independent and has maximal possible cardinality. If there are no such sections the assertion is immediate. Otherwise $t \leq \operatorname{rk} \mathcal{V}$. Let $\mathcal{W} \subset \mathcal{V}$ be the saturation of the image of the corresponding direct sum of line bundles. Every further section lies in $\mathcal{W}$ generically, by maximality, and hence everywhere as a map into this saturated subsheaf.

A further nonzero section s of twist k replaces at least one s_i while preserving generic independence. The two determinants are nonzero sections, respectively, of

$$\det \mathcal{W} \otimes A^{\sum k_j} \quad \text{and} \quad \det \mathcal{W} \otimes A^{\sum k_j + k - k_i},$$

where determinants are reflexive. Their ratio is a nonzero meromorphic section of A^{k-k_i} . For $k > \max_i k_i$, this is a positive multiple of L . After clearing denominators in C , multiplication by its

canonical meromorphic divisor section converts it into a meromorphic section of a positive multiple of K_M , contradicting (94). This proves the eventual vanishing without choosing a meromorphic frame of A .

By Lemma 5.1, L has a semipositive metric with zero Lelong numbers everywhere. The multiplier ideal of every positive integral multiple of this metric is trivial by Skoda integrability [70]. The hard Lefschetz theorem with multiplier ideals [28, Theorem 0.1] therefore gives, for every $0 \leq j \leq 4$, a surjection

$$H^0(M, \Omega_M^{4-j} \otimes \mathcal{O}_M(mL)) \longrightarrow H^j(M, \mathcal{O}_M(K_M + mL))$$

for divisible $m > 0$. The source is zero for all sufficiently large such m . Riemann–Roch makes $\chi(M, \mathcal{O}_M(K_M + mL))$ a polynomial in m ; its vanishing on all these multiples implies that its constant term $\chi(M, \mathcal{O}_M(K_M))$ is zero. Serre duality in dimension four yields $\chi(M, \mathcal{O}_M) = 0$.

Equation (94) gives $h^{4,0} = 0$, and Lemma 6.2 gives $h^{1,0} = 0$. If $h^{2,0} = 0$, then all of $H^2(M, \mathbb{R})$ is of type $(1, 1)$. Approximating a Kähler class by a rational one and applying the Kodaira embedding theorem would make M projective, contrary to $a(M) = 0$. Thus $h^{2,0} \geq 1$. Finally, Hodge symmetry and the Euler characteristic identity give $h^{3,0} = 1 + h^{2,0} \geq 2$. $\square$

6.2 A saturated integrable conormal line

Write $K = K_M$. Choose $0 \neq \sigma \in H^0(M, \Omega_M^2)$. Holomorphic forms on M are closed, and $\sigma^2 = 0$ because it is a canonical section. Thus σ has rank two wherever it is nonzero. The kernel of σ is integrable there, and its saturated conormal is a rank-two subsheaf

$$F \subset \Omega_M^1, \quad G = \Omega_M^1 / F.$$

All determinants and line factors below are taken reflexively. The form σ is a nonzero section of $\det F$, so

$$\det F = \mathcal{O}_M(H), \quad \det G = \mathcal{O}_M(K - H) \tag{99}$$

for an effective divisor H .

We use the natural identification $\Omega_M^3 \simeq K \otimes T_M$, obtained by contraction with a local volume form. Global linearly independent three-forms are generically independent: coefficients of a dependence relative to a maximal generically independent tuple are global meromorphic functions, hence constants. The same observation applies to their projections into $F^* \otimes K$.

If v and w are the K -valued vector fields associated with two three-forms, then $\sigma(v, w)$ is a section of $2K$, so it is zero. Since the form induced by σ on the rank-two quotient of T_M is nondegenerate at a general point, the projections of all these vector fields to $F^* \otimes K$ have rank at most one. On the other hand, two generically independent tangent vector fields would have a nonzero wedge in

$$2K \otimes (\det G)^* = \mathcal{O}_M(K + H).$$

Removing the divisor factor would give a prohibited meromorphic canonical section. The space of global three-forms has dimension at least two, so we can choose associated vector fields v_1, v_2 such that v_1 is nonzero and tangent to the kernel of σ , while v_2 is not tangent. Denote their three-forms by β_1, β_2 .

Let $E \subset F$ be the saturation of the line defined by the nonzero K -valued one-form

$$\eta = \iota_{v_2} \sigma.$$

It follows that

$$E = \mathcal{O}_M(-K + e) \tag{100}$$

for a divisor e (its divisorial zero divisor). Saturation makes E a reflexive rank-one sheaf, hence a line bundle on M . Its inclusion in Ω_M^1 is a subbundle away from a set of codimension at least two. Likewise, all the rank-two sheaves and quotients just used are vector bundles at general points of every prime divisor.

The rank-two identity

$$K \otimes G^* \simeq \mathcal{O}_M(H) \otimes G$$

turns v_1 into a section of $\mathcal{O}_M(H) \otimes G$. Saturating its image produces a line factor $\mathcal{O}_M(j - H) \subset G$, where j is a divisor; by the determinant identity the other factor is $\mathcal{O}_M(K - j)$.

Lemma 6.4. *The line E is an integrable saturated conormal. On a dense open set, its line is the intersection of the degree-one wedge annihilators of σ and β_2 .*

Proof. Work first on the complement of a codimension-at-least-two set where F, G, E and the line factors above are bundles. Let $D \subset T_M$ be the integrable distribution annihilated by F . Lie derivative along D defines the partial Bott connection on F . Its second fundamental form for E is a morphism

$$B_E: E \longrightarrow (F/E) \otimes D^* = (F/E) \otimes G. \quad (101)$$

For clarity, this is tensorial in both variables: for $w \in D$ and a local section s of E , the terms introduced by replacing s by fs are multiples of s , while the additional term in $\mathcal{L}_{fw}s$ is $s(w)df = 0$.

If the composite of B_E with the second line factor of G is nonzero, it is a nonzero section of the line bundle of class

$$(H - E) + (K - j) - E = 3K + H - j - 2e.$$

If that composite vanishes and $B_E \neq 0$, it factors through the first line factor and gives a nonzero section of class

$$(H - E) + (j - H) - E = 2K + j - 2e.$$

These sections extend across the omitted set by reflexivity and Hartogs' theorem. Removing their divisor factors gives a nonzero meromorphic section of $3K$ or $2K$, respectively, contradicting (94). Thus $B_E = 0$.

In local Frobenius coordinates for D , write its transverse coordinates as (z_1, z_2) and a frame of E as $adz_1 + b dz_2$. Vanishing of B_E says that the ratio a/b , where defined, is constant along the plaques of D . After multiplication by an invertible local function, the frame is therefore a one-form on the two-dimensional transversal. Every rank-one conormal on a smooth surface is integrable. The resulting Frobenius identity extends to the entire regular locus of E by holomorphic equality, including general points of divisorial zeros of the original forms.

The wedge annihilator of σ in degree one is F . The annihilator of β_2 is the hyperplane of one-forms vanishing on v_2 . Since v_2 is not tangent to D , their intersection is a line. It contains $\iota_{v_2}\sigma$, which both wedges to zero with σ and evaluates to zero on v_2 . Hence that line is E . $\square$

Lemma 6.5. *For every closed positive $(1, 1)$ -current T in the class α , one has*

$$T \wedge \sigma \wedge \bar{\sigma} = 0, \quad T \wedge \beta_2 \wedge \bar{\beta}_2 = 0 \quad (102)$$

on the open subset where p is an isomorphism onto a smooth open subset of Y . In particular,

$$T \wedge \boldsymbol{\eta} = 0 \quad (103)$$

on a dense open subset with analytic complement.

Proof. Off the singular set of the saturated codimension-one foliation, whose codimension is at least two, holomorphic first integrals give frames dz for E . If $dz_a = g_{ab}dz_b$ on an overlap, then $d \log g_{ab}$ belongs to E : differentiating the transition identity gives $dg_{ab} \wedge dz_b = 0$. Consequently the first Chern cocycle wedges to zero with both σ and β_2 there.

These vanishings extend in Dolbeault cohomology to all of M . Indeed, for a locally free sheaf $\mathcal{V}$ on a smooth manifold and an analytic set A of codimension at least two, $\mathcal{H}_A^j(\mathcal{V}) = 0$ for $j = 0, 1$, by depth and Hartogs' theorem. The local-to-global sequence for cohomology with support therefore gives an injection $H^1(M, \mathcal{V}) \rightarrow H^1(M \setminus A, \mathcal{V})$. Apply it to Ω_M^3 and Ω_M^4 . We obtain

$$c_1(E) \wedge [\sigma] = 0, \quad c_1(E) \wedge [\beta_2] = 0. \quad (104)$$

We next dispose of the exceptional divisor terms. Fix a Kähler form ω_Y on Y . For every exceptional prime P on M ,

$$\int_P \sigma \wedge \bar{\sigma} \wedge p^* \omega_Y = 0, \quad \int_P \sqrt{-1} \beta_2 \wedge \bar{\beta}_2 = 0. \quad (105)$$

Here and below integration on a prime divisor can be computed on a smooth resolution. To see the assertion, rationality of klt singularities gives $Rp_* \mathcal{O}_M = \mathcal{O}_Y$, hence $H^k(Y, \mathcal{O}_Y) \simeq H^k(M, \mathcal{O}_M)$. Resolve P and the image $p(P)$, and resolve the graph so that the resulting smooth compact Kähler space $\hat{P}$ maps to a smooth model B of $p(P)$. Functoriality of restriction makes the classes of $\bar{\sigma}|_{\hat{P}}$ and $\bar{\beta}_2|_{\hat{P}}$ pullbacks from $H^2(B, \mathcal{O}_B)$ and $H^3(B, \mathcal{O}_B)$, respectively. Conjugation and compact Kähler Hodge theory give the corresponding pullback assertions for their de Rham classes. The class of $p^* \omega_Y$ restricts from B as well. Since $\dim B \leq 2$, the degree-six products in (105) vanish.

By (100) and (95), $c_1(E) = -c_1(L)$ modulo exceptional divisor classes. Combining (104) and (105) gives

$$\int_M T \wedge \sigma \wedge \bar{\sigma} \wedge p^* \omega_Y = 0, \quad \int_M T \wedge \sqrt{-1} \beta_2 \wedge \bar{\beta}_2 = 0.$$

The integrands are nonnegative: σ is decomposable wherever nonzero, and every three-form on a four-dimensional vector space is decomposable. The form $p^* \omega_Y$ is strictly positive on the isomorphism locus. Thus the zero-mass equalities imply (102) there.

This positivity argument applies to currents as well as smooth forms. Locally, write the coefficient matrix of T as a positive semidefinite matrix of Radon–Nikodym derivatives with respect to its trace measure. The two wedge equalities imply, almost everywhere for that measure, that each one-form direction of the matrix annihilates both σ and β_2 by wedge product. Lemma 6.4 identifies their intersection with E . This proves (103). $\square$

6.3 Index charts and a closed transverse current

We will repeatedly use the following local integrability fact. All integrals in its proof are computed in fixed smooth coordinate volumes.

Lemma 6.6. *Let $f: V \rightarrow U$ be a proper generically finite holomorphic map between smooth complex manifolds of the same dimension. Let ϕ be a plurisubharmonic function on U such that $e^{-s\phi} \in L_{\text{loc}}^1(U)$ for every $s > 0$. Then*

$$e^{-s(\phi \circ f)} \in L_{\text{loc}}^1(V) \quad (s > 0).$$

The assertion is locally uniform for a family of functions for which all the indicated exponential integrals downstairs are locally uniformly bounded.

Proof. Take relatively compact coordinate charts upstairs and downstairs so that f maps the former into the latter. Let J be the local holomorphic Jacobian determinant; it is not identically zero. A sufficiently small negative power of $|J|$ is locally integrable. Choose $P > 1$ large enough that $|J|^{-2/(P-1)}$ is integrable on the chosen compact set. Hölder's inequality gives

$$\int e^{-s\phi \circ f} \leq \left(\int e^{-sP\phi \circ f} |J|^2 \right)^{1/P} \left(\int |J|^{-2/(P-1)} \right)^{(P-1)/P}. \quad (106)$$

Change of variables bounds the first factor by the degree of f times the corresponding exponential integral downstairs, up to fixed coordinate-volume constants. Covering compact sets by finitely many charts proves both assertions. $\square$

A positive closed $(1, 1)$ -current with zero Lelong numbers has no mass on any proper analytic subset. We recall precisely the form used here. The Skoda–El Mir extension theorem says that restricting such a current off an analytic set and then extending by zero gives a closed current. The difference is a positive closed current supported on that set. The support theorem reduces it to its divisorial components, whose coefficients are the corresponding generic Lelong numbers, so the difference is zero. These current-theoretic facts, together with Skoda integrability and Siu decomposition below, are used in their ordinary smooth-manifold forms; see [25].

Lemma 6.7. *Let T be a positive current in α with zero Lelong numbers everywhere. Choose local plurisubharmonic metric weights ϕ for T on L . On the isomorphism locus of p , write $\eta = \eta \otimes \tau$ in the corresponding local canonical frame τ . The expression*

$$\Theta_T = \sqrt{-1} e^{-\phi} \eta \wedge \bar{\eta} \quad (107)$$

has a canonical extension to a nonzero positive closed $(1, 1)$ -current on M . This extension has no mass on proper analytic subsets and satisfies

$$\alpha \smile [\Theta_T] = 0. \quad (108)$$

The weights are understood with a fixed global additive normalization when Θ_T is compared for different currents.

Proof. We first specify the extension through the singular model. Cover Y by neighborhoods U on which a Cartier pluricanonical power has a nowhere vanishing frame. Take the ordinary local index cover $\widehat{U} \rightarrow U$ defined by a root of that frame, and normalize. The cover is quasi-étale and klt, by the index-cover discrepancy formula; in particular it has rational singularities. On its smooth locus the root is a canonical frame.

Resolve a main component of $\widehat{U} \times_U p^{-1}(U)$, obtaining a diagram

$$\begin{array}{ccc} \widetilde{U} & \longrightarrow & \widehat{U} \\ f \downarrow & & \downarrow \\ p^{-1}(U) & \longrightarrow & U. \end{array}$$

Here f is proper and generically finite. The resolution is chosen isomorphic over the dense smooth open where the index cover is unramified and p is an isomorphism. The coefficient η in the root frame is an ordinary holomorphic one-form on that open. It extends reflexively to $\widehat{U}$ by Hartogs' theorem, since the complement of the relevant smooth locus downstairs has codimension at least two. The extension theorem for rational complex spaces then extends it holomorphically to $\widetilde{U}$; we

use [50, Corollary 1.8]. We continue to write η for this holomorphic form and ϕ for the pulled metric weight in the root frame.

Zero Lelong numbers downstairs give $e^{-s\phi} \in L_{\text{loc}}^1$ for every $s > 0$. By Lemma 6.6, the same is true on $\tilde{U}$. In particular the pulled weight has zero Lelong numbers, and the pulled curvature has no analytic-subset mass. The upstairs expression in (107) has L_{loc}^r coefficients for every finite r . This holds also after multiplication by ϕ : on a compact set ϕ is bounded above, while $|\phi|e^{-\phi}$ is bounded by a constant times $1 + e^{-2\phi}$.

Define Θ_T on $p^{-1}(U)$ as f_* of this expression divided by $\deg f$. On the good open this is the original formula. It is independent of the root frame: if $\tau' = g\tau$, then $\eta' = g^{-1}\eta$ and $\phi' = \phi - \log |g|^2$. A pushforward of the indicated L^r coefficient forms has no mass on the analytic complement, because its inverse image is a proper analytic subset and hence has zero smooth volume upstairs. Consequently these local pushforwards agree on overlaps and define a global positive current. It is nonzero since η is generically nonzero.

We now prove closedness, rather than assuming that a transverse-looking density is closed. On $\tilde{U}$, the saturation of the line defined by η gives an integrable codimension-one conormal. Integrability holds first on the good open by Lemma 6.4 and then everywhere by holomorphic equality. Its singular set A has codimension at least two. On a regular foliated chart, including one through a divisorial zero of η , write

$$\eta = f_0 dz \quad (109)$$

with z a holomorphic submersion and f_0 holomorphic. Lemma 6.5 gives $\text{dd}^c \phi \wedge dz = 0$ on the good part where $f_0 \neq 0$. It holds on the whole chart because the positive closed curvature has no mass on the remaining analytic set.

In coordinates (z, w_1, w_2, w_3) , positivity shows that the only coefficient of $\text{dd}^c \phi$ is its $\sqrt{-1} dz \wedge d\bar{z}$ coefficient. Closedness of that current makes this coefficient independent, as a distribution, of every w_j and $\bar{w}_j$. Thus it is the pullback of a positive measure on the z -disc. Taking a one-variable subharmonic potential and then solving the pluriharmonic difference on a smaller polydisc gives

$$\phi = \phi_0(z) + g + \bar{g}, \quad g \text{ holomorphic.} \quad (110)$$

With $h = f_0 e^{-g}$, the upstairs current becomes

$$\sqrt{-1} e^{-\phi_0(z)} |h(z, w)|^2 dz \wedge d\bar{z}.$$

Its dd^c is positive. More explicitly, only differentiation in the w directions survives wedging with $dz \wedge d\bar{z}$, and

$$\text{dd}^c \Theta_T = e^{-\phi_0(z)} \sqrt{-1} \partial_w h \wedge \bar{\partial}_w \bar{h} \wedge \sqrt{-1} dz \wedge d\bar{z} \geq 0 \quad (111)$$

on the regular foliated chart. This is a distributional identity; the one-variable coefficient is locally integrable, and no transverse derivative of it contributes to the displayed wedge.

We give the extension estimate across A . On a relatively compact coordinate ball, choose smooth cutoffs χ_ϵ equal to zero on an ϵ -tube around A and equal to one outside a 3ϵ -tube, with

$$|\nabla \chi_\epsilon| \leq C\epsilon^{-1}, \quad |\nabla^2 \chi_\epsilon| \leq C\epsilon^{-2}.$$

They may be constructed by smoothing the distance cutoffs. The tubes have volume $O(\epsilon^4)$. To justify the bound also when A is singular, use locally finite area for its analytic components and monotonicity to bound the number of disjoint balls of radius $\epsilon/4$ centered on each component by $C\epsilon^{-2d}$, where $d \leq 2$ is its complex dimension. Enlarging this cover to the tube gives volume $O(\epsilon^{8-2d}) = O(\epsilon^4)$; a compact subset meets finitely many local components.

If B denotes either Θ_T or $\phi\Theta_T$ upstairs and $r > 2$, then for any fixed smooth test form the second-derivative cutoff terms are bounded by

$$C\epsilon^{-2}\|B\|_{L^r} \text{vol}(N_{3\epsilon}(A))^{1-1/r} \leq C_r\|B\|_{L^r}\epsilon^{2-4/r} \longrightarrow 0. \quad (112)$$

Terms with only one derivative are smaller. Testing (111) against χ_ϵ times a positive test form and passing to the limit therefore proves $\text{dd}^c\Theta_T \geq 0$ on all of $\tilde{U}$.

Proper pushforward commutes with dd^c , so $\text{dd}^c\Theta_T$ is a global positive exact $(2, 2)$ -current on M . Pairing with the square of a Kähler form gives zero; positivity then makes it the zero current. On the good open the chart maps are local biholomorphisms. All branch contributions to the pushforward are positive, so each upstairs Hessian in (111) vanishes there separately. Since $e^{-\phi_0} > 0$ almost everywhere, the holomorphic derivatives $\partial_w h$ vanish there, and hence on every regular foliated chart by holomorphic continuation. Thus

$$h = f_0 e^{-g} \text{ depends only on } z. \quad (113)$$

Equations (110) and (113) imply, on these charts,

$$\text{d}\Theta_T = 0, \quad \text{dd}^c(\phi\Theta_T) = 0.$$

For the second identity, the coefficient depending only on z causes no derivative after wedging with $\text{d}z \wedge \text{d}\bar{z}$, while the remaining $g + \bar{g}$ is pluriharmonic in the leaf directions. The first- and second-derivative versions of (112) extend both identities across A .

Finally choose a smooth reference metric on L , with local weights ψ and curvature θ . The difference $u = \phi - \psi$ is a global function downstairs. The local pushforwards defining $u\Theta_T$ agree: upstairs they are locally integrable by the exponential estimates, they agree on the good open, and they have no analytic-subset mass. Closedness and the last displayed identities give the global current identity

$$\text{dd}^c(u\Theta_T) = -\theta \wedge \Theta_T. \quad (114)$$

It proves (108). $\square$

6.4 Hodge index and the space of positive currents

Lemma 6.8. *Let X be a smooth compact Kähler fourfold, with Kähler form ω , and let R be a positive closed $(1, 1)$ -current with no divisorial mass. Then*

$$\int_X [R]^2 [\omega]^2 \geq 0.$$

Proof. Regularization with analytic singularities gives currents $R_k \in [R]$ and numbers $\epsilon_k \downarrow 0$ such that $R_k \geq -\epsilon_k \omega$ and $\nu(R_k, x) \leq \nu(R, x)$ at every point; see [11, Theorem 2.1(ii)]. In particular the analytic pole sets of R_k have codimension at least two. Resolve these pole sets by a projective modification $f_k: X_k \rightarrow X$. For $\beta_k = [R] + \epsilon_k [\omega]$, the pulled positive current has a decomposition

$$f_k^* \beta_k = P_k + [D_k],$$

where D_k is an effective exceptional real divisor and P_k is nef. With the usual smooth-remainder version of analytic regularization, P_k is represented by a smooth semipositive form. The bounded-remainder version gives a positive residual current with bounded potentials, which is nef by a further regularization.

Since $f_{k*}[D_k] = 0$, the projection formula gives

$$\int_{X_k} f_k^* \beta_k [D_k] (f_k^* [\omega])^2 = 0.$$

Consequently

$$\int_X \beta_k^2 [\omega]^2 = \int_{X_k} f_k^* \beta_k P_k (f_k^* [\omega])^2 \geq 0. \quad (115)$$

The inequality pairs a pseudo-effective class with three nef classes. It follows by approximating the nef factors by Kähler classes and pairing with a positive current. Passing to the limit proves the lemma. In particular, no positivity of the self-intersection of the exceptional divisor has been used. $\square$

For a Kähler form ω on M , write

$$Q_\omega(\xi, \zeta) = \int_M \xi \zeta [\omega]^2 \quad (\xi, \zeta \in H^{1,1}(M, \mathbb{R})).$$

The Hodge index theorem says that this quadratic form has signature $(1, h^{1,1} - 1)$. We use its following elementary consequence: two nonzero classes with nonnegative square, positive pairing with $[\omega]$, and zero mutual pairing must both lie on the same isotropic ray.

Choose the zero-Lelong current supplied by Lemma 5.1, and let Θ be the current of Lemma 6.7. The nef class α has nonnegative square, and Lemma 6.8 applies to Θ . Both classes have positive mass against ω^3 ; (97) is used here. Their mutual pairing is zero by (108). Thus

$$Q_\omega(\alpha, \alpha) = 0, \quad [\Theta] = c\alpha \quad \text{for some } c > 0. \quad (116)$$

In fact,

$$\alpha^2 = 0 \quad \text{in } H^4(M, \mathbb{R}). \quad (117)$$

To check the stronger assertion, choose Kähler representatives of $\alpha + \epsilon[\omega]$. Their positive squares have mass against ω^2 tending to zero by (116). Their cohomology classes tend to α^2 , whereas their currents tend weakly to zero, because their positive masses tend to zero. Thus the limiting cohomology class is zero.

Lemma 6.9. *Every positive closed current in α has zero Lelong numbers everywhere.*

Proof. Suppose $T \in \alpha$ has a positive Lelong number at a point, and let $f: X \rightarrow M$ be its blowup. The pullback has a positive generic Lelong number on the exceptional divisor. Write its Siu decomposition as

$$f^*T = R + \sum_j a_j [D_j], \quad a_j > 0, \quad (118)$$

where R has no divisorial mass and the sum is not zero. Fix a Kähler form ω_X on X , and set $\alpha_X = f^*\alpha$. The class α_X is nef, nonzero, and $\alpha_X^2 = 0$. Pairing (118) with $\alpha_X [\omega_X]^2$ gives zero. All terms are nonnegative, since a nef class paired with a positive current and two Kähler classes has nonnegative intersection. Therefore

$$Q_{\omega_X}(\alpha_X, [R]) = 0, \quad Q_{\omega_X}(\alpha_X, [D_j]) = 0 \quad \text{for every } j.$$

Lemma 6.8 and Hodge index give $[R] = b\alpha_X$ with $b \geq 0$; this includes $b = 0$ when $R = 0$. Comparing masses in (118) gives $b < 1$, because the divisorial sum has positive mass. Hence

$$(1 - b)\alpha_X = \sum_j a_j c_1(\mathcal{O}_X(D_j)). \quad (119)$$

The real span of the integral classes $c_1(\mathcal{O}_X(D_j))$ is a finite-dimensional, hence closed, subspace. The convergent series in (119) therefore places the rational class $c_1(f^*L) = \alpha_X/(2\pi)$ in that real span. Choose a finite subset of these integral classes forming a basis of the span. Solving the corresponding rational linear system shows that a rational vector in their real span is in their rational span. Thus, for some rational divisor D with finite support,

$$c_1(f^*L) = c_1(\mathcal{O}_X(D)) \quad \text{in } H^2(X, \mathbb{Q}).$$

After clearing denominators and integral torsion, the difference line bundle has zero integral first Chern class. By Lemma 6.2, $q(X) = 0$, so the exponential sequence makes this difference line bundle trivial. A positive multiple of f^*L consequently has a nonzero meromorphic section. The discrepancy identity for $X \rightarrow M \rightarrow Y$ converts it to a meromorphic section of a positive multiple of K_X , contradicting Lemma 6.2. $\square$

The normalized-current fixed-point construction is inspired by Touzet's method [73]. The singular extension, including continuity on the higher charts, is proved here.

Lemma 6.10. *There is a positive current $T \in \alpha$ such that, on all the smooth index charts constructed in Lemma 6.7,*

$$\sqrt{-1}\partial\bar{\partial}\phi = \sqrt{-1}be^{-\phi}\eta \wedge \bar{\eta} \quad (120)$$

for one constant $b > 0$. On these charts ϕ is smooth and

$$d\eta = \partial\phi \wedge \eta. \quad (121)$$

Proof. Fix a smooth reference weight for L , with curvature θ , and a smooth probability volume on M . The set

$$\mathcal{C}_\alpha = \{T \geq 0 : T \text{ closed of type } (1,1), [T] = \alpha\}$$

is nonempty, compact, and convex in the weak topology of currents. Its mass against ω^3 is fixed. Write $T = \theta + dd^c u_T$, uniquely normalized by $\int_M u_T dV = 0$. Thus all local weights used for T have a fixed common additive normalization.

By Lemma 6.9, the construction of Lemma 6.7 applies to every $T \in \mathcal{C}_\alpha$. Applying Hodge index as in (116) shows that $[\Theta_T]$ is a positive multiple of α . Define

$$\mathcal{F}(T) = \frac{\int_M \alpha[\omega]^3}{\int_M \Theta_T \wedge \omega^3} \Theta_T. \quad (122)$$

This is a selfmap of $\mathcal{C}_\alpha$.

We verify continuity, including on the higher charts. If $T_j \rightarrow T$ weakly, compactness for quasi-plurisubharmonic functions with fixed lower curvature bound and the chosen normalization gives $u_{T_j} \rightarrow u_T$ in L^1 . Indeed every subsequential limit has the same dd^c and normalization as u_T . On a coordinate chart the metric weights are plurisubharmonic and converge in L^1 . All complex singularity exponents of the limit are infinite, by Lemma 6.9 and Skoda integrability. The effective semicontinuity theorem [26, Theorem 0.2(2)] consequently gives locally uniform bounds for $e^{-s\phi_j}$ for every fixed $s > 0$, and in fact convergence of these exponentials in L^1 on smaller neighborhoods.

Apply (106) on the fixed proper chart diagrams. It gives locally uniform upstairs bounds for every inverse exponential power as well. After extraction, the weights converge almost everywhere off the exceptional and critical analytic sets. Pullback preserves almost-everywhere convergence there, because the chart maps are local biholomorphisms; the omitted analytic sets have measure zero. Uniform L^r bounds for some $r > 1$ then give convergence in L^1 of the densities defining

Θ_{T_j} upstairs. Proper pushforward gives $\Theta_{T_j} \rightarrow \Theta_T$ downstairs. The argument applies to each subsequence and so proves continuity for the whole sequence. Their masses also converge, and the limiting mass is positive. Thus $\mathcal{F}$ is continuous. The Schauder–Tychonoff fixed-point theorem yields a fixed point.

Let b be the positive normalization factor in (122) at this fixed point. Its equality $T = b\Theta_T$ holds upstairs on the good locus, giving (120). Both sides extend with no analytic-subset mass, so the equality holds on each entire smooth higher chart.

Taking a smooth Euclidean trace on a smaller coordinate ball gives a scalar Poisson equation whose right side belongs to every finite L^r , because η is holomorphic and all inverse exponential moments of ϕ are finite. Interior elliptic regularity gives $\phi \in W_{\text{loc}}^{2,r}$ for every finite r . Choosing $r > 8$ gives continuous first derivatives, after which the semilinear equation and ordinary elliptic bootstrap make ϕ smooth. On a regular foliated chart, (110) and (113) give $\eta = e^g h(z) dz$. Therefore $d\eta = dg \wedge \eta = \partial\phi \wedge \eta$ there. Both sides are now smooth on the entire higher chart, and the identity extends by density. This proves (121). $\square$

6.5 Spherical developing maps and boundary meridians

Fix the current given by Lemma 6.10. On a smooth higher chart put

$$\gamma = \sqrt{b/2} e^{-\phi/2} \eta, \quad a = \frac{\partial\phi - \bar{\partial}\phi}{2}. \quad (123)$$

Equations (120) and (121) imply

$$d\gamma = a \wedge \gamma, \quad da = -2\gamma \wedge \bar{\gamma}. \quad (124)$$

Indeed differentiating $e^{-\phi/2} \eta$ gives the first identity, while $da = -\partial\bar{\partial}\phi$ gives the second. Consequently the matrix-valued one-form

$$A = \begin{pmatrix} a/2 & \bar{\gamma} \\ -\gamma & -a/2 \end{pmatrix} \quad (125)$$

is skew-Hermitian, trace-free, and satisfies $dA + A \wedge A = 0$. On a simply connected small ball it therefore has the form $U^{-1}dU$ with U valued in $\text{SU}(2)$.

The map

$$F: x \mapsto U(x)\mathbb{C}e_1 \in \mathbb{P}^1$$

is holomorphic. In fact, $\bar{\partial}(Ue_1) = \frac{1}{2}a^{0,1}Ue_1$, because γ has type $(1,0)$; hence the line spanned by Ue_1 is a holomorphic line. This argument remains valid at zeros of η . For the Fubini–Study normalization $\omega_{\text{FS}} = \sqrt{-1}\partial\bar{\partial}\log(1+|z|^2)$, the usual unitary-frame computation gives $F^*\omega_{\text{FS}} = \sqrt{-1}\gamma \wedge \bar{\gamma}$. Thus, with $\omega_{\text{sph}} = 2\omega_{\text{FS}}$,

$$F^*\omega_{\text{sph}} = T \quad (126)$$

on the higher chart, where the right side is pulled back from M .

Choose a connected dense open subset $S \subset M$ with analytic complement such that p identifies S with a smooth open subset of Y , the relevant forms have their generic ranks, and η is nowhere zero on S . The index charts are unramified over S , and their resolutions can be chosen isomorphic there. Equation (126) gives local holomorphic submersions from S to $\mathbb{P}^1$. Two such submersions with the same pullback form differ locally by a constant element of $\text{PU}(2)$. Indeed their kernels agree, so one factors locally through the other on a transverse disc; the resulting local holomorphic isometry of the round sphere is the restriction of a projective unitary transformation. The transformation is unique on each connected overlap because the image of a submersion contains an open subset.

These local maps and their constant transitions give a developing map and a monodromy homomorphism

$$\text{dev}: \tilde{S} \rightarrow \mathbb{P}^1, \quad \rho: \pi_1(S) \rightarrow \text{PU}(2). \quad (127)$$

The developing map is nonconstant and is equivariant for ρ .

Lemma 6.11. *Let X be a smooth space with a proper generically finite map to M that is unramified over S , and let S_X be the inverse image of S . After any further modification supported outside S_X , the monodromy of a small meridian about every prime divisor in the complement of S_X has finite order.*

Proof. The assertion is local at a general point of a boundary prime D . Choose a small disc Δ transverse to D at a smooth point not on another boundary component, with $\Delta^* = \Delta \setminus \{0\}$ contained in S_X . Choose an original proper higher chart over a neighborhood of the image point in M , take its fiber product with X , and resolve a main component. The resulting map $g: W \rightarrow X$ is proper and generically finite, and is unramified over S_X . Near every point of W there is an ambient holomorphic map to $\mathbb{P}^1$: compose its map to the original higher chart with the map constructed in (125). Over the good open these maps agree with the developing germs up to constant projective unitary transformations.

A component of the inverse image of Δ dominating Δ is a curve, finite over Δ^* . Its normalization has a point over 0, and its local map to Δ can be written, after changing a parameter, as $t \mapsto t^r$ for some positive integer r . For a sufficiently small circle in the t -disc, the lifted loop stays inside one neighborhood on which an ambient map to $\mathbb{P}^1$ is defined. Continuing this ambient map around the lifted loop returns the same germ. On the punctured part, g is locally biholomorphic, so this is the analytic continuation of the original developing germ around the r -th power of the meridian.

A projective unitary transformation fixing that germ must be the identity, since it is a submersive ambient germ on the good open. Thus the meridian has monodromy whose r -th power is the identity. It is essential here to use the ambient germ: even if the transverse disc happens to be tangent to the foliation, the argument does not infer identity of holonomy merely from its action on the values of the map along that disc. The same fiber-product construction applies after the further modifications specified in the statement. $\square$

We record the compactification fact in the precise analytic category needed here.

Lemma 6.12. *Let X be a smooth compact Kähler manifold and D a simple normal crossing divisor. Every finite unramified cover of $X \setminus D$ extends to a finite normal cover of X . It has a smooth compact Kähler resolution which is unchanged over the original cover, and whose complement of that cover can be made a simple normal crossing divisor.*

Proof. Near a point of D , choose a polydisc on which its complement is $(\Delta^*)^r \times \Delta^{n-r}$. Each connected finite cover is described by a finite-index subgroup of its fundamental group $\mathbb{Z}^r$. Such a subgroup contains $N\mathbb{Z}^r$ for some $N > 0$. The cover is consequently dominated by the coordinate power cover

$$(t_1, \dots, t_r, w) \mapsto (t_1^N, \dots, t_r^N, w).$$

It extends across the coordinate hyperplanes as the normal quotient of the full polydisc power cover by the corresponding finite subgroup of its deck group. The normal finite extensions glue uniquely: an isomorphism on the dense punctured locus extends between the normal finite algebras, equivalently by their integral closures. This gives a finite normal cover $\bar{X} \rightarrow X$.

For completeness, a finite source over a compact Kähler space is Kähler in the local-embedding sense. Over finitely many small base neighborhoods choose relative embeddings of the finite source

into products with affine spaces, with relative coordinates w_a . Choose a smooth partition of unity λ_a on the base and put

$$h = \sum_a (\lambda_a \circ f) |w_a|^2$$

on the source, extending each summand by zero outside its support. This is a smooth function on the complex space. At a point where $\lambda_a > 0$, use the corresponding relative embedding. The other relative coordinate functions have local holomorphic extensions to its ambient space. On vertical tangent directions the Levi form of h is $\sum_a \lambda_a |dw_a|^2$ and is strictly positive. The terms arising from derivatives of the partition have a base-direction factor. A sufficiently large multiple of the pulled base Kähler form dominates their horizontal and mixed contributions. Compactness allows one such multiple on the finite cover. Thus

$$Cf^*\omega_X + dd^c h$$

has strictly plurisubharmonic local potentials in the chosen ambient embeddings and defines a Kähler form on the normal source. Finally take a projective resolution and then an embedded resolution of the boundary, both unchanged over the smooth covered open. Projective modifications of compact Kähler spaces remain Kähler, by the usual relative ample curvature construction. This proves the claim. $\square$

Resolve $M \setminus S$ by a projective modification, unchanged on S , so that the complement is a simple normal crossing divisor. The fundamental group of S is finitely generated: a complement of this kind in a compact smooth manifold has the homotopy type of a finite complex, or one may retract onto a compact manifold with corners obtained by removing sufficiently small tubular neighborhoods.

The group $\mathrm{PU}(2)$ is linear, for example through the adjoint embedding into $\mathrm{GL}_3(\mathbb{C})$. Selberg's lemma therefore gives a finite-index torsion-free subgroup of $\rho(\pi_1(S))$. Take the connected finite unramified cover $S_0 \rightarrow S$ associated with its inverse image in $\pi_1(S)$. By Lemma 6.12, it is contained in a smooth compact Kähler manifold M_0 , with simple normal crossing complement, and there is a proper generically finite morphism $M_0 \rightarrow M$. Its monodromy image is torsion-free.

Every boundary meridian of $M_0 \setminus S_0$ has finite-order monodromy by Lemma 6.11, hence trivial monodromy. The inclusion $S_0 \hookrightarrow M_0$ induces a surjection of fundamental groups whose kernel is normally generated by those meridians. One way to see both assertions is to put loops and homotopies in general position relative to the normal crossing divisor: loops avoid it, and a homotopy meets its smooth part in finitely many transverse points, each contributing a meridian; it avoids the intersections of components. Consequently the representation descends to

$$\rho_0: \pi_1(M_0) \rightarrow \mathrm{PU}(2). \tag{128}$$

6.6 Finite holonomy and the contradiction

Lemma 6.13. *Let X be a connected smooth compact Kähler manifold such that $a(X) = 0$ and $q(X') = 0$ for every connected finite étale cover $X' \rightarrow X$. Every representation $\pi_1(X) \rightarrow \mathrm{PU}(2)$ with torsion-free image has trivial image.*

Proof. Let Γ be the image, and take its complex Zariski closure in $\mathrm{PSL}_2(\mathbb{C})$, regarded as a linear algebraic group through the adjoint representation.

If this closure is all of $\mathrm{PSL}_2(\mathbb{C})$, apply the semisimple Shafarevich theorem [15, Theorem 1]. Its hypotheses are exactly a smooth compact Kähler source and a representation Zariski dense in a semisimple linear group; for torsion-free image the Shafarevich base is normal projective of general type, and the representation factors through a smooth model of that base. Algebraic dimension zero forces this base to be a point: a positive-dimensional projective base would supply a nonconstant

meromorphic function on X . Factorization through a point makes Γ trivial, contradicting its supposed Zariski density.

If the Zariski closure is proper, its identity component is solvable, by the classification of proper connected algebraic subgroups of $\mathrm{PSL}_2(\mathbb{C})$. Thus Γ is virtually solvable. Pass to a finite-index subgroup whose image Γ_0 is solvable and let $X_0 \rightarrow X$ be the corresponding finite étale cover. For every further finite-index subgroup H of $\pi_1(X_0)$, its abelianization is finite: it is the first integral homology of a compact Kähler finite cover with $q = 0$, hence is finitely generated of rank $2q = 0$.

Now induct along the derived series of Γ_0 . The quotient Γ_0/Γ'_0 is an image of $\pi_1(X_0)^{\mathrm{ab}}$, so it is finite. Its preimage has finite index in $\pi_1(X_0)$; that subgroup again has finite abelianization, making Γ'_0/Γ''_0 finite. Repeating gives finite successive indices down to the identity, because the derived series has finite length. Thus Γ_0 , and then Γ , is finite. A finite torsion-free group is trivial. $\square$

Lemma 6.2 supplies the hypotheses of Lemma 6.13 for M_0 and all its finite étale covers. The representation (128) is therefore trivial. The developing map descends to a single-valued nonconstant holomorphic map

$$F_0: S_0 \longrightarrow \mathbb{P}^1.$$

We finish by proving its meromorphic extension, rather than appealing to compactness of the target. Over a neighborhood in M_0 , form the proper generically finite higher charts used in Lemma 6.11. Each small ball upstairs has an ambient sphere map obtained by composition from (125). On the good part of the ball this map and the lift of F_0 have the same transverse round metric, so they differ by a constant projective unitary transformation. The good part, being the complement of a proper analytic subset in a ball, is connected; uniqueness on submersive germs makes that transformation constant throughout it. After this transformation the ambient map extends the lift of F_0 . The extensions agree on intersections by density and glue to a holomorphic map on the higher chart.

Choose a scalar projective coordinate on $\mathbb{P}^1$. Composing with these extensions gives a meromorphic function on each proper generically finite chart. Its meromorphic trace, divided by the chart degree, is a meromorphic function on the base neighborhood. To define the trace, first descend through the proper modification in the chart's Stein factorization and then take the ordinary trace of the finite normal map. On the good open every branch is the same pullback of the chosen coordinate of F_0 , so its trace is exactly the degree times that coordinate. The resulting local meromorphic functions on M_0 consequently agree on overlaps and give a global meromorphic extension of the coordinate of F_0 . It is nonconstant, contradicting $a(M_0) = 0$ from Lemma 6.2.

This contradiction proves Proposition 6.1.

7 Birational constructions and reduced-boundary models

This section supplies the ordinary dlt constructions needed in the algebraic-dimension-zero argument. We prove special termination through dimension four, using arbitrary klt termination through dimension three, and then apply Assumption 2.3 only to ordinary effective klt fourfold pairs. In particular, arbitrary dlt fourfold termination is not an input.

7.1 Conventions, comparison, and integral descent

All pairs in this section are ordinary rational pairs with effective boundary and rationally invertible adjoint. Auxiliary programs start on normal globally strongly $\mathbb{Q}$ -factorial compact Kähler spaces. Thus every global coherent rank-one reflexive sheaf has an invertible reflexive power. This condition will be used for global sheaves; no claim of $\mathbb{Q}$ -factoriality on arbitrary analytic open subsets is

made. Divisors over a space mean divisorial places represented on proper modifications, identified on common higher models. This convention does not distinguish places by their action on global meromorphic functions.

Definition 7.1 (Elementary birational steps). An elementary step for an adjoint $J = K_T + B$ is either a nontrivial projective bimeromorphic divisorial contraction, with the pair pushed forward, or a diagram

$$T \xrightarrow{f} Z \xleftarrow{f^+} T^+, \quad (129)$$

whose morphisms are projective, bimeromorphic, and small, with strict transform boundary on T^+ . The base is normal compact Kähler, and the source and next space are normal globally strongly $\mathbb{Q}$ -factorial compact Kähler spaces. All curves contracted by the negative morphism span one nonzero ray for degrees of global rational line bundles. The line $-J$ is relatively ample; in a small step J^+ is relatively ample. No positive-side Bott–Chern rank condition is imposed in this definition.

A projective morphism here has a relatively ample holomorphic line bundle. In the relative constructions, “relatively nef” means nonnegative degree on curves in its fibers. Absolute nefness continues to mean analytic nefness.

We use common projective resolutions and the natural meromorphic comparisons of canonical bundles, defined by Jacobians in local canonical frames, together with boundary pullback. No global meromorphic canonical frame is presumed. A global rank-one reflexive sheaf is transported through a bimeromorphic correspondence by pullback to a common resolution, proper direct image, and double dual. For a small correspondence this is the unique reflexive strict transform.

Lemma 7.2 (Exceptional comparison of global sheaves). *Let $h : Y \rightarrow T$ be a projective modification of normal spaces and let $\mathcal{P}$ be a global rank-one reflexive sheaf on Y . Suppose that $(h_*\mathcal{P})^{**}$ is rationally invertible. Then, for some positive integer l , an invertible sheaf M on T and an integral h -exceptional Weil divisor E satisfy*

$$\mathcal{P}^{[l]} \simeq (h^*M \otimes \mathcal{O}_Y(E))^{**}. \quad (130)$$

Proof. Choose l so that the corresponding reflexive power of $(h_*\mathcal{P})^{**}$ is invertible. This is $(h_*\mathcal{P}^{[l]})^{**}$, since the two sheaves agree where h is an isomorphism, including the general points of all prime divisors of T . Call this line M . Evaluation of local sections of $h_*\mathcal{P}^{[l]}$, and their images in M , gives meromorphic comparisons with h^*M . These comparisons are the same on their common domain and hence define a divisorial comparison globally. Its orders vanish away from the exceptional primes. The resulting integral exceptional divisor gives (130) in codimension one and therefore everywhere by reflexivity. Local meromorphic generators are enough for this argument; it does not represent an arbitrary global line bundle by a global divisor. $\square$

We shall use the projective analytic negativity lemma [31, 56]: if a rational Cartier divisor on a projective bimeromorphic morphism is relatively nef and has nonpositive pushforward, it is nonpositive. The local-over-target proof of [56, Lemma 3.39] applies in this setting.

Lemma 7.3 (Comparison in an elementary step). *On a common smooth projective resolution of an elementary step, with projections p to the negative model and q to the next model, the natural adjoint comparison is*

$$p^*J = q^*J' + F, \quad F \geq 0, \quad q_*F = 0. \quad (131)$$

Log discrepancies do not decrease. They increase strictly for a place whose center on either side is contained over the non-isomorphism locus in the contraction base. Consequently an elementary step preserves klt, respectively dlt, singularities.

Proof. The codimension-one comparison gives $q_*F = 0$. The signs of the two adjoints make $-F$ nef over the contraction base, and hence over the target of q . Negativity gives $F \geq 0$.

In a small diagram the non-isomorphism sets in Z are the same on both sides. Otherwise, over a region where one side is an isomorphism, smallness identifies the other adjoint with a pullback, contradicting its relative ample sign on a contracted curve. A non-isomorphism fiber is positive-dimensional by normality. Over any such point, lift a negative curve to the common resolution. Its F -degree is strictly negative, so the fiber meets $\text{Supp } F$. A connected projective fiber meeting the support of an effective relatively anti-nef Cartier divisor is contained in that support: if a component outside the support met it, a curve section through an intersection point would have positive intersection. Clearing denominators gives the same statement for F . Thus $\text{Supp } F$ contains the full fibers in question. Pullback to any higher model now proves strict increase at each specified place.

For dlt preservation use the characterization by log canonicity and an SNC open set meeting every lc center. A new discrepancy-zero place was already a zero place, and strictness keeps the general point of its center out of the surgery. The SNC characterization, checked on global log resolutions, is therefore preserved. The klt assertion is immediate. Finally, on a globally strongly $\mathbb{Q}$ -factorial dlt pair, slightly decreasing the coefficients of the floor gives a klt pair: the floor is effective rational Cartier, and every discrepancy-zero place has center in it and positive order on its pullback. $\square$

We recall several analytic projective facts used in these arguments. Projective modifications and projective spaces over compact Kähler bases are Kähler, also for singular spaces. To see the needed positivity directly, give a relatively ample line a metric by local relative embeddings and Fubini–Study metrics on a finite base cover, scaling back from the relatively very ample powers. Patch weights on the actual line by a partition from the base. In a local embedding, the active coordinates and frame changes lift to ambient holomorphic germs. Base cutoffs have zero first and second derivatives in vertical directions, so the patched weight has positive Levi form on vertical Zariski tangent vectors. Add a large multiple of a pulled-back base Kähler potential. One multiplier works on compact regions: otherwise a sequence of unit tangent vectors on shrinking compact charts would limit to a vertical tangent vector contradicting the strict vertical positivity. The Zariski tangent spaces vary in a closed set in a fixed local embedding. Squared absolute values of defining equations adjust the extensions to strict ambient positivity. This also treats finite morphisms, for which the trivial line is relatively ample; compare [74].

Common resolutions in projective diagrams are obtained by graph or fiber products followed by projective log resolution. Relative ampleness composes after adding sufficiently large pullback multiples from below, locally over compacta. A curve on the base of a projective surjection has a curve lift: pull back to its normalization, a projective curve, and use relative generation and base twists to make the total space projective; then take curve sections. This justifies all curve lifts above and below. Individual projective fibers permit the usual curve sections and the projective Kleiman criterion.

The established local inputs are the relative klt cone and base point free theorems for projective analytic morphisms over Stein neighborhoods of compacta satisfying property (P), and the big-klt adjoint finite-generation theorem. We can take arbitrarily small compact coordinate neighborhoods cut out by balls or polydiscs, with Stein ambient neighborhoods and the required finite-component intersection property. We use the cone theorem with its finite negative-ray decomposition after a positive relatively ample truncation. We use base point freeness in the following integral form: if L is relatively nef Cartier and $aL - (K + \Gamma)$ is relatively ample for some positive integer a , then every sufficiently high integral power of L is relatively generated after shrinking. See [31, Theorems 6.2,

6.5, and 7.2].

Lemma 7.4 (Integral klt descent). *Let $f : T \rightarrow Z$ be projective bimeromorphic with normal target. Suppose an ordinary klt adjoint is f -antiample. If an integral line bundle L has degree zero on every contracted curve, then f_*L is a line bundle and evaluation is an isomorphism*

$$f^*(f_*L) \simeq L. \quad (132)$$

Proof. The line L is relatively nef, and the base point free hypothesis holds by fiberwise ampleness. Locally over a smaller base neighborhood, all sufficiently high powers are generated. The induced maps are constant on each connected fiber, since their tautological lines have degree zero on every curve there. They factor through Z : the graph projection is finite bimeromorphic onto the normal base. The pulled-back tautological lines descend two consecutive powers of L , and their quotient descends L itself. The projection formula identifies the descent with f_*L and the pullback map with evaluation. These intrinsic identifications agree on overlaps. $\square$

7.2 Finite generation, nonextraction, and continuation

The finite-generation result we use is [19, Theorem 3.1]: on a smooth space projective over the indicated analytic base, multigraded rings of rational adjoints with simultaneous SNC effective subunit boundaries and a common relatively ample rational part are locally finitely generated, after clearing denominators. The relevant sums are klt. All applications here are over bimeromorphic projective bases, so the relative bigness conditions hold. The analytic projective big-klt framework and finiteness of models are also described in [31, Theorem E]. Neither citation is used as an arbitrary termination theorem over a nonprojective base.

We spell out the reduction to that theorem. Over a Stein base in the bimeromorphic case, the direct image of either sign of an actual line bundle is a coherent sheaf of generic rank one. Cartan generation therefore supplies a nonzero section after shrinking. Thus both signs have local-over-base effective meromorphic representatives. Local canonical representatives can be obtained in the same way, or from forms pulled back from general local projections downstairs.

For finitely many effective rational klt boundaries containing a common positive relatively ample part, take a simultaneous projective log resolution $a : S \rightarrow T$. Add effective exceptional corrections to their log pullbacks so that the resulting SNC boundaries Γ'_j are effective and have exceptional coefficients strictly between zero and one. Choose an effective exceptional divisor E with $-E$ resolution-ample, by composing the exceptional antiample choices for the successive blowups. If H is the common effective ample part below, then $a^*H - \beta E$ is relatively ample for sufficiently small $\beta > 0$. A common small multiple can be removed from each Γ'_j while preserving effectiveness and subunit coefficients: the strict transforms contain the common part, and the exceptional coefficients have positive margins. Relatively ample summands may also be represented by divided free general divisors after relative generation, Stein sections, and analytic Bertini. The smooth theorem applies. Effective exceptional corrections leave the cleared adjoint rings unchanged by projection to a normal space. The finitely many actual bundle identifications can be powered and tensored simultaneously, so they identify the multigraded rings multiplicatively.

In particular, a single rational klt adjoint has locally finitely generated ring in this setting. Add a sufficiently small positive rational multiple of the sum of effective representatives of opposite relatively ample integral bundles. Their sum is actually linearly equivalent to zero. The addition supplies the common ample part and preserves klt on a fixed resolution near the compactum. This replacement uses the projective bimeromorphic Stein setting essentially.

Lemma 7.5 (Relative Proj extracts no divisors). *Let X_0 be normal and projective bimeromorphic over a normal compact base T_0 . Let L be a rational line whose cleared relative ring is locally finitely generated. The normalized main component Y of the relative Proj is projective bimeromorphic over T_0 ; $X_0 \dashrightarrow Y$ extracts no divisors, and the reflexive trace of L on Y is an actual relatively ample rational line.*

Proof. Compactness permits a common divisible integer m such that the relative ring of mL is generated in degree one. Resolve its base ideal and graph, and denote the maps to X_0 and Y by p and q . The map to Proj lifts to the normalization. There is an actual moving/fixed decomposition

$$p^*(mL) = q^*H + G, \quad G \geq 0, \quad (133)$$

where H is the relatively ample tautological line on Y . Degree-one generation and projection show that every relative section of $p^*(kmL)$ has vanishing at least kG .

First, G is q -exceptional. If a component mapped to a prime on Y , then q would be an isomorphism at its general point. For large k , relative ample generation of $q_*\mathcal{O}(G) \otimes H^{\otimes k}$ gives a local-over-base section having a pole at that prime relative to $H^{\otimes k}$. Multiplying its pullback by the section of $(k-1)G$ gives a section of $p^*(kmL)$ whose vanishing is less than kG , a contradiction.

Second, every p -exceptional prime P is q -exceptional. Otherwise relative generation of $q_*\mathcal{O}(P) \otimes H^{\otimes k}$ gives a section with a pole at the image prime. Add kG to obtain a section of $p^*(kmL) + P$. Since $p_*\mathcal{O}(P) = \mathcal{O}_{X_0}$ by normality, it comes from the original ring. As a section of the enlarged line it must vanish once along P , whereas the chosen pole prevents that vanishing. Here the coefficient of G at P is zero by the first part. This is again a contradiction. Thus no divisor is extracted. Taking the trace of (133) gives $L_Y = H/m$ as actual rational lines. $\square$

Lemma 7.6 (Finitely many marked models). *For a rational polytope of actual adjoints satisfying the preceding local multigraded finite-generation hypotheses, only finitely many marked normalized main relative Proj models occur at rational parameters.*

Proof. After a common Veronese and shrinking, choose finitely many homogeneous generators of the multigraded ring. Diagonal rings on rational degree rays are finitely generated by the semigroup argument for monomial degrees. Their Proj charts can be taken with homogeneous monomials as denominators. The subsets of generators that occur as supports of monomials on the specified positive degree ray form a finite pattern. Localization at such a monomial inverts exactly its support, and the multidegree-zero subring is the degree-zero localization of the diagonal ring. Intersections use unions of supports with the canonical localization maps. Thus parameters with the same pattern have the same charts and gluing. Taking the main component and normalization preserves finiteness.

The base identification fixes the marking on the common bimeromorphic open. A finite collection of the smaller neighborhoods covers the compact base. If two marked models agree on each neighborhood, their identifications over the base agree on the common dense open, hence everywhere, and glue uniquely. There are consequently only finitely many global marked possibilities. $\square$

Proposition 7.7 (Relative continuation in the global strong category). *Let (T, B) be an effective rational klt or dlt pair satisfying the conventions above, and suppose T is projective bimeromorphic over a normal compact Kähler space V . If $K_T + B$ is not curve-nef over V , there is an elementary negative step over V . Its next space remains projective over V , compact Kähler, and globally strongly $\mathbb{Q}$ -factorial, and the appropriate singularity type is preserved.*

Proof. Use the finite-dimensional space of degrees of global rational line bundles on curves contracted over V ; finite dimensionality follows from first Chern classes in finite-dimensional cohomology. For

a fixed relatively ample H , the closed curve cone has a compact slice of H -degree one. Indeed, for every global line L , both $kH + L$ and $kH - L$ are relatively ample for sufficiently large k , so all coordinates on the normalized slice are bounded.

Choose a negative extremal ray. For dlt input decrease the floor coefficients rationally just enough to obtain a klt adjoint J_0 still negative on this ray; for klt input let $J_0 = K_T + B$. Cover the base by interiors of finitely many Stein compacta as above. Projecting the local cone decompositions to global degrees shows that, for each positive rational δ , the normalized slice lies in the convex hull of its compact part $J_0 + \delta H \geq 0$ and finitely many points of actual contracted curves. A projected remainder stays in the global closed cone, and a remainder of zero H -degree contributes zero. The stated convex hull is compact.

Choose δ so that the selected ray is strictly in the truncated negative region. It is therefore represented by an actual curve. Separating its point on the slice from the compact hull of the remaining generators and remainder gives a supporting nef class vanishing only on this ray. In the open set of such supports one can choose a rational global line N in the rational annihilator of the ray, so that a positive multiple of N minus J_0 is relatively ample. Strict positivity on the compact slice and fiberwise ampleness give this last assertion. Local klt base point freeness on the finite cover generates a common multiple of N . Its section morphism and Stein factorization give a projective bimeromorphic contraction $f : T \rightarrow Z$ over V , with normal target and connected fibers, contracting precisely the ray. The line $-J_0$ is f -ample by fiberwise ampleness.

If an exceptional prime E exists, it is rational Cartier and negativity gives E strictly negative degree on the ray. All contracted curves lie in E ; they cover the nontrivial fibers. There can be no second exceptional prime, since its negative ray degree would force the same curves, and hence E , into it. For a global rank-one reflexive sheaf on Z , take its reflexive pullback to T . It is rationally invertible. Add a rational multiple of E to kill its ray degree and apply Lemma 7.4 after clearing denominators. The descended rational line agrees with the given sheaf off codimension two on Z and hence everywhere by reflexivity. Thus Z has the required global strong property.

If f is small, apply the preceding finite generation over Z to J_0 . Lemma 7.5 gives a projective positive model with no extracted divisors; it is therefore small over Z . The transformed adjoint is relatively ample and rationally invertible. For any global line L on T , killing the ray degree and applying integral descent gives an actual rational identity

$$L = cJ_0 + f^*M, \quad c \in \mathbb{Q}. \quad (134)$$

Smallness gives the corresponding identity on the positive model. Any global rank-one reflexive sheaf there first transports to T , where a power is such an L . Transforming back and using (134) proves the global strong property on the positive model. The positive morphism cannot be an isomorphism, since that would make J_0 a pullback from Z .

For dlt input the unperturbed adjoint has negative degree on the ray. Its version of (134) has $c > 0$, so its transform has the positive sign as well. Lemma 7.3 preserves the required singularities. All resulting spaces are projective bimeromorphic over V and hence compact Kähler, so the construction can continue whenever relative curve-nefness fails. $\square$

7.3 Two finiteness counts and transversal adjunction

Lemma 7.8 (Divisorial count). *A sequence of bimeromorphic transformations extracting no divisors between normal irreducible compact Kähler n -folds contracts divisors only finitely often.*

Proof. Count the dimension of the span of prime $(n-1)$ -cycle classes in $H_{2n-2}(-, \mathbb{R})$. This is a finite nonnegative integer, since compact analytic spaces are triangulable and have finite-dimensional

homology. Nonextraction provides common isomorphic opens whose complement in the target has dimension at most $n - 2$. The localization sequence identifies the target's degree- $2n - 2$ homology with the Borel–Moore homology of that open. Restriction from the source maps its prime-cycle span onto the target's span by strict transforms, killing the classes of lost primes. Each lost prime has nonzero class by its positive Kähler volume, so the count strictly drops.

For completeness, that volume detects a topological class also on a normal singular space. Kähler potentials with pluriharmonic differences determine a class in $H^2(-, \mathbb{R})$ through the real-part sequence $0 \rightarrow \sqrt{-1}\mathbb{R} \rightarrow \mathcal{O} \rightarrow \mathcal{PH} \rightarrow 0$, with a fixed normalization. If pluriharmonicity is initially known only on the regular locus, pull the difference to a resolution. Its smooth pullback is pluriharmonic. Near the compact fiber over a point, choose holomorphic real-part primitives and adjust imaginary constants to make their values on that fiber agree. The real part is constant there, and each irreducible fiber germ forces the holomorphic primitive to be constant there. The primitives therefore agree on overlaps near the fiber. A finite cover and smaller neighborhoods glue them on a neighborhood of the whole fiber, and properness and normality descend them. Thus the original difference is locally a holomorphic real part. Resolving a compact prime cycle now evaluates the corresponding class power as the strictly positive integral of the pulled-back Kähler form. Its fundamental class pushes to the stated cycle class. $\square$

Lemma 7.9 (Strict low-discrepancy counts). *For a compact ordinary rational klt pair there exists $0 < \epsilon \leq 1$ such that every discrepancy is at least ϵ and only finitely many exceptional places have log discrepancy strictly less than $1 + \epsilon$. In particular all exceptional places of discrepancy at most one can be realized on a single projective SNC resolution. For a terminal pair with largest boundary coefficient b_0 (zero for empty boundary), the exceptional places of discrepancy strictly less than $2 - b_0$ are finite. For an lc pair the first assertion has the same form when restricted to centers not contained in its non-klt locus.*

Proof. On an SNC resolution let the log-pullback coefficients be $d_j < 1$, put $w_j = 1 - d_j$, and take $\epsilon = \min(1, \min_j w_j)$, with value one if the list is empty. For a place still exceptional over this resolution, choose at the general point of its center normal coordinates $x_1, \dots, x_s$, the first b defining the boundary components through the center. The top exterior Jacobian calculation gives

$$a_E \geq \sum_{j=1}^b w_j \operatorname{ord}_E(x_j) + \sum_{j=b+1}^s \operatorname{ord}_E(x_j). \quad (135)$$

At most one differential saves one order by normal differentiation along the place. This proves the lower bound. A place in the strict cutoff must have center a stratum: an additional normal coordinate would give at least $1 + \epsilon$ (or at least two if no boundary is present).

Blow up that closed stratum. Its new SNC weight is the sum of the passing weights, and the next center lies in the new exceptional component. Until the place is divisorial, only strata of codimension at least two can occur; each next weight increases by at least ϵ . The fixed strict cutoff bounds the chain length, and there are finitely many strata at each stage. The finitely many exceptionals already on the starting resolution complete the count. All these blowups can be made globally and projectively.

For a terminal pair, exceptional log-pullback coefficients are negative. The same calculation uses the strict cutoff $2 - b_0$; non-stratum centers and centers involving such exceptional components cannot contribute new places below it. For the lc version take the minimum of one and the strictly positive weights. A center not contained in the non-klt locus meets no zero-weight component generically, so the same argument applies. $\square$

Lemma 7.10 (Transversal surface calculation). *Let a normal analytic space carry an ordinary rational Weil boundary with rationally Cartier adjoint. At an analytically general point of a codimension-two irreducible locus, a general transversal surface cut is normal and the crepant formula on a simultaneous log resolution restricts to its exact surface crepant formula. For a coefficient-one prime, normalized divisorial adjunction is computed by normalized curve adjunction on this cut.*

Proof. Choose local parameters $t_1, \dots, t_{n-2}$ along the locus from an embedding. Take a sufficiently general nearby common value, regular on the smooth locus, upstairs on a projective log resolution, and on all relevant smooth strata; discard images of nondominating strata. The cut upstairs is a smooth surface, with the required SNC support, and no component is exceptional. Codimension-two bad loci are cut to isolated points.

The surface downstairs has dimension two and is regular in codimension one. It is Cohen–Macaulay as well. Indeed the non-Cohen–Macaulay locus of a normal analytic space has codimension at least three, by the local depth/coherent Ext criterion and normality at height at most two. Thus at the chosen general point the parameters are a regular sequence. The cut is generically reduced and Cohen–Macaulay, hence reduced; Serre’s criterion gives normality. Read the restricted boundary transversely as a cycle and take its closure. Complete intersection adjunction off the isolated bad set, followed by reflexive extension, identifies the restricted rational adjoint with $K_S + B_S$, using division by the same parameter volume form upstairs and downstairs. The restriction of the crepant formula has the correct nonexceptional coefficients. Any difference from the surface crepant formula is exceptional and relatively numerically zero, hence zero by both signs of negativity. This also restricts klt formulas, or dlt formulas from resolutions preserving an SNC good open, with their stated discrepancies.

For a coefficient-one prime, first subtract its smooth strict transform in the resolved formula, then take residue and push to its normalization. At general points over the chosen codimension-two locus, the cut of that strict transform is a smooth curve germ, finite and generically an isomorphism onto its branch of the reduced surface curve. It is therefore the curve normalization. Transversality preserves coefficient orders. Restricting first to the surface and then taking residue gives the same result. Only the indicated sums need be rational Cartier on the normalization; the individual canonical and boundary terms need not be. $\square$

7.4 Extraction of prescribed low places

Proposition 7.11 (Low extraction). *Let (T, B) be an ordinary rational klt pair on a normal globally strongly $\mathbb{Q}$ -factorial compact Kähler space. Any specified set of exceptional places of log discrepancy at most one can be extracted, and no other exceptional prime extracted, by a projective crepant morphism $Y \rightarrow T$ with effective klt boundary, where Y is normal, compact Kähler, and globally strongly $\mathbb{Q}$ -factorial.*

Proof. The set is finite by Lemma 7.9. Choose a projective SNC resolution $h : R \rightarrow T$ carrying it. Put an effective SNC klt boundary Θ on R , retaining the strict boundary and the crepant coefficients of the specified primes, and choosing every other exceptional coefficient strictly above its crepant value. With $J = K_T + B$ this gives

$$D(0) = K_R + \Theta = h^*J + P, \quad P \geq 0, \quad (136)$$

where P is supported exactly on the undesired exceptionals.

Fix an h -ample line H_0 and let $E_1, \dots, E_N$ be all exceptional primes of R . On a small full-dimensional rational polytope in formal coefficient space consider

$$D(u) = D(0) + u_0 H_0 + \sum_{i=1}^N u_i E_i, \quad u_0 \geq 0, \quad |u_i| \leq \eta u_0. \quad (137)$$

Choose $\eta > 0$ small enough that the perturbation is relatively ample at every nonzero parameter, and then make the polytope small enough. On the finitely many Stein neighborhoods choose effective representatives of $H_0, -H_0$, and $\pm E_i$. A small common positive multiple of the effective sum representing $H_0 - H_0 \sim 0$ supplies a common ample part. Taking the other coefficients sufficiently small on a simultaneous resolution gives equivalent ordinary klt adjoints at the vertices. Lemmas 7.5 and 7.6 give finitely many marked normal relatively ample models over T for all rational parameters.

For each model occurring arbitrarily near zero, take the affine spans of its parameter sets in successively smaller punctured neighborhoods. These nested affine spaces eventually stabilize. Discard models not accumulating at zero. If every eventual span were proper, finitely many proper affine subspaces would cover all sufficiently small rational points in the full-dimensional cone, which is impossible. Thus one marked model Y occurs arbitrarily near zero with full eventual affine span.

Choose affinely independent rational parameters for this model. Their actual ample adjoint traces solve a rational linear system for the traces of $D(0), H_0, E_1, \dots, E_N$ in the group of rank-one reflexive sheaves tensored with $\mathbb{Q}$. All these traces are therefore actual rational line bundles. A sequence of its parameters tending to zero makes $D(0)_Y$ curve-nef over T . Taking (136) in codimension one gives the actual identity $D(0)_Y = h_Y^* J + P_Y$, with P_Y effective exceptional. Negativity forces $P_Y = 0$.

No specified prime is lost. At a nonzero parameter defining Y , the relative system of a divisible power of $D(u)$ contains the system of the ample perturbation, multiplied by the section of the corresponding power of P and a base pullback. It embeds relatively at general points outside $\text{Supp } P$. Every specified prime has generic point there, including a discrepancy-one prime whose crepant boundary coefficient is zero. It cannot be contracted. Thus $Y \rightarrow T$ is crepant with precisely the prescribed exceptional primes and effective klt boundary.

Each surviving exceptional prime is rational Cartier by the traces of the E_i . For an arbitrary global rank-one reflexive sheaf on Y , its trace on T has an invertible power by the global strong hypothesis. Lemma 7.2 expresses the corresponding power upstairs as a pullback line with an integral exceptional correction. Clearing the rational Cartier denominators of that correction makes a further power invertible. This proves the global strong property. Projectivity over the compact Kähler base gives the remaining category assertions. $\square$

7.5 Arbitrary klt birational termination through dimension three

Proposition 7.12 (Terminal threefold termination). *An arbitrary sequence of elementary birational steps for an effective rational terminal pair of dimension at most three is finite. Here terminal means that all exceptional log discrepancies are greater than one. No pseudo-effectivity hypothesis is required.*

Proof. In dimensions at most two there are no nontrivial small diagrams, and Lemma 7.8 handles divisorial contractions. In dimension three discard a finite prefix to make all steps small. Terminality persists by Lemma 7.3. The underlying threefold singularities are ordinary terminal: deleting the effective rational Cartier boundary does not decrease discrepancies. In particular they are smooth at general curve points. This conclusion also holds for local analytic places. On a global resolution the exceptional coefficients for the empty boundary are negative; local places still exceptional over that smooth resolution have ordinary log discrepancy at least two.

Let b be the largest boundary coefficient, or zero for empty boundary. The coefficient set is fixed under small steps. If $b > 0$, test each step whose positive side has a flipped curve contained in a coefficient- b component. If $b = 0$, test every step with any flipped curve; a nontrivial positive side exists by the ample-sign argument in Lemma 7.3. Blowing up the generic point of that curve on the positive side gives an exceptional place with log discrepancy

$$t = 2 - \sum_j m_j b_j \leq 2 - b, \quad (138)$$

where m_j are nonnegative integral multiplicities. One can realize this place by the normalized blowup of the whole curve. Positivity and the fixed boundary denominators make the possible t a finite set. For each such t , the number of exceptional places with discrepancy strictly below t is finite by Lemma 7.9. These counts do not increase. At a tested step its tested place had discrepancy strictly below t before the step and equals t afterwards, so the corresponding count drops. Only finitely many tested steps occur.

If $b > 0$, consider the normalizations of the finitely many coefficient- b surfaces on the remaining tail. Both sides map bimeromorphically to the normalization of their common image in the contraction base. The positive map contracts no curve and is therefore an isomorphism. The negative normalization consequently maps projectively bimeromorphically to the next normalization, contracting a curve whenever the corresponding surface contains a flipping curve. These are compact Kähler surfaces. The cycle count leaves a tail on which no maximal-coefficient component contains a contracted curve on either side.

Remove all maximal-coefficient components from the boundary on this tail. The new adjoint is still negative, because an effective rational Cartier divisor has nonnegative degree on a curve not contained in it. Its one-ray identity with the original adjoint, obtained by Lemma 7.4, has a positive rational coefficient. Smallness then makes its transform relatively positive. Terminality persists after decreasing the boundary. Induct on the number of distinct positive boundary coefficients. The empty-boundary case was handled by testing every step, so the sequence is finite. $\square$

We require a uniform index statement to pass from klt to terminal pairs. The local terminal-point results used are the classical complex analytic threefold theorems of Mori–Reid and Kawamata. If a terminal point has canonical index $r > 1$, there is an exceptional place centered there with ordinary log discrepancy $1 + 1/r$; see [49]. Also, the index of every integral $\mathbb{Q}$ -Cartier Weil divisor germ divides r , including when $r = 1$. For the latter statement, the analytic index-one cover is smooth or an isolated cDV hypersurface, with simply connected punctured small neighborhood. The connectivity theorem for isolated hypersurface links and the analytic Kummer sequence make its germ divisor class group torsion-free. Pulling up a $\mathbb{Q}$ -Cartier divisor therefore makes it principal, and the norm gives the divisibility downstairs. The small-discrepancy place can be detected globally on a compact terminal space: on a global resolution the empty-boundary exceptional coefficients are negative, so a local place still exceptional over the resolution has discrepancy at least two. A place of discrepancy $1 + 1/r < 2$ must already be a component of its restricted exceptional divisor and hence a global exceptional place.

Lemma 7.13 (A surface estimate for a single extraction). *Let $h : U \rightarrow T$ extract only a prime P from an effective rational klt threefold pair as in Proposition 7.11. Write its crepant boundary as $B^{\text{str}} + (1 - a)P$, where $0 < a \leq 1$. There is a contracted curve $C \subset P$, not contained in $\text{Supp } B^{\text{str}}$, such that*

$$(K_U + P) \cdot C \geq -3. \quad (139)$$

Proof. First $-P$ is h -ample. Compare any h -ample line with its rationally invertible trace downstairs. Their difference is a rational multiple of the unique exceptional prime P , by Lemma 7.2. Negativity makes that multiple strictly negative, proving the assertion.

Divisorial adjunction of $K_U + P$ to the normalization P^ν gives an actual rational line of the form $K_{P^\nu} + \Delta$, with $\Delta \geq 0$, even though (U, P) need not be lc. We justify the sign, without requiring the individual terms on P^ν to be rational Cartier. Take residue along the smooth strict prime on a log resolution and push its restricted crepant boundary to P^ν . This identifies the actual restricted line in codimension one, then by reflexivity. Lemma 7.10 computes a tested coefficient on a klt normal surface germ with the reduced curve of all sliced branches of P . Such a surface germ is a quotient of a smooth germ by a small finite group, by the analytic klt surface classification; see [56, Chapter 4].

On the smooth chart, adjunction to a normalized branch of a reduced plane curve has effective different: the conductor contribution from normalization and the intersections with other branches are nonnegative. More explicitly, hypersurface adjunction makes the plane curve dualizing sheaf a line, and finite duality identifies the normalization's dualizing sheaf with its pullback multiplied by the conductor ideal. The quotient pullback of the reduced divisor is reduced. Frames of a Cartier log-adjoint power pull to the corresponding frames by the codimension-one étale property, and their meromorphic residues are related by pluricanonical pullback. Therefore the different downstairs pulls to the different upstairs plus the ramification divisor on the normalized branch. This proves $\Delta \geq 0$.

On a minimal smooth resolution $\pi : S \rightarrow P^\nu$ the restricted adjoint is $K_S + \Delta_S$ with $\Delta_S \geq 0$. Indeed K_S is relatively nef by curve adjunction, negative definiteness, and the absence of exceptional smooth rational (-1) -curves. The correction $\Delta_S = \pi^*(K_{P^\nu} + \Delta) - K_S$ is relatively anti-nef with effective pushforward; negativity applied to $-\Delta_S$ gives the sign.

If P maps to a curve, take a general smooth fiber on S , after Stein factorization. Its K_S -degree is at least -2 . If P maps to a point, P and S are projective. The classification of smooth projective surfaces supplies moving curves covering S with K_S -degree at least -3 : use ample curves if the minimal canonical class is nef, ruling fibers in the ruled case, or lines on $\mathbb{P}^2$, and general strict transforms; see [4]. In either case choose a curve not contained in the correction or in the finitely many excluded curves, including the inverse image of $\text{Supp } B^{\text{str}} \cap P$. It maps birationally to a curve C and the effective correction has nonnegative degree there. This proves (139). $\square$

Proposition 7.14 (Uniform global reflexive index). *Fix $0 < \epsilon \leq 1$ and an integer $n \geq 0$. For effective rational klt threefold pairs on normal globally strongly $\mathbb{Q}$ -factorial compact Kähler spaces, suppose every discrepancy is at least ϵ , at most n exceptional places have discrepancy at most one, and every exceptional discrepancy greater than one is at least $1 + \epsilon$. There is an integer $I(n, \epsilon) > 0$ such that $\mathcal{F}^{[I(n, \epsilon)]}$ is invertible for every global rank-one reflexive sheaf $\mathcal{F}$ on each such space.*

Proof. For $n = 0$ the underlying space is terminal with the ordinary exceptional discrepancy gap $1 + \epsilon$. The terminal-point statement above bounds every canonical index by $\lfloor 1/\epsilon \rfloor$. A global rank-one reflexive sheaf has $\mathbb{Q}$ -Cartier germs by the global strong hypothesis, so its local indices divide those canonical indices. Their bounded common multiple gives $I(0, \epsilon)$.

Induct on n , seeking a multiple of previous bounds. If there is no low place use the previous case. Otherwise extract only a place P of discrepancy $a \in [\epsilon, 1]$ by Proposition 7.11. The pair on U has effective crepant boundary $B^{\text{str}} + (1 - a)P$. Its exceptional places are exceptional downstairs, except that P is no longer counted; all hypotheses hold with $n - 1$. Set $I' = I(n - 1, \epsilon)$.

For the curve in Lemma 7.13, crepancy and effectiveness of B^{str} give

$$-3 \leq (K_U + P + B^{\text{str}}) \cdot C = aP \cdot C < 0. \quad (140)$$

Let $\mathcal{F}$ be a global rank-one reflexive sheaf on T , and let $\mathcal{F}_U$ denote its reflexive sheaf pullback. Distinguish this from the rational line pullback defined by an invertible power of $\mathcal{F}$. Their meromorphic comparison has the actual rational-line form

$$h^*\mathcal{F} = \mathcal{F}_U + sP. \quad (141)$$

Both $I'\mathcal{F}_U$ and $I'P$ are integral lines. Degree zero of the left side on C gives

$$s = \frac{I'\mathcal{F}_U \cdot C}{-I'P \cdot C}, \quad 1 \leq -I'P \cdot C \leq \frac{3I'}{\epsilon}. \quad (142)$$

The numerator and denominator are integers. Put

$$I = I' \operatorname{lcm}\{1, \dots, \lceil 3I'/\epsilon \rceil\}. \quad (143)$$

Then I and Is are multiples of I' , so $I\mathcal{F}_U + (Is)P$ is an integral line of zero degree on all contracted curves. Increase the crepant coefficient of P by a sufficiently small positive rational number. The pair remains klt, and its adjoint is relatively antiample because $-P$ is ample. Lemma 7.4 descends this integral line. Its descent agrees with $\mathcal{F}^{[I]}$ away from the codimension-two center of the extraction, hence everywhere by reflexivity. This proves the induction. $\square$

Theorem 7.15 (Klt termination through dimension three). *Every arbitrary sequence of elementary birational steps for effective rational klt pairs of dimension at most three is finite, without a pseudo-effectivity hypothesis.*

Proof. Only dimension three remains. Suppose there is an infinite sequence; after the cycle count all steps are small. The finite sets of exceptional places of discrepancy at most one are nonincreasing, so stabilize to a set $\mathcal{E}$. There is a uniform positive ϵ of the type in Proposition 7.14 on this tail. Indeed start with Lemma 7.9 on its first model. Only finitely many places lie below its strict cutoff $1 + \epsilon_0$; outside $\mathcal{E}$ their discrepancies are greater than one, with a positive minimum gap. Take the minimum of this gap, ϵ_0 , and the positive starting discrepancy lower bound. All subsequent comparisons are nondecreasing.

The boundary denominators are fixed. Proposition 7.14 gives a common denominator for the actual adjoints on all models of the tail, and hence for all discrepancies by the crepant formulas. The nondecreasing values on $\mathcal{E}$, bounded above by one, therefore eventually become constant. Over the first model of this stabilized tail extract exactly $\mathcal{E}$ crepantly. The resulting pair is terminal with effective boundary and is globally strongly $\mathbb{Q}$ -factorial.

Lift a flip $X_- \dashrightarrow X_+$ over Z from its extracted terminal model $Y_0 \rightarrow X_-$. Run relative elementary steps over Z using Proposition 7.7. On a common projective resolution of any finite prefix ending at Y , denote the pulled-back adjoints of Y_0, Y, X_+ by P_0, P, P_+ respectively. The comparisons give

$$P_0 = P + G = P_+ + F, \quad G, F \geq 0, \quad (144)$$

where G is exceptional over Y and F is exceptional over X_+ . The line P_+ is nef over Z . Over Y the divisor $G - F = P_+ - P$ is nef and has nonpositive pushforward; negativity gives $G \leq F$, before any termination is known. A lost prime of Y_0 would have positive coefficient in G , but zero coefficient in F : this follows from smallness below for old primes and from constancy of discrepancies on $\mathcal{E}$ for the extracted ones. Thus no prime is lost. The run stays small and terminal, so Proposition 7.12 makes it finite. Its endpoint has P nef over Z .

At the endpoint negativity over X_+ applied to $F - G = P - P_+$ gives the opposite inequality, hence $P = P_+$. Since the adjoint on X_+ is relatively ample, its projection from the common

resolution is constant on the fibers over Y : otherwise a curve in a connected projective fiber would have positive pullback degree. Factoring the graph over the normal space Y gives a projective crepant morphism $Y \rightarrow X_+$. It extracts the same places, by smallness below and the absence of lost primes, so the construction repeats.

Each nontrivial flip below forces a negative step in its lift, by lifting a negative curve. Infinitely many flips would concatenate to an infinite small terminal sequence with effective strictly transformed boundary, contrary to Proposition 7.12. This proves the theorem. $\square$

7.6 Adjunction and special termination for dlt pairs

Lemma 7.16 (Actual adjunction on normalized strata). *Let (V, A) be an ordinary effective rational dlt pair, and let T be the normalization of an lc center, with map $\nu : T \rightarrow V$. A chain of coefficient-one primes through its generic SNC stratum gives an effective dlt adjunction pair (T, A_T) and an actual rational-line identification*

$$J_T = K_T + A_T = \nu^*(K_V + A). \quad (145)$$

Its lc centers map to proper lc subcenters of the given center. If the ambient coefficients belong to a DCC subset of $[0, 1]$, the adjunction coefficients belong to a DCC set depending only on that set and the chain length. In sufficiently divisible even powers, the identification is the canonical meromorphic residue identification, independent of the resolution and of the order of the generic residues. No global strong $\mathbb{Q}$ -factoriality of T is asserted.

Proof. The lc centers are the images of the finitely many coefficient-one strata on a log resolution, and each is generically an SNC floor stratum. Order the floor components defining the chosen generic chain. Take a projective log resolution isomorphic over an SNC open meeting all lc centers. Successively restrict its crepant formula to the smooth strict strata by residue, then push the adjunction boundaries in codimension one to their normalizations. At each stage the SNC log pullback has coefficients at most one, and its rational adjoint is the restriction of the ambient actual line. Reflexive extension gives (145); any exceptional difference from the crepant comparison is numerically zero and vanishes by both signs of negativity. A discrepancy-zero stratum of a restricted formula comes from an ambient coefficient-one stratum whose image meets the good open. This gives the sub-dlt discrepancy and good-open properties, and identifies nested lc centers by the generic SNC calculation.

There is also a canonical meromorphic comparison, not merely an abstract equality of line classes. On a smooth strict stratum the residue of the ambient log-pullback frame has exactly the pole and zero orders of the remaining crepant boundary. It therefore gives the frame of the pushed adjunction in codimension one on the normalization. Extending the line identification reflexively and composing with the natural embedding of its divisorial adjoint sheaf into meromorphic pluricanonical tensors gives the claimed map. Different resolutions give the same map on the dense generic SNC stratum, and hence everywhere meromorphically. Permuting the residues changes only signs, removed in even degree. Sequential adjunction through normalized intermediate strata agrees with this strict-chain calculation by the same dense-open test. In particular all codimension-one boundary orders agree.

We prove effectivity and the DCC assertion at one restriction step; iteration then proves the statement. By Lemma 7.10, a tested different coefficient is a normalized-curve coefficient on a normal dlt surface with a coefficient-one branch. If the tested point is an lc center, the surface pair is SNC there. Indeed on a sliced log resolution preserving the good open, a zero center must lift to the intersection of two strict floor curves, with no exceptional locus through it; the resolution is an isomorphism there.

Otherwise deleting the boundary leaves a numerically klt surface germ. The Mumford numerical pullbacks of effective terms are effective by negative definiteness. We recall why this is ordinary klt even if rational Cartierness of the separate canonical term has not yet been established. On the minimal resolution of the singular germ, write

$$M = (E_i \cdot E_j) < 0, \quad K_S - \pi_{\text{num}}^* K = \sum_i a_i E_i.$$

The numerical klt condition gives $a_i > -1$. Relative nefness of K_S , from minimality and curve adjunction, gives $Ma \geq 0$ and hence $a_i \leq 0$. For any nonzero effective integral exceptional cycle $Y = \sum n_i E_i$, set $c_i = n_i + a_i > 0$ on its support and $c_i = 0$ elsewhere. Off-diagonal entries of M are nonnegative, so

$$c^\dagger M(n + a) \leq c^\dagger M c < 0. \quad (146)$$

Thus some supported E_i has $(K_S + Y) \cdot E_i < 0$. Consequently $\mathcal{O}_{E_i}(-Y + E_i)$ has degree greater than $2p_a(E_i) - 2$ and has zero H^1 by curve duality. Peeling off such components with the cycle exact sequences proves $H^1(\mathcal{O}_Y) = 0$ for every exceptional cycle Y . Grauert formal functions gives rationality; multiples of the full exceptional curve are cofinal with the maximal-ideal thickenings.

An integral multiple of the numerical pullback of any Weil divisor is now a line-bundle divisor on the resolution with all exceptional degrees zero. Its class restricts to zero in H^2 of the exceptional curve, by normalization and the degree description of the curve's top cohomology. Continuity around the compact fiber, or topological proper base change, makes that class zero after shrinking. On the preimage of a small Stein neighborhood, $H^1(\mathcal{O}) = 0$ by rationality. The exponential sequence makes this line actually trivial there. Pushing in codimension one proves that the original Weil divisor is rational Cartier; in particular the canonical divisor is. The numerical klt test is therefore the ordinary one.

The analytic log-terminal surface classification now realizes the germ as a smooth germ modulo a small finite group [56, Chapter 4]. Pull up the full boundary by canonical pullback, which is étale in codimension one. No exceptional place over the origin on the chart has nonpositive discrepancy, by finite discrepancy comparison with positive ramification factor. The point blowup therefore tests total boundary multiplicity strictly below two. There is exactly one smooth coefficient-one branch. The finite group preserves it and acts faithfully on its tangent: finite actions linearize, and a nonidentity element with trivial tangential eigenvalue would be a quasi-reflection. Thus the group is cyclic of order r , also the branch ramification index; $r = 1$ is allowed.

Residue and ramification give $1 - 1/r$ for the branch alone. Other components of coefficients d_j contribute intersection multiplicities $k_j \geq 0$ on the chart. The different coefficient is consequently

$$1 - \frac{1}{r} + \frac{\sum_j k_j d_j}{r}, \quad 0 \leq \sum_j k_j d_j \leq 1. \quad (147)$$

The upper bound follows from the sub-lc resolution formula. This proves effectivity and hence the full dlt condition at this step. Positive elements of a nonnegative DCC set have a positive minimum, if any occur. The sums in (147) thus have a bounded number of positive terms, counted with multiplicity, and satisfy DCC. In a decreasing sequence of coefficients below one, r is bounded as well, since $1 - 1/r$ is a lower bound. This proves DCC for each restriction and for its iterations. $\square$

Lemma 7.17 (Strict comparison on strata). *Suppose a small dlt elementary step preserves the generic points of an lc center and its chosen adjunction chain. The normalized strata on both sides map projectively bimeromorphically to the normalization S of their common image in the contraction*

base. Their adjoints are relatively antiample and ample, respectively. On a common higher model their canonical pullback comparison satisfies

$$P_T - P_{T^+} \geq 0. \quad (148)$$

The resulting discrepancy increase is strictly positive at any place whose center on either stratum maps into the ambient exceptional locus on that side.

Proof. The morphisms are the restrictions of the ambient projective morphisms followed through finite normalizations. The relative ample signs restrict as stated. Choose a common ambient resolution preserving the generic SNC chain. By Lemma 7.3 its effective pullback difference has support containing the full fibers over the non-isomorphism set. Restrict the two crepant formulas along the common strict chain. The same coefficient-one terms are subtracted, and Lemma 7.16 identifies the remaining canonical comparisons. Their difference is precisely the restriction of that effective divisor. The stratum itself is not contained in its support, since its generic point is preserved. The full inverse image of any indicated center is contained in the support, so pullback gives strictly positive multiplicity at each such place. This proves both assertions for actual crepant boundaries, rather than only for restricted first Chern classes. $\square$

Theorem 7.18 (Dlt special termination and modifications). *The following hold for ordinary rational pairs in the analytic projective setting above.*

- (i) *In dimension $d \leq 4$, any dlt elementary sequence has a tail whose exceptional loci on both sides are disjoint from every lc center.*
- (ii) *An ordinary rational lc pair of dimension $d \leq 4$ on a normal compact Kähler space, with rationally invertible adjoint, has a projective crepant dlt modification whose total space is globally strongly $\mathbb{Q}$ -factorial and compact Kähler. The boundary is effective, and every exceptional prime has coefficient one.*
- (iii) *For $d \leq 3$, every dlt elementary sequence is finite.*

The bases of the elementary steps may vary. Assertion (iii) is not asserted in dimension four.

Proof. We induct simultaneously on dimension. The order within a dimension is (i), then (ii), then (iii) when $d \leq 3$. Dimension zero is immediate. In proving (i) in dimension d , discard a prefix so that all steps are small, by Lemma 7.8. A zero-discrepancy place on a new space was already a zero place. Strictness in Lemma 7.3 excludes containment of its center in the exceptional locus. The finite lists of lc centers therefore stabilize, and the generic point of every remaining center is preserved at every subsequent step.

Induct increasingly on the dimension e of a surviving center; the point case is already settled by preservation of its generic point. Normalize its successive transforms and use the same generic adjunction chain. By the smaller-center conclusion, the stratum diagrams and adjunction pairs are isomorphisms near their non-klt loci on a tail. Their discrepancies are nondecreasing by Lemma 7.17.

If a prime is extracted by a stratum transformation, it lies on the new side over the ambient exceptional locus. Its earlier discrepancy is strictly smaller than its new discrepancy, the latter being at most one by effectivity. At the start of this tail it therefore had discrepancy less than one and center outside the non-klt locus, since the diagrams are unchanged near that locus. Lemma 7.9 gives finitely many possible exceptional places; the possible nonexceptional positive-boundary primes add only finitely many. For each such place, repeated extraction would give a strictly decreasing sequence of boundary coefficients, by the strict comparisons. The fixed adjunction DCC set of

Lemma 7.16 excludes infinitely many such occurrences. After discarding a prefix there are no stratum extractions.

The cycle count then leaves no prime contractions by stratum transformations either; for normal curves the diagrams are already isomorphisms. The primes now match. Their coefficients do not increase, and finite positive support and DCC stabilize the whole boundary. Neither morphism of a stratum diagram to S can contract a prime on this tail: its generic point would lie over the ambient exceptional locus, and strict discrepancy increase would contradict the matched coefficients. Thus the stratum diagrams are small, allowing isomorphisms, and their effective pullback differences are exceptional over the positive stratum. If the stratum transformation is an isomorphism, its discrepancies agree, and strictness excludes any ambient exceptional intersection with the stratum. One may test a divisor over a point of such an intersection, or the point itself for a curve.

For each remaining nontrivial stratum diagram, its dimension is $e < d$. Use the modification assertion already proved in dimension e to start on a projective globally strong crepant dlt model of the negative stratum. Run relative elementary steps over S by Proposition 7.7. The lower-dimensional termination assertion makes this run finite, with a relatively nef endpoint. On a common higher model write

$$P_0 = P + G = P_+ + F, \quad (149)$$

where P_0, P, P_+ are the initial, endpoint, and positive-stratum adjoint pullbacks. The divisors G, F are effective and exceptional over the respective endpoint sides. Both P and P_+ are relatively nef, so negativity in both directions gives $G = F$ and $P = P_+$. Relative ampleness on the positive stratum makes its projection constant on the connected projective fibers over the endpoint, as in the proof of Theorem 7.15. The graph therefore factors into a projective crepant morphism to that stratum. Remaining exceptional primes came from old exceptional primes, since there was no extraction upstairs and the stratum transformation was small; they retain coefficient one. This endpoint can start the next lift.

Each nontrivial stratum surgery forces at least one negative step in its lift. Its negative morphism is nontrivial as well, by smallness, matched boundary data, and the ample signs; lift a negative curve to see the assertion. Infinitely many surgeries would concatenate to an infinite lower-dimensional dlt elementary sequence across possibly varying bases, contrary to (iii) in dimension e . This proves disjointness for the chosen center. There are finitely many centers, completing (i) in dimension d .

For (ii), take a projective SNC resolution of the lc pair with boundary equal to the strict boundary plus the full reduced exceptional divisor. Its adjoint is the pullback of the original adjoint plus an effective exceptional correction, by log canonicity; the correction is supported in the floor. Run relative elementary steps over the original space. At every stage the same identity holds with the pushed-forward correction. If the run were infinite, (i) and the cycle count would leave a tail disjoint from the floor on both sides. The current adjoint could not have negative degree on a contracted curve there, since its correction is supported in the floor and its other term is a base pullback. Thus the run ends. Relative nefness and negativity kill its effective exceptional correction. The endpoint is crepant dlt and globally strongly $\mathbb{Q}$ -factorial by continuation. Every remaining exceptional prime came from an exceptional prime on the resolution and still has coefficient one. This proves (ii).

Finally, if $d \leq 3$, an infinite dlt sequence would, by (i) and the cycle count, have a small tail disjoint from the floor on both sides. Delete the floor. The resulting pair is klt, using the global strong condition and the dlt discrepancy characterization. The ample signs are unchanged by disjointness on both sides. This contradicts Theorem 7.15, proving (iii) and completing the induction. $\square$

7.7 A reduced-boundary model with a nef klt interval

Lemma 7.19 (Transport of canonical pseudo-effectivity). *Suppose a bimeromorphic transformation of normal globally strongly $\mathbb{Q}$ -factorial spaces extracts no divisors, and a Cartier power of the canonical line on its source has a semipositive singular metric. Then a Cartier power of the canonical line on its target has such a metric as well.*

Proof. Choose a common canonical power on the two spaces. On their isomorphic open, the actual canonical identification transports the metric. Nonextraction makes the complement of this open in the target have codimension at least two. In a local frame of the target line we must extend a psh weight over that complement.

Here is the local upper bound needed for this Hartogs argument on a normal germ. Take a finite local projection to a ball. Off the branch locus and the image of the missing set, the maximum of the weight on the finite fibers is psh. It extends across the branch locus while the fibers still avoid the missing set, by local upper boundedness. The bad image has codimension at least two, so psh Hartogs extension on the smooth ball extends this maximum there as well. It bounds the original weight on a dense analytic complement. The bound holds on its whole original domain by the disc test, using discs generically outside the excluded analytic sets through any tested point. Normality and the psh Riemann extension theorem on locally irreducible spaces now give the psh extension by upper regularization. These extensions respect the transition functions of the actual line. They define the required semipositive metric. For a related actual-line statement under rational singularities, see [46, Lemma 3.6]. $\square$

Proposition 7.20 (Reduced-boundary model). *Assume Assumption 2.3. Let (T_0, G_0) be an ordinary rational dlt fourfold pair with reduced boundary, on a normal irreducible globally strongly $\mathbb{Q}$ -factorial compact Kähler space. Assume that K_{T_0} has a semipositive singular metric on a Cartier power. The boundary may be empty. There is a nonextracting bimeromorphic map $T_0 \dashrightarrow T$ such that:*

- (i) *T is normal, compact Kähler, globally strongly $\mathbb{Q}$ -factorial, and klt for zero boundary; K_T remains pseudo-effective with such a semipositive metric.*
- (ii) *The pushforward G is reduced, (T, G) is lc, and the actual rational line $A = K_T + G$ is analytically nef. There is a rational $t_0 \in [0, 1)$ such that (T, tG) is klt and $K_T + tG$ is analytically nef for every rational $t \in [t_0, 1)$.*
- (iii) *There is a projective crepant dlt modification*

$$h : (V, D) \longrightarrow (T, G), \quad J := K_V + D = h^* A, \quad (150)$$

where V is normal, compact Kähler, and globally strongly $\mathbb{Q}$ -factorial, and D is reduced. The line J is analytically nef. If $G \neq 0$, then $D \neq 0$.

- (iv) *On a simultaneous smooth compact Kähler projective resolution $\mu : M \rightarrow V$ of the displayed models and boundaries, the line $\mu^* J$ has a semipositive metric with minimal singularities and zero Lelong numbers everywhere.*

The construction uses the fourfold MMP assumption only for ordinary effective klt pairs with pseudo-effective adjoint.

Proof. We give the two phases separately. In the first phase the current pair (T_i, G_i) stays dlt with reduced boundary, and we make a step whenever $K_{T_i} + G_i$ has a negative extremal ray of the cone $\overline{\text{NA}}(T_i)$ specified in Assumption 2.3. The underlying zero-boundary pair is klt by the dlt floor perturbation argument, and canonical pseudo-effectivity is retained by Lemma 7.19.

For a chosen negative ray, take a rational $t < 1$ sufficiently close to one that it stays negative for $K_{T_i} + tG_i$. This is an effective klt pair, and its adjoint is pseudo-effective because K_{T_i} is. Assumption 2.3 supplies the projective bimeromorphic ray step with its prescribed face, negative-side Bott–Chern rank, global strong condition, and Kähler category. The full adjoint has negative degree on this ray, hence the negative relative ample sign. Its one-ray degree-zero adjustment and Lemma 7.4, applied to the klt contraction, give the positive relative ample sign on a flip. Thus it is an elementary dlt step for the full adjoint, and Lemma 7.3 keeps the transformed pair dlt.

This first phase is finite. Otherwise the cycle count and Theorem 7.18(i) leave a small tail disjoint from the boundary on both sides. It is then a genuine zero-boundary K -negative klt program. The same rays are K -negative, as tested by their actual contracted curves; the contraction faces and negative-side Bott–Chern conditions are the ones already supplied. Both relative canonical signs are unchanged because the floor is disjoint from all fibers involved in the surgery. Moreover these are the actual canonical flips stipulated by the assumption: under a small map, direct images of common Cartier canonical powers agree as reflexive sheaves, and the positive canonical powers are relatively ample. They give the required relative canonical algebra and tautological line. The supporting-class existence for these zero-boundary negative rays is also supplied by the same assumption. This infinite tail would therefore be one of the programs excluded by its arbitrary termination clause, starting from its first zero-boundary klt model. No dlt fourfold termination assertion has been used.

At the end of phase one write $A_i = K_i + G_i$. It has no negative extremal ray and, in particular, is nonnegative on curves. We justify this conclusion without asserting an absolute curve criterion for nefness. The normalized Kähler-mass slice of the positive closed current cone defining $\overline{\text{NA}}(T_i)$ is compact for smooth Bott–Chern pairings. Indeed positive closed currents of unit mass have locally bounded order-zero coefficients by positivity and the Kähler mass; weak limits remain positive and closed. Their pairing image is compact in the product of the pairing coordinates, or in the finite-dimensional realization when used. Its positive cone with zero is closed, since the mass coordinate and normalized coordinates control every convergent sequence. Thus this is the full cone in the definition. A negative value of A_i would give a negative minimum face of the compact convex slice and hence an extreme point, contradicting the absence of a negative extremal ray. Only curve nonnegativity is needed at this stage.

In the second phase we make A_i -trivial steps of a single zero-boundary K program. Maintain the properties that A_i is nonnegative on curves, (T_i, G_i) is lc, $(T_i, 0)$ is klt, and K_i is pseudo-effective. If some rational $t_0 \in [0, 1)$ already has $K_i + t_0 G_i$ analytically nef, every rational $t \in [t_0, 1)$ does as well. Indeed these pairs are klt by interpolation between the zero-boundary klt pair and (T_i, G_i) , and their adjoints are pseudo-effective. Their degrees are nonnegative by convexity between $K_i + t_0 G_i$ and the curve-nonnegative A_i . A non-nef one would, by Assumption 2.3, have a negative ray with a projective contraction and an actual negative contracted curve, a contradiction. Closedness of the analytic nef cone then makes A_i analytically nef as t tends to one.

Suppose instead that every rational $K_i + tG_i$, $0 \leq t < 1$, is non-nef. Choose current positive integers m, r with mA_i and rK_i Cartier. Choose a rational t so close to one that

$$\frac{m(1-t)}{t} < \frac{1}{r(\dim T_i + 1)}. \quad (151)$$

The klt assumption gives a negative extremal ray R for

$$K_i + tG_i = tA_i + (1-t)K_i.$$

Its contraction supplies an actual curve class. Since A_i is nonnegative on curves, R is K_i -negative. Choose the prescribed contraction and step for this same ray now from the zero-boundary application of Assumption 2.3.

We claim $A_i \cdot R = 0$. If this degree were positive, the Cartier line mA_i would be relatively ample on that contraction, since its fiber-curve degrees are a positive multiple of those of $-K_i$. Work over a Stein neighborhood with a property-(P) compactum containing a point with nontrivial fiber. The local projective analytic rationality theorem for the ample Cartier line mA_i and the klt canonical adjoint gives denominator at most $r(\dim T_i + 1)$ for its positive finite nef threshold; see [32, Theorem 4.3.1 and Remark 4.3.4]. All contracted curve degrees are proportional, so for a curve C of the ray that threshold is exactly

$$\frac{mA_i \cdot C}{-K_i \cdot C} \geq \frac{1}{r(\dim T_i + 1)}. \quad (152)$$

The actual line data and canonical representatives needed for the theorem are available locally over this bimeromorphic Stein base, as in Section 7.2. On the other hand, negativity for the test boundary gives

$$\frac{mA_i \cdot C}{-K_i \cdot C} < \frac{m(1-t)}{t},$$

contradicting (151). This proves the claim using only current-stage indices, with no uniform index assumption on the fourfold sequence.

By Lemma 7.4, a Cartier multiple of A_i descends to an actual line on the contraction base. The pushed adjoint on a divisorial step, or the transformed adjoint on a small positive model, is the corresponding pullback by codimension-one identification and reflexivity. On a common projective model any remaining exceptional comparison is relatively numerically zero, and hence zero by negativity. Thus the step is crepant for the actual adjoint A_i , and the pair stays lc. Curve nonnegativity on the next model follows by lifting each curve to a common model and using the equality of pullbacks. The zero-boundary klt and canonical pseudo-effective hypotheses persist by Assumption 2.3 and Lemma 7.19.

Repeat this procedure whenever no rational nef parameter exists. Every step is a prescribed K_i -negative step of the same zero-boundary program. Arbitrary termination in Assumption 2.3 excludes indefinite repetition. We therefore reach the asserted T, G, A and nef klt interval. Neither phase extracts divisors, so their composite is nonextracting.

Apply Theorem 7.18(ii) to (T, G) to obtain (150). The boundary D is the strict transform of G plus coefficient-one exceptional primes, hence reduced. If G is nonzero its strict transform is nonzero. Analytic nefness pulls back, giving that of J .

Finally take the stated simultaneous resolution and write $\rho = h \circ \mu : M \rightarrow T$. For every rational t in the nef klt interval, Lemma 5.1 applied to (T, tG) gives a semipositive metric with minimal singularities and zero Lelong numbers on $\rho^*(K_T + tG)$. Add the metric of the effective rational divisor $(1-t)\rho^*G$. This is a semipositive metric on the fixed actual line

$$\rho^*A = \rho^*(K_T + tG) + (1-t)\rho^*G = \mu^*J. \quad (153)$$

A metric with minimal singularities on that line is no more singular than each of these metrics, up to an additive constant in weights. At every $x \in M$ its Lelong number is therefore at most $(1-t)\nu([\rho^*G], x)$. The latter is finite and tends to zero as rational t tends to one. All Lelong numbers of the minimal metric vanish. This proves (iv) and the proposition. $\square$

8 A section on the entire reduced boundary

This section proves the adjunction statement needed on the reduced-boundary model. The distinction between a section on one component and a section on the whole reduced boundary is essential. In particular, none of the gluing below follows merely from abundance on the normal components.

Proposition 8.1 (Whole-boundary nonvanishing). *Let (V, D) be the compact Kähler dlt fourfold supplied by Proposition 7.20, where V is globally strongly $\mathbb{Q}$ -factorial and $D \neq 0$ is reduced. Put*

$$J = K_V + D.$$

If the actual rational adjoint line bundle J is analytically nef, then, for some positive integer m for which mJ is Cartier on V ,

$$H^0(D, \mathcal{O}_D(mJ)) \neq 0. \quad (154)$$

We prove the proposition by constructing sections on the normalized components with matching canonical residues. All adjunction boundaries in the argument are the full effective rational differentials; their fractional parts are never discarded. All sufficiently divisible degrees will also be even. Evenness removes the sign obtained by interchanging two residue operations, but does not permit a change of the actual adjunction line bundle.

8.1 Adjunction and the descent locus

Write $D = \sum_i D_i$ and let $Y_i = D_i^\vee$ denote the normalization of a component. Dlt chain adjunction gives an effective rational dlt pair (Y_i, Γ_i) and an equality of actual rational line bundles

$$J_i := J|_{Y_i} = K_{Y_i} + \Gamma_i. \quad (155)$$

Here and below restriction includes pullback by the indicated normalization. Further adjunction is always performed with the full boundary in (155).

We use the good SNC opens and chain-adjunction construction of Lemma 7.16. Every lc center is generically a stratum of distinct coefficient-one primes. In the particular modification used here, this can also be seen by following a discrepancy-zero place from the starting SNC resolution: strict discrepancy increase over a nonisomorphism locus, as in Lemma 7.17, forces the general point of its center to remain unchanged at every step. The distinct primes at that stratum consequently remain distinct.

Keeping only one component D_i in the ambient boundary gives a plt pair. Indeed, any exceptional place with zero log discrepancy for (V, D_i) would also have zero log discrepancy for (V, D) and would have its center generically in one of these unchanged SNC strata. For a single smooth coefficient-one branch there is no such exceptional zero-discrepancy place. Comparisons are legitimate because the components are effective rationally Cartier divisors. Adjunction for (V, D_i) therefore supplies an effective rational klt pair on Y_i . In particular, Y_i has rational singularities. A Moishezon Y_i is projective by Namikawa's theorem, since it is also compact Kähler [60].

Lemma 8.2 (Codimension-one matching suffices). *The reduced space D is S_2 and has only smooth points and ordinary double crossings in codimension one. Its codimension-one conductor branches are the normalizations of the primes of $[\Gamma_i]$. They are paired between distinct components D_i, D_j . For a sufficiently divisible even m , a tuple*

$$s_i \in H^0(Y_i, \mathcal{O}_{Y_i}(mJ_i))$$

descends to $H^0(D, \mathcal{O}_D(mJ))$ if its restrictions agree on every paired normalized conductor surface under the canonical adjunction identifications.

Proof. First, both $\mathcal{O}_V$ and $\mathcal{O}_V(-D)$ are Cohen–Macaulay. The first assertion follows from klt rationality: decreasing the reduced dlt boundary makes the zero-boundary space klt. For the second, work locally and trivialize a Cartier multiple of the integral Weil divisor D . The resulting

normal cyclic index cover is *étale in codimension one*: divisorial orders in the defining equation are multiples of the index. The cover is klt by the finite canonical pullback and discrepancy formula, hence Cohen–Macaulay. The desired divisorial sheaf is an eigensummand of the pushforward of its structure sheaf, so is Cohen–Macaulay as well. Normality identifies the reduced ideal of D with $\mathcal{O}_V(-D)$. The sequence

$$0 \longrightarrow \mathcal{O}_V(-D) \longrightarrow \mathcal{O}_V \longrightarrow \mathcal{O}_D \longrightarrow 0$$

now shows that D is Cohen–Macaulay, in particular S_2 .

For the codimension-one description, take general transverse surface germs in the dlt adjunction calculation. At a single coefficient-one branch the plt local model is a cyclic quotient of a smooth pair, and its boundary curve is smooth. At two coefficient-one branches, the different and the sub-lc condition give an ordinary double SNC point; these are precisely the generic deeper lc strata described above. The branches come from distinct primes. To lift the assertion about a single-branch transverse curve to regularity at the corresponding point of D , note that the transverse boundary curve is generically reduced and has no embedded points by Cohen–Macaulayness. Its regularity lifts by lifting generators of the maximal ideal through the transverse parameters. This proves the asserted codimension-one description.

Let S be a normalized conductor surface on Y_i . Chain adjunction gives

$$J_S = K_S + \Xi_S = J|_S. \quad (156)$$

The matching branch on Y_j is the normalization of the same image, and the two normalizations agree. At a general SNC point the two iterated residue maps differ by the sign from interchanging two differentials. Their even powers are equal. On a common smooth model the two invertible pullback subsheaves of meromorphic pluricanonical forms are therefore equal: they have the same invertible domain and the same meromorphic map on a dense open. Thus this is an equality of the actual residue lines, not only a numerical comparison. The same argument applies to further normalized lc strata. Proper subcenters are generically deeper SNC strata, so the chains on either side identify the same further centers and the same even residue maps.

At an ordinary double point the usual two-branch local calculation glues sections precisely when their values agree on the intersection. Hence a matching tuple is a section of the ambient invertible restriction away from a subset of codimension at least two in D . For a reduced S_2 analytic ring, regularity of a meromorphic element is tested in its height-one localizations. Applied after trivializing $\mathcal{O}_D(mJ)$, this extends the section across the omitted subset. Equivalently, the class of the normalization-pushforward section modulo the invertible sheaf has support of codimension at least two and vanishes by the S_2 Hartogs criterion. The residue comparisons used here are the comparisons induced by the one ambient line bundle; no arbitrary scalar choices in adjunction enter the descent. $\square$

Lemma 8.3 (Systems on the normal components). *Each J_i is semiample as an actual rational line bundle. There is a holomorphic connected-fiber map to a normal projective variety and an ample rational line bundle such that*

$$f_i : Y_i \longrightarrow Z_i, \quad J_i = f_i^* H_i. \quad (157)$$

Proof. The normal compact Kähler threefold log-abundance theorem applies to the effective rational adjunction pair (155): its actual adjoint is analytically nef. One may first pass to a projective crepant dlt modification and use the $\mathbb{Q}$ -factorial dlt form of the theorem. We use here Das–Ou’s normal threefold theorem, together with their dlt-modification theorem [21, Corollary 1.3] [20, Theorem 5.2]. Generation on the modification descends by projection to the normal target. These inputs concern normal pairs; no nonnormal gluing assertion is being invoked.

For completeness, the passage to divisor representatives in these theorems does not impose an additional global-frame hypothesis. One can first produce a section of a power of J_i as follows. If Y_i is projective, use projective threefold log abundance. Otherwise resolve it by a smooth compact Kähler space. If that space is non-uniruled, smooth threefold canonical nonvanishing gives a section which pushes to J_i , since $\Gamma_i \geq 0$. If it is uniruled and non-Moishezon, take a resolved rational quotient. Its smooth non-uniruled compact Kähler base has nonnegative Kodaira dimension, and its very general smooth quotient fibers are projective of dimension at most two. On a simultaneous resolution of the pair, use the strict boundary and all reduced exceptional divisors as an effective SNC boundary $\tilde{\Gamma}$. Log canonicity gives

$$K_{\tilde{Y}} + \tilde{\Gamma} = p^* J_i + E, \quad E \geq 0 \quad p\text{-exceptional}.$$

This adjoint restricts pseudo-effectively on a very general quotient fiber. Projective log nonvanishing in dimensions one and two supplies fiberwise sections. Assumption 2.2, including unit coefficients and its invariant-base interpretation, now supplies a section upstairs exactly as in the rational-quotient reduction. It pushes to a section of an actual power of J_i by normality and reflexivity. The rational-quotient and uniruledness inputs are used in their smooth compact Kähler scopes [14, 64].

Cancelling the boundary contribution in such a section gives a meromorphic pluricanonical tensor. Its divisor divided by its degree is a rational canonical representative, compatible with pullback to resolutions. The discrepancies are the local canonical discrepancies, and the divisible systems are still the actual systems of J_i . Thus the divisor formulation of threefold abundance gives precisely the claimed semiample. The Stein factorization of a sufficiently high generated system yields (157). $\square$

8.2 Dominant clusters and a finite-product construction

Set $r_i = \dim Z_i$ and $r = \max_i r_i$. At a vertex with $r_i = r$, call a normalized conductor surface S *dominant* when it surjects onto Z_i . By (156) and (157), this is equivalent to $\kappa(S, J_S) = r$. On the opposite side of a matched conductor surface, the same equality forces $r_j = r$, and the surface is dominant there too. Thus dominance is symmetric.

Form the finite graph whose vertices have $r_i = r$ and whose edges are the dominant conductor surfaces. Fix a connected component of this graph, called a *cluster*. It is enough to construct a nonzero matching tuple on this cluster which vanishes on every nondominant branch, including nondominant edges within the cluster; we then put zero on every other component. The finitely many nondominant images are proper subvarieties of the relevant Z_i . An ample power has a nonzero section vanishing on their union. An isolated vertex is therefore immediate. If $r = 0$, there are no nonempty nondominant branch images. If the cluster has an edge, then $r \leq 2$, since an edge is a surface. This also settles any vertex with $r = 3$.

We shall use normalized strata with the full chain adjunction

$$(L, \Delta_L), \quad J_L = K_L + \Delta_L = J|_L.$$

Each chosen object L dominates the corresponding Z_i and has semiample J_L . An arrow between objects means a B -bimeromorphic comparison: on a common resolution, the actual adjunction lines and their meromorphic pluricanonical pullback maps agree. In sufficiently divisible even degrees, arrows consequently give isomorphisms of section spaces. These isomorphisms commute with multiplication.

Lemma 8.4 (Finite products of transported sections). *Consider finitely many such objects and invertible arrows. Suppose that in one sufficiently divisible even degree m the image of every group*

of loops on its section space is finite. At each object choose a nonzero section in that degree, obtained by restricting a base section which vanishes on the prescribed nondominant images. Then, in a common higher degree, there are nonzero sections at all objects, invariant under every arrow, and retaining the prescribed vanishing.

Proof. Fix one orbit of objects. At an object L , take the set of all sections transported to L from all chosen initial sections in that orbit, along all paths. The set is finite: there are finitely many starting objects, and two paths from the same object differ by a loop, whose image is finite. Arrows biject these finite sets. Multiply all their distinct members. The resulting section is nonzero because the object is irreducible, and the products are carried to one another because pullback is multiplicative. They have equal degree within the orbit. The initial section at L occurs among its factors, so its prescribed zeros remain. Taking further powers makes the degrees equal across the finitely many orbits. $\square$

In each application with $r > 0$ we shall verify that the invariant sections on the objects descend to sections of the node's base polarization. This will also preserve the required vanishing: a dominating object's section contains the pullback of the initial vanishing base section as a factor.

We record the residue calculation used to construct internal arrows. Suppose a log rational-curve fibration has, on its general fiber, two distinct coefficient-one points. The total horizontal weighted degree of the boundary is two, so there are no other horizontal markings. After ordering the points, choose a rational coordinate x with them as zero and infinity. A local base frame of an adjoint power has, in relative log differentials, the form

$$a(d\log x)^{\otimes m}, \quad (158)$$

where a is constant on the compact general rational fiber. The residues at zero and infinity differ by $(-1)^m$, and hence agree for even m . A change $x \mapsto cx$ adds a base differential, which disappears in the top-form expression. This proves that pairing the two marks preserves restrictions of base sections. For two degree-one branches it gives a bimeromorphic map between their normalizations; for one degree-two branch it gives the involution exchanging the two generic sheets. The latter is defined by normalization of the other main component of the relative fiber square. Since the lines on a common resolution pull back the same base line, equality of the meromorphic residue maps on this dense open proves the full B -crepant comparison.

8.3 Clusters containing a non-Moishezon vertex

We first describe the geometry of such a vertex; as before we omit its subscript and write (Y, Γ) , $J_Y = f^*H$, and Z .

Lemma 8.5 (The rational-quotient surface). *At a non-Moishezon vertex with a dominant conductor branch there is a diagram of compact Kähler spaces*

$$\begin{array}{ccc} W & \xrightarrow{g} & P \\ p \downarrow & & \downarrow u \\ Y & \xrightarrow{f} & Z \end{array} \quad (159)$$

in which W is smooth and resolves the pair, P is a smooth nonprojective compact Kähler surface, both g and u have connected fibers, the very general g -fiber is $\mathbb{P}^1$, and every p -exceptional divisor is vertical over P . The horizontal crepant boundary on W is effective, with weighted degree two on a general g -fiber. Moreover $r \leq 1$ and $H^0(P, K_P) \neq 0$.

Proof. On an initial smooth projective log resolution of Y , restrict the crepant equality

$$K_W + \Gamma_W = p^* J_Y$$

to a very general smooth connected fiber of the map to Z . Its right side is trivial. Pair with the appropriate power of the pullback of a Kähler class from Y . Exceptional terms have zero pairing, because their images have codimension at least two on this general fiber, or they do not meet it. Strict boundary terms have nonnegative pairing; a dominant coefficient-one branch gives a strictly positive term. Hence the canonical class of this smooth fiber is not pseudo-effective. Ou's theorem implies that the fiber is uniruled [64]. For $r = 0$ this argument is applied to the total space.

Take the almost holomorphic rational quotient of the smooth Kähler resolution. Its fiber has positive dimension, since the rational curves through very general points of the fibers just considered are quotient-contracted [14]. Its base cannot have dimension at most one. Indeed, rational connectedness of the general quotient fibers and the differential sequence would then force every holomorphic two-form on a resolved total space to vanish. The Kähler projectivity criterion would make that total space projective, a contradiction. The quotient base is therefore a surface, and the general quotient fiber is a smooth rational curve. Those curves are vertical over Z , so f factors meromorphically through the quotient. Resolving the graphs and the base gives (159). Stein factorization and very general connectedness give connected g -fibers; the fibers of u are connected as images of the connected fibers of fp .

If P were projective, the rational-curve fibration would have Moishezon total space. For example, coherence and GAGA give meromorphic sections of $g_* \mathcal{O}_W(-K_{W/P})$ generating its general fibers. Evaluation embeds the general smooth rational fiber, and these sections together with base functions give full algebraic dimension. Thus P is nonprojective.

Every exceptional prime of the initial projective resolution of Y is projective over a compact curve or a point, hence Moishezon. It cannot dominate P : a dominant generically finite map of surfaces preserves algebraic dimension, by the norm or characteristic-polynomial argument after finite Stein factorization. The strict transform of a Moishezon prime remains Moishezon. The further graph resolutions can be made isomorphisms over a general quotient-base open, so their new exceptional primes are vertical too. Consequently all horizontal crepant boundary terms are strict transforms of effective boundary components. Adjunction on the general rational fiber makes their weighted degrees sum to two.

A nonprojective compact Kähler surface has a nonzero holomorphic two-form, again by the Kähler projectivity criterion. Finally, if $r = 2$, then $P \rightarrow Z$ would be generically finite over a projective surface, forcing P to be Moishezon and projective. Thus $r \leq 1$. $\square$

We use two elementary consequences of nonprojectivity for surfaces. If Q is a smooth nonprojective compact Kähler surface, then the intersection form on

$$N_Q = \text{NS}(Q)_{\mathbb{R}}$$

is nonpositive. Indeed, a positive-square vector in this rational subspace can be approximated by a rational one, with sign chosen to have positive Kähler degree. The corresponding line bundle has quadratic section growth by Riemann–Roch: its high powers have no top cohomology by Serre duality and the Kähler degree test. This would make Q Moishezon and hence projective. Moreover, if $Q \rightarrow C$ is a map to a projective curve, Q has no multisection. For a curve B dominating C , the class $B + tF$, with F the rational fiber class and $t \gg 0$, would have positive square.

Lemma 8.6 (Horizontality, including fractional boundaries). *Every dominant conductor branch at the non-Moishezon vertex of Lemma 8.5 is horizontal over P . A cluster containing such a vertex*

consists entirely of non-Moishezon vertices. At a vertex there are at most two dominant branches, counted with their generic degrees over P .

Proof. If $r = 1$, a vertical surface which dominates Z would have image a curve in P dominating Z . This contradicts the preceding no-multisection observation. It remains to prove horizontality when $r = 0$, so that $J_Y \sim_{\mathbb{Q}} 0$.

Choose a nonzero $\eta \in H^0(P, K_P)$. On the general rational fiber write the distinct horizontal marked points and their coefficients as

$$b_1, \dots, b_q \in \mathbb{Q}, \quad 0 < b_j \leq 1, \quad \sum_{j=1}^q b_j = 2.$$

Take a smooth compact Kähler generically finite base change $b : P^+ \rightarrow P$ splitting and ordering the markings. It can be constructed from a main component of a fiber product of the horizontal prime normalizations over P , on the finite *étale* locus parametrizing an ordering of the distinct marks, and then resolved. Resolve the main component of the pulled-back rational fibration, obtaining

$$h : W^+ \longrightarrow W, \quad g^+ : W^+ \longrightarrow P^+.$$

Let H_j be the prime closures of the ordered generic sections. The very general rational fibers are unchanged by these resolutions.

For $j \neq k$ there is a meromorphic top form

$$\beta_{jk} = d \log x_{jk} \wedge (g^+)^* b^* \eta, \tag{160}$$

where x_{jk} has horizontal divisor $H_j - H_k$. This description is valid meromorphically even across degenerations. To see its local construction on the base, the coherent sheaf $(g^+)_* \mathcal{O}_{W^+}(H_j - H_k)$ has generic rank one, because its restriction to a general rational fiber has degree zero and is trivial. Divide the canonical meromorphic section by the evaluation of a locally chosen generically nonzero direct-image section. Different choices change x_{jk} by a base function on the general fibers. Its logarithmic differential wedges to zero with the pulled-back top form of P^+ , so (160) is independent of the choices. Reversing j, k changes its sign.

The rational vector (b_j) lies in the convex hull of the vectors $\mathbf{e}_j + \mathbf{e}_k$ for $j < k$. These are exactly the vertices of the rational polytope

$$\{(x_j) : 0 \leq x_j \leq 1, \sum_j x_j = 2\}.$$

Average a rational convex decomposition over the permutations which preserve the coefficients. After clearing denominators and multiplying by an even integer, we obtain nonnegative even integers l_{jk} and an integer $m > 0$ such that

$$\sum_{j < k} l_{jk} = m, \quad \sum_{k \neq j} l_{\min\{j,k\}, \max\{j,k\}} = mb_j. \tag{161}$$

We also make m divisible by all adjunction indices. The tensor

$$\beta = \bigotimes_{j < k} \beta_{jk}^{\otimes l_{jk}} \tag{162}$$

is a nonzero meromorphic m -canonical tensor, invariant under the permutations of the ordered marks. Evenness removes the signs from reversed pairs. It therefore descends to W . More explicitly,

relative to the pullback of a local canonical power frame, its coefficient is the same on all generic sheets. Factoring h through its finite Stein part, normalized trace descends this meromorphic coefficient, and modifications do not change meromorphic functions. Formula (161) shows that the descended tensor has exactly the allowed horizontal poles, of orders mb_j .

We must also control *every* vertical prime, including a prime whose image on P is a point. Let E be a vertical prime of W , and choose a prime E^+ above it, with ramification index e under h . At its general point the tangential map is generically finite and separable, so the canonical Jacobian has order $e - 1$. Near a general point of the image in P , take coordinates t, s with t vanishing on that image. Whether the image is a curve or a point,

$$\text{ord}_{E^+}(h^*g^*t) \geq e.$$

In logarithmic differential coordinates at E^+ , the differential of this function retains that order. Both $d \log x_{jk}$ and the pullback of ds are at most logarithmic. A logarithmic top form has at most one simple pole. Writing η in the $dt \wedge ds$ frame, with its holomorphic coefficient, gives

$$\text{ord}_{E^+}(\beta_{jk}) \geq e - 1. \tag{163}$$

Thus (162) has order at least $m(e - 1)$, exactly paying the canonical ramification in descending an m -canonical tensor. The descended tensor on W is regular at E .

Push it to Y . The horizontal pole orders are allowed by the effective boundary, and at vertical primes it is regular; hence normality gives a nonzero section of mJ_Y . Since J_Y is torsion, this section has no zeros: after trivializing a torsion power, a nonzero section is a nonzero holomorphic function on a connected compact normal space. A vertical floor prime would give a positive zero in the adjoint section, since its coefficient is one and the canonical tensor is regular there. This is impossible. Horizontality follows also for $r = 0$.

A horizontal conductor surface is generically finite over the nonprojective surface P , so is non-Moishezon. A Moishezon neighbor would be projective, and its normalized conductor surfaces would be projective. Thus every vertex reached across a dominant edge remains non-Moishezon. Finally, each dominant branch has coefficient one and contributes its generic degree to the horizontal degree sum two. This proves the last claim. $\square$

At a non-Moishezon node, if there are two degree-one branches or one degree-two branch, we consequently have the internal arrows from (158), pairing the two marks over P . They are over Z and preserve the restrictions of base sections. One degree-one branch requires no internal arrow.

Lemma 8.7 (Finite loop action for a surface with a curve system). *Let S be a normal non-Moishezon conductor surface in a cluster with $r = 1$. A group of B -bimeromorphic loops on its semiample adjunction pair acts through a finite group on every sufficiently divisible adjoint section space.*

Proof. Let C be the normal connected-fiber system base of J_S . It is a polarized projective curve. A sufficiently divisible system embeds C , and the loop action induces automorphisms of C preserving a fixed ample power. Take the smooth minimal compact Kähler surface Q in the bimeromorphic class of S . Classical compact-surface theory will be used in its Kähler and elliptic-surface forms [4]. Here $a(Q) = 1$ and $H^0(Q, K_Q) \neq 0$; bimeromorphic self-maps of this minimal surface of nonnegative Kodaira dimension are automorphisms. The map to C is holomorphic on Q . Indeed, on a common resolved model no curve can dominate C by the nonpositive Néron–Severi test, so the blowdowns to Q factor the map. Its general fibers are connected. Every meromorphic function on Q comes from C : the function field has transcendence degree one and is algebraic over that of C , so the function is constant on connected general smooth fibers and descends meromorphically.

Let F be the general fiber class. It is nonzero, rational and isotropic. Since the intersection form on $\mathrm{NS}(Q)_{\mathbb{R}}$ is nonpositive, $K_Q \cdot F = 0$. Adjunction gives genus one for a general smooth fiber. The induced cohomology group preserves the rational line $\mathbb{Q}F$. On

$$F^{\perp}/\mathbb{Q}F \tag{164}$$

it preserves an integral lattice and the Hodge summands. The form is positive definite on real $(2, 0) + (0, 2)$ and negative definite on the remaining $(1, 1)$ quotient by the Hodge index theorem. Reversing the latter sign gives an invariant positive definite norm. Its integral isometry group is finite. In particular the action on $H^{2,0}(Q)$ is finite.

The action on the base is also finite. For genus at least two this follows from finiteness of the curve automorphism group; for genus one the subgroup preserving a fixed ample line bundle is finite. Suppose $C = \mathbb{P}^1$. We claim $q(Q) = 0$. Restriction of holomorphic one-forms to a general smooth elliptic fiber is injective. Indeed, a form vanishing on one general fiber vanishes on all general fibers by constancy of periods, since holomorphic forms on a compact Kähler space are closed. It is then pulled back from a one-form on the smooth base open. This base form extends at a critical value: at a general point of a fiber divisor the local parameter is $t = w^d$, and a pole or a nonremovable singularity cannot pull back to a regular form. There is no nonzero holomorphic one-form on $\mathbb{P}^1$. Thus $q(Q) \leq 1$. If equality held, the elliptic Albanese map would be nonconstant on a general elliptic fiber. A general Albanese fiber would then contain a curve dominating $\mathbb{P}^1$, contradicting the absence of multisections. This proves the claim.

It follows that

$$\chi(\mathcal{O}_Q) = 1 + h^{2,0}(Q) \geq 2, \quad e(Q) = 12\chi(\mathcal{O}_Q) - K_Q^2 \geq 24. \tag{165}$$

There are at least three critical values of the relatively minimal elliptic fibration. Otherwise the smooth base is $\mathbb{P}^1$ with at most two points removed. Its universal cover is compact or parabolic, so the genus-one period map to the upper half-plane is constant. The integral linear monodromy is then finite. Kodaira's singular-fiber and monodromy classification, including multiple fibers, bounds the Euler number of a fiber with finite linear monodromy by ten. The types I_n and their multiples for $n > 0$, and I_n^* for $n > 0$, have infinite monodromy. Euler addition with at most two critical values would therefore give $e(Q) \leq 20$, contrary to (165). The finite set of critical values is preserved by the base action, and an automorphism of $\mathbb{P}^1$ fixing three points is the identity. The base image is finite in this final case too.

Pass to the finite-index kernel fixing both C and all holomorphic top forms. Choose a nonzero holomorphic top form η on Q . Any meromorphic m -canonical tensor, divided by $\eta^{\otimes m}$, is a meromorphic function and hence comes from C . The kernel fixes it. It therefore fixes all the adjoint section spaces under their meromorphic canonical realizations. The original loop action has finite image. $\square$

Lemma 8.8 (Matching in non-Moishezon clusters with $r = 1$). *Such a cluster admits the required nonzero matching tuple, with zero restriction on every nondominant branch.*

Proof. Use the dominant surfaces as objects, with conductor matchings and the internal arrows just constructed. Lemma 8.7 supplies loop finiteness, so Lemma 8.4 gives invariant nonzero sections in a common degree m' .

We verify their descent at each node. A degree-one branch S is bimeromorphic to P , so its general fibers over Z are connected. Its section of $m'J_S$ is therefore pulled back from $m'H$ on Z . When there are two such branches, the internal arrow makes these base sections identical. For a degree-two branch, resolve $S \dashrightarrow P$. Over a general $z \in Z$, every component of the fiber dominates

the connected smooth curve P_z , with total degree two. Indeed, an exceptional or bad curve on P cannot dominate Z , by the no-multisection observation. Divide the section by the pullback of a local frame of $m'H$. This holomorphic function is constant on each compact connected component of the fiber. Invariance under the sheet-exchange arrow equates the two generic values. It descends meromorphically to Z , and then holomorphically: a meromorphic function on a normal base whose pullback by a proper surjection is regular has no pole, as can also be checked through finite Stein factorization.

Pull these base sections back to Y . Their restrictions match along dominant edges because the object sections do. They vanish on the prescribed nondominant images because the invariant object sections retain the initial vanishing factor and dominate Z . This gives the desired tuple. $\square$

Lemma 8.9 (Matching in non-Moishezon clusters with $r = 0$). *Such a cluster also admits a nonzero matching tuple.*

Proof. All node adjoints are torsion. Choose a common sufficiently divisible even degree and a trivializing section at each node. Conductor comparison across an edge is multiplication by a nonzero constant on these sections. A graph with no cycles permits all comparisons to be solved by rescaling. In general it is enough to prove that the oriented product of these constants around each cycle is a root of unity, and then to take one further common power.

The graph has degree at most two by Lemma 8.6. At a node on a cycle, including a cycle formed by parallel edges, the two distinct branches must each have degree one over its P_i . The section from the proof of Lemma 8.6 is now a power of $d \log x \wedge \eta_i$: the two unit points exhaust the horizontal weighted degree. Its even residues on the two branches are the corresponding powers of the base top form η_i . Following the cycle gives bimeromorphic identifications of the P_i and hence isomorphisms of their smooth minimal models Q_i , since they have nonnegative Kodaira dimension. The cycle constant is the scalar on $\eta^{\otimes m}$, or its inverse, for the resulting automorphism ϕ of one minimal model Q .

The existence of a single ambient Kähler class restricts this automorphism. Fix a Kähler form ω on V . At the two branches of a cycle node, choose smooth branch resolutions and maps

$$r_k : B_k \longrightarrow W_i, \quad \alpha_k : B_k \longrightarrow Q_i, \quad k = 1, 2,$$

where α_k is bimeromorphic. Put

$$v_k = (\alpha_k)_* r_k^* p_i^* [\omega] \in H^{1,1}(Q_i, \mathbb{R}).$$

Each v_k has positive square. In fact, the pulled-back ambient class β_k on B_k has positive square because the branch maps generically birationally onto its surface image in V . Write $\beta_k = \alpha_k^* v_k + e_k$ with e_k exceptional. Orthogonality and negative definiteness of exceptional surface classes give

$$v_k^2 = \beta_k^2 - e_k^2 > 0. \tag{166}$$

Furthermore,

$$v_1 - v_2 \in \text{NS}(Q_i)_{\mathbb{R}}. \tag{167}$$

Indeed, the rational Hodge morphism

$$(\alpha_1)_* r_1^* - (\alpha_2)_* r_2^* : H^2(W_i, \mathbb{Q}) \longrightarrow H^2(Q_i, \mathbb{Q})$$

kills $H^{2,0}$ and $H^{0,2}$. Holomorphic two-forms are basic on the smooth rational-fiber open, by the differential sequence and the absence of fiberwise one-forms; their restrictions to the two birational

sections agree. Hence the rational image of this Hodge morphism has type $(1, 1)$, and is contained in the rational Néron–Severi space. Apply it to the ambient real class to obtain (167). Across a conductor edge the corresponding branch maps to V agree on a common higher model. The transported v_k therefore agree across that edge. Going around the cycle yields

$$\phi^*v - v \in N := \text{NS}(Q)_{\mathbb{R}}, \quad v^2 > 0. \quad (168)$$

The form on N is nonpositive. If it is negative definite (including $N = 0$), project v orthogonally to $N^\perp$. The result w has positive square and is fixed by ϕ^* , by (168). The Hodge index theorem makes the $(1, 1)$ complement of w negative definite; the real $(2, 0) + (0, 2)$ summand is positive definite. Together these give a positive definite real norm on $H^2(Q, \mathbb{R})$ preserved by the integral action. The cyclic image is finite, so in particular its eigenvalues on $H^{2,0}(Q)$ are roots of unity. If N has a radical, the Hodge index theorem makes it a rational invariant isotropic line. Its orthogonal quotient has the definite Hodge summands and integral lattice used in (164); the image on $H^{2,0}$ is again finite.

These exhaust the possibilities for N . Since $\phi^*\eta^{\otimes m}$ is proportional to $\eta^{\otimes m}$, the nonzero form η itself is an eigenform: its meromorphic ratio to its pullback has an m th power which is constant, and hence is constant. Its eigenvalue, and thus the cycle constant, is a root of unity. Taking a common power over the finitely many cycles and then rescaling the node sections solves all conductor comparisons. $\square$

8.4 Projective clusters and log-trivial linking

Every remaining cluster consists of projective vertices. Indeed, Lemma 8.6 excludes a dominant edge from a non-Moishezon vertex to a Moishezon one, and a Moishezon vertex is projective by the klt-type and Namikawa argument above. All the strata and birational diagrams used in such a cluster are algebraic in characteristic zero.

For loop finiteness we use the projective B -pluricanonical-representation theorem: for a projective lc pair with effective rational boundary and semiample rational adjoint, its B -birational group acts with finite image on sufficiently divisible log-pluricanonical systems [33, Theorem 1.1]. This applies to our normal projective stratum objects with their full adjunction boundaries. The remaining issue is to specify enough internal arrows so that the invariant object sections descend to the nodes. We give the required linking arguments.

Lemma 8.10 (Connected components of the non-klt locus). *Let two effective projective lc pairs be crepant bimeromorphic, and suppose each is klt after decreasing its floor. On a common projective log resolution write their common crepant subboundary as Θ , and put*

$$T_\Theta = \Theta^{\leq 1}, \quad N = -\lfloor \Theta^{\leq 1} \rfloor.$$

Then N is effective and exceptional for each projection, and T_Θ maps with connected fibers onto the non-klt locus of either pair. In particular their floor supports have the same number of connected components.

Proof. Fix either projection d . Nonexceptional coefficients are those of an effective subboundary, so the negative part contributing to N is exceptional. The divisor

$$N - T_\Theta = -\lfloor \Theta \rfloor$$

differs from the smooth klt adjoint $K + \{\Theta\}$ by $-(K + \Theta)$, a pullback from the base. It is thus relatively nef and big; relative bigness here uses that d is birational. Relative Kawamata–Viehweg vanishing gives

$$R^1 d_* \mathcal{O}(N - T_\Theta) = 0 \quad (169)$$

[48]. Normality gives $d_* \mathcal{O}(N) = \mathcal{O}$. The divisor sequence and (169) therefore give a surjection

$$\mathcal{O} \longrightarrow d_* \mathcal{O}_{T_\Theta}(N).$$

It factors through $d_* \mathcal{O}_{T_\Theta}$, which injects into $d_* \mathcal{O}_{T_\Theta}(N)$ because N has no component in T_Θ . Thus $d_* \mathcal{O}_{T_\Theta}$ is the structure sheaf of the reduced image of T_Θ . This image is the non-klt locus. Stein factorization, or the absence of nontrivial fiberwise idempotents in this equality, shows that the fibers are connected. A proper surjection with connected fibers induces a bijection on connected components. The common T_Θ gives the assertion for both projections. Since the pairs become klt after decreasing the floor, their non-klt loci are exactly their floor supports. $\square$

We also use the following elementary consequence of relative degree zero. Let $E \geq 0$ be a rationally Cartier divisor on a connected projective fiber of a contraction, with $E \cdot C = 0$ for every contracted curve C . If the fiber meets E , its whole support is contained in E . Otherwise a chain of projective curves joining the part in E to a point outside it contains a curve not in E which meets E . Its intersection with E is positive, a contradiction. We shall refer to this as the *whole-fiber property*. In particular, a vertical floor prime of degree zero which meets a fiber meets every horizontal floor component, since each horizontal component surjects onto the contraction base.

Lemma 8.11 (Log-trivial internal linking in dimension at most three). *Let (Y, Γ) be a connected normal projective dlt pair of dimension at most three, with effective rational boundary, nonempty floor, and $K_Y + \Gamma \sim_{\mathbb{Q}} 0$. Suppose the underlying space admits an effective rational klt pair. Then the normalized minimal lc centers, with chain adjunction, can be connected by B -birational comparisons preserving the canonical residues of a common trivializing log-pluricanonical section in sufficiently divisible even degrees.*

Proof. Take a projective small $\mathbb{Q}$ -factorialization for the underlying klt pair [9]. It is crepant for the full adjunction pair and isomorphic at general SNC stratum points. The full pair remains dlt, its centers correspond bimeromorphically, and the chain adjunctions and even residue maps correspond on common resolutions. We may therefore work on this $\mathbb{Q}$ -factorial model.

Proceed by dimension. In dimension one, a unit marking on a log-trivial curve forces a rational curve; there are at most two unit markings. For two markings the even residues of the logarithmic generator agree by (158). A single marking requires no comparison.

Suppose first that $[\Gamma]$ is connected. Each normalized floor component with full adjunction again has a dlt log-trivial pair, and it admits a klt pair by adjunction keeping only that component in the ambient boundary. Within it use induction if there are proper lc subcenters; otherwise that component is itself the minimal lc center. For two successive intersecting floor components choose a minimal lc center in their intersection. Such centers exist by ordinary dlt stratum adjunction; intersecting distinct unit primes meet in lc strata, as is also seen on a general transverse surface. The two even chain restrictions through this shared center agree. Connectedness of the floor then links all the minimal centers.

Suppose instead that the floor is disconnected. Decrease all its coefficients by one fixed small positive rational ϵ . The resulting pair is klt, and its adjoint is $\mathbb{Q}$ -linearly equivalent to $-\epsilon[\Gamma]$, so is not pseudo-effective. The projective klt MMP in dimension at most three ends in a $\mathbb{Q}$ -factorial elementary Mori fiber space

$$Y' \longrightarrow P$$

[53, 9]. These steps are crepant for the full log-trivial pair. One may see this by transporting a trivializing pluricanonical tensor: there is no extraction, its divisor remains the negative of the corresponding multiple of the pushed full boundary, and the resulting pullback formulas agree. Equivalently, the exceptional comparison of the full, relatively numerically trivial adjoints is zero by negativity.

The pushed floor is relatively ample and supports the entire non-klt locus, because its decrease is klt. Lemma 8.10 shows that it is still disconnected. Some floor component is horizontal by relative ampleness. A vertical floor prime has degree zero on the Mori ray, since it misses a general fiber. By the whole-fiber property it contains every fiber it meets, and so meets every horizontal floor component. Every other vertical prime is likewise connected to the horizontal components. The presence of any vertical floor would consequently connect the entire floor. There are therefore no vertical floor primes.

Disconnectedness is now visible on a general fiber, since every component is horizontal. That fiber must be a curve. Indeed, on a normal projective Cohen–Macaulay fiber of dimension at least two an effective ample divisor has connected support. To check this, take a large Cartier multiple A of it. Serre duality and Serre vanishing give $H^1(\mathcal{O}(-kA)) = 0$ for $k \gg 0$; the divisor sequence then gives $H^0(\mathcal{O}_{kA}) = \mathbb{C}$, implying connected support. The general fiber here is of klt type and hence Cohen–Macaulay, and the floor restricts to an effective ample divisor. Its restriction is lc and becomes klt after decreasing the floor, as can be checked on the restricted common log resolution.

The general fiber is consequently a rational curve with exactly two degree-one unit markings: it has a nonempty boundary and log-trivial adjoint, the total weighted degree is two, and disconnectedness gives two different horizontal components. Over a smooth projective model R of P we obtain a bimeromorphic model $\mathbb{P}^1 \times R$ with these marks as zero and infinity. The trivializing tensor is, over the function field and hence meromorphically, a power of the two-pointed $d \log$ generator times a meromorphic pluricanonical tensor on R . Resolving its divisor on R , the crepant subboundary on the product is therefore

$$\{0\} \times R + \{\infty\} \times R + \mathrm{pr}_R^* B_R, \quad (170)$$

where B_R may be signed, is log smooth, and has all coefficients at most one.

No vertical coefficient in (170) can equal one. Such a prime would connect the two sections by an SNC path of unit components. This path remains connected through unit components on a common resolution mapping to Y' . Explicitly, when a smooth blowup separates two adjacent unit components at their general codimension-two crossing, its exceptional divisor has coefficient one and joins their strict transforms. Blowups not separating that crossing leave the connection. The image of the path would lie in the non-klt locus of Y' and meet both disconnected floor components, a contradiction.

All vertical coefficients are therefore strictly below one. The log-smooth discrepancy formula now says that the only zero-log-discrepancy places are the two horizontal sections themselves. In fact zero places of a log-smooth subboundary with coefficients at most one are generated by its unit strata; here the two unit divisors are disjoint and there are no deeper unit strata. Both section valuations occur as divisors on the original pair, since its floor is disconnected. They are exactly its minimal lc centers. Pair them bimeromorphically through R . Their residue restrictions from the common generator agree in even powers by (158). Since these restrictions trivialize their adjunction lines, equality as meromorphic tensors on a common resolution gives B -crepancy. This finishes the disconnected case and the induction. $\square$

Lemma 8.12 (Matching in projective clusters with $r = 0$). *Every projective cluster with $r = 0$ has a nonzero matching tuple.*

Proof. Use as objects the normalized minimal lc centers of the nodes, with their iterated full adjunction pairs. Their adjoints are torsion. Lemma 8.11 supplies internal arrows connecting all objects at a node and preserving even restrictions from a trivializing node section.

For a matched conductor surface choose a minimal lc center inside it. More precisely, choose a minimal ambient lc stratum contained in the double stratum. It is a minimal center for the chains on both sides: any further subcenter would again be a nested ambient stratum, by the generic SNC description. Its shared normalization gives an arrow between the two node objects, with the symmetric even residue comparison from V .

The projective representation theorem gives finite loop images. Lemma 8.4 therefore provides nonzero invariant sections in a common degree. They are trivializing sections on the objects. At a fixed node, their scalar ratios to restrictions of one trivializing node section agree by the internal arrows, so one node section induces all of them. Across a conductor surface the two node restrictions are trivializations of the same torsion line. Their ratio is constant on that connected normal surface. It is one on the chosen minimal center by the even residue comparison, so it is one everywhere. The node sections thus match on every conductor surface. $\square$

The preceding linking argument also explains the usual construction of pre-admissible sections; compare [37, Section 5] and [54]. We have given it here to retain the actual residue comparisons needed for the present ambient line.

Lemma 8.13 (Matching in projective clusters with $r = 2$). *Every projective cluster with $r = 2$ admits a nonzero matching tuple vanishing on all nondominant branches.*

Proof. At a node with a dominant branch, the general f_i -fiber is a smooth connected rational curve. Indeed it has $K_F + \Gamma_F \sim_{\mathbb{Q}} 0$ and a unit marking, so adjunction forces genus zero and horizontal weighted boundary degree two. Use dominant surface branches as objects. If there are two degree-one unit branches, or one degree-two unit branch, the two-mark calculation (158) gives their internal comparison or sheet-exchange involution over Z_i . These are B -crepant and preserve restrictions of base sections. A single degree-one unit branch needs no internal comparison, even if other horizontal markings have fractional coefficients.

Combine these arrows with conductor matchings and apply the projective representation theorem and Lemma 8.4. At a degree-one object, a section descends to the base by the birational comparison. At a degree-two object, divide by a local base frame. Invariance under the involution equates the two generic values, so the ratio is a meromorphic base function; normality and properness give holomorphic descent. For two degree-one objects the internal arrow equates their descended sections. Pulling back gives the desired node sections. Conductor matchings and the retained initial vanishing factors give exactly the required matching and nondominant vanishing. $\square$

8.5 Projective clusters with a curve system

It remains to treat $r = 1$. Fix a projective node (Y, Γ) with system map $f : Y \rightarrow Z$, where Z is a projective curve. On sufficiently general fibers, including fibers on simultaneous log resolutions, we have a connected normal projective dlt log surface

$$(F, \Gamma_F), \quad K_F + \Gamma_F \sim_{\mathbb{Q}} 0, \quad \Sigma_F := \lfloor \Gamma_F \rfloor \neq 0. \quad (171)$$

Normality follows from the general-slice and Cohen–Macaulay properties, and dlt follows from the restricted discrepancy formula and general transversality. Horizontal boundary components restrict reduced, while vertical components are absent on this general fiber. The floor curves are smooth and their intersections are ordinary double lc points. Effective curve adjunction shows that a floor

curve meeting another is rational and meets the other floor curves at at most two points. If there are two points, its entire log boundary is precisely these two unit markings.

Lemma 8.14 (Disconnected floor on a general surface fiber). *If Σ_F in (171) is disconnected, it consists of exactly two disjoint floor curves and there is no deeper lc center in F . The corresponding dominant floor data on Y admit an internal pairing over an intermediate projective surface $P \rightarrow Z$ with connected general fibers. Generically they are the two unit markings of log rational-curve fibers on a model crepant over Z . The two curves over a general $z \in Z$ are bimeromorphic to the same smooth connected curve P_z . Globally the pairing is either between two surfaces or an involution of one surface.*

Proof. Start on a projective crepant $\mathbb{Q}$ -factorial dlt model of Y . Decrease all floor coefficients by a fixed small rational $\epsilon > 0$ and run the projective klt threefold MMP over Z . The decreased adjoint restricts to a negative multiple of the nonzero effective floor on a general fiber, so it is not relatively pseudo-effective. The program ends in a relatively elementary Mori fiber contraction

$$Y' \longrightarrow P \longrightarrow Z, \quad (172)$$

with Y' $\mathbb{Q}$ -factorial. The full adjoint remains the actual pullback of H at every step: all steps are over Z and extract no divisors, and negativity kills the exceptional relatively numerically trivial comparison. Thus the full lc data are crepant throughout. The transformed floor Σ' is relatively ample over P and supports the full non-klt locus, since its decrease is klt. Both maps in (172) have connected fibers; for the second this also follows by pushing the structure sheaf of the connected system map through the first.

On a common smooth projective resolution of the general fibers F and $F' = Y'_z$, the crepant subboundary is the same. Lemma 8.10, applied to these surfaces, shows that the floor support on F' remains disconnected. This application does not require that the transformed full pair be dlt: it is lc, and its decrease is klt, which are the hypotheses used in that lemma.

If $\dim P = 1$, the floor on F' would be ample and connected. One may use the connected-support argument in the proof of Lemma 8.11, or the surface Hodge index theorem: a disjoint splitting of an ample rational divisor would give orthogonal classes with positive square. Consequently $\dim P = 2$. A general Mori fiber over P is rational, with total horizontal weighted boundary degree two and at least one unit marking. A vertical floor prime has degree zero on the Mori ray. If it dominates Z , the whole-fiber property gives, on a general F' , a full fiber over a point of P_z , meeting every horizontal floor curve. Every other vertical floor contribution likewise joins the horizontal ones. All components restricted from a horizontal surface dominate P_z for general z , after discarding the finitely many nongeneral base values. Thus such a vertical floor prime would connect the entire floor of F' , which is impossible.

There are therefore no vertical floor contributions on general F' . Disconnectedness and the degree sum two force exactly two disjoint degree-one floor curves over P_z . The curve P_z is smooth and connected for general z : the normal surface base has only isolated singularities, and generic smoothness and connected fibers apply.

We also need to exclude hidden deeper zero-discrepancy places. Choose the rational fiber coordinate x over P_z with these two curves as zero and infinity. This gives a bimeromorphic product model $\mathbb{P}^1 \times P_z$ of F' . A divisible pluricanonical tensor trivializing its full log adjoint is, in relative notation,

$$(d \log x)^{\otimes m} \otimes \gamma, \quad (173)$$

where γ is a meromorphic m -canonical tensor on P_z . There are no other horizontal boundary terms because the two unit marks exhaust the degree. The crepant subboundary on the product is the two unit sections and vertical fibers with coefficients $-\text{ord}_p(\gamma)/m \leq 1$.

A coefficient-one vertical fiber would join the two sections by an SNC unit path. Successive point blowups preserve this path through unit components: a blowup at a crossing of two unit components inserts a unit exceptional curve. On a common resolution its image in F' would join the two disconnected floor curves inside the non-klt locus. Thus every vertical coefficient is strictly below one. The log-smooth discrepancy calculation now leaves only the two horizontal sections as zero-log-discrepancy places. They must both appear as divisors on F , since its floor is disconnected. Hence its floor consists of those two curves and has no deeper lc center.

In particular, no zero-discrepancy place horizontal over Z is extracted or lost between Y and Y' : restricting such a place to a general surface fiber would give another zero-discrepancy place there. The two Mori marks consequently correspond bimeromorphically to the original dominant floor data. Pair them over P , using normalization of the degree-two cover when the two marks belong globally to one surface. The full crepant equalities over Z transport local base frames, and the even d log-residue calculation gives the asserted internal B -comparison. $\square$

Classify a node as *type I* if it has a deeper curve lc center dominating Z , and as *type II* otherwise. In type I the floor of the general surface fiber is connected and has crossings, by Lemma 8.14. Every dominant floor surface contains a deeper dominant center: every vertex of the connected floor graph meets another, and its crossings trace such centers. In type II the floor is either a single smooth curve, or the disconnected pair of that lemma. These types are constant across dominant edges. A shared surface has the same full adjunction on both sides; a further lc center which dominates the system curve on one side does so on the other, equivalently because its curve adjoint, the restriction of J , has positive degree. The generic SNC nesting identifies the further center on both chains. Since every branch of a type I node contains such a center, the type propagates through the cluster.

Lemma 8.15 (Type II matching). *A type II projective cluster with $r = 1$ admits a nonzero matching tuple with the required nondominant vanishing.*

Proof. Use dominant surface branches as objects. For a single smooth floor curve on a general fiber, the corresponding surface has connected general fibers over Z , so its sections come from Z . In the disconnected case use the internal arrows of Lemma 8.14. Over a general z , the two curves are bimeromorphic to the connected curve P_z . The values of a section relative to a local base frame are constant on each of them, and internal invariance equates the constants. This remains true when the two curves are in one global surface and the arrow is an involution. Thus invariant sections descend meromorphically and then holomorphically to Z , using normality and properness.

The projective representation theorem and Lemma 8.4 provide the invariant nonzero object sections. Their descended base sections pull back to node sections. Matching follows from the conductor arrows, and vanishing from the retained initial base factors. $\square$

Lemma 8.16 (Type I matching by flagged curves). *A type I projective cluster with $r = 1$ also admits a nonzero matching tuple with the required nondominant vanishing.*

Proof. The objects now are normalized curve strata, *flagged* by the node Y_i , a dominant surface branch S at that node, and a further normalized curve branch on S dominating Z_i . The flag records the actual chain of residue maps. Give each curve its full effective adjunction boundary and the actual restricted line J_L .

There are three kinds of arrows. First, identify the corresponding flags across a shared conductor surface. Second, identify the flags through the two surface branches at a deeper curve center within one node. The generic SNC calculation and even residues identify these chains. Third, consider a fixed S and the curve Stein factor of $S \rightarrow Z_i$. If a general connected fiber has two floor crossings, pair these two markings over that Stein curve. They give a bimeromorphic map between curve

objects or an involution of one object. By the surface-fiber description preceding Lemma 8.14, this fiber is log rational with exactly these two unit points. Thus (158), now applied to the full adjunction on S , gives a B -crepant arrow preserving restrictions of base sections. A floor fiber curve with only one crossing requires no internal pairing of points.

These arrows connect all flagged crossing points over a general $z \in Z_i$ within a node. Indeed, the whole floor graph of F is connected, each floor curve meets another at one or two points, the two crossings on a curve are paired by the third kind of arrow, and the two flags at a crossing are paired by the second. This connects the flags along every path in the connected graph.

All curve objects are projective lc pairs with semiample adjoints, so the projective representation theorem gives finite loop images. Apply Lemma 8.4, with initial sections pulled from the respective node bases and vanishing on the required nondominant images. In the common final degree, divide the invariant curve sections by a local pulled-back base frame. The preceding connectivity says that their values coincide on all flag sheets over a general z . They therefore define one meromorphic base section. It is holomorphic, since its pullbacks to the finite dominating curve covers are holomorphic and the base is normal. Pulling it back gives a section on the node.

Across a dominant conductor surface S , equality on a chosen dominant further curve implies equality of the two node restrictions on all of S . To see this precisely, both restrictions are sections of the semiample actual line mJ_S , hence are pulled from its connected-fiber Stein system base. That base is a curve, and the chosen curve stratum dominates it, since it dominates Z_i . Pullback of sections to that stratum is injective. The canonical even adjunction compares these pullbacks, which agree by the first kind of arrow; hence the original restrictions agree. Finally, the initial vanishing factors and dominance of the objects show that the node sections vanish on every nondominant branch. This proves the claim. $\square$

8.6 Completion of whole-boundary nonvanishing

Proof of Proposition 8.1. Take a cluster of vertices of maximal system dimension r . If it is isolated, choose a nonzero high base section vanishing on all nondominant branch images, as explained above. Otherwise $r \leq 2$. A cluster containing a non-Moishezon vertex is entirely non-Moishezon and has $r \leq 1$; Lemmas 8.8 and 8.9 handle its two possibilities. Every other cluster is projective. Lemmas 8.12, 8.13, 8.15, and 8.16 cover all its possibilities.

In each case we have, in one sufficiently divisible even degree, a nonzero tuple of node sections matching on dominant edges and vanishing on all nondominant branches. Put zero on components outside the chosen cluster and take a further common power if necessary so that mJ is Cartier on the ambient V . At every conductor surface either the two sections match by construction or both restrictions are zero. Lemma 8.2 therefore descends the tuple to an actual section of $\mathcal{O}_D(mJ)$. It is nonzero because its restriction to a node in the chosen cluster is nonzero. This proves (154) on the whole reduced boundary. $\square$

9 Extension and completion in algebraic dimension zero

We now combine the preceding constructions. The point of using a *reduced* boundary is that its whole floor has a section by Proposition 8.1, while the zero-Lelong metric makes the obstruction to extending that section vanish if the ambient adjoint has no sections. We first isolate the elementary finiteness argument needed for this extension.

9.1 Finite divisorial support and eventual vanishing

Lemma 9.1. *Let M be a smooth compact Kähler manifold with $a(M) = q(M) = 0$. Then M has finitely many prime divisors, and their cohomology classes are linearly independent over $\mathbb{R}$.*

Proof. Suppose a finite collection of distinct prime divisors satisfies a nonzero rational relation in $H^2(M, \mathbb{Q})$. Clearing denominators and then torsion gives a nonzero integral divisor E with $c_1(\mathcal{O}_M(E)) = 0$ in integral cohomology. The exponential sequence and $H^1(M, \mathcal{O}_M) = 0$ imply $\mathcal{O}_M(E) \simeq \mathcal{O}_M$. Its canonical meromorphic section is then a meromorphic function with divisor E . Since $a(M) = 0$, every meromorphic function is constant, so $E = 0$, a contradiction. The classes are rational, so a real dependence would give a rational dependence by linear algebra. They are therefore linearly independent over $\mathbb{R}$; their number is bounded by $\dim H^2(M, \mathbb{R})$. $\square$

Lemma 9.2 (Eventual vanishing). *Let M be as in Lemma 9.1, and let L be a rational holomorphic line bundle with $\kappa(M, L) = -\infty$. For every fixed holomorphic vector bundle $\mathcal{E}$,*

$$H^0(M, \mathcal{E} \otimes \mathcal{O}_M(mL)) = 0 \quad (174)$$

for all sufficiently large m in any fixed sufficiently divisible sequence of integral multiples.

Proof. Assume the contrary. Interpret a section as a morphism $\mathcal{O}_M(-mL) \rightarrow \mathcal{E}$; this requires no meromorphic frame of L . Among all such morphisms choose a tuple $s_1, \dots, s_k$, with respective indices $m_1, \dots, m_k$, whose generic rank is maximal. Let $\mathcal{H} \subset \mathcal{E}$ be the saturation of the image of their direct sum. Every further such morphism has image in $\mathcal{H}$: otherwise adding it would increase generic rank. A nonzero further section can replace at least one of the s_j while preserving generic rank.

There are infinitely many increasing indices m with a nonzero section, so one fixed replacement index j works for infinitely many of them. Taking the nonzero determinant gives effective integral divisors in the actual line bundles

$$\det \mathcal{H} \otimes \mathcal{O}_M((c + m)L), \quad c = \sum_{i \neq j} m_i. \quad (175)$$

The determinant is interpreted reflexively; on the smooth space its double dual is a line bundle, and the wedge morphism extends to it.

By Lemma 9.1, all the effective divisors in (175) have the same finite set of possible prime components. Their coefficient vectors lie in $\mathbb{Z}_{\geq 0}^N$. Such an infinite sequence has an infinite componentwise nondecreasing subsequence: successively pass to a constant or increasing subsequence in each coordinate. Choose two members with indices $m' < m''$. Subtracting their divisors gives an effective divisor in the actual line $(m'' - m')L$, contrary to $\kappa(M, L) = -\infty$. This determinant argument is related to the nonvanishing method in [46]. $\square$

9.2 Extension from a nonempty reduced boundary

Let T_0 be a smooth compact Kähler non-uniruled fourfold with $a(T_0) = q(T_0) = 0$, and let G_0 be a reduced SNC divisor. Apply Proposition 7.20. Write its nonextracting output as (T, G) , with

$$A = K_T + G, \quad h : (V, D) \longrightarrow (T, G), \quad J = K_V + D = h^*A.$$

Here T is globally strongly $\mathbb{Q}$ -factorial and klt, K_T remains pseudo-effective, A is analytically nef, and (V, D) is the projective crepant dlt auxiliary model with reduced boundary. The pullback of J to a smooth resolution has a semipositive singular metric with zero Lelong numbers.

Proposition 9.3. *In this situation, $G \neq 0$ implies $\kappa(T, A) \geq 0$.*

Proof. If $G \neq 0$, then $D \neq 0$. By Proposition 8.1, a positive Cartier multiple of $J|_D$ has a nonzero section on the whole reduced divisor D . Increase this multiple to clear the ambient index. We show that, under the contrary assumption $\kappa(T, A) = -\infty$, high powers of this section extend to V .

Take a simultaneous projective compact Kähler log resolution $p : M \rightarrow V$ also resolving the map to (T, G) , and put

$$L = p^* J, \quad K_M + F \sim_{\mathbb{Q}} L. \quad (176)$$

The support of F is SNC and every coefficient is at most one; negative coefficients are allowed. All comparisons here are actual rational line-bundle comparisons. Normality and projection give $\kappa(M, L) = \kappa(V, J) = \kappa(T, A)$. The space M still has $a = q = 0$.

For every sufficiently divisible m ,

$$p_* \mathcal{O}_M(mL - \lfloor F \rfloor) = \mathcal{I}_D \otimes \mathcal{O}_V(mJ). \quad (177)$$

Indeed, the nonexceptional coefficient-one components of F are the strict transforms of the primes of D . Every other coefficient-one component has center in D , since (V, D) is dlt and is klt away from D . A holomorphic function vanishing on the reduced D pulls back with order at least one along all of these components. Conversely, any poles allowed by negative coefficients of $\lfloor F \rfloor$ are exceptional. Normality and Hartogs extension remove them downstairs, while vanishing is required along each prime of D . This proves (177), after the actual pullback line is removed by projection.

The low-degree Leray sequence therefore gives an injection

$$H^1(V, \mathcal{I}_D \otimes \mathcal{O}_V(mJ)) \hookrightarrow H^1(M, \mathcal{O}_M(mL - \lfloor F \rfloor)). \quad (178)$$

No vanishing of a higher direct image is needed for this injection. Write the line on the right as $K_M + \mathcal{L}_m$, where

$$\mathcal{L}_m = \mathcal{O}_M(-K_M - \lfloor F \rfloor + mL) \sim_{\mathbb{Q}} (m-1)L + \{F\}. \quad (179)$$

For $m \geq 1$, give $\mathcal{L}_m$ the metric obtained from the zero-Lelong semipositive metric on L and the divisor metric of $\{F\}$, taking roots of identified powers if necessary. Its curvature is semipositive and its multiplier ideal is trivial. To see the latter directly, the zero-Lelong weight has every finite local exponential integrability exponent by Skoda's theorem. The SNC coefficients of $\{F\}$ are strictly below one, and hence their divisor weight has an integrability margin. Hölder's inequality combines that margin with a sufficiently high finite exponent for the zero-Lelong weight.

Hard Lefschetz with multiplier ideals [28, Theorem 0.1] gives a surjection

$$H^0(M, \Omega_M^3 \otimes \mathcal{L}_m) \longrightarrow H^1(M, K_M + \mathcal{L}_m).$$

Its source vanishes for all large divisible m by Lemma 9.2, applied to the fixed bundle $\Omega_M^3 \otimes \mathcal{O}_M(-K_M - \lfloor F \rfloor)$. It follows from (178) that

$$H^1(V, \mathcal{I}_D \otimes \mathcal{O}_V(mJ)) = 0.$$

The exact sequence for the reduced divisor D now makes restriction of sections of $\mathcal{O}_V(mJ)$ onto D surjective. High divisible powers of the nonzero floor section remain nonzero because D is reduced, and therefore extend. Their extensions push to T , since $J = h^* A$ and $h_* \mathcal{O}_V = \mathcal{O}_T$. This contradicts $\kappa(T, A) = -\infty$. $\square$

9.3 Canonical nonvanishing

Theorem 9.4. *Assume Assumptions 2.2, 2.3, and 2.4. If W is a smooth compact Kähler non-uniruled fourfold with $a(W) = 0$, then $\kappa(W, K_W) \geq 0$.*

Proof. Suppose $\kappa(W, K_W) = -\infty$. The Albanese argument of Lemma 3.4 gives $q(W) = 0$. Both $a = q = 0$ persist on every smooth compact Kähler model used below. Their canonical classes are pseudo-effective by non-uniruledness [64]. We distinguish the existence of a nonzero meromorphic section of a positive canonical power. That property is birationally invariant on smooth spaces: Jacobian comparison gives it on a common resolution, and meromorphic tensors extend over codimension two.

No signed canonical frame. Suppose first that no positive power of K_W has a nonzero meromorphic section. Resolve the finitely many prime divisors on W , and on the resulting smooth model T_0 let G_0 be the reduced union of their strict transforms and all exceptional divisors, with SNC support. This union contains every prime of T_0 : each such prime is either exceptional or maps to a prime on W . The union may be empty.

Apply Proposition 7.20. A section of a positive multiple of $A = K_T + G$ would, after pullback to a resolution and division by the meromorphic divisor factors, give a signed meromorphic pluricanonical tensor. Here G is rational Cartier by the global strong condition, and the canonical comparison has a signed rational exceptional divisor. This is impossible. By Proposition 9.3, it follows that $G = 0$.

The space T has no prime divisors: its map from T_0 extracts none, and the full prime support was included in G_0 . It is klt and has nef $K_T = A$. Its smooth resolution has algebraic dimension zero, pseudo-effective canonical class, and no meromorphic pluricanonical tensor. These are precisely the hypotheses of Proposition 6.1, which excludes this case.

A signed canonical frame. Otherwise resolve the signed divisor of a meromorphic section of mK_W . On the resulting smooth model write the actual identity

$$K_{T_0} \sim_{\mathbb{Q}} P - N, \tag{180}$$

where $P, N \geq 0$ have SNC total support and no common prime. The pullback tensor includes the resolution discrepancy, so (180) is an actual canonical identity. Set $G_0 = (\text{Supp } P)_{\text{red}}$, and again apply Proposition 7.20. Nonextraction and reflexive extension preserve

$$K_T \sim_{\mathbb{Q}} P_T - N_T, \quad G = (\text{Supp } P_T)_{\text{red}}.$$

If $G = 0$, then $P_T = 0$, and pseudo-effectivity of K_T forces $N_T = 0$. Indeed, after pullback to a resolution a nonzero effective rational Cartier divisor has strictly positive Kähler mass, whereas its negative cannot represent a pseudo-effective class.

If $G \neq 0$, Proposition 9.3 gives a section of a multiple of A . The signed divisor $P_T + G - N_T$ representing A is unique: two meromorphic sections of the same power differ by a meromorphic function, and $a(T) = 0$. The section's divisor is effective. Since $P_T + G$ and N_T have disjoint support, we again obtain $N_T = 0$. Thus in both cases K_T has nonvanishing.

Run Assumption 2.3 on the exact ordinary klt pair $(T, 0)$, and denote its nef endpoint by T' . Its input is globally strongly $\mathbb{Q}$ -factorial and its canonical class is pseudo-effective, as required. Canonical sections push forward through the nonextracting program. Applying Assumption 2.4 therefore makes $K_{T'}$ semiample. Algebraic dimension zero forces the resulting map to have a point image, so $K_{T'}$ is actually torsion.

It remains to return to a *smooth* canonical bundle. On a smooth compact Kähler resolution $\mu : M' \rightarrow T'$,

$$K_{M'} \sim_{\mathbb{Q}} \mu^* K_{T'} + E \sim_{\mathbb{Q}} E \quad (181)$$

for a possibly signed exceptional rational divisor E . Let Θ be a positive current in the pseudo-effective class $c_1(K_{M'})$. If $\omega_{T'}$ is a Kähler form on T' , exceptionality gives

$$\int_{M'} \Theta \wedge \mu^* \omega_{T'}^3 = 0.$$

The measure is nonnegative and $\mu^* \omega_{T'}$ is strictly positive on the isomorphism locus. Hence Θ is supported in the analytic exceptional/non-isomorphism locus. The support theorem for positive closed $(1,1)$ -currents expresses it as an effective real sum of divisorial currents [25]. By Lemma 9.1, the prime classes on M' are linearly independent. Comparison with (181) therefore forces every coefficient of E to be nonnegative. The actual identity $K_{M'} \sim_{\mathbb{Q}} E \geq 0$ contradicts smooth birational invariance of Kodaira dimension and completes the proof. $\square$

9.4 The good minimal model

Proof of Theorems 1.2 and 1.1. The reduction in Proposition 3.5 treats the projective, uniruled and irregular cases and leaves smooth non-uniruled non-Moishezon fourfolds with $q = 0$. Positive algebraic dimension is handled by Proposition 4.1; algebraic dimension zero is handled by Theorem 9.4. The sections push to the actual adjoint of the nef pair exactly as in the reduction. This proves Theorem 1.2.

Finally start Assumption 2.3 from the original pair (X, B) . Its endpoint (Y, B_Y) is in the required global strong category, with nef actual adjoint, no extraction, and all the stated discrepancy inequalities. Theorem 1.2 gives a first section there. Assumption 2.4 then makes a positive Cartier multiple of that *same endpoint* adjoint globally generated. These are all the assertions of Theorem 1.1. $\square$

Part II

Projective abundance from logarithmic subadditivity

A Conditional projective log abundance

Appendices A–I establish the conditional projective abundance theorem used in the main proof. Its logarithmic-Iitaka premise, Assumption A.1, follows from the reduction in Lemma 3.1. We give the complete argument, including its signed-boundary, current-theoretic, and Frobenius constructions.

The log abundance conjecture asserts that a nef log canonical divisor on a projective log canonical pair is semiample. Its conclusion turns a numerical positivity condition into a morphism defined by pluricanonical sections. In the minimal model program, abundance complements the existence of minimal models: a nef adjoint should determine the appropriate canonical fibration. The existence and finite-generation theorems for big klt adjoints are central foundations of this program [9]. The lower-dimensional reductions relevant here are developed in [44, 35].

These appendices prove the full rational-boundary assertion under one explicit subadditivity assumption. The result is a positive *conditional* resolution of the log abundance conjecture. It is not an unconditional abundance theorem.

A.1 The assumption and the theorem

Assumption A.1 (Logarithmic Iitaka subadditivity). Let $f : X \rightarrow Y$ be a surjective morphism with connected fibers between smooth connected projective complex varieties. Let D_X, D_Y be reduced effective simple normal crossing divisors, allowing zero boundaries, such that

$$\mathrm{Supp}(f^*D_Y) \subseteq \mathrm{Supp}(D_X).$$

For a very general smooth fiber F , put $D_F = D_X|_F$. Then

$$\kappa(X, K_X + D_X) \geq \kappa(F, K_F + D_F) + \kappa(Y, K_Y + D_Y).$$

Here D_F is reduced with simple normal crossings and $K_F + D_F \sim (K_X + D_X)|_F$. The convention $(-\infty) + b = -\infty$ applies also when $b = -\infty$, and the zero divisor on a point has Iitaka dimension zero.

Assumption A.1 requires no nefness, abundance, bigness, or good-model hypothesis, and does not require f to be smooth away from the boundary. The divisor D_X may contain additional components. No fractional-boundary, orbifold, nonprojective, Hodge-conjectural, or arithmetic extension is assumed. In fact, the proof uses only $D_X = D_Y = 0$, in the Albanese reduction in Appendix F.

Theorem A.2 (Conditional log abundance). *Assume Assumption A.1. Let k be an algebraically closed field of characteristic zero, and let (X, B) be a normal projective log canonical pair over k , where B is an effective $\mathbb{Q}$ -divisor and $K_X + B$ is $\mathbb{Q}$ -Cartier. If $K_X + B$ is nef, then it is semiample: for some integer $m > 0$, the divisor $m(K_X + B)$ is Cartier and $\mathcal{O}_X(m(K_X + B))$ is generated by its global sections.*

A.2 Main constructions

The proof proceeds by simultaneous induction on smooth nonvanishing and existence of good log minimal models, as set out in Appendix B. Its principal constructions are as follows.

First, Appendices C–E prove an abundance criterion for a nef reduced dlt adjoint that has a *signed* rational representative supported on its boundary. Semiampleness on that boundary is supplied inductively. A normalized root construction separates its positive and negative parts. An infinitesimal obstruction is placed in the lowest piece of a projective Hodge-module direct image. Negative-ample vanishing on a finite cover of a root gerbe kills the obstruction. Algebraization then produces compact numerically trivial subvarieties, and nef reduction yields rational-linear descent to a big divisor on a base.

Second, Appendix F subjects a hypothetical smooth nonvanishing counterexample to geometric exclusions. Algebraic webs and positive currents force very-general proper subvarieties to be of general type. An ascending chain condition for the Lelong numbers of a fixed current at valuations of bounded discrepancy converts asymptotically small intersection gaps into exact zero gaps. This rules out canonical-class positive currents that are pulled back from a lower-dimensional base on a dense open, as well as moving curves of bounded normalized degree and genus.

Third, Appendices G and H compare two jet estimates. Moving base-locus estimates on a projective bundle over a product create a large Seshadri constant while the corresponding one-factor section orders remain small. After all geometric choices have been fixed, reduction modulo large primes converts these two estimates into incompatible determinant multiplicities. The Frobenius diagonal filtration and a uniform curve-slope bound are the numerical ingredients of this final comparison.

These constructions are proved below; they are not additional parts of Assumption A.1. Established results from the minimal model program, Hodge modules, and positivity theory are invoked with their hypotheses at the point of use. The order of the induction is important: the proof of smooth nonvanishing in dimension n uses good models only in dimensions below n . Appendix I closes the induction and transfers the complex conclusion to arbitrary algebraically closed fields of characteristic zero.

A.3 Conventions

A variety is integral. Canonical divisors are chosen compatibly on birational models. For a divisor E on a smooth model $p : W \rightarrow X$, our log discrepancy is

$$a(E; X, B) = 1 + \text{ord}_E(K_W - p^*(K_X + B)).$$

Log canonicity means nonnegative log discrepancies, and klt means strictly positive log discrepancies. We use the usual dlt and slc conventions. A $\mathbb{Q}$ -Cartier divisor L is nef if $L \cdot C \geq 0$ for every integral curve C . Its numerical dimension, when nef, is

$$\nu(L) = \max\{j : L^j H^{\dim X - j} > 0\},$$

with H ample. For non-nef pseudo-effective divisors, statements about numerical dimension use Nakayama's κ_σ , as specified in the relevant reduction. These notions are not interchanged without a hypothesis that justifies doing so.

The Iitaka dimension is the maximum dimension of the images of the complete systems of integral multiples, and is $-\infty$ when all these systems are empty. Numerical and rational-linear equivalence are denoted by $\equiv$ and $\sim_{\mathbb{Q}}$, respectively. Unless a different base field is specified, the constructions are over $\mathbb{C}$. A very general point avoids a countable union of proper closed subvarieties.

The transfer to arbitrary algebraically closed characteristic-zero fields is made only after the complex argument has been completed.

B The inductive framework

All varieties in this section are projective over $\mathbb{C}$. We first isolate the established minimal-model results that will be used, and explain exactly where the new signed-representative theorem enters the induction.

For an effective real boundary Δ , a good log minimal model of (X, Δ) is a log minimal model on which the adjoint divisor is semiample. For real divisors, semiamplicity means real linear equivalence to the pullback of an ample real divisor by a contraction. We allow the usual definition of a log minimal model in which extracted prime divisors are included in its boundary with coefficient one. For a pseudo-effective divisor A , we use $\kappa_\sigma(X, A)$ for Nakayama's numerical dimension. When A is nef this agrees with the intersection-theoretic numerical dimension $\nu(X, A)$.

Fix an integer $n \geq 1$.

Assumption B.1 (Lower-dimensional good models). Every projective log canonical pair of dimension less than n , with real boundary and pseudo-effective adjoint divisor, has a good log minimal model.

The dimension-zero case is immediate. Our inductive step will first prove smooth nonvanishing in dimension n under Assumption B.1, and then apply Proposition B.6 below. The propositions on boundary restrictions and special termination preceding that reduction do not assume smooth nonvanishing in dimension n .

B.1 Comparison and boundary restrictions

Lemma B.2 (Comparison with a nef model). *Let (X, Δ) be log canonical and let (X', Δ') be a log minimal model. On a common resolution*

$$X \xleftarrow{u} W \xrightarrow{v} X'$$

there is an effective v -exceptional divisor F such that

$$u^*(K_X + \Delta) = v^*(K_{X'} + \Delta') + F. \quad (182)$$

If $K_X + \Delta$ is nef, then $F = 0$. Consequently a nef rational adjoint is semiample whenever it has a good log minimal model. If (X, Δ) is klt, its log minimal model is klt and extracts no divisors.

Proof. Put $F = u^*(K_X + \Delta) - v^*(K_{X'} + \Delta')$. Discrepancy improvement on divisors of X gives $u_*F \geq 0$, while $-F$ is u -nef because $K_{X'} + \Delta'$ is nef. The negativity lemma therefore gives $F \geq 0$. At a prime of W that is not v -exceptional, the coefficient of F is zero unless that prime is extracted on X' . In the latter case it equals the negative of its log discrepancy over (X, Δ) . It is therefore nonpositive, and hence zero. Thus F is v -exceptional. If the source adjoint is nef, then F is also v -nef, so the negativity lemma gives $F = 0$.

Equality of the pullbacks transfers semiamplicity, since a proper birational morphism onto a normal variety has direct image of its structure sheaf equal to the structure sheaf. More explicitly, global sections of a Cartier multiple and its pullback coincide; generation upstairs implies generation downstairs. Finally, for a klt source the coefficient at an extracted prime would be strictly negative, which is impossible. The remaining discrepancy inequalities show that the model is klt. $\square$

Proposition B.3 (Semiampleness on the reduced boundary). *Assume Assumption B.1. Let (V, D) be a projective $\mathbb{Q}$ -factorial dlt n -fold with D reduced and $M = K_V + D$ nef. Then $M|_D$ is semiample as a rational line bundle on the reduced scheme D .*

Proof. The ambient variety V is klt. The reduced floor of a $\mathbb{Q}$ -factorial dlt pair is S_2 , and it has ordinary double crossings in codimension one; its irreducible components are normal. For the S_2 assertion one can work locally with the cyclic class cover of the $\mathbb{Q}$ -Cartier divisor D . This cover is quasi-étale and klt, hence Cohen–Macaulay. The eigensheaf $\mathcal{O}_V(-D)$, being a direct summand of its finite direct image, is Cohen–Macaulay. The divisor sequence then shows that D is Cohen–Macaulay.

Divisorial adjunction on the normalization $\coprod D_i \rightarrow D$ gives lc pairs $(D_i, \text{Diff}_{D_i}(D - D_i))$. The different includes the conductor with coefficient one. Removing that conductor contribution defines an effective boundary on D ; together with D it is an slc pair whose adjoint line bundle is $M|_D$. The divisible residue identifications hold in codimension one, with even powers eliminating residue signs at double crossings, and extend by S_2 .

Each normalized adjoint is nef and semiample by Assumption B.1 and Lemma B.2. Semiampleness descends from the normalization by the slc gluing theorem [33, Theorem 1.5, equivalently Theorem 4.3]. This gives the required statement on the whole reduced scheme, not merely on its individual components. $\square$

B.2 The special termination needed below

Proposition B.4 (Special termination over a point). *Assume Assumption B.1. Let (V, Δ) be a projective $\mathbb{Q}$ -factorial dlt n -fold with rational boundary and pseudo-effective adjoint. Choose an effective ample rational divisor A such that $(V, \Delta + A)$ is dlt and $K_V + \Delta + A$ is nef. The LMMP for $K_V + \Delta$ with scaling of A has special termination: if the program is infinite, its flipping loci are eventually disjoint from the round-down of the boundary.*

In particular, if at every stage the adjoint is rationally linearly equivalent to a signed divisor supported on that round-down, the program terminates at a log minimal model.

Proof. In an infinite ample-scaling program the scaling limit is zero: a positive limit is covered by the termination theorem for a dlt pair with an ample summand in the scaling divisor [8, Theorem 1.9(ii), equivalently Theorem 4.1(ii)]. Discard the finitely many divisorial contractions and write the flips as $V_i \dashrightarrow V_{i+1}/Z_i$, with scaling numbers $\lambda_i > 0$ tending to zero. Fix a component S_1 of $\lfloor \Delta_1 \rfloor$, write S_i for its normal strict transform, and let T_i be the normalization of its image in Z_i . Since the ambient contraction is small, $S_i \rightarrow T_i$ is projective birational. The standard discrepancy argument makes $S_i \dashrightarrow S_{i+1}$ an isomorphism in codimension one after finitely many steps [6, proof of Lemma 3.6].

Here are the precise auxiliary relative programs in that argument. Take a small projective $\mathbb{Q}$ -factorialization $p_i : S'_i \rightarrow S_i$, and put

$$D_i = K_{S'_i} + B_i = p_i^*((K_{V_i} + \Delta_i)|_{S_i}), \quad C_i = p_i^*(A_i|_{S_i}).$$

Adjunction gives a dlt pair (S'_i, B_i) , with $B_i, C_i \geq 0$, and $(S'_i, B_i + \lambda_i C_i)$ is lc. The scaling divisor has no floor component, so its restriction is effective. The number λ_i is rational, being the ratio of intersections of rational divisors on the contracted rational curve. The globally nef divisor $L_i = K_{V_i} + \Delta_i + \lambda_i A_i$ descends rationally linearly across $V_i \rightarrow Z_i$. Indeed, for sufficiently small rational $\delta > 0$, the pair $(V_i, (1 - \delta)\Delta_i)$ is klt and its negative adjoint is ample over Z_i . Relative basepoint freeness applies to a Cartier multiple of L_i ; its numerical triviality and the connected fibers give descent. Restricting and pulling back yields

$$D_i + \lambda_i C_i \sim_{\mathbb{Q}} 0/T_i.$$

By [8, Theorem 1.1(3)], a D_i -LMMP over T_i with scaling of a fresh ample divisor terminates. The theorem applies to the effective rational lc boundary $B_i + \lambda_i C_i$, with $\lambda_i C_i$ rational Cartier and the driving pair $\mathbb{Q}$ -factorial dlt. Since the base map is birational, the endpoint is a log minimal model, not a Mori fiber space. Comparison with the relatively ample model S_{i+1} identifies this endpoint with a small $\mathbb{Q}$ -factorialization S'_{i+1} , as in [8, Remarks 2.9–2.10].

These finite relative programs are genuine pieces of an absolute program with scaling of the transforms of C_1 . Throughout the i -th piece, $D_i + \lambda_i C_i$ remains the pullback of a nef divisor on T_i , hence is globally nef. Every contracted negative ray has positive C_i -degree, so its global scaling threshold is exactly λ_i . The relative cone is a face of the absolute cone, cut out by the pullback of an ample divisor on the projective base T_i ; its extremal rays are therefore absolute extremal rays. Thus the pieces concatenate as asserted in the cited remarks. Rescaling the initial scaling divisor by λ_1 , if necessary, makes its sum with the initial boundary lc and nef.

If the concatenation were infinite, its positive scaling numbers would tend to zero. Its initial adjoint is pseudo-effective: for each late scaling value, pull back the corresponding nef scaled adjoint and use the effective comparison divisor for the preceding nonpositive steps, then take the limit. Assumption B.1 therefore gives an absolute log minimal model for that initial pair. The concatenation terminates by [8, Theorem 1.9(iii)], since its zero limit is never attained. The floor-component argument of [6, Lemma 3.6] now gives special termination. No effective representative in dimension n , nor any arbitrary relative good-model assertion, has been used.

For the final assertion, a curve disjoint from the round-down has degree zero against any signed divisor supported there. It therefore cannot generate an adjoint-negative flipping ray. Special termination rules out all sufficiently late flips. There are only finitely many divisorial contractions, since each lowers the Picard number, and pseudo-effectivity excludes a Mori fiber space as the endpoint. Thus the endpoint is nef. $\square$

B.3 A uniform trivial-perturbation argument

Lemma B.5. *Let (Z, Δ) be a projective $\mathbb{Q}$ -factorial dlt rational pair, and suppose $L = K_Z + \Delta$ is nef. Set $P = \Delta$ if the pair is klt and $P = \lfloor \Delta \rfloor$ otherwise. Fix $m > 0$ such that mL is Cartier. For every rational*

$$0 < \epsilon < \frac{1}{4nm + 2},$$

every step of an LMMP for $L - \epsilon P$ is L -trivial. On all its models, the transform of L is nef and its multiple by the same integer m is Cartier.

Proof. Both $\Delta - \epsilon P$ and $\Delta - \frac{1}{2}P$ are effective klt boundaries. If R is a ray negative for $L - \epsilon P$, nefness of L shows that R is also negative for $L - \frac{1}{2}P$. Choose a rational curve C spanning this ray and satisfying the length bound

$$-(L - \tfrac{1}{2}P) \cdot C \leq 2n.$$

Writing $a = L \cdot C \geq 0$ and $p = P \cdot C$, we have

$$p \leq 2a + 4n, \quad a < \epsilon p, \quad (1 - 2\epsilon)a < 4n\epsilon.$$

Our choice of ϵ gives $a < 1/m$. Since ma is a nonnegative integer, $a = 0$.

For the driving klt contraction, the line bundle $\mathcal{O}_Z(mL)$ is numerically trivial over the contraction base. The line-bundle clause of the contraction theorem gives its descent as a line bundle, with no change of m [29, Theorem 3.74(3)]. One may also see the unchanged index directly from relative basepoint freeness: two consecutive sufficiently large powers descend, so their quotient descends.

For a flip, pull this descended line bundle back on the other side. The transformed L is nef, and mL remains Cartier.

The driving boundary remains klt throughout its LMMP. The transformed $\Delta - \frac{1}{2}P$ remains effective and is smaller than the driving boundary, so it too is klt. The same estimate and the same integer m apply inductively at every subsequent step. $\square$

B.4 Completion of the good-model step

Proposition B.6 (Inductive reduction to smooth nonvanishing). *Assume Assumption B.1. Assume in addition that every smooth projective n -fold with pseudo-effective canonical divisor has nonnegative Kodaira dimension. Then every projective lc n -fold with real boundary and pseudo-effective adjoint has a good log minimal model. In particular, every nef rational lc adjoint in dimension n is semiample.*

Proof. Hashizume's reduction gives nonvanishing and log minimal models for projective lc pairs with real boundary in dimensions at most n [44, Theorem 1.4]. Thus we may pass to a $\mathbb{Q}$ -factorial dlt log minimal model. Fix the support of its boundary, impose the rational affine constraints that its coefficient-one components remain equal to one, and take a sufficiently small rational polytope around the given boundary within those constraints. On a fixed log resolution witnessing dlt, the strict discrepancy inequalities remain strict in this neighborhood; thus its boundaries remain dlt. The rational-polytope theorem for nef adjoints [7, Remark 3.1 and Proposition 3.2(3)], applied to all extremal rays and intersected with this neighborhood, expresses the given real boundary in a finite rational simplex of dlt boundaries with nef adjoints. It is enough to prove semiampleness for these rational adjoints. Their positive real combinations are semiample, using the product of the associated contractions. For a rational adjoint, real semiampleness can also be rationalized: the finite linear equations expressing its divisor and principal-divisor coefficients have rational data, so a real solution with positive coefficients has a rational solution with the same positivity.

We therefore work with a rational dlt pair (Z, Δ) and $L = K_Z + \Delta$ nef. Real nonvanishing gives rational nonvanishing in this case: retain the finitely many divisors and principal divisors in an effective real representative and solve the resulting rational linear system with nonnegative rational coefficients. If $\kappa(Z, L) \geq 1$, lower-dimensional good models give a good minimal model by [35, Lemma 3.5]. Lemma B.2 transfers semiampleness back to Z . It remains to treat

$$\kappa(Z, L) = 0.$$

The non-pseudo-effective perturbation. Put $P = \Delta$ in the klt case and $P = \lfloor \Delta \rfloor$ in the other case. Suppose $L - \epsilon P$ is not pseudo-effective for every sufficiently small $\epsilon > 0$. Choose rational ϵ as in Lemma B.5, and run the klt LMMP with scaling to a Mori fiber space

$$(Z, \Delta - \epsilon P) \dashrightarrow (Z', \Delta' - \epsilon P') \xrightarrow{f} T.$$

Existence and termination in the non-pseudo-effective case are the established klt results of [9]. The program is crepant for L . The line-bundle descent in Lemma B.5, applied also to the final contraction, gives a nef rational divisor A on T with

$$K_{Z'} + \Delta' \sim_{\mathbb{Q}} f^* A.$$

The base has dimension less than n .

If (Z, Δ) is klt, the transformed original pair (Z', Δ') is klt as well. Ambro's descent theorem [1, Theorem 0.2] gives an effective rational boundary Δ_T such that (T, Δ_T) is klt and

$$A \sim_{\mathbb{Q}} K_T + \Delta_T.$$

All its hypotheses hold: the total pair is globally klt and effective, f is a projective contraction, and its adjoint is rationally linearly pulled back. In particular, this use requires no conjecture about semiampleness of a moduli divisor. Assumption B.1 and Lemma B.2 make A semiample.

Otherwise P' is effective and relatively ample, since $K_{Z'} + \Delta' - \epsilon P'$ is relatively antiample. Some component of the floor therefore dominates T . On a dlt blowup of the original lc pair (Z', Δ') , choose the strict transform S of such a component. Adjunction produces an lc pair on S whose nef adjoint is the pullback of A . It is semiample by the lower-dimensional hypothesis. Semiampleness of a rational line bundle descends along a proper surjection: use the equality of sections for its connected-fiber Stein factor, and norms for the remaining finite morphism. For the latter assertion, a basepoint-free system has a section nonzero at every point of a chosen finite fiber; its norm is nonzero at the point downstairs. Hence A is semiample also in this case. Crepancy transfers the conclusion back to L .

The klt pseudo-effective perturbation. We next settle the klt case when $L - \epsilon\Delta$ is pseudo-effective for some small rational $\epsilon > 0$. Nonvanishing gives

$$L \sim_{\mathbb{Q}} G_0 := H_0 + \epsilon\Delta, \quad H_0 \geq 0.$$

Take a resolution $p: W \rightarrow Z$ with reduced SNC divisor D consisting of all exceptionals and the strict support of $H_0 + \Delta$. We have

$$K_W + D = p^*L + R, \quad R \geq 0.$$

Here every component of D has positive coefficient in

$$G_W := p^*G_0 + R \sim_{\mathbb{Q}} K_W + D.$$

Indeed, at strict boundary components the difference between coefficient one and a klt boundary coefficient is positive, and at exceptional components the coefficient contributed by the adjoint comparison is the positive log discrepancy. The pushforward p_*G_W is bounded coefficientwise by a fixed multiple of G_0 . Section spaces inject under birational pushforward, so

$$0 \leq \kappa(W, K_W + D) \leq \kappa(Z, G_0) = 0.$$

Take a $\mathbb{Q}$ -factorial dlt log minimal model (V, D_V) of (W, D) . Its boundary is reduced. On a common resolution, the comparison divisor in Lemma B.2 is exceptional over V . Pushing the pullback of G_W to V therefore gives

$$K_V + D_V \sim_{\mathbb{Q}} G_V = \sum_i b_i D_i, \quad b_i > 0, \quad \text{Supp } G_V = D_V.$$

For completeness, positivity at a component extracted on V also holds: its log discrepancy over (W, D) is zero, its center lies in D , and the pullback of the full-support effective G_W has positive order there. There is no comparison-divisor coefficient at a prime retained on V . Effectivity and exceptionality in the comparison preserve the adjoint section ring, so the nef adjoint on V still has Iitaka dimension zero.

The small-perturbation hypothesis of Theorem C.1 is now explicit. If $D_V \neq 0$, choose rational c with $0 < c < \min_i b_i$. Then

$$K_V + (1 - c)D_V \sim_{\mathbb{Q}} \sum_i (b_i - c)D_i \geq 0.$$

If $D_V = 0$, pseudo-effectivity is immediate. Proposition B.3 supplies the required semiample on D_V . Theorem C.1 thus makes $K_V + D_V$ abundant. Its numerical dimension is zero, and its effective full-support representative forces $G_V = 0$, hence $D_V = 0$.

On a common resolution of $W \dashrightarrow V$, the effective pullback of p^*G_0 is now exceptional over V . It is nef because it is rationally linearly equivalent to the pullback of L . The negativity lemma forces this divisor to vanish. Therefore $G_0 = 0$ and $L \sim_{\mathbb{Q}} 0$. Together with the preceding cases, this proves the klt good-model assertion in dimension n : for a non-nef pseudo-effective klt adjoint, first take its klt log minimal model and apply what was just proved.

The remaining non-klt case. Finally suppose the original pair is not klt and a small floor perturbation is pseudo-effective. Choose rational $\epsilon > 0$ so that both

$$L - \epsilon P, \quad L - 2\epsilon P, \quad P = \lfloor \Delta \rfloor,$$

are pseudo-effective klt adjoints. Their Iitaka dimensions are zero: nonvanishing gives the lower bound, and adding the effective perturbation bounds each by $\kappa(Z, L) = 0$. The klt result just proved gives good models for both. Consequently their Nakayama numerical dimensions are zero. Since

$$2(L - \epsilon P) = L + (L - 2\epsilon P)$$

and the last summand has an effective rational representative, monotonicity and homogeneity of κ_σ give

$$\kappa_\sigma(Z, L) \leq \kappa_\sigma(Z, 2(L - \epsilon P)) = 0.$$

The nef divisor L is therefore numerically trivial. Nonvanishing is already available in this finishing step: write $L \sim_{\mathbb{Q}} E \geq 0$. For an ample divisor H , the equality $E \cdot H^{n-1} = 0$ forces $E = 0$. Thus $L \sim_{\mathbb{Q}} 0$, without any additional nonvanishing or abundance premise.

All rational nef dlt adjoints are now semiample. The polytope reduction gives the real-boundary good-model assertion, and Lemma B.2 gives the final statement for every nef rational lc adjoint. $\square$

C Geometry of a signed boundary representative

The next statement isolates the extension argument needed in the induction. Its boundary is reduced, but its prescribed representative need not be effective.

Theorem C.1 (Signed representative). *Let (X, D) be a projective $\mathbb{Q}$ -factorial dlt pair over $\mathbb{C}$, with $D = \sum_i D_i$ reduced. Suppose that*

$$L = K_X + D \quad \text{is nef}, \quad L - cD \quad \text{is pseudo-effective for some } c > 0,$$

and that

$$L \sim_{\mathbb{Q}} G = \sum_i a_i D_i, \quad a_i \in \mathbb{Q}.$$

If $L|_D$ is semiample as a rational line bundle on the reduced scheme D , then L is abundant:

$$\kappa(X, L) = \nu(X, L).$$

The proof occupies this section, the Hodge-theoretic lifting argument of Proposition D.3, and the descent argument in Appendix E. No Iitaka subadditivity assumption is used in this theorem.

C.1 The positive boundary and its small intersections

Write

$$G = G_+ - G_-, \quad D_0 = \sum_{a_i=0} D_i,$$

where G_+ and G_- are effective with disjoint prime supports. Choose a positive integer m such that $mG_\pm$, mD_0 , and mL are integral Cartier divisors and

$$N_0 = \mathcal{O}_X(mG) \simeq \mathcal{O}_X(mL).$$

Increase m so that $N_0|_D$ is generated by global sections. Fix the resulting morphism

$$\phi : D \longrightarrow P = \mathbb{P}^s, \quad N_0|_D \simeq \phi^* \mathcal{O}_P(1),$$

and let s_0 be the rational section of N_0 with divisor mG . Set $n = \dim X$ and $v = \nu(X, L)$, and fix an ample Cartier divisor H .

Lemma C.2. *If $v = n$, then L is big. If $v = 0$, then $D = 0$ and $L \sim_{\mathbb{Q}} 0$. In the remaining case $0 < v < n$, put $r = v - 1$. Every component of D has ϕ -image of dimension at most r , and some component of G_+ has image of dimension r . Moreover,*

$$\dim \phi(\text{Supp } G_+ \cap (\text{Supp } G_- \cup D_0)) < r;$$

when $r = 0$, the intersection in this formula is empty.

Proof. The assertion for $v = n$ is the numerical criterion for bigness of a nef divisor. Suppose $v < n$. Intersecting the pseudo-effective class $L - cD$ with a product of nef classes gives

$$0 \leq (L - cD)L^v H^{n-v-1} = -c \sum_i D_i L^v H^{n-v-1}.$$

Every summand on the right is nonnegative. Thus each is zero. For $v = 0$, ampleness gives $D = 0$, and the signed representative gives $L \sim_{\mathbb{Q}} 0$.

Now suppose $v > 0$. On a component of D , the numerical dimension of $L|_D$ equals its image dimension under ϕ . The preceding vanishing gives the upper bound r . On the other hand,

$$0 < L^v H^{n-v} = \sum_i a_i D_i L^r H^{n-v},$$

so a positive-coefficient component attains that bound.

Consider the symmetric matrix

$$Q_{ij} = D_i D_j L^r H^{n-r-2}.$$

Its off-diagonal entries are nonnegative: distinct effective $\mathbb{Q}$ -Cartier divisors have effective intersection cycles. The mixed Hodge index theorem, obtained by approximation with ample classes and the ordinary Hodge index theorem on complete-intersection surfaces, says that the ambient intersection form has at most one positive direction. In this form, L is isotropic, is orthogonal to every D_i , and has strictly positive pairing with H . Its orthogonal space therefore has negative semidefinite form. The same applies after pullback to a resolution, so this argument does not require X to be smooth.

The signed relation implies $Q(a_i) = 0$. Write $a = a^+ - a^-$ temporarily for the positive and negative coefficient vectors. Then

$$Q(a^+, a^+) = Q(a^+, a^-) \geq 0.$$

Negative semidefiniteness makes both sides zero and gives $Qa^+ = 0$. The vanishing rows belonging to negative or zero coefficients show that each intersection of a positive component with a negative or zero component has zero L^r -degree against H^{n-r-2} . Semiampleness on D gives image dimension less than r . If $r = 0$, an effective nonzero intersection has positive ample degree, proving emptiness. $\square$

We henceforth work in the case $0 < v < n$, and set $N = n - 1$.

C.2 A geometric package for infinitesimal lifting

The root gerbe

$$\mathcal{P} = \sqrt[\ell]{\mathcal{O}_P(1)}$$

parametrizes ℓ -th roots of the indicated line bundle, without a chosen section. Its tautological line is denoted by C , so C^ℓ is the pullback of $\mathcal{O}_P(1)$. We use the same notation for further pullbacks of C .

Proposition C.3. *Choose ℓ divisible by m and every nonzero integer $|ma_i|$, and set $a = \ell/m$. There is a normal tame Deligne–Mumford stack $\mathcal{X}$, considered on a neighborhood of a reduced Cartier divisor S of pure dimension $N = n - 1$, together with a morphism to X and a reduced boundary T having no component in common with S , with the following properties.*

- (i) *The pair $(\mathcal{X}, S + T)$ is log canonical, and a fixed isomorphism of reflexive sheaves is given by*

$$\omega_{\mathcal{X}}(S + T) \simeq \mathcal{O}_{\mathcal{X}}(aS). \quad (*_{\text{adj}})$$

The support of T is locally set-theoretically principal. The ambient space and S are Cohen–Macaulay on scheme charts. Off T , the ambient space is Gorenstein and the displayed isomorphism is ordinary Cartier adjunction data.

- (ii) *Put $J = (S \cap T)_{\text{red}}$. Both S and J are Du Bois on scheme charts, the ideal $\mathcal{I}_{J/S}$ is maximal Cohen–Macaulay, and*

$$\mathcal{B} := \mathcal{H}om_S(\mathcal{I}_{J/S}, \omega_S) \simeq \mathcal{O}_S(aS).$$

*The identification agrees off J with the residue convention specified by $(*_{\text{adj}})$.*

- (iii) *There is a finite representable morphism*

$$S \longrightarrow D \times_P \mathcal{P}$$

whose image covers $\text{Supp } G_+$. The line $\mathcal{O}_S(S)$ is the pullback of C . The image of J in P has dimension less than r , whereas the image of S has dimension r .

- (iv) *There is a smooth projective bundle $p : \mathcal{Y} \rightarrow \mathcal{P}$ and a representable projective morphism $g : S \rightarrow \mathcal{Y}$. Writing $h = p \circ g$ and $\mathcal{T} = \text{Spec}_{\mathcal{P}} h_* \mathcal{O}_S$, the morphism g factors through a closed embedding $\mathcal{T} \hookrightarrow \mathcal{Y}$, and*

$$g_* \mathcal{O}_S = \mathcal{O}_{\mathcal{T}}, \quad \mathcal{O}_S(S) = g^* C, \quad \mathcal{B} = g^* C^a.$$

- (v) *For a separated quasi-projective scheme chart $U \rightarrow \mathcal{Y}$ that is etale and of finite type, put $S_U = S \times_{\mathcal{Y}} U$. The etale morphism $S_U \rightarrow S$ extends compatibly to every nilpotent thickening qS in $\mathcal{X}$. The resulting qS_U are quasi-projective schemes, separated and quasi-finite over X .*

All the adjunction and duality identifications are compatible with changes of etale chart.

Proof. Start with the normalized simultaneous root stack of the Cartier divisors mG_+ , mG_- , and mD_0 , taking an ℓ -th root of each divisor with its section. Empty divisors cause no difficulty. Write S_+ , S_- , S_0 for the tautological root divisors. To check their reducedness, work at a generic prime where the downstairs multiplicity is b . A normalized chart of $y^\ell = x^b$ has ramification index ℓ/b , and y has order one. This applies to D_0 because m divides ℓ . The divisors are Cartier; normality and their generic reducedness imply that they are reduced.

Log ramification gives a log canonical pair with this full reduced boundary and

$$K + S_+ + S_- + S_0 \sim_{\mathbb{Q}} a(S_+ - S_-).$$

Its ambient charts are klt. Indeed they are klt off the boundary, and decreasing all boundary coefficients slightly removes all zero-discrepancy places, as can be checked on a log resolution. The integral Weil class given by the difference of the two sides is torsion. Take the finite representable cover formed from its reflexive powers, with a chosen periodicity isomorphism, and normalize. On codimension-one charts this is the usual cover of a torsion line bundle, hence is étale there. The adjoint relation becomes a fixed linear equivalence; log canonicity, klt ambient singularities, and reduced Cartier boundary divisors persist. More explicitly, before taking this cover write $B = S_+ + S_- + S_0$. The torsion reflexive sheaf is

$$\mathcal{F} = (\omega(B) \otimes \mathcal{O}(-a(S_+ - S_-)))^{**}.$$

The relative spectrum of its reflexive-power algebra, with a chosen periodicity $\mathcal{F}^{[q]} \simeq \mathcal{O}$, has a tautological evaluation trivializing the pulled-back $\mathcal{F}$ in codimension one. This is a morphism of sheaves on the cover stack itself. The resulting adjoint isomorphism is therefore equivariant on every atlas, rather than a separately chosen nonequivariant trivialization.

Blow up the ideal of $S_+ \cap S_-$ and normalize. Locally the ideal has two generators, so the ordinary blowup embeds in a relative projective line. Its fibers, and those of its finite normalization, have dimension at most one. Every exceptional divisor therefore lies over a codimension-two component of the intersection. Such a component is a dlt stratum downstairs and is generically SNC. Separate normalized Kummer charts, followed by purity for the index cover at the smooth generic locus, give the same description upstairs.

It follows that the tautological exceptional divisor E is reduced Cartier and

$$S'_+ = p_1^* S_+ - E, \quad S'_- = p_1^* S_- - E$$

are disjoint reduced Cartier divisors, where p_1 denotes this normalized blowup. No codimension-two two-branch stratum is contained in S_0 , so $p_1^* S_0$ is reduced Cartier without an exceptional component. The boundary

$$S'_+ + S'_- + E + p_1^* S_0$$

is crepant and log canonical. Write B' for this total boundary on a chart V' . The crepant equality establishes log canonicity for every valuation. Outside B' , the blowup is an isomorphism to the klt complement of the old boundary. Thus a divisor F with $a(F; V', B') = 0$ has center in $\text{Supp } B'$. As B' is effective Cartier, $\text{ord}_F(B') > 0$, and

$$a(F; V', 0) = a(F; V', B') + \text{ord}_F(B') > 0.$$

Divisors with positive pair discrepancy remain positive after dropping the boundary. This proves that V' is klt, including at higher-codimension singular loci. Set $S = S'_+$ and $T = S'_- + E + p_1^* S_0$

for the moment. Near S , the disjoint divisor S'_- contributes its unit section, and the fixed linear equivalence becomes

$$\omega(S + T) \simeq \mathcal{O}(aS).$$

The ambient charts remain klt and hence Cohen–Macaulay. Since all displayed boundary divisors are Cartier, this isomorphism also makes those charts Gorenstein.

The strict divisor S is finite over S_+ . Before normalization it lies in the zero section of the projective ratio coordinate: its fibers over S_+ have at most one point. The morphism is proper, and normalization is finite. In particular a codimension-one point of $S \cap T$ lies over a codimension-two intersection downstairs. The generic SNC description above shows that $T|_S$ is generically reduced. It is Cartier on the Cohen–Macaulay scheme S , so it is reduced everywhere.

We next remove the root characters that will not be used. The line $\mathcal{O}(S - S'_-)$ has ℓ -th power equal to the pullback of N_0 . It defines a morphism to the gerbe of ℓ -th roots of N_0 on X . Take the relative coarse space over this gerbe and call it $\mathcal{X}$. On a trivializing chart this means quotienting by the kernel of the finite group character acting on the indicated root line. These are ordinary quasi-projective quotient schemes: the preceding covers are finite, the blowup is projective, and a finite quotient preserves quasi-projectivity. The chartwise construction patches.

Retain S, T for their reduced images. Near S , the pole section is a unit, so both the line of S and its section descend through this kernel quotient. Thus S remains Cartier. A finite-group norm of a local equation for the upstairs T shows that its image is set-theoretically principal; we do not need reduced T to be Cartier. The only divisorial ramification is on the boundary. Log ramification therefore preserves the lc pair and descends the equivariant adjoint isomorphism to $(*_\text{adj})$. Cohen–Macaulayness of the ambient chart and of S follows from the finite CM covers and their invariant direct summands. For clarity, the coarse kernel acts trivially on the retained root line, and hence on $\mathcal{O}(aS)$. Equivariance of the fixed adjoint isomorphism gives the same kernel character on $\omega(S + T)$. Taking invariants therefore descends the isomorphism, with log ramification identifying the invariant log dualizing sheaf. Off T the descended adjoint isomorphism makes the canonical sheaf invertible, giving the claimed Gorenstein property there.

The reduced supports S and J are unions of log canonical centers: intersections of lc centers are again unions of lc centers. Hence both are Du Bois by [55, Theorems 1.4 and 1.7]. For the more precise coherent statement, on one quotient chart write $S^\text{up} \rightarrow S$ for the finite cover used above. Since $T^\text{up}|_{S^\text{up}}$ is reduced,

$$\mathcal{I}_{J/S} = (f_* \mathcal{O}_{S^\text{up}}(-T^\text{up}|_{S^\text{up}}))^{\text{inv}}.$$

Indeed an invariant function vanishes on the reduced quotient image precisely when its pullback vanishes on this reduced preimage. The sheaf on the right is maximal Cohen–Macaulay: upstairs it is invertible on a CM scheme, finite pushforward has the same depth over the quotient, and invariants are a direct summand in characteristic zero.

Finite-map duality with trace, followed by invariants, gives

$$\text{Hom}_S(\mathcal{I}_{J/S}, \omega_S) = (f_* \omega_{S^\text{up}}(T^\text{up}|_{S^\text{up}}))^{\text{inv}} \simeq \mathcal{O}_S(aS).$$

This use of duality allows ramification and does not assert that ω_S itself is invertible. The last isomorphism is upstairs Cartier adjunction followed by descent of the S line. Off J it is the ordinary residue determined by the fixed ambient isomorphism. Normalized trace preserves that convention, proving compatibility on overlaps.

Restricted to S , the root gerbe of N_0 is $D \times_P \mathcal{P}$. The strict-divisor finiteness already proved, together with removal of the relative inertia kernel, makes $S \rightarrow D \times_P \mathcal{P}$ finite and representable. Its image covers the positive boundary, and its root line is $\mathcal{O}_S(S) = C$. Lemma C.2 gives all the asserted

image-dimension bounds, including the bound for J . The resulting $h : S \rightarrow \mathcal{P}$ is representable projective.

Its Stein algebra defines a finite stack $\mathcal{T} = \mathrm{Spec}_{\mathcal{P}} h_* \mathcal{O}_S$ over $\mathcal{P}$. There are enough vector bundles on $\mathcal{P}$ to generate this coherent algebra as a module. Namely, decompose by the finitely many inertia characters, twist each summand by a power of C to descend it to P , and apply Serre generation. A vector-bundle surjection gives an embedding of $\mathcal{T}$ in a vector bundle over $\mathcal{P}$. Its closure in the projective completion is still $\mathcal{T}$, since it is proper over the base. This gives $\mathcal{Y}$ and g . The Stein construction supplies $g_* \mathcal{O}_S = \mathcal{O}_{\mathcal{T}}$ and the required line identities.

Finally, etale morphisms extend uniquely over nilpotent thickenings, compatibly as q varies. The reduction S_U is a scheme. The extension qS_U has trivial inertia and is an algebraic space. It is separated over X : the chosen chart is separated, the morphism to the root gerbe is finite, and that gerbe has finite diagonal. Its finite-type geometric fibers over X are zero-dimensional, and this is unchanged by nilpotents. Thus $qS_U \rightarrow X$ is separated and quasi-finite. Zariski's main theorem embeds it in a scheme finite over X , proving that it is a quasi-projective scheme. This completes the construction. $\square$

D Hodge theory and formal lifting

We retain the geometric package of Proposition C.3. In particular, $\mathcal{P} = \sqrt[\ell]{\mathcal{O}_{\mathbb{P}^s}(1)}$ is a root gerbe, C is its tautological line bundle, and

$$g : S \longrightarrow \mathcal{Y} \xrightarrow{p} \mathcal{P}$$

is a representable projective morphism followed by a smooth projective bundle. The Stein image $\mathcal{T} \subset \mathcal{Y}$ is finite over $\mathcal{P}$, and $g_* \mathcal{O}_S = \mathcal{O}_{\mathcal{T}}$. The reduced Cartier divisor $S \subset \mathcal{X}$ has dimension $N = n - 1$. If $J = (S \cap T)_{\mathrm{red}}$, then S, J are Du Bois, $\mathcal{I}_{J/S}$ is maximal Cohen–Macaulay, and

$$\mathcal{B} := \mathrm{Hom}_S(\mathcal{I}_{J/S}, \omega_S) \simeq \mathcal{O}_S(aS) = g^* C^a, \quad \mathcal{O}_S(S) = g^* C, \quad (183)$$

where $a > 0$ is an integer. Off T , the fixed ambient isomorphism $\omega_{\mathcal{X}}(S) \simeq \mathcal{O}_{\mathcal{X}}(aS)$ supplies these identifications by adjunction. We prove that functions and transverse parameters can be lifted through every infinitesimal neighborhood of S .

D.1 A global vanishing statement

We use graded-polarizable mixed Hodge modules on ordinary complex algebraic varieties. The needed established results are projective strictness and the usual functorial operations [66, Theorem 2.14 and Section 4], the comparison with the Du Bois complex and its coherent dual [67, Theorem 0.2 and Corollary 0.3], and Kodaira–Saito vanishing [66, Proposition 2.33]. No Hodge-module theory on stacks is assumed. Our objects on smooth stacks will be compatible systems on scheme etale charts, whose filtered differential modules descend.

Throughout this section, differential modules are right modules with increasing filtration. The module itself is in degree zero in its Spencer de Rham complex. Thus, on a smooth chart V , the term in degree $-i$ of $\mathrm{Gr}_k^F \mathrm{DR}_V(M)$ is

$$\mathrm{Gr}_{k-i}^F M \otimes_{\mathcal{O}_V} \bigwedge^i T_V.$$

In particular, if $F_{<0} M = 0$, the lowest graded de Rham complex is the sheaf $F_0 M$ in degree zero.

Lemma D.1. *Let M be a compatible system of mixed Hodge modules in perverse degree zero on the scheme etale charts of $\mathcal{Y}$. Suppose its support is finite over $\mathcal{P}$. Then, for every integer k , every positive integer b , and every $j < 0$,*

$$\mathbb{H}^j(\mathcal{Y}, \mathrm{Gr}_k^F \mathrm{DR}_{\mathcal{Y}}(M) \otimes p^* C^{-b}) = 0.$$

Here the graded de Rham complex is the descended coherent complex.

Proof. Because p is finite on the support, its direct image has only perverse cohomology in degree zero. Denote that system on $\mathcal{P}$ by M' . Projective strictness on the ordinary scheme charts gives

$$Rp_* \mathrm{Gr}_k^F \mathrm{DR}_{\mathcal{Y}}(M) \simeq \mathrm{Gr}_k^F \mathrm{DR}_{\mathcal{P}}(M'). \quad (184)$$

We first explain why this is a global identity of descended complexes, rather than a claim of local vanishing.

For right differential modules, the direct image is formed from the filtered transfer bimodule

$$\mathcal{D}_{\mathcal{Y} \rightarrow \mathcal{P}} = \mathcal{O}_{\mathcal{Y}} \otimes_{p^{-1}\mathcal{O}_{\mathcal{P}}} p^{-1}\mathcal{D}_{\mathcal{P}},$$

with its order filtration, by derived tensor over $\mathcal{D}_{\mathcal{Y}}$ and derived direct image. These operations may equivalently be performed on Rees modules. Applying de Rham on the base means further derived tensor with $\mathcal{O}_{\mathcal{P}}$, whose filtration begins in degree zero. The filtered Spencer resolution, associativity of derived tensor, and projection formula identify the result with the direct image of de Rham upstairs: the transfer module tensored over the base differential operators with its structure sheaf is $\mathcal{O}_{\mathcal{Y}}$. These are canonical sheaf constructions on the etale site; the Spencer resolutions are locally free over differential operators. They therefore respect chart changes before passing to associated graded. Strictness and the concentration in perverse degree zero can be checked on the base charts by the ordinary projective direct-image theorem. Coherent cohomology on those charts is unchanged by using their etale sites. This proves Equation (184) as an identity that computes global coherent hypercohomology. The same argument applies to the representable finite morphism used next.

There is a representable finite surjective morphism

$$v : \mathbb{P}_z^s \longrightarrow \mathcal{P}$$

defined by the power map $[z_0 : \cdots : z_s] \mapsto [z_0^\ell : \cdots : z_s^\ell]$ and the root $\mathcal{O}_{\mathbb{P}_z^s}(1)$. In particular, $v^*C = \mathcal{O}_{\mathbb{P}_z^s}(1)$. On the standard affine opens of $\mathbb{P}^s$ choose the trivial root, obtaining scheme etale charts $\check{U}_i \rightarrow \mathcal{P}$. On $z_i \neq 0$, the morphism v factors through $\check{U}_i$. Pull back M' on these charts and take perverse cohomology in degree zero. Functoriality and the chosen root identify these objects on overlaps, giving a system H on the Zariski opens of $\mathbb{P}_z^s$.

For completeness, this is a global graded-polarizable mixed Hodge module on $\mathbb{P}_z^s$. The perverse, filtered, and weight data glue. The pure weight graded have strict-support decompositions, which glue by uniqueness. On a smooth connected dense stratum of an irreducible support, a polarization from a nonempty Zariski open extends as a flat pairing: the fundamental group of that open surjects onto the fundamental group of the stratum. Its Hodge compatibility extends by continuity; nondegeneracy and positivity persist for the extended flat pairing. Quasi-unipotence at the boundary is already supplied by the local Hodge modules. Saito's extension theorem for polarizable variations [66, Section 3.b] and uniqueness of strict-support extension identify this global pure object with the glued one. There is no additional boundary at infinity, since $\mathbb{P}_z^s$ is projective. The weight extensions then give the stated mixed object.

The system M' is a retract of v_+H . To see this on a chart, use the *whole* base-changed finite cover. Over $\check{U}_i$ it is a disjoint union of copies of the corresponding affine coordinate chart of $\mathbb{P}_z^s$. The unit on unshifted constants and its dual trace, using smooth duality in equal dimensions, have composition multiplication by the degree. On the finite etale locus the trace is summation over sheets. On each connected smooth base chart, the shifted constant Hodge module is the rank-one intersection complex and has endomorphism ring $\mathbb{Q}$; its endomorphisms are determined on a dense open. Thus the composition equals the degree globally, with no additional term supported on the branch locus. Ramification is therefore allowed. Tensoring with M' , applying proper projection formula, and taking perverse cohomology in degree zero gives the claimed retraction, since finite direct image is perverse exact. Divide the trace by the degree. All maps are canonical under etale base change, so the retraction descends also on filtered differential modules.

Now apply the finite version of Equation (184) and projection formula. The desired hypercohomology is a direct summand of

$$\mathbb{H}^j\left(\mathbb{P}_z^s, \mathrm{Gr}_k^F \mathrm{DR}(H) \otimes \mathcal{O}_{\mathbb{P}_z^s}(-b)\right).$$

This is zero for $j < 0$ by ordinary negative-ample Kodaira–Saito vanishing. One may apply it to the pure weight gradeds and then use the exact sequences of the weight filtration; the Hodge filtrations of those sequences are strict. $\square$

D.2 The lowest Hodge piece

Take a separated quasi-projective scheme etale chart $U \rightarrow \mathcal{Y}$, of finite type. Write $S_U = S \times_{\mathcal{Y}} U$, $J_U = J \times_{\mathcal{Y}} U$, and again $g : S_U \rightarrow U$ for the induced projective morphism. For $j : S_U \setminus J_U \hookrightarrow S_U$, put

$$A_U^\bullet = \mathbf{D}\left(j_! \mathbb{Q}_{S_U \setminus J_U}^H[N]\right), \quad A_{0,U} = {}^p H^0 A_U^\bullet, \quad M_U = {}^p H^1 g_* A_{0,U}.$$

All these constructions commute with etale restriction. The M_U form a system supported on $\mathcal{T}$, to which Lemma D.1 applies.

Lemma D.2. *The following identifications hold compatibly on the charts:*

$$F_{<0} A_{0,U} = 0, \quad F_0 A_{0,U} = \mathcal{B}|_{S_U}, \quad F_{<0} M_U = 0, \quad F_0 M_U = R^1 g_*(\mathcal{B}|_{S_U}).$$

In a smooth ambient embedding $S_U \hookrightarrow V$ of codimension c , the underlying module of $A_{0,U}$ is $\mathcal{H}_{S_U}^c(\omega_V)$, localized off J_U . The lowest-piece inclusion is the adjunction, or Ext-to-support, inclusion off J_U , extended by meromorphic localization.

Proof. The difference triangle for the constant objects of S_U and J_U , together with Du Bois comparison, identifies its degree-zero graded de Rham complex with $\mathcal{I}_{J_U/S_U}$. Coherent duality and the maximal-Cohen–Macaulay property therefore give

$$\mathrm{Gr}_k^F \mathrm{DR}(A_U^\bullet) = 0 \quad (k < 0), \quad \mathrm{Gr}_0^F \mathrm{DR}(A_U^\bullet) = \mathcal{B}|_{S_U}[0]. \quad (185)$$

The shift $[N]$ cancels the dimension shift in the dualizing complex. In the smooth case this convention says $\mathbf{D}(\mathbb{Q}^H[N]) = \mathbb{Q}^HN$; its right differential module has lowest piece ω at index zero.

We spell out the passage from the complex to perverse cohomology. Let q be the smallest filtration index occurring in any perverse cohomology object of $A_U^\bullet$, locally on a fixed chart. Such a lower bound exists because the complex is bounded and its filtrations are good. At index q , the

graded Spencer complex of every perverse cohomology object has only its degree-zero term. The spectral sequence from perverse truncation therefore has a single nonzero row and gives

$$\mathcal{H}^i \mathrm{Gr}_q^F \mathrm{DR}(A_U^\bullet) = F_q {}^p H^i A_U^\bullet.$$

If $q < 0$, this contradicts Equation (185). Thus all these objects have $F_{<0} = 0$. Applying the same argument at index zero gives $F_0 A_{0,U} = \mathcal{B}|_{S_U}$, and gives zero for the lowest piece of the other perverse cohomology objects. No perversity assertion about the shifted constant complex is required.

For the unfiltered description, duality identifies the dual of the reduced constant complex, in the ambient V , with

$$R\Gamma_{[S_U]}(\omega_V)[c].$$

Its degree-zero module is the first nonzero support cohomology $\mathcal{H}_{S_U}^c(\omega_V)$. The support of J_U is locally set-theoretically principal in S_U , by the construction. Invert a local defining function, lifted to V ; this exact meromorphic localization realizes j_* on the underlying module. It yields the asserted description of $A_{0,U}$.

We also need compatibility of the two descriptions of its lowest piece. On the smooth locus of $S_U \setminus J_U$, filtered duality and smooth graph pushforward identify it with the ordinary dualizing sheaf included by residue. This is the Ext-to-support inclusion, with the normalization fixed by the ambient adjunction isomorphism. Naturality in smooth etale coordinates fixes the same scalar on every chart. Agreement extends over $S_U \setminus J_U$: the first support-cohomology module has no sections supported on a smaller-dimensional subset. For example, this follows from the Cousin resolution of the smooth dualizing sheaf, whose first term supported on S_U is a sum of injective modules at its codimension- c generic points. It extends across J_U by localization. Hence the inclusion is the one stated in the lemma, not merely an abstract isomorphism of coherent sheaves.

Finally use a graph embedding followed by a smooth projection to compute g_+ . At filtration zero the relative Spencer complex has only its top term, namely the direct image of $\mathcal{B}|_{S_U}$ on the graph. Projective strictness gives

$$F_{<0} M_U = 0, \quad F_0 M_U = R^1 g_*(\mathcal{B}|_{S_U}),$$

and identifies this sheaf with its actual submodule in the unfiltered degree-one direct image. The constructions used here are canonical on etale changes of charts. $\square$

D.3 Lifting through the infinitesimal neighborhoods

Let $I = \mathcal{O}_{\mathcal{X}}(-S)$. The etale object $S_U \rightarrow S$ extends uniquely and compatibly over every nilpotent thickening qS ; denote its extension by qS_U . These are quasi-projective schemes by Proposition C.3: they are separated and quasi-finite over X , hence are open in schemes finite over the projective variety X . Powers and quotients of I below are pulled back to the respective thickenings.

Proposition D.3. *For every such chart U , every $j \geq 0$, and every $k \geq 1$, the transition*

$$g_*(I^j / I^{j+k+1}) \longrightarrow g_*(I^j / I^{j+k})$$

is surjective as a map of sheaves of abelian groups. Its kernel is $\mathcal{O}_{\mathcal{T}_U} \otimes C^{-j-k}$, where $\mathcal{T}_U = \mathcal{T} \times_Y U$. The statements are compatible with further etale changes of charts. On an affine chart, these transitions are also surjective on global sections.

Proof. Define

$$E_{j,k} = g_*(I^j / I^{j+k}), \quad j \geq 0, \quad k \geq 1.$$

Here g_* uses the underlying topological map. We do not assume that a thickening already has a ringed-space map to U . The first graded layer is

$$E_{j,1} = \mathcal{O}_{\mathcal{T}_U} \otimes C^{-j}.$$

The exact sequence for adjacent lengths has kernel $g_*\mathcal{O}_{S_U} \otimes C^{-j-k}$, and its connecting homomorphism takes values in

$$R^1 g_*\mathcal{O}_{S_U} \otimes C^{-j-k}.$$

Induct on k , proving the assertion simultaneously for all j . Assume all shorter transitions are surjective. A section of $E_{j,k}$ with zero length-one term comes from $E_{j+1,k-1}$ when $k > 1$. By induction it locally lifts from $E_{j+1,k}$, so its connecting class is zero. Also $E_{j,k} \rightarrow E_{j,1}$ is surjective. For $k = 1$ the factorization is immediate. The obstruction therefore factors through a map D_k on the graded algebra of length-one layers, with graded shift k . Choose local lifts and take their Čech differences. The difference of a product is the sum of the two first-order differences; products of the errors vanish in the layer at issue. Thus D_k is a $\mathbb{C}$ -derivation from this graded algebra to the graded $R^1 g_*\mathcal{O}_{S_U}$ -module.

In degree zero, restrict it along $\mathcal{O}_U \rightarrow \mathcal{O}_{\mathcal{T}_U}$. The resulting derivation factors through Ω_U^1 , hence defines a section of $T_U \otimes R^1 g_*\mathcal{O}_{S_U} \otimes C^{-k}$. These local sections descend to

$$e \in H^0(\mathcal{Y}, T_{\mathcal{Y}} \otimes R^1 g_*\mathcal{O}_S \otimes C^{-k}) = H^0(\mathcal{Y}, T_{\mathcal{Y}} \otimes F_0 M \otimes C^{-(a+k)}). \quad (186)$$

The equality uses Equation (183), Lemma D.2, and projection formula. For descent, lifts of functions pull back on the unique étale thickenings. Their Čech classes pull back by ordinary flat base change for the coherent obstruction sheaves on the reductions. Differentials pull back and generate under an étale map, proving the asserted compatibility.

Shrink U and take étale coordinates $t_1, \dots, t_d$, where $d = \dim U$, and a frame of C^{-1} . Its pullback is a conormal frame, denoted by y . By induction lift the coordinates along g modulo I^k , and lift y to I/I^{k+1} . On an open cover of S_U , lift these one step further, obtaining functions $h_{i,\lambda}$ modulo I^{k+1} and generators y_i modulo I^{k+2} . Formal étaleness of the coordinate map makes each h_i a local map from $(k+1)S_U$ to U . On overlaps write

$$h_{j,\lambda} - h_{i,\lambda} = e_{ij,\lambda} y_i^k, \quad y_j - y_i = \eta_{ij} y_i^{k+1}. \quad (187)$$

The reduced coefficients are Čech cocycles representing $D_k(t_\lambda)$ and $D_k(y)$ in the chosen frames. Let $\sigma = y^{-a}$ denote the induced frame of $\mathcal{B}|_{S_U}$.

Off J_U , the fixed adjunction isomorphism gives

$$\omega_{(k+1)S_U} \simeq \mathcal{O}_{\mathcal{X}}((a+k)S)|_{(k+1)S_U}.$$

It therefore defines local dualizing sections $b_i = y_i^{-(a+k)}$. Under the trace inclusion $\omega_{S_U} \hookrightarrow \omega_{(k+1)S_U}$, the identities

$$b_j - b_i = -(a+k)\eta_{ij}\sigma, \quad y_i^k b_i = \sigma, \quad y_i^{k+1} b_i = 0 \quad (188)$$

hold when the dualizing sheaves are realized in support cohomology. Indeed the first identity is the binomial expansion of $(1 + \eta_{ij} y_i^k)^{-(a+k)}$; its quadratic terms vanish because $2k \geq k+1$. The other two identities describe Cartier trace and the annihilator of the thickening.

Embed $(k+1)S_U$ as a closed subscheme of a smooth open $Z \subset \overline{Z} = \mathbb{P}^m$. The graph of the reduced map g is closed in $\overline{Z} \times U$, by its properness over U . Use this reduced graph for an ambient

realization of $A_{0,U}$. Each local graph lift h_i maps b_i , by Ext-to-support cohomology, to a section $\delta_i(b_i)$ of the underlying graph module. We claim that

$$\delta_j(b_j) - \delta_i(b_i) = -(a+k)\eta_{ij}\sigma + \sum_{\lambda=1}^d (e_{ij,\lambda}\sigma) \cdot \partial_{t_\lambda}. \quad (189)$$

The reduced dualizing sections on the right use the inclusion of Lemma D.2.

Here is the local calculation, including the non-smooth case. Put $c_0 = \dim Z - N$. The Ext-to-support map identifies the dualizing sheaf of the thickening with its annihilator submodule in $\mathcal{H}_{S_U}^{c_0}(\omega_Z)$. One can obtain this from the support-cohomology spectral sequence: there is no support cohomology below c_0 , so in total degree c_0 the Ext group is the homomorphisms from the thickening's structure sheaf into that first support-cohomology module. Thus the calculation does not require S_U to be a local complete intersection in Z .

Lift $h_{i,\lambda}$ locally to functions on Z . In $Z \times U$, the graph inclusion on a thickness-dualizing section b is the generalized fraction

$$\frac{b \, dt_1 \wedge \cdots \wedge dt_d}{\prod_{\lambda=1}^d (t_\lambda - h_{i,\lambda})}.$$

This is successive support cohomology in the additional coordinate equations, or equivalently the Koszul residue for the graph. It may be computed etale-locally near the graph branch. After first taking support cohomology from Z , the translated new coordinates form a regular sequence, so their support cohomology is concentrated in their top number. Localization in either set of translated coordinates gives the same localization on this support-torsion module: the two translations differ nilpotently on every section.

Changing from h_i to h_j therefore gives a finite Taylor expansion on b_j . Products of two differences in Equation (187) annihilate b_j , again because $2k \geq k+1$. Its linear term uses

$$(h_{j,\lambda} - h_{i,\lambda})b_j = e_{ij,\lambda}\sigma.$$

For a top differential form the right action satisfies

$$\left(\frac{dt_\lambda}{t_\lambda - h} \right) \cdot \partial_{t_\lambda} = \frac{dt_\lambda}{(t_\lambda - h)^2}.$$

Consequently the doubled pole has the positive sign displayed in Equation (189). Together with Equation (188), this proves that identity. The calculation initially takes place off J_U . The sections and identities extend meromorphically across J_U in the *unfiltered* localized module; arbitrary finite pole orders there are allowed.

Now push by $\bar{Z} \times U \rightarrow U$, using relative Spencer. If i denotes the closed reduced graph embedding and π the projection, closed direct image is perverse exact and $\pi_+ i_+ A_{0,U} = g_+ A_{0,U}$. Thus degree-one holonomic differential-module cohomology of this Spencer direct image is precisely the underlying module of $M_U = {}^p H^1 g_* A_{0,U}$. The cochain $(\delta_i(b_i))$ lies in its top, degree-zero term, which has no outgoing relative Spencer differential. Its Cech difference is therefore a boundary in total degree one. A sufficiently refined ambient cover around the closed graph computes this assertion; all sheaves involved are supported there. The reduced cocycles in Equation (189) represent their $R^1 g_* \mathcal{B}$ classes inside $F_0 M$, by Lemma D.2. Right differentiation in the U -coordinates acts on the pushforward complex. Hence the right side of Equation (189) represents zero in the unfiltered module M .

That right side lies in $F_1 M$; the term involving η lies in $F_0 M$. Since $F_1 M$ is an actual subsheaf of M , its vanishing in M is vanishing in $F_1 M$. Taking the order-one symbol thus says that the global section e of Equation (186) is killed by

$$T_{\mathcal{Y}} \otimes F_0 M \longrightarrow \mathrm{Gr}_1^F M, \quad (190)$$

after twisting by $C^{-(a+k)}$. This reasoning places no Hodge-filtration bound on the auxiliary cochain (b_i) .

There is no term preceding degree -1 in $\mathrm{Gr}_1^F \mathrm{DR}(M)$, because $F_{<0}M = 0$. Its degree- -1 hypercohomology after the indicated twist is therefore precisely the space of global sections of the kernel of Equation (190). Since $a + k > 0$, Lemma D.1 makes that space zero. We conclude that $e = 0$.

In a local coordinate and line trivialization, this says that every class $[e_{ij,\lambda}\sigma]$ is already zero in F_0M , hence in M . Its actual differential-operator image is consequently zero, before taking symbols. The unfiltered relation now gives

$$(a + k)[\eta_{ij}\sigma] = 0.$$

Lowest-piece injectivity and $a + k > 0$ imply $D_k(y) = 0$. The degree-zero derivation is zero as well: it vanishes on $\mathcal{O}_U$, which surjects onto $\mathcal{O}_{\mathcal{T}_U}$. Locally the whole graded length-one algebra is generated by that coefficient algebra and the conormal frame y . Thus $D_k = 0$ in every degree, proving the induction and all sheaf transitions.

Their kernels are the coherent sheaves $\mathcal{O}_{\mathcal{T}_U} \otimes C^{-j-k}$. On an affine U these have no first cohomology. The exact sequences of sheaves of abelian groups therefore also give surjectivity on global sections. Choosing lifts successively gives compatible formal lifts of any chosen functions on $\mathcal{T}_U$, and, when C is trivialized, of its transverse conormal frame. $\square$

E Compact null families and descent

We return to the setting of Theorem C.1, with $0 < v < n$, $r = v - 1$, and the geometric construction of Proposition C.3. The infinitesimal lifting result now turns a boundary fiber into a compact subvariety outside the entire boundary.

Proposition E.1. *Assume the transition surjectivity of Proposition D.3. There is a dominating algebraic family of integral projective subvarieties of X of dimension*

$$d = n - 1 - r = n - v$$

whose general members avoid D and on which L is numerically trivial.

Proof. Choose a separated affine étale chart $U \rightarrow \mathcal{Y}$ around a general point of a top-dimensional component of $\mathcal{T}_U$. Shrink it so that C is trivial, this component of $\mathcal{T}_U$ is smooth of pure dimension r , and S_U is flat over it. We may also ensure that S_U does not meet J_U , because the image of J in P has dimension less than r . Choose a closed point t of this open set and regular parameters $t_1, \dots, t_r$ on $\mathcal{T}_U$ cutting out t alone after further shrinking. Then

$$F = S_{U,t}$$

is projective of pure dimension d , and the pulled-back parameters form a regular sequence along F .

Put $I = \mathcal{O}_{\mathcal{X}}(-S)$. Proposition D.3 gives surjections between all the sheaves $g_*(I^j/I^{j+k})$ as the length k increases. Their transition kernels are coherent sheaves of the form $g_*\mathcal{O}_{S_U} \otimes C^{-j-k}$. Since U is affine, the surjections hold on global sections as well. We may therefore lift the parameters compatibly to all thickenings and lift the chosen conormal frame to a compatible element $\tilde{y}$ generating I formally.

On qS_U , cut out the lifted r parameters and denote the resulting closed subscheme by Z_q . Give it the structure of a scheme over

$$R_q = \mathbb{C}[s]/(s^q), \quad s \longmapsto \tilde{y}.$$

These schemes are compatible under reduction in q . Their closed fiber is F , and they are flat over R_q . Indeed the powers of the Cartier generator identify the successive s -layers of qS_U with $\mathcal{O}_{S_U}$, proving flatness before cutting. The local flatness criterion then shows that quotienting by lifts of a closed-fiber regular sequence preserves flatness.

The map $Z_q \rightarrow X \times \operatorname{Spec} R_q$ is quasi-finite. It is proper as well: its reduction is the map from the projective scheme F , and properness of a finite-type morphism is unchanged by nilpotent thickenings. Hence this map is finite. Let $R = \mathbb{C}[[s]]$. Projective Grothendieck existence, including its full faithfulness, algebraizes the compatible finite algebra sheaves on $X \times \operatorname{Spec} R_q$ to a finite algebra on $X_R = X \times \operatorname{Spec} R$; multiplication and the unit algebraize by full faithfulness [72, Lemmas 30.24.1 and 30.24.3]. Its relative spectrum is a finite morphism

$$Z \longrightarrow X_R$$

with the prescribed reductions. The scheme Z is projective over R . Formal flatness proves flatness along the closed fiber, and flatness is automatic on the generic fiber over $\mathbb{C}((s))$, so Z is flat over R .

The images of the pole and coefficient-zero boundary parts do not meet F . Their inverse images in Z are closed and proper over R , so they do not meet Z : a nonempty closed subset of a proper R -scheme has specialization in the closed fiber. Away from these parts and the graph exceptional divisor, the root construction identifies the divisor of s_0 with ℓS . The formal generator $\tilde{y}$ therefore supplies compatible trivializations in which

$$N_0|_{\widehat{Z}} \simeq \mathcal{O}_{\widehat{Z}}, \quad s_0|_{\widehat{Z}} = s^\ell.$$

Full faithfulness algebraizes this trivialization and the identity. Thus the generic fiber of Z avoids also the positive part of D , and hence all of D . Its finite image in $X_{\mathbb{C}((s))}$ has pure dimension d . After a finite extension of $\mathbb{C}((s))$, a component can be reduced and chosen geometrically integral.

It remains to show that these compact subvarieties cover X , rather than merely a neighborhood of one point. Choose a positive component D_i with image dimension r , and a component of S covering it. As the general chart and the point t vary, images of points of F contain a dense open subset of D_i . Every such point is in the specialization of a d -dimensional generic finite image constructed above. Indeed flatness over the discrete valuation ring rules out vertical components of Z , and the dimension formula shows that the closures of its generic components have special-fiber components of dimension d . Their finite images retain that dimension.

Consider now all Hilbert schemes of d -dimensional subvarieties of X . Their geometrically integral loci whose members avoid D admit a countable stratification by integral parameter spaces with irreducible universal families. Let W_α be the irreducible closures in X of the corresponding evaluation images. A generic image over a field extension determines a Hilbert point of one of these strata. Properness of the Hilbert scheme gives its specialization, so every point of the dense open subset of D_i just described lies in some W_α .

Over the uncountable field $\mathbb{C}$, an irreducible variety cannot have a dense open covered by countably many proper closed subsets. Therefore one W_α contains D_i . It also contains points outside D , by its definition. Since D_i is a prime divisor in the integral variety X , the only proper irreducible closed subset containing it is D_i itself. Consequently $W_\alpha = X$, giving a dominating family. On every member of this family, the rational section s_0 is regular and nowhere zero, since the member avoids D . It trivializes N_0 , so L is numerically trivial there. $\square$

E.1 Descent along the nef reduction

Lemma E.2. *Let $f : Y \rightarrow B$ be a surjective projective morphism from a normal integral variety to a smooth projective variety, with geometrically connected integral generic fiber. Suppose all fibers have dimension at most $\dim Y - \dim B$. Let G_Y be a vertical $\mathbb{Q}$ -Cartier divisor that is relatively nef. Then*

$$G_Y = f^*G_B$$

for a $\mathbb{Q}$ -divisor G_B on B .

Proof. If B is a point, a vertical divisor is zero. If the relative dimension is zero, the fiber bound makes f finite; its geometrically integral connected generic fiber makes it birational, and normality of B makes it an isomorphism. Both cases are immediate. Assume henceforth that the base and the relative dimension are positive.

The fiber bound implies that every vertical prime divisor maps onto a prime divisor of B . For each such base prime P , write the full pullback as $f^*P = \sum_j m_j E_j$ and let b_j be the coefficient of G_Y along E_j , including zero coefficients. Give P coefficient $\min_j (b_j/m_j)$ in G_B . Only finitely many coefficients are nonzero. Then

$$R' = G_Y - f^*G_B$$

is effective, vertical, $\mathbb{Q}$ -Cartier, relatively nef, and misses at least one component over each base prime. We prove that $R' = 0$.

If $R' \neq 0$, choose a base prime below its support. Take a general complete-intersection curve in B meeting that prime at a general point; when $\dim B = 1$ use B itself. Its inverse image in Y is normal by normal Bertini. It is integral: generic geometric integrality gives the dominating component, and the fiber bound excludes additional vertical components. Next take $\dim Y - \dim B - 1$ general very ample hyperplanes upstairs. This produces a normal integral surface over the base curve with geometrically connected generic fiber. At the chosen general base point, the cuts retain curves in the residual support and in a missing component of the full pullback; these curves are distinct because the original components were distinct at the generic point of the base prime.

Resolve the surface. The pulled-back residual divisor is effective, vertical, and relatively nef. Its intersection with the full fiber is zero, whereas its intersection with each fiber component is nonnegative. Every one of the latter intersections is therefore zero. The intersection matrix of a connected surface fiber is negative semidefinite with kernel generated by the full fiber. Thus the residual part at the chosen point is a multiple of that full fiber. A missing component forces this multiple to be zero, contradicting the component in its support. Connectedness of the fiber follows from generic connectedness and Stein factorization over the normal base curve. $\square$

Completion of the proof of Theorem C.1. Lemma C.2 handles $v = 0$ and $v = n$. In the remaining case, combine Propositions C.3, D.3, and E.1.

Apply the nef-reduction theorem to a Cartier multiple of L [5, Theorem 2.1]. It gives an almost holomorphic rational map with connected fibers, with L numerically trivial on its compact general fibers, and with positive L -degree on every noncontracted curve through a very general point of X . Let its base dimension be b . A compact L -trivial d -fold through such a point is contracted: otherwise general ample slices through the point yield a noncontracted curve of L -degree zero. Hence

$$b \leq n - d = v.$$

Resolve the graph, flatten the main component over a modification of the base, resolve that

base, and normalize the main transform. We obtain projective morphisms

$$\begin{array}{ccc} Y & \xrightarrow{\pi} & X \\ f \downarrow & & \\ B & & \end{array}$$

with π birational, B smooth, Y normal, and geometrically connected integral generic fiber of f . All fibers have dimension at most $n - b$. Indeed the flat main transform remains integral after the base modification by flatness and generic integrality, and finite normalization preserves the fiber-dimension bound. Normalization need not preserve flatness; only this bound is used.

The pullback π^*D has no horizontal component. To see this, restrict to a very general fiber of f . The class of π^*L is numerically trivial there, and $\pi^*(L - cD)$ restricts to a pseudo-effective class. The latter assertion follows by restricting effective approximants, avoiding the countably many fibers contained in their supports. Its restricted class is $-c\pi^*D$. A negative nonzero effective divisor on a projective variety cannot be pseudo-effective, as intersection with an ample power shows. Thus the restriction of π^*D is zero. In particular $b = 0$ would force $D = 0$, contrary to $v > 0$.

Consequently $G_Y = \pi^*G$ is vertical and is relatively nef, since $G_Y \sim_{\mathbb{Q}} \pi^*L$. Lemma E.2 gives

$$\pi^*L \sim_{\mathbb{Q}} f^*G_B$$

for a $\mathbb{Q}$ -divisor G_B on the smooth base. It is nef: every curve of B is dominated by a curve of Y , on which the pullback has nonnegative degree. The intersection formula for a pullback gives $\nu(X, L) \leq b$. Combined with $b \leq v$, this yields $b = v$ and $G_B^b > 0$. Therefore G_B is big. Pulling back sections gives $\kappa(X, L) \geq b = v$, and the general inequality $\kappa(X, L) \leq \nu(X, L)$ for a nef divisor proves abundance. $\square$

F Geometric exclusions for a nonvanishing counterexample

Throughout this section we assume the lower-dimensional good-model hypothesis of Assumption B.1. We suppose, towards a contradiction, that X is smooth projective of dimension $n > 0$ and

$$K_X \text{ is pseudo-effective,} \quad \kappa(X, K_X) = -\infty. \quad (191)$$

These properties persist on smooth projective birational models. All varieties in this section are complex. Numerical dimension for a pseudo-effective divisor means Nakayama's numerical dimension κ_{σ} ; for a nef divisor it agrees with the intersection definition used in Theorem C.1.

F.1 The Albanese reduction and algebraic webs

Lemma F.1. *Every smooth projective birational model of X has irregularity zero. Consequently, on such a model, or on a projective $\mathbb{Q}$ -factorial terminal birational model, numerical equivalence of rational divisors implies their $\mathbb{Q}$ -linear equivalence.*

Proof. If $q(X) > 0$, resolve the Stein factorization of the Albanese map to obtain a morphism $f : W \rightarrow B$ with connected fibers, with W, B smooth projective and $\dim B > 0$. The map from B to a subvariety of $\text{Alb}(X)$ is generically finite. Wedges of invariant one-forms on the abelian variety therefore give a nonzero section of K_B : at the generic point, choose $\dim B$ independent pulled-back one-forms. Thus $\kappa(B, K_B) \geq 0$.

For a very general smooth fiber F , the restriction of the pseudo-effective class K_W is pseudo-effective. For example, restrict a positive current representing it to almost every fiber, or use effective approximations with arbitrarily small ample error. Adjunction identifies this restriction with K_F . If F has positive dimension, the induction hypothesis gives $\kappa(F, K_F) \geq 0$; for a point the same assertion holds by convention. Applying Assumption A.1 with both boundaries empty gives $\kappa(W, K_W) \geq 0$, contradicting (191).

This is the only use of Assumption A.1 in the proof. Irregularity is a smooth birational invariant. Finally, when $\text{Pic}^0 = 0$, a numerically trivial line bundle is torsion, since the group of numerically trivial line bundles modulo Pic^0 is finite. Clear denominators for rational divisors. On a terminal $\mathbb{Q}$ -factorial model, pull back to a smooth resolution and then descend the rational linear equivalence. $\square$

We use covering families of proper subvarieties through very general points. They can be parameterized in countably many algebraic families, using Hilbert schemes followed by resolution and stratification. After shrinking a parameter space, the domains form a smooth projective family with integral fibers and have a dominant evaluation map. If the fiber maps are generically finite onto their images, general ample cuts of the parameter space make the total evaluation generically finite and still dominant. We always resolve and compactify this evaluation when applying ramification. Invariance of plurigena for smooth projective families [65, Theorem 1] permits the same construction for the Iitaka fibers of general members: choose a divisible pluricanonical system on the general member, spread it by base change, and resolve its relative rational map.

Lemma F.2 (Algebraic web quotient). *Let a smooth projective variety be dominated generically finitely by the total space of a family with smooth integral projective general fibers. On a dense regular open, form the distribution generated by the tangent spaces of the fiber images, taking spans, saturation, and Lie brackets. Its general leaves are dense opens of algebraic subvarieties. Their closures are the general fibers of a rational map, which has connected general fiber after Stein factorization.*

Proof. Shrink to an open U where the evaluation has finitely many etale sheets, the parameter map is smooth, and the generated involutive distribution $\mathcal{F}$ has constant rank. Its construction is algebraic: spans and brackets stabilize at the generic point after finitely many operations, and descend from the etale sheets. The nonempty open parts of the integral parameter fibers are connected. We use only chains of fiber-image steps lying in U .

For $x \in U$, endpoints of chains of at most k steps form a constructible set, by the algebraic fiber-product construction and Chevalley's theorem. Every endpoint lies in the analytic leaf through x . A locally closed smooth variety contained in an analytic leaf has dimension at most $\text{rk } \mathcal{F}$. Indeed, in a Frobenius chart the leaf meets at most countably many plaques: use finite plaque chains in a countable foliation atlas. The transverse coordinate map on any connected smooth piece then has countable image and is constant.

There is therefore a maximal dimension among the closures of all such endpoint loci; it is attained for some finite k . The loci are nested because we allow at most k steps, including the identity step. Choose an irreducible component V of maximal dimension and a dense open $O \subset V$ of reachable points. For each etale sheet, restrict its fiber-equivalence relation to first coordinate in O , and take the component through the diagonal section. Its second-image closure contains O , hence V , while its second image consists of endpoints reachable in at most $k + 1$ steps. Maximality forces that closure to equal V . At a general diagonal point, smoothness of the parameter map identifies its vertical tangent with the corresponding web tangent. Every web tangent is consequently tangent to V , and so is every bracket. Hence $\dim V \geq \text{rk } \mathcal{F}$, while the reverse inequality was proved above. A

plaque is thus open in V . The equations of V , pulled back to the connected immersed leaf, vanish on a nonempty open and hence on the entire leaf. Its closure is exactly V .

These rank-dimensional invariant subvarieties have countably many Hilbert or Chow parameter spaces. Tangency on the regular locus is an algebraic condition after stratification, so one such family dominates. An invariant irreducible variety meeting U contains the leaf through any of its points in U : the tangent vector fields preserve its reduced ideal, first on its smooth locus and then everywhere by closure. A rank-dimensional leaf closure is therefore unique through a general point. Assigning that closure gives the desired rational map. Resolve its graph and take the Stein factorization. $\square$

For positive closed $(1, 1)$ -currents we shall use this quotient in the following way. If a current vanishes on almost every parameter fiber away from a fixed removed divisor, it annihilates the corresponding fiber tangents on a dense open. In submersion coordinates, choose locally integrable plurisubharmonic potentials. Fubini's theorem makes the pure fiber Hessian zero as a distribution; positivity then makes the mixed coefficients in fiber directions zero as well. A generically étale evaluation transfers this assertion to the target. Annihilation passes to brackets by the identities

$$\mathcal{L}_v T = d(\iota_v T) + \iota_v(dT), \quad \iota_{[v,w]} T = \mathcal{L}_v(\iota_w T) - \iota_w(\mathcal{L}_v T).$$

Pullbacks by dominant morphisms and restrictions to almost every smooth parameter fiber are defined by the same local potentials.

F.2 The general-type exclusion

Proposition F.3. *Every positive-dimensional proper subvariety through a very general point of X is of general type on resolution. The assertion holds also on every projective birational model.*

Proof. Otherwise take a covering family of nongeneral-type subvarieties, with smooth domains V , and slice its parameters as above. Restricting the ramification formula of its generically finite total evaluation shows that K_V dominates the restriction of the pulled-back K_X by an effective divisor. Thus K_V is pseudo-effective. The induction hypothesis gives a good minimal model of V , so $\kappa(V) \geq 0$. Since V is not of general type, its general Iitaka fibers have positive dimension and Kodaira dimension zero. Spreading these fibers gives another covering family with smooth domains G , of dimension less than n , and $\kappa(G) = 0$. Each G has a good minimal model and $\kappa_\sigma(K_G) = 0$.

Fix a positive current T representing K_X . After slicing and resolving the total evaluation of the G 's, its restriction to almost every parameter fiber, plus the effective restricted ramification divisor, is a positive current in K_G . Every positive current in that class is supported on the fixed canonical divisor of G . To see this, take a common resolution with its good minimal model. The latter has torsion canonical divisor, so the canonical divisor upstairs is effective exceptional. Intersection with a pullback of an ample class to the power $\dim G - 1$ forces any such positive current to vanish off the exceptional locus. The support theorem makes it divisorial, and independence of exceptional divisor classes, by negativity, fixes its coefficients. Pushing down proves the assertion on G . The excluded support is algebraic in the family: it is the fixed divisor of a sufficiently divisible relative pluricanonical system after shrinking the parameter space.

It follows that T annihilates the web distribution of the G 's on a dense open. If that distribution has full rank, T is supported on a proper algebraic set. The support theorem for positive closed $(1, 1)$ -currents expresses it as a finite nonnegative real combination of prime divisors [69, 24]. Its rational cohomology class then has a nonnegative rational representative: solve the finite rational linear system for the coefficients on the same face of the nonnegative orthant. Lemma F.1 converts numerical effectivity to $\mathbb{Q}$ -linear effectivity, a contradiction.

Otherwise Lemma F.2 gives a rational quotient with general smooth fiber F on a smooth resolved graph, where $0 < \dim F < n$. Its canonical divisor is pseudo-effective. The moving G 's still generate its tangent space at general points. Indeed the quotient is constant on every general web member, hence induces a rational map on their parameter space; restricting to a general quotient fiber preserves dominance of evaluation. A pluricanonical rational map of F is constant along each such G . Slice the parameters inside F to make the total evaluation generically finite: ramification injects the restricted pluricanonical sections into sections of a multiple of K_G , and these span a space of dimension at most one. Consequently their ratios are constant on G . Their differentials vanish on the web and its brackets, so the pluricanonical map is constant on F . Lower-dimensional nonvanishing therefore gives $\kappa(F) = 0$, and its good minimal model gives $\kappa_\sigma(K_F) = 0$.

Now apply the numerical-zero-fiber reduction of Gongyo–Lehmann [38, Theorem 1.3 and Corollary 4.5]. For a projective $\mathbb{Q}$ -factorial klt rational pair with a connected-fiber morphism and numerical dimension zero on the general fiber, that theorem produces a klt pair on a smooth birational base whose good-minimal-model existence is equivalent. Here the total space is the smooth graph, the boundary is zero, and the base has dimension less than n . The required numerical dimension is κ_σ , following the convention in [30, Section 3 and Theorem 3.2]. It is zero by the good model of F , so the numerical-dimension hypothesis holds. The induction hypothesis supplies the base good model, hence nonvanishing upstairs, contradicting (191). $\square$

Corollary F.4. *Let Y be a projective $\mathbb{Q}$ -factorial terminal birational model of X . If a rational divisor P has $\kappa(Y, P) \geq 1$, then $K_Y + cP$ is big for every rational $c > 0$. Consequently, if $\alpha \neq 0$ is a supporting functional of the pseudo-effective cone with $\alpha(K_Y) = 0$, then $\alpha(P) > 0$.*

Proof. Resolve a moving subsystem of a multiple of P . If its map is generically finite, P is big and the assertion follows from pseudoeffectivity of K_Y . Otherwise its general fiber is a proper positive-dimensional subvariety through very general points and is of general type by Proposition F.3. The canonical divisor of the smooth resolution is relatively big, so adding a sufficiently large ample pullback from the image makes it big. Since that pullback is bounded by a multiple of the resolved moving subsystem, pushforward gives $K_Y + aP$ big for some $a > 0$. Convexity with the pseudo-effective K_Y , and then addition of the pseudo-effective P , gives the assertion for every $c > 0$. A nonzero nonnegative functional on a closed convex cone is strictly positive on its interior. Applying it to $K_Y + cP$ proves the last assertion. $\square$

F.3 Valuations and positive currents

For a positive closed $(1, 1)$ -current T on a smooth projective variety V , let $\nu_E(T)$ denote the generic Lelong number of its pullback at a divisorial valuation E . The normalization is such that a reduced smooth divisor has generic number one. We write $A_V(E) = a(E; V, 0)$ for log discrepancy.

Lemma F.5. *For a fixed T and a fixed M , the set*

$$\{\nu_E(T) : A_V(E) \leq M\}$$

satisfies the ascending chain condition.

Proof. We induct on the integer part of M ; discrepancies over a smooth variety are positive integers. Suppose that a strictly increasing sequence exists, and discard initial terms so its members exceed a fixed positive number c . Skoda integrability and change of variables give

$$\nu_E(T) \leq A_V(E)\nu_x(T) \tag{192}$$

at a general point x of the center. Indeed every exponent smaller than $1/\nu_x(T)$ is locally integrable, whereas integrability after pullback imposes the corresponding discrepancy bound. Hence all centers lie in the fixed proper Siu locus $\{\nu_x(T) \geq c/M\}$, which is algebraic by Siu analyticity and projectivity [69].

If, after passage to a subsequence, the centers lie in a fixed codimension-at-least-two subvariety, principalize its ideal. All lifted centers lie in the exceptional locus, where the relative canonical divisor has positive integral order. The discrepancy bound for the new smooth ambient space has therefore decreased by at least one. Pull back T and apply induction.

Otherwise the centers lie in a fixed prime divisor D of the Siu locus. Write $T = c_D[D] + T'$ with T' positive and with zero generic Lelong number along D . The integers $\text{ord}_E(D)$ are bounded: the fixed effective divisor D has positive log canonical threshold, and $\text{lct}(D) \text{ord}_E(D) \leq A_V(E)$. Pass to a constant order. The residual numbers $\nu_E(T')$ are still strictly increasing, and after discarding a term have a positive lower bound. Equation (192) places their centers in a fixed Siu locus of T' , which does not contain D . Their intersection with D has codimension at least two, so the preceding case applies. $\square$

We also record the multiplier-ideal approximation used below [24, Theorems 5.11, 6.27, and 14.2]. On a smooth projective variety, if $\{T\}$ is a real algebraic class, one can choose integral line bundles L_k such that $c_1(L_k) - k\{T\}$ stays in a bounded, uniformly sufficiently ample set and $L_k \otimes \mathcal{J}(kT)$ is globally generated. Round the coefficients of $k\{T\}$ in a fixed integral basis and add a fixed sufficiently positive integral class. Nadel vanishing and Castelnuovo–Mumford regularity give generation. For every divisorial valuation,

$$\text{ord}_E \mathcal{J}(kT) \leq k\nu_E(T). \quad (193)$$

The local Bergman weight is bounded below by the original weight up to a constant; this inequality persists after pullback and gives (193). These are statements for a fixed current, not uniform assertions over all currents.

Proposition F.6. *Let Y be a fixed projective $\mathbb{Q}$ -factorial terminal birational model of X , with K_Y nonbig. A positive current in the pullback class of K_Y on a smooth resolution cannot, on a dense regular Zariski open, annihilate the tangent spaces of the general fibers of a rational fibration of positive relative dimension.*

Proof. Suppose otherwise, and choose a connected-fiber quotient of minimal base dimension b . If $b = 0$, the current vanishes on a dense open; the divisorial-current argument in Proposition F.3 contradicts (191). Thus $0 < b < n$. Resolve the graph and the space carrying the current:

$$\begin{array}{ccc} W & \xrightarrow{p} & Y \\ h \downarrow & & \\ B & & \end{array}$$

Here W, B are smooth projective and h has connected general fiber. Flatten the main component over a modification of the base, resolve that base, normalize the main transform, and then resolve upstairs. Every vertical prime on the flat transform maps to a divisor of the base: a prime over base codimension c would have generic fiber dimension at least $n - b + c - 1$, whereas all fibers have dimension at most $n - b$. Consequently every vertical divisor on W over base codimension at least two is exceptional over Y . This property persists under later resolutions and base modifications: a nonexceptional prime already maps onto a base prime and its generic image is unchanged.

For divisors and numerical classes put

$$T_B = p_* h^*.$$

This strict pullback is well defined on numerical classes because Y is $\mathbb{Q}$ -factorial: numerical classes upstairs decompose into pulled-back classes from Y and exceptional classes. It preserves pseudoeffectivity. For a higher base model $r : B' \rightarrow B$, its analogue satisfies $T_{B'} r^* = T_B$, by computing on a common graph. It kills divisors exceptional over B : otherwise a prime in such a pullback that dominates a divisor of Y would have center of codimension at least two on B , contrary to the property just established.

Descent of the current. Let T also denote the pulled-back current on W . Shrink to a smooth open B^0 so that h is smooth proper, the removed set contains no vertical divisor over B^0 , and the remaining open in every fiber is nonempty and connected. Away from that fixed removed algebraic set upstairs, T descends to a positive current S^0 . Indeed in submersion coordinates the horizontal coefficient distributions are independent of fiber variables by closedness, and the open parts of general fibers are connected. They therefore glue to S^0 on B^0 . The difference $T - h^* S^0$ is closed of order zero and supported on the removed algebraic set. The support theorem expresses it as a signed divisor sum; a closed order-zero $(1, 1)$ -current supported in codimension at least two is zero. Its horizontal coefficients are nonnegative, since a pullback has zero generic Lelong number along a horizontal divisor.

Subtract those finitely many global Siu components from T , obtaining a positive current T' that equals $h^* S^0$ on all of $h^{-1}(B^0)$. Choose a Kähler form ω representing an ample algebraic class and put $d = n - b$. Fiber integration gives $h_*(\omega^d) = c > 0$, constant on B^0 , by closedness. The projection formula therefore gives

$$c^{-1} h_*(T' \wedge \omega^d)|_{B^0} = S^0.$$

The left side defines a global positive closed extension. Its class is real algebraic, since $\{T'\}$ is the algebraic class $p^* K_Y$ minus real divisor classes and algebraic intersection and pushforward preserve such classes. Remove its divisorial parts along $B \setminus B^0$, and call the result S . On a dense open $T = h^* S$. Their difference is again closed of order zero, so the same support theorem leaves only a signed divisor sum. It has nonnegative coefficients at every prime dominating a base prime. For the latter assertion, the generic local map is a transverse power map times a submersion. A current with zero generic Lelong number along the base prime has zero generic number at such an upstairs prime: local target balls of a radius equal to a fixed power of the source radius give the usual Lelong comparison. The only possible negative coefficients lie over base codimension at least two and are exceptional over Y . It follows that

$$K_Y - T_B \{S\} \quad \text{is pseudo-effective.} \tag{194}$$

Horizontal parts subtracted above merely contribute effective divisors to this difference.

A negative canonical direction on the base. Choose a nonzero supporting functional α of the pseudo-effective cone at K_Y , and put $\eta = T_B^* \alpha$. The corresponding functionals η' on higher smooth base models are movable classes by pseudo-effective/movable duality [12, Theorem 0.2]. They annihilate base-exceptional divisors. Equation (194) gives $\eta \cdot \{S\} = 0$.

Only finitely many prime divisors on B have positive divisorial Lelong coefficient for S . Otherwise their nonzero strict pullbacks are effective divisors on Y killed by α , with pairwise disjoint prime supports. The supports are disjoint because a nonexceptional prime of Y cannot map into the intersection of two base primes. Only finitely many base primes have zero strict pullback, since

their total pullbacks are supported in the finite exceptional locus of p . In the finite-dimensional rational space of divisor classes, infinitely many remaining strict pullbacks have a rational relation. Its positive and negative sides are nonzero effective divisors with disjoint support; Lemma F.1 makes the relation $\mathbb{Q}$ -linear. They define a moving pencil killed by α , contradicting Corollary F.4.

We claim that

$$K_B \cdot \eta < 0. \quad (195)$$

The smooth general fiber F of h is of general type by Proposition F.3. The relative-positivity theorem of Kovács–Patakfalvi [57, Theorem 9.9] applies to a log canonical fiber space with smooth base, SNC total pair, and log-general-type geometric generic fiber; for a rational base divisor M with $\kappa(M) \geq 0$, it gives relative subadditivity with the term h^*M . Here both spaces are smooth projective, the boundary is zero, and we take $M = 0$. Invariance of plurigenera on the smooth locus identifies the generic-fiber Kodaira dimension with that of F . Thus

$$\kappa(W, K_W - h^*K_B) \geq \kappa(F, K_F) = n - b > 0. \quad (196)$$

This is an established general-type-fiber theorem, not another use or strengthening of Assumption A.1. Choose compatible canonical divisors and set $P = K_Y - T_B(K_B) = p_*(K_W - h^*K_B)$. For every sufficiently divisible m , pushing forward an effective divisor $\text{div}_W(\varphi) + m(K_W - h^*K_B)$ gives $\text{div}_Y(\varphi) + mP \geq 0$. The resulting injection of sections preserves their ratios, so $\kappa(Y, P) \geq 1$. Corollary F.4 now gives $0 < \alpha(P) = -K_B \cdot \eta$, proving (195).

Rational curves and an exact zero gap. Apply the multiplier approximation to S on B , and principalize $\mathcal{J}(kS)$ by $r_k : B_k \rightarrow B$. If F_k is its divisor, the class

$$P_k = \frac{r_k^*L_k - F_k}{k}$$

is nef. Put $\eta_k = T_{B_k}^* \alpha$. Since it kills exceptional divisors and $L_k - k\{S\}$ stays bounded,

$$0 \leq P_k \cdot \eta_k = O(k^{-1}), \quad K_{B_k} \cdot \eta_k = K_B \cdot \eta < 0.$$

For a fixed ample H on B , polarize by $P_k + k^{-1}r_k^*H$ plus a sufficiently small rational ample class on B_k . Approximate η_k by positive sums of covering-curve classes. Discard terms with nonnegative canonical degree. Some remaining covering curve has the ratio of polarization degree to anticanonical degree $O(k^{-1})$. Quantitative bend-and-break [59, Theorem 5] supplies rational curves through its general points with polarization degree at most $2 \dim B$ times that ratio. Parameterization and uncountability give a covering family of rational curves R_k satisfying

$$P_k \cdot R_k = O(k^{-1}), \quad H \cdot r_{k*}R_k = O(1). \quad (197)$$

The general parametrized member is free in characteristic zero. Consequently $K_{B_k} \cdot R_k \leq -2$, and

$$0 \leq K_{B_k/B} \cdot R_k = K_{B_k} \cdot R_k - K_B \cdot r_{k*}R_k = O(1). \quad (198)$$

Here the upper bound follows from the fixed ample-degree bound in (197).

Write $\bar{R}_k = r_{k*}R_k$. Every positive contact with an exceptional prime contributes a positive integer discrepancy times a positive integer contact degree to (198). Thus the number of such contacts, their discrepancies, and their contact degrees are uniformly bounded. Contacts with the finitely many strict transforms of divisorial Lelong components of S are bounded by their fixed

degrees against $\bar{R}_k$. Subtracting these divisorial parts from r_k^*S leaves a positive current. Using (193) and (197) gives

$$\begin{aligned} 0 &\leq \{S\} \cdot \bar{R}_k - \sum_E \nu_E(S) E \cdot R_k \\ &\leq \{S\} \cdot \bar{R}_k - (F_k/k) \cdot R_k \leq O(k^{-1}). \end{aligned} \tag{199}$$

The sum includes exceptional primes and those strict divisorial components; terms with zero contact are irrelevant.

Bounded ample degree gives only finitely many numerical classes of the integral cycles $\bar{R}_k$. Pass to one class. By Lemma F.5, the sums in (199) belong to an ACC set: there are a bounded number of terms, their nonnegative integral multiplicities are bounded, and all relevant discrepancies are bounded. Finite sums of this kind preserve ACC, by successively taking nonincreasing subsequences of their entries. The nonnegative gaps in (199) must therefore equal zero for some covering families. Otherwise a sequence tending to zero has a strictly decreasing positive subsequence, giving a strictly increasing sequence of the complementary sums.

For such a family the residual positive current restricts to degree zero on almost every R_k , and hence vanishes there. Off its removed divisors it is the original current. The discussion following Lemma F.2 and that lemma itself yield a positive-relative-dimensional rational quotient of B whose vertical directions are annihilated by S on an open. Composing with h gives a quotient of smaller base dimension whose vertical directions are annihilated by T , contradicting the choice of b . $\square$

F.4 Excluding normalized bounded curves

Run a canonical LMMP on X with scaling of a fixed very ample rational divisor A . For rational $t > 0$, its positive-threshold stages give terminal $\mathbb{Q}$ -factorial models Y_t on which $K_t + tA_t$ is nef, big, and semiample [9]. There are only finitely many divisorial contractions, since each lowers Picard number. Fix a late model Y after the last such contraction. All subsequent maps $Y \dashrightarrow Y_t$ are small, and are canonical nonpositive birational maps. If the program terminates, use the last model repeatedly. Define

$$\mu_t = \text{vol}(K_X + tA)^{1/n}. \tag{200}$$

Since K_X is not big, $\mu_t \rightarrow 0$. In particular K_Y is nonbig. The transform A_t is effective up to rational linear equivalence; all intersections below use general covering curves, which are not contained in a chosen such representative.

Lemma F.7 (Exceptional current comparison). *Fix a positive current T on a smooth resolution in the pullback class of K_Y . On a common smooth resolution of $Y \dashrightarrow Y_t$, write*

$$p^*K_Y = p_t^*K_t + J_t, \quad J_t \geq 0,$$

where J_t is exceptional over Y_t . Then $\nu_E(T) \geq \text{coeff}_E J_t$ for every prime on that resolution.

Proof. Pass to a common smooth model V also dominating the fixed resolution carrying T . Choose on the fixed resolution a sufficiently positive line bundle H . For sufficiently divisible k , multiplier approximation gives a globally generated $\mathcal{O}(kp^*K_Y + H) \otimes \mathcal{J}(kT)$. Pull this system to V . A general member D_k of its underlying effective divisor system has, at every specified prime E , multiplicity $\text{ord}_E \mathcal{J}(kT)$; global generation after removal of the ideal ensures no additional generic vanishing. Here the notation H also denotes its pullback to V .

Put $D'_k = p_{t*}D_k$ and $H_t = p_{t*}H$. The first is effective, and both are $\mathbb{Q}$ -Cartier because Y_t is $\mathbb{Q}$ -factorial. Pushing the rational linear relation gives

$$D'_k \sim_{\mathbb{Q}} kK_t + H_t.$$

Thus the exceptional divisor $D_k - p_t^*D'_k$ is rationally equivalent to

$$kJ_t + H - p_t^*H_t.$$

The two exceptional divisors are equal: their difference is numerically trivial and exceptional, so the negativity lemma applied with both signs makes it zero. Since $p_t^*D'_k$ is effective,

$$\text{coeff}_E D_k \geq k \text{coeff}_E J_t + \text{coeff}_E (H - p_t^*H_t).$$

The last coefficient is fixed for this model and t . Combining with (193), dividing by k , and letting $k \rightarrow \infty$ proves the assertion. The same proof can be made on each common model, so the comparison applies to all divisorial valuations needed later. No uniformity in t of the fixed error is required. $\square$

Proposition F.8. *There are no sequences $t_j \downarrow 0$ and covering families of curves on Y_{t_j} , with smooth projective domains C_j , for which both*

$$g(C_j), \quad \frac{(K_{t_j} + t_j A_{t_j}) \cdot C_j}{\mu_{t_j}}$$

are bounded by a fixed constant. Degrees mean degrees on the domains, or equivalently intersection with their pushforward cycles.

Proof. Suppose such families exist. Slice their parameters and resolve total evaluation, the map to the fixed Y , the fixed smooth resolution carrying T , and the common models comparing Y and Y_t . Denote the resulting generically finite dominant morphism by $q_t : U_t \rightarrow Y$. The source is smooth near its complete general parameter fiber and has dimension n , with parameter space of dimension $n - 1$. Rational maps from this smooth source to each required proper target extend at codimension-one points by the valuative criterion of properness. Their indeterminacy loci consequently have codimension at least two, and their images cannot dominate the parameter space. Shrink that space to avoid these finitely many images, and resolve and compactify preserving the remaining open. Equivalently, nontrivial smooth-center blowups have codimension-at-least-two centers; a codimension-one Cartier-center blowup is the identity. Thus the complete smooth general parameter fiber is still C_t , unchanged as a curve, and its genus is unchanged.

For a prime Z of U_t exceptional over Y , the restricted valuation of its function field is $r_Z \text{ord}_E$ for a divisorial valuation E over Y . The coefficient of Z in the ramification difference is

$$\text{coeff}_Z(K_{U_t} - q_t^*K_Y) = r_Z a(E; Y, 0) - 1. \quad (201)$$

This follows by factoring through a model extracting E and calculating the tame ramification of DVRs. It is positive and bounded away from zero: Y is terminal, so $a(E; Y, 0) > 1$, and a fixed Cartier index of K_Y puts discrepancies in a fixed rational lattice. The remaining ramification coefficients are nonnegative. On the general parameter fiber,

$$K_{U_t} \cdot C_t = 2g(C_t) - 2, \quad q_t^*K_Y \cdot C_t \geq 0.$$

The latter inequality follows from pseudoeffectivity and the covering property. Therefore the total ramification cost is bounded. Formula (201) bounds the number of positive exceptional contacts,

their discrepancies, their ramification indices, and their integral contact degrees. Moreover $q_t^* K_Y \cdot C_t$ lies in a fixed rational lattice inside a bounded interval, so takes only finitely many values.

The generic Lelong number of $q_t^* T$ at Z is $r_Z \nu_E(T)$. Extract E , remove its generic divisorial part, and use the transverse-power-map comparison for the zero-generic-number remainder. Lemma F.7 shows that these coefficients dominate the coefficients of the pulled-back canonical difference J_t . Its support is exceptional over Y as well as Y_t , since the map between these models is small. Subtracting all exceptional divisorial parts of $q_t^* T$ leaves a positive current. Consequently

$$\begin{aligned} 0 &\leq q_t^* K_Y \cdot C_t - \sum_{Z \text{ exceptional}} r_Z \nu_E(T) Z \cdot C_t \\ &\leq K_t \cdot C_t \leq (K_t + tA_t) \cdot C_t = O(\mu_t). \end{aligned} \tag{202}$$

All divisor contacts used here are nonnegative, since the curves cover the total space.

On the fixed smooth resolution $W \rightarrow Y$, one has

$$a(E; W, 0) = a(E; Y, 0) - \text{ord}_E(K_W - p^* K_Y) \leq a(E; Y, 0),$$

because the relative canonical divisor is effective. Thus Lemma F.5 applies with a fixed bound. The sums in (202) form an ACC set, their number of terms and integral multiplicities being bounded. Pass to a fixed value of $q_t^* K_Y \cdot C_t$. Since $\mu_t \rightarrow 0$, the exact-zero argument of (199) gives a family for which the left gap is zero.

The residual positive current then restricts to zero on almost every curve in that family. On the dense open away from the removed exceptional divisors it agrees with the original current. Its curve tangents therefore generate, by Lemma F.2, an annihilated rational fibration of positive relative dimension on Y . This contradicts Proposition F.6. $\square$

F.5 The signed alternative

Proposition F.9. *On any model Y_t , let a signed rational divisor represent K_t up to $\mathbb{Q}$ -linear equivalence. On a log resolution $p : W \rightarrow Y_t$, let D be the reduced SNC divisor consisting of its strict support and all exceptional divisors. Then $K_W + D$ is big.*

Proof. Put $P = K_W + D$, and suppose it is not big. It cannot have Iitaka dimension at least one: by Corollary F.4, $K_W + cP$ would be big, and $(1 + c)P = (K_W + cP) + D$ would then be big as well. Hence $\kappa(W, P) \leq 0$.

Run the dlt LMMP for P with ample scaling. The divisor P has a signed rational representative supported on the floor D , and its transforms retain that property. Every negative flip must meet the floor: on its complement the representing rational section is nowhere vanishing, so the adjoint has zero degree on every complete curve there. By Proposition B.4, an infinite sequence would eventually have all flips disjoint from the floor, a contradiction. Divisorial contractions are finite in number. Thus the program terminates at a $\mathbb{Q}$ -factorial dlt log minimal model (Z, D_Z) .

The boundary is reduced, K_Z is pseudo-effective as the birational pushforward of K_W , and the nef adjoint has a signed rational representative supported on D_Z . Its restriction to D_Z is semiample by Proposition B.3. Theorem C.1 applies, with the pseudo-effective perturbation K_Z , and gives abundance. Since the Iitaka dimension is at most zero, the numerical dimension is zero. Intersecting $K_Z + D_Z \equiv 0$ with an ample $(n - 1)$ -fold product, and using pseudoeffectivity of K_Z , forces $D_Z = 0$. The signed relation then gives $K_Z \sim_{\mathbb{Q}} 0$.

On a common resolution, the pullback of K_W is rationally equivalent to a signed divisor exceptional over Z . It is pseudo-effective. A positive current in that class has zero intersection with

a pullback of an ample class on Z to the power $n - 1$, so is supported on the exceptional locus. The support theorem makes it effective divisorial, and independence of exceptional divisor classes by negativity says that the original signed coefficients are these nonnegative coefficients. Thus K_W is $\mathbb{Q}$ -linearly effective, contradicting (191). $\square$

G Very-general jet estimates

We retain the counterexample in Equation (191) and the induction hypothesis of Assumption B.1. Thus X is smooth projective, K_X is pseudo-effective, and $\kappa(X, K_X) = -\infty$, whereas good minimal models exist in smaller dimensions. We use Propositions F.3, F.8, and F.9. Dimension one is impossible, since a smooth curve with pseudo-effective canonical divisor has genus at least one. Hence $n = \dim X \geq 2$.

Fix the late terminal $\mathbb{Q}$ -factorial model Y from Proposition F.8. Write $K = K_Y$ and $A = A_Y$. The subsequent scaling models Y_t are small modifications of Y ; the transforms A_t are effective up to $\mathbb{Q}$ -linear equivalence, and $K_t + tA_t$ is nef, big, and semiample. Put

$$\mu_t = \text{vol}(X, K_X + tA)^{1/n}.$$

Here, in the volume on X , A denotes the original ample divisor. We have $\mu_t \rightarrow 0$. Choose an integer $m > 0$ with mK Cartier.

G.1 Normalization and two local tools

Fix a small rational number $\epsilon > 0$. An integer $q > 0$ will be chosen after ϵ and before t . As rational $t \downarrow 0$, choose positive integers $r = r(t)$ with

$$r\mu_t \rightarrow \epsilon, \quad L = r(K + tA), \quad L_b = L + bmK \quad (0 \leq b \leq q). \quad (203)$$

Only rational weights b are used for model constructions; volume functions in integrals are extended continuously to real weights. The positive part of L_b is the nef semiample class

$$N_b = (r + bm)(K_s + sA_s) \quad \text{on } Y_s, \quad s = \frac{rt}{r + bm}.$$

When emphasizing the weight, denote this axis model by $Y(b) = Y_s$ and write A_b for its transform of A . All references to positive parts below mean these classes on their scaling models, or their pullbacks to common resolutions. Monotonicity of volume in pseudo-effective order gives

$$\text{vol}(L) \leq \text{vol}(L_b) \leq \left(1 + \frac{qm}{r}\right)^n \text{vol}(L). \quad (204)$$

Indeed $L_b - L = bmK$ is pseudo-effective, and $(1 + bm/r)L - L_b = bmtA$ is effective up to equivalence. Consequently N_b^n is bounded above and bounded away from zero, uniformly for $b \in [0, q]$, and tends uniformly to ϵ^n . More explicitly, for every fixed q , once t is sufficiently small depending on q ,

$$\frac{\epsilon^n}{2} \leq N_b^n \leq 2\epsilon^n \quad (0 \leq b \leq q). \quad (205)$$

The displayed constants are independent of q . All unspecified constants in this section may depend on n, ϵ, q, Y, A, m , but not on sufficiently small t or on b .

A family of curves with uniformly bounded geometric genus and N_b -degree cannot cover the corresponding Y_s for arbitrarily small t . In fact $s/t \rightarrow 1$ uniformly in b , and

$$(r + bm)\mu_s = \text{vol}(L_b)^{1/n}$$

is bounded above and away from zero. Thus such curves would have uniformly bounded $(K_s + sA_s)$ -degree divided by μ_s , contrary to Proposition F.8. We call this consequence the *normalized curve exclusion*.

The following numerical form of subadjunction is useful because an auxiliary pair need only be lc at the generic point of its center. All intersections with a divisor on a possibly nonnormal center mean intersections after normalization and resolution.

Lemma G.1 (Numerical subadjunction). *Let Z be projective and $\mathbb{Q}$ -factorial klt, let $\Theta \geq 0$ be rational, and let V be an lc center of (Z, Θ) at whose generic point the pair is lc. If $f = \dim V > 0$, P is a nef $\mathbb{Q}$ -Cartier class on V , and $V^* \rightarrow V$ is a resolution, then*

$$K_{V^*} P^{f-1} \leq (K_Z + \Theta)|_V P^{f-1}. \quad (206)$$

Proof. The dlt modification for an arbitrary effective boundary [36, Theorem 2.10], applied after truncating coefficients exceeding one, gives

$$K_Q + B + E = p^*(K_Z + \Theta),$$

where Q is $\mathbb{Q}$ -factorial, (Q, B) is dlt, and $E \geq 0$ is supported above the non-lc locus. Choose, over the generic point of V , a minimal lc stratum, and denote its closure by S . The divisor E does not contain S . Iterated dlt adjunction, including the effective restriction of E , produces an effective divisor B_S with

$$K_S + B_S \sim_{\mathbb{Q}} (p|_S)^*((K_Z + \Theta)|_V).$$

The pair is klt over the generic point of V ; singularities elsewhere are not asserted to be klt.

Factor $S \rightarrow V^\nu$ through its Stein base V' . This is a klt-trivial fibration: generic klt, effectivity of the boundary, and connected fibers give the rank-one condition. The canonical bundle formula, in precisely this generically subklt form, gives a discriminant and a b-nef moduli b-divisor [34, Definition 3.1 and Theorem 3.5]. The discriminant trace on V' is effective. To see the sign, over a generic base prime a divisor in its inverse image has multiplicity at least one and a nonnegative boundary coefficient; the lc threshold of the fiber is therefore at most one. Coefficients exceeding one in the discriminant cause no difficulty for the present inequality.

Intersect the resulting base formula with the pullback of P^{f-1} . The moduli term is nonnegative: on a model where it is nef this is a nef intersection, and exceptional terms vanish on pushing down. The effective discriminant is also nonnegative. Finally the codimension-one ramification formula for the finite map $V' \rightarrow V^\nu$, followed by projection, can only increase the canonical intersection. Discrepancy terms on resolutions have images of codimension at least two and pair trivially with the pulled-back P^{f-1} . Dividing by the finite degree gives Equation (206). $\square$

Lemma G.2 (Tracking a moving base component). *Let P be nef, semiample, and big on a projective d -fold Z . Work at very general smooth marked points x . Choose $d + 1$ levels*

$$\tau_0 < \tau_1 < \cdots < \tau_d, \quad \tau_{i+1} - \tau_i = \delta > 0.$$

Suppose, in sufficiently large divisible degree k , all the spaces

$$\mathcal{V}_{i,x} = H^0(Z, kP \otimes \mathfrak{m}_x^{[k\tau_i]})$$

are nonzero, and the base locus of $\mathcal{V}_{0,x}$ has a positive-dimensional component through x . Then one obtains an algebraic family of integral components $V = V_x$ sweeping Z , of dimension $0 < f < d$, and a step for which V is a component of both successive base loci. At its generic point the higher subseries has an isolated base component V and vanishes to order at least $k\delta/2$, once k is sufficiently large. Moreover

$$\begin{aligned} P^f.V &\leq (2/\delta)^{d-f} P^d, \\ K_{V^*}P^{f-1} &\leq (K_Z + cP)|_V P^{f-1}, \quad 0 < c \leq 2d/\delta, \end{aligned} \tag{207}$$

whenever Z is $\mathbb{Q}$ -factorial klt.

Proof. The base loci increase with the level. Following containing irreducible components through the marked point produces a nested chain of proper positive-dimensional subvarieties. There are only $d - 1$ possible dimensions, so two successive components agree. For fixed degree, jet kernels form vector bundles after shrinking the marked-point parameter space. Components of their base loci spread after a finite parameter extension and further shrinking. We may fix the chosen step, dimension, and component on an irreducible parameter space T . Its incidence $\Gamma \subset T \times Z$ dominates Z , because it contains the varying marked point.

Here is the multiplicity argument. Regard a local section of the higher kernel bundle as a varying element of the fixed vector space $H^0(Z, kP)$. A parameter derivative of order j loses at most j orders of vanishing along the moving diagonal. Thus, for $j < k\delta - O(1)$, every such derivative belongs to the lower kernel and vanishes on the selected incidence Γ . At a general smooth point of Γ , the projection $\Gamma \rightarrow Z$ is smooth. Therefore the pure parameter directions, together with TT , span $T(T \times Z)$. In local coordinates over Z , this says that normal coordinates to Γ can be chosen among parameter coordinates. Vanishing of all parameter derivatives of orders below h is consequently equivalent there to membership in $\mathcal{I}_\Gamma^h$. Restricting to a parameter fiber proves generic multiplicity at least $k\delta - O(1)$ along V , hence at least $k\delta/2$. This applies to a local basis of the higher kernel bundle, and therefore to every section of the higher subseries.

Put $a = d - f$ and $h = \lceil k\delta/2 \rceil$; the stronger bound $k\delta - O(1)$ permits this choice for sufficiently large k . At the smooth generic point of V , the base ideal is maximal-ideal primary and lies in the h -th power of that maximal ideal. Cut by a general members of the subseries. Their local intersection multiplicity along V is at least h^a . For completeness, global excess base components do not invalidate this bound: at each cut discard components wholly contained in the base locus before the next intersection. None contains the generic point of V , since V is a base-locus component. The discarded cycles have nonnegative P -degree by nefness. The remaining proper intersections therefore have total P^f -degree at most $k^a P^d$. Since $h \geq k\delta/2$, this gives the first inequality.

The local lc threshold of this primary ideal is at most a/h : the exceptional divisor of the blowup of its smooth closed point in the a -dimensional transverse local scheme gives this bound. On a log resolution, a sufficiently long average of general members realizes the ideal threshold by an effective rational divisor $\Theta \sim_{\mathbb{Q}} cP$. It is lc at the generic point of V , with V an lc center, and $c \leq ka/h \leq 2d/\delta$. Lemma G.1 gives the second inequality. $\square$

We record how these estimates give curves rather than merely bounded intersection numbers. Suppose a moving f -fold V is of general type, $P|_V$ is nef and big, and

$$0 < P^f.V \leq C_0, \quad K_{V^*}P^{f-1} \leq C_1 P^f.V. \tag{208}$$

Effective birationality gives a uniform pluricanonical degree whose moving part, on a further resolution, is a big basepoint-free Cartier divisor H defining a birational morphism [42, Theorem 4.0.1].

Thus $HP^{f-1} \leq C_2 P^f.V$. Mixed Hodge index gives

$$H^j P^{f-j} \leq C_2^j P^f.V \quad (0 \leq j \leq f);$$

no lower bound on $P^f.V$ is needed here. For $f > 1$, cut by $f - 1$ general members of $|H|$. The resulting integral curves have bounded P -degree and are birational to linear curve sections of a projective variety of bounded degree. Generic plane projection bounds their geometric genus. For $f = 1$, the canonical-degree bound already bounds the genus. The same argument works for a log smooth pair of log general type with reduced boundary, using coefficients in the fixed set $\{1\}$ in effective birationality.

If such components sweep an ambient space mapping to Y , and their curves project nonconstantly with N_b -degree bounded by their P -degree, we obtain forbidden covering curves on Y_s . Nonconstant projection follows by choosing the complete intersections generally for the big birational system. The curves can be parameterized in covering algebraic families: the components sweep, the choices can avoid any prescribed countable exceptional union, and Hilbert schemes and spaces of maps have only countably many components. We will use this consequence of Equation (208) repeatedly.

Proposition G.3 (Scalar jet bound). *For sufficiently small t , let P be the positive part of $2r(K + 2tA)$, and put $\rho = (P^n)^{1/n}$. At a very general point, every nonzero section of every divisible multiple kP has order at most $4k\rho$. The same statement holds on a common resolution after adding an effective exceptional divisor that does not change the section spaces. Moreover*

$$2r\mu_t \leq \rho \leq 4r\mu_t,$$

so in particular $\rho \leq 8\epsilon$ for small t .

Proof. The volume bounds follow from pseudo-effectivity of K :

$$\text{vol}(K + tA) \leq \text{vol}(K + 2tA) \leq 2^n \text{vol}(K + tA).$$

Suppose the asserted order bound fails for arbitrarily small t . Taking powers of the offending section permits arbitrarily large divisible degrees. Choose $n + 1$ levels strictly between ρ and 4ρ , with equal gaps comparable to ρ . All corresponding subseries are nonzero. The base point at the lowest level cannot be isolated: local Bezout for n general members would exceed the total intersection $k^n P^n$. Lemma G.2 therefore supplies a moving proper positive-dimensional component V .

The component is of general type by Proposition F.3. On the scaling model for $K + 2tA$, the effective transform of A does not contain a general such component, and $K_Z|_V \leq (2r)^{-1}P|_V$ in effective order. Equations (207) consequently imply Equation (208) with constants uniform in t ; the gap is bounded above and away from zero because ρ is comparable to the fixed ϵ . The resulting bounded-genus, bounded-normalized-degree covering curves contradict Proposition F.8. Exceptional additions preserve sections and their orders at general points, proving the additional assertion. $\square$

G.2 The two-slot bundle

On $Y \times Y$, let

$$Z_0 = \mathbb{P}(mK_1 \oplus mK_2), \quad \xi = \mathcal{O}_{Z_0}(1), \quad M = L_1 + L_2 + q\xi. \quad (209)$$

We use the convention that sections of $k\xi$ are symmetric powers of the indicated direct sum. Fixing one base slot gives the one-slot bundle $\mathbb{P}(\mathcal{O}_Y \oplus mK)$, with class $L + q\xi$, up to a constant line from the fixed slot. We call it a *slice*. The torus open in either bundle is the complement of its two axes.

Lemma G.4 (Models, volume, and weights). *Both the total class M and the slice class have big semiample positive parts P on $\mathbb{Q}$ -factorial klt models Z , with*

$$K_Z + \Delta \sim_{\mathbb{Q}} c_t P, \quad \Delta \geq 0,$$

where c_t is bounded independently of small t . Their volumes satisfy, respectively,

$$\text{vol}(M) = \frac{(2n+1)!}{(n!)^2} \int_0^q \text{vol}(L_b) \text{vol}(L_{q-b}) db, \quad (210)$$

$$\text{vol}(L + q\xi) = (n+1) \int_0^q \text{vol}(L_b) db. \quad (211)$$

In particular both volumes are bounded above and below by positive constants times q . For each rational weight, on common resolutions,

$$P \sim_{\mathbb{Q}} N_{b,1} + N_{q-b,2} + E_b, \quad E_b \geq 0, \quad (212)$$

with the analogous one-term formula on slices. At a generic point of a base divisor of a slice, and on a general torus fiber, the full system has no fixed subtraction.

Proof. The product $Y \times Y$ is $\mathbb{Q}$ -factorial. Indeed on a product resolution the Picard group has no cross term because the smooth models have irregularity zero; the two factorwise $\mathbb{Q}$ -factorial descent statements then apply. Products of canonical singularities are canonical, and the projective bundles are klt. The two disjoint Cartier axes form a plt boundary D , and

$$K_{Z_0} + D = K_{\text{sum}}, \quad D \sim 2\xi - mK_{\text{sum}}.$$

The notation K_{sum} means $K_1 + K_2$, or K on a slice.

For $0 < \sigma \leq 1$, put

$$\gamma = 2\sigma + \frac{q(1+\sigma m)}{r}, \quad \Xi_t \sim_{\mathbb{Q}} \xi + \frac{(1+\sigma m)t}{\gamma} A_{\text{sum}}.$$

Both endpoint weight systems are big, so there are effective rational representatives of Ξ_t containing neither axis. We first obtain a threshold bound independent of σ . Restrict to $0 < t \leq \sigma/(1+m)$. Since $\gamma \geq 2\sigma$, the coefficient

$$a = \frac{(1+\sigma m)t}{\gamma}$$

lies in $(0, 1]$. Thus the numerical classes $\xi + aA_{\text{sum}}$, and their restrictions to either fixed axis, range over bounded segments independent of σ, q, r, t . On a fixed smooth resolution their effective pullbacks have uniformly bounded degree against a fixed very ample class. That degree bounds their multiplicity at every point, including the coefficients of exceptional components. Rational denominators do not enter this estimate. The smooth lc threshold is at least the reciprocal of maximal multiplicity.

On the fixed klt space, write the crepant boundary on its resolution as an SNC divisor with coefficients below one. The minimum of one and the positive numbers one minus its positive coefficients bounds its log discrepancies below by a fixed positive multiple of smooth log discrepancies. Therefore the preceding smooth bound gives a common threshold $\eta > 0$ on the original space. The same reasoning on the axes gives, after decreasing η , the same bound for both restricted divisors.

Now choose rational $0 < \sigma < \min\{1, \eta/4\}$, independently of q, t , and then take t small enough that $t \leq \sigma/(1+m)$ and $q(1+\sigma m)/r < \sigma$. It follows that $\gamma < 3\sigma < \eta$. Inversion of adjunction with

the disjoint coefficient-one axes gives plt near them; away from them the threshold bound gives klt. Lowering their coefficients to $1 - \sigma$ consequently shows that

$$(Z_0, (1 - \sigma)D + \gamma\Xi_t)$$

is klt. This choice respects the order: first ϵ , then q , then sufficiently small t with r as in Equation (203). A direct calculation gives its adjoint class

$$K_{Z_0} + (1 - \sigma)D + \gamma\Xi_t \sim_{\mathbb{Q}} \frac{1 + \sigma m}{r} M.$$

The big klt adjoint has a good minimal model by [9]. Pushing the boundary to this model gives the stated Δ and $c_t = (1 + \sigma m)/r$.

For the volumes, the weight- l summand in degree k has base classes $kL_{l/k}$ and $kL_{q-l/k}$; on a slice there is one such factor. The section decomposition and asymptotic Riemann–Roch give Equations (210)–(211) as Riemann sums. One can justify passage to the integral without assuming uniform asymptotic Riemann–Roch: partition $[0, q]$ into rational bins, compare weight classes in each bin by adding and subtracting the bin width times a fixed very ample divisor dominating both mK and $-mK$, take divisible-degree limits, and then shrink the bins. Volume continuity gives the formulas. Equation (204) supplies their uniform bounds and, in particular, proves bigness used above. In the range of Equation (205), the bounds needed when choosing q are explicitly

$$\text{vol}(M) \geq \frac{(2n+1)!}{4(n!)^2} q\epsilon^{2n}, \quad \text{vol}(L + q\xi) \leq 2(n+1)q\epsilon^n.$$

Their coefficients are independent of q ; the smallness threshold for t is allowed to depend on q .

Compare a complete system on a common resolution with the subsystem at a fixed rational weight. The latter has fixed divisor consisting of its toric monomial and the base-model fixed differences. Subtracting the fixed divisor of the full system leaves an effective divisor E_b ; its moving class is precisely the sum of the pullbacks of the indicated N_b 's. This proves Equation (212). The same comparison in individual sufficiently divisible degrees shows that every section at that weight, after the full fixed subtraction, contains the corresponding multiple of E_b .

At a generic base-divisor point the small base-model maps are isomorphisms. The two endpoint systems are free there in divisible degree, so together they have no common zero on the projective-line fiber. The full system therefore has no fixed subtraction there. The same holds on a general fiber. Finally, when restricting Equation (212) to a component sweeping the torus open, choose a general component not contained in $\text{Supp } E_b$. There are only countably many rational weights, so these effective restrictions can be required simultaneously. $\square$

For a nef $\mathbb{Q}$ -Cartier class P on a projective variety Z and a smooth point $x \in Z$, its Seshadri constant is

$$\varepsilon(P; x) = \inf_{\substack{C \subset Z \\ \text{integral curve} \\ x \in C}} \frac{P \cdot C}{\text{mult}_x C}.$$

Proposition G.5 (Large two-slot Seshadri constant). *For each fixed sufficiently small $\epsilon > 0$, there is an integer $q > 0$ such that, for all sufficiently small rational $t > 0$ and r as in Equation (203), the positive part P of Equation (209) has*

$$\varepsilon(P; x) > 2n + 2$$

at a very general smooth point of its torus open. In particular, for some sufficiently divisible integer $k > 0$, the complete system $|kP|$ separates jets of order strictly greater than $(2n + 2)k$ at such a point. The section spaces and these general-point jets agree on the original bundle and on common birational resolutions.

Proof. Write $d = 2n + 1$. Choose q sufficiently large that the volume root in Equation (210) admits $d + 1$ levels above $2n + 3$, with equal gap δ as large as needed below. This is possible uniformly for small t , since the volume is bounded below by a positive constant times q . Assume that the Seshadri assertion fails for arbitrarily small t . Jet counts show that the systems at all these levels are nonzero in sufficiently large divisible degree. A curve through the marked point with degree-to-multiplicity ratio below the lowest level is contained in that system's base locus. Such a curve exists from the assumed Seshadri bound, whether or not the infimum defining the constant is attained. Lemma G.2 supplies a moving proper component V and both estimates (207).

For all moving components under consideration, the effective boundary Δ of Lemma G.4 does not contain the component. Thus $K_Z|_V \leq c_t P|_V$ in effective order. Whenever V is of general type, the curve construction following Equation (208) and weight domination give a contradiction. We must also handle components that are not generically finite over a proper base image.

Restriction to a slice. Fix a general value of one slot projection of V , and suppose the corresponding fiber component F has $0 < \dim F < n + 1$. It is an isolated high base component for the restricted subseries on the opposite slice.

We give the local justification. On a common isomorphic torus open, let I be the total high-series base ideal. Remove the other base components from a dense open V° of V . On an ambient neighborhood U of its generic point, $\text{Supp}(\mathcal{O}_U/I) = V \cap U$. After restricting V and its image to smooth opens, generic smoothness makes the general projection fiber generically reduced. A general fiber component F meets V° , and the ideal of V restricts to the ideal of F at its generic point. Near that point, restriction to the slice therefore has support exactly F ; the restricted ideal is maximal-ideal primary there. The inclusion in the h -th power of the generic ideal of V restricts to the h -th power of the generic ideal of F . The fixed differences of the full and slice models are units on a further common open, so dividing them out does not alter this statement. The restricted global sections are a subseries of the complete slice system after trivializing the fixed-slot lines. Thus the proof of Lemma G.2 applies to F , using slice volume, with multiplicity at least $k\delta/2$.

These F 's may be chosen in families sweeping the opposite slice. Indeed the incidence of the V -family dominates the total torus. Choose a general incidence point, not necessarily the original marked point, and then a general fiber of its first slot evaluation. Dominance gives the required dominant evaluation onto the opposite slice. Generic flatness permits the component choices just made. For each fixed q, t , very general choices also avoid the exceptional sets for all rational weights.

Proper base images on a slice. If F is generically finite over a proper positive-dimensional image W in the remaining base, then W , and hence F , is of general type by Proposition F.3. The slice estimates, subadjunction, and weight domination give forbidden bounded curves.

Otherwise F is saturated over its base image W : on the original bundle it is the full projective-line bundle over W . Put $w = \dim W = \dim F - 1$. The nef weight comparison gives

$$qN_b^w.W \leq P^{w+1}.F. \quad (213)$$

Indeed $P - \pi^*N_b$ is effective on the resolved graph, so successive mixed nef intersections give $P^{w+1}.F \geq P(\pi^*N_b)^w.F$. The latter is $qN_b^w.W$, since endpoint freeness gives fiber degree q . If W is a point, the lower bound q contradicts the slice Bezout upper bound $2(n+1)q\epsilon^n(2/\delta)^n$, after q has been chosen to make δ large. In particular the coefficient used in this choice does not depend on q .

Suppose $w > 0$. We construct a small adjoint on W , since the ruled F itself need not be of general type. In degree k , expand every section of the high-vanishing slice subseries into its toric weights. At the generic point of W , let I_l be the ideal generated by the coefficients of the nonzero

weight l over all these sections. Put $h = \lceil k\delta/2 \rceil$. All I_l lie in $\mathfrak{m}_W^h$: vanishing along the generic torus fiber says that the coefficient of every Laurent monomial vanishes to this order. Their sum is $\mathfrak{m}_W$ -primary. Otherwise its larger common zero germ, times the generic torus fiber, would contradict isolation of F .

Work in the regular local ring at the generic point of W , of dimension $c = \text{codim } W$. Resolve the finitely many ideals simultaneously and consider the rational lc polytope

$$u_l \geq 0, \quad \sum_l u_l \text{ord}_E(I_l) \leq a(E; Y, 0).$$

Include the exceptional divisor of the blowup of the closed point. Since every $I_l \subset \mathfrak{m}_W^h$, it gives $\sum_l u_l \leq c/h$. The polytope is compact and has a positive rational maximum for $\sum_l u_l$. At a maximizing point, each coordinate participates with positive order in an active inequality; otherwise that coordinate could be increased. The corresponding lc center C_l is contained in $V(I_l)$ and contains the generic point of W . Their intersection is contained in $V(\sum_l I_l)$, which is W locally. The intersection property for lc centers [55, Theorem 1.7], applied to the identity morphism on the lc open, therefore makes W an lc center generically. To realize the ideals, choose an integer $M_0 > \max_l u_l$ and M_0 general coefficient divisors $D_{l,j}$ from each system. Set $\Theta = \sum_l (u_l/M_0) \sum_{j=1}^{M_0} D_{l,j}$. On the simultaneous resolution its fixed crepant coefficients are at most one, and its general moving divisors meet transversely with coefficients below one. Thus the pair is lc near the generic point of W , and every active valuation remains an lc place. The intersection argument therefore applies to this actual effective rational divisor. It also applies when some maximizing coordinate u_l is zero: the active valuation blocking that coordinate still has center in $V(I_l)$.

Set

$$b = \frac{\sum_l u_l(l/k)}{\sum_l u_l}, \quad c' = k \sum_l u_l \leq \frac{kc}{h}.$$

Its class on the base is $c'L_b$. Transform to the small scaling model for L_b , which is unchanged at the generic point under consideration. There its class is $c'N_b$. Lemma G.1, effective A_b , and Equation (213) give bounded adjoint intersection and bounded N_b^w -degree for W . Since c' is bounded by a dimension-dependent multiple of $1/\delta$, these bounds are uniform in t . The variety W is of general type by Proposition F.3; normalized curve exclusion again gives a contradiction.

A slice multisection. The remaining proper positive-dimensional slice case is $\dim F = n$, with F generically finite of degree e over the whole base. Weight comparison and the slice degree bound give

$$eN_b^n \leq P^n.F,$$

so e is bounded uniformly in t . The closure of F on the original bundle is neither axis. Its intersections with the two axes push to effective integral Weil divisors D_1, D_2 on Y , and the projective-bundle relation gives

$$D_1 - D_2 \sim_{\mathbb{Q}} \pm emK. \tag{214}$$

At the appropriate endpoint weight $b \in \{0, q\}$, the difference $E_b|_F$ contains the corresponding toric-axis intersection with coefficient q . There is no full fixed subtraction above generic base-divisor points by Lemma G.4. Intersecting with N_b^{n-1} and using mixed nef comparison yields

$$qN_b^{n-1}.D_i \leq P^n.F.$$

The other divisor has bounded degree for the same N_b , because of Equation (214) and

$$0 \leq K_{Y(b)}N_b^{n-1} \leq \frac{N_b^n}{r + bm}.$$

On a log resolution add the reduced strict support of $D_1 + D_2$ and all exceptional divisors to the canonical divisor. This log canonical divisor is big by Proposition F.9. Its intersection with the pulled-back N_b^{n-1} is bounded: exceptionals vanish under projection, and reduced support has degree no larger than $D_1 + D_2$. Log effective birationality and the curve construction therefore contradict normalized curve exclusion on the base itself.

Reduction to a correspondence. We have excluded fiber dimensions strictly between zero and $n + 1$ over either slot. A full $(n + 1)$ -dimensional fiber over one slot would mean that V contains the whole opposite slice over its first image W . Its fiber dimension over the other slot would then be $\dim W + 1$, strictly between zero and $n + 1$, unless V were the full total space. That is impossible. Hence V projects generically finitely in both slots and $\dim V \leq n$. If either image is proper, general type and the total-space estimates give the previous curve contradiction. The same is true if V is of general type. Only the case $\dim V = n$, dominant with bounded degree over both copies of Y , remains.

Moving branch divisors. Take the dominating algebraic family of these correspondences, resolving maps after shrinking the parameter space. Suppose a divisorial branch component of one projection moves. On a resolution V^* , choose a ramified divisor R above its generic point. There is a rational weight b for which the effective difference $E_b|_{V^*}$ does not contain R . Indeed take a fixed free divisible degree of P ; some section does not vanish generically on R . Monomial-weight sections span that degree, so one of them does not vanish there. Only finitely many weights are tested in this fixed family; their model comparisons can be resolved simultaneously.

Let J be the branch divisor on the corresponding small axis model. Nef comparison on R , ramification into a terminal target, and pseudo-effectivity of its canonical divisor give

$$N_b^{n-1}.J \leq P^{n-1}.R \leq K_{V^*}P^{n-1} \leq C_q. \quad (215)$$

The ramification coefficient of R is a positive integer, so is at least one; the other ramification and exceptional terms are nonnegative. The last bound is the total-space subadjunction estimate.

We spell out the adjunction needed for J , without assuming $(Y(b), J)$ globally lc. Terminal $Y(b)$ is smooth in codimension two. Thus normalization and conductor adjunction along J give

$$K_{J^\nu} + C = (K_{Y(b)} + J)|_{J^\nu}, \quad C \geq 0$$

in codimension one. On resolving J^ν , exceptional divisors map to codimension at least two and pair trivially with the pulled-back N_b^{n-2} . Consequently, with $d_J = N_b^{n-1}.J$,

$$\begin{aligned} K_{J^*}N_b^{n-2} &\leq (K_{Y(b)} + J)JN_b^{n-2} \\ &\leq \frac{d_J}{r + bm} + \frac{d_J^2}{N_b^n} \leq C'_q d_J. \end{aligned} \quad (216)$$

For the second line, a general moving J is not a component of a fixed effective representative of A_b , giving the first term; mixed Hodge index gives the second. The lower bound for N_b^n and Equation (215) give the final uniform constant.

A moving J sweeps very general base points and is of general type by Proposition F.3. Equations (215)–(216) therefore give forbidden bounded curves. Branch components can be named after a finite parameter extension and shrinking. It follows that, for all sufficiently small t , all divisorial branch components in the chosen family are fixed.

Fixed covers. For each such fixed t , remove the fixed branch divisors and the singular locus from Y . Normalization and purity identify the covers over the resulting smooth open with finite étale covers. Their degrees are bounded. The topological fundamental group of a smooth complex quasi-projective variety is finitely generated, so it has only finitely many subgroups of bounded index. There are therefore only finitely many possible covers in each slot, up to isomorphism; their finite normalizations over Y are determined by these restrictions.

Some algebraic family of graphs of birational isomorphisms between two fixed covers must still dominate the base product. To justify the family assertion, parameterize the graphs by their Hilbert schemes, shrink for flatness and birationality of the two projections, and use countability of these parameter spaces together with the finite cover choices. Identify the two covers by one such birational isomorphism. The resulting birational selfmaps of a fixed cover move a general point densely through that cover, since their projected graphs dominate $Y \times Y$.

The cover is non-uniruled, being generically finite over the non-uniruled Y . Hanamura's non-uniruled birational-group theorem gives a smooth projective birational model for which the reduced birational group is a group scheme, locally of finite type, and its identity component is an abelian variety [43, Theorems 2.1–2.2]; see also [10, Proposition 3.7 and Theorems 3.8–3.9]. After conjugating the maps to that model, shrink the irreducible parameter variety anew so that their graph closures are flat and both graph projections remain birational on every fiber. They then define a morphism to the represented birational scheme; because the parameter variety is reduced, this morphism factors through its reduction. Its connected image lies in a single component, hence in a translate of the identity component. Dominance of evaluation makes the corresponding abelian variety action generically transitive. Its orbit is an abelian-variety quotient, so the fixed cover is birational to an abelian variety.

This is impossible for the original counterexample. Resolve the generically finite rational map from that abelian variety to X : on a smooth model U , ramification gives

$$K_U = f^*K_X + R, \quad R \geq 0,$$

whereas K_U is an effective divisor exceptional over the abelian variety. Let T be a positive current in K_X . The current $f^*T + [R]$ is positive and represents K_U . Intersecting with powers of the ample class pulled back from the abelian variety shows that this sum is supported on the exceptional locus. Its positive summand f^*T is therefore supported there as well, and the support theorem makes f^*T divisorial. Push forward this summand alone: $f_*(f^*T)$ is a positive divisorial current whose class is $\deg(f)c_1(K_X)$ by the projection formula. Rationality of the numerical class supplies a rational effective representative, and irregularity zero turns numerical effectivity into nonvanishing, as in Lemma F.1. This contradicts $\kappa(X, K_X) = -\infty$.

All possible V have now been excluded. This proves the Seshadri assertion. At a very general point the big semiample contraction is an isomorphism onto its image near that point. The jet interpretation of the Seshadri constant for the ample class downstairs gives the stated divisible jet-separating multiple. Complete systems agree under the model comparisons, so this conclusion transfers to the original torus open and to common resolutions. $\square$

H Frobenius comparison and smooth nonvanishing

We now turn the two opposite jet estimates into a contradiction. The reduction to positive characteristic is used only for this comparison: no minimal-model statement or pseudo-effectivity assertion will be specialized to positive characteristic.

Theorem H.1 (The smooth nonvanishing step). *Assume Assumption A.1 and the lower-dimensional real-boundary good-model hypothesis of Assumption B.1. If X is a smooth projective complex variety of dimension n and K_X is pseudo-effective, then $\kappa(X, K_X) \geq 0$.*

The assertion is immediate in dimension zero. On a smooth curve, pseudo-effectivity of K_X gives $g(X) \geq 1$, hence $h^0(X, K_X) = g(X) > 0$. We therefore suppose $n \geq 2$ and argue by contradiction. We use the late terminal model Y , the divisors $K = K_Y$ and $A = A_Y$, and the scaling models of the preceding section. Fix $m > 0$ with mK Cartier, and write

$$\mu_t = \text{vol}(K_X + tA_X)^{1/n}, \quad L = r(K + tA).$$

As before, r is a positive integer chosen so that $r\mu_t \rightarrow \epsilon$ as $t \downarrow 0$. Choose the rational number $\epsilon > 0$ sufficiently small for Propositions G.3 and G.5, and also so that

$$(14\epsilon)^n < \frac{1}{4}, \quad 80\epsilon < \frac{3}{4}. \quad (217)$$

The inequalities are strict. We next fix a sufficiently large integer q as in Proposition G.5, and then fix a sufficiently small positive rational t . In particular, we may require $r > qm + 1$. All subsequent characteristic-zero choices will be made with these data fixed.

H.1 A small polarization and an actual moving curve

Choose a smooth common resolution W of Y and the scaling model for $K + tA$. In this section K, A, L also denote their pullbacks to W , and

$$K_W = K + E, \quad E \geq 0.$$

Let H_0 be the pullback of the nef semiample positive part N_0 of L on its scaling model. The comparison of canonical pullbacks and moving parts gives

$$H_0^n \rightarrow \epsilon^n, \quad LH_0^{n-1} = H_0^n, \quad K_W H_0^{n-1} = K H_0^{n-1} \leq \frac{H_0^n}{r}. \quad (218)$$

The maps $Y \dashrightarrow Y_t$ are small by the choice of the late model, so every divisor exceptional over Y is also exceptional over Y_t . Indeed, the differences being discarded are exceptional over the scaling model, and the transform of A is effective up to rational linear equivalence. Their intersections with H_0^{n-1} therefore give the displayed equalities and inequality. For a fixed ample divisor H_{amp} , choose a sufficiently small positive rational η and put $H = H_0 + \eta H_{\text{amp}}$. By continuity, after the preceding choices of t and r , we have

$$H^n \leq (2\epsilon)^n, \quad K_W H^{n-1} \leq \frac{2H^n}{r}, \quad (2L + qmK)H^{n-1} \leq 4H^n. \quad (219)$$

The canonical divisor of W is pseudo-effective. Generic semipositivity, applied with empty boundary [18, Theorem 2.1], and restriction to a sufficiently general complete-intersection curve give the following fixed data. Choose $l > 0$ such that $h = lH$ is integral and sufficiently very ample, and choose a smooth complete-intersection flag

$$W = W_n \supset W_{n-1} \supset \cdots \supset W_1 = C, \quad W_{i-1} \in |h|_{W_i},$$

for which

$$\mu_{\min}(\Omega_W^1|_C) \geq 0. \quad (220)$$

Here and below slopes on C mean degree divided by rank. One may obtain the flag by applying restriction to the finitely many Harder–Narasimhan quotients of Ω_W^1 .

Choose also a very general point $x \in W$, away from the exceptional loci, and let $\pi_x : \widehat{W} \rightarrow W$ be its blowup, with exceptional divisor J . Set

$$D^+ = 2r(K + 2tA) + 2E.$$

This divisor is big. Its sections, pushed to Y , are the scalar sections in Proposition G.3; the added exceptional part does not change them. If $\rho = \text{vol}(2r(K + 2tA))^{1/n}$, then

$$\rho \leq 4r\mu_t \leq 8\epsilon$$

for our sufficiently small t . Thus every section of a sufficiently divisible multiple of D^+ has normalized order at x at most $4\rho \leq 32\epsilon$.

It follows that $\pi_x^*D^+ - 40\epsilon J$ is not pseudo-effective. Otherwise its convex combination with the big class $\pi_x^*D^+$ would make $\pi_x^*D^+ - cJ$ big for some rational $32\epsilon < c < 40\epsilon$, giving a section of excessive order. Movable-curve duality [12, Theorem 2.2] therefore supplies an actual covering curve class γ on $\widehat{W}$ such that

$$(\pi_x^*D^+) \cdot \gamma < 40\epsilon J \cdot \gamma, \quad J \cdot \gamma > 0. \quad (221)$$

For precision, the strict negative pairing can first be detected by a strongly movable curve: take the pushforward of a general complete intersection of very ample divisors on one fixed smooth birational model of $\widehat{W}$. Fix this model, its divisors, and the resulting covering family. This choice is finite algebraic data, rather than a limiting real movable class. Positivity of $J \cdot \gamma$ follows from the strict inequality and pseudo-effectivity of $\pi_x^*D^+$.

H.2 Fixed jets and reduction modulo primes

On $W \times W$, let

$$Z = \mathbb{P}(mK_1 \oplus mK_2), \quad \xi = \mathcal{O}_Z(1), \quad M = L_1 + L_2 + q\xi,$$

with the symmetric-power convention for the projective bundle. Proposition G.5 gives a semiample big model whose Seshadri constant exceeds $2n + 2$ at a very general torus point. On the open set where its birational contraction is an isomorphism, the jet interpretation of the Seshadri constant consequently gives an integer $k_0 > 0$ such that $|k_0 M|$ generates jets of order at least $(2n + 2)k_0$ at a fixed smooth torus point $z_* \in Z$. Indeed, on the ample model choose a rational number c strictly between $2n + 2$ and its Seshadri constant. On the blowup of the selected smooth point, the pullback of the ample class minus c times the exceptional divisor is ample. Serre vanishing and the exceptional-divisor sequence then give jets of order $kc - 1$ for all sufficiently divisible large k . Pulling sections back on the isomorphic open gives the asserted bound. Enlarge k_0 to clear every denominator, in particular that of L , and fix finitely many sections realizing this jet surjection.

Spread the varieties, maps, divisors, flag, points, covering family, and these finitely many sections over an integral finitely generated $\mathbb{Z}$ -algebra of characteristic zero. Shrink its spectrum so that the relevant fibers and flag are smooth, the maps defining the covering family remain dominant, and the fixed jet evaluation remains surjective. Spread also the Harder–Narasimhan filtration of $\Omega_W^1|_C$. Its quotients are vector bundles on the curve; their semistability is open, and their degrees are constant in the family. After another shrinking, Equation (220) holds on each reduction. Such an open set has closed points in arbitrarily large prime characteristics. We work over algebraic closures of their residue fields, retaining the notation W, C, h, x, γ .

In particular, all numbers in Equations (219) and (221) are unchanged. We use the spread covering family to test effective divisors after reduction. We do not assert that the characteristic-zero pseudo-effective cones, or the scaling models, specialize.

Choose a fixed sufficiently ample integral divisor T_0 on W , also spread with suitable sections, and for a prime p put

$$N_p = \left\lfloor \frac{p}{k_0} \right\rfloor, \quad B_p = k_0 N_p L + T_0. \quad (222)$$

The difference $B_p - pL$ belongs to a fixed finite set of rational divisor classes. The system

$$|pq\xi + B_{p,1} + B_{p,2}|$$

contains products of N_p members of $|k_0 M|$, multiplied by a filler nonvanishing at z_* . To see that one T_0 suffices, write $p = k_0 N_p + d$, where $0 \leq d < k_0$. The remainder systems have fiber degrees dq ; their finitely many monomial weights require only the finitely many line bundles $T_0 + a mK$ for $0 \leq a \leq (k_0 - 1)q$ to have sections nonvanishing at the selected base points. A sufficiently ample T_0 has this property, and the chosen fillers can be spread at the same time.

Products of the fixed jet sections generate jets of order $(2n + 2)k_0 N_p$. Indeed each monomial of that degree or less is a product of N_p monomials of degree at most $(2n + 2)k_0$; multiply sections with these leading monomials and then eliminate successive higher-order terms. No factorial is divided out. Since

$$(2n + 2)k_0 N_p > (2n + 1)(p - 1)$$

for all sufficiently large p , the same system surjects onto the quotient of the local ring at z_* by the p -th powers of its $2n + 1$ regular parameters.

H.3 Full rank away from the diagonal

Let $F : W \rightarrow W'$ be relative Frobenius, and put

$$S_0 = mK, \quad \mathcal{E} = F_* \mathcal{O}_W(B_p), \quad s = \text{rk } \mathcal{E} = p^n, \quad U_a = H^0(W, \mathcal{O}_W(B_p + paS_0)) \quad (0 \leq a \leq q).$$

Primes on line bundles indicate the base twist, so $F^* S'_0 = pS_0$. Projection formula gives an evaluation map

$$\bigoplus_{a+b=q} U_a \otimes U_b \otimes \mathcal{O}_{W' \times W'}(-aS'_{0,1} - bS'_{0,2}) \longrightarrow \mathcal{E} \boxtimes \mathcal{E}. \quad (223)$$

Its generic rank is s^2 .

Here is a coordinate verification of this assertion. Choose local frames of the projective-bundle axes and let z be a torus coordinate at z_* , with value $z_0 \neq 0$. On its Frobenius fiber the relation is $z^p = z_0^p$. Over the two base Frobenius fibers, the fiber-coordinate part of the quotient is free with basis $1, z, \dots, z^{p-1}$. Project the jet surjection just obtained to its coefficient of 1. The monomial z^j survives precisely when p divides j , in which case it becomes the nonzero scalar z_0^j . The global section formula for the projective bundle has weights $0 \leq j \leq pq$. Its surviving terms $j = pa$ are exactly

$$H^0(W, B_p + paS_0) \otimes H^0(W, B_p + p(q - a)S_0).$$

They therefore span the tensor product of the two base Frobenius fibers. These have dimensions s each, proving the rank assertion. The two base points used for this argument need not coincide with x .

H.4 A small rank on the diagonal

Let

$$\Delta_F = W \times_{W'} W = (F \times F)^{-1}(\Delta_{W'}).$$

On this scheme the two copies of pS_0 are canonically identified, since both descend from S'_0 on the diagonal. All source twists in Equation (223) restrict on $\Delta_{W'}$ to $\mathcal{O}_{W'}(-qS'_0)$, with these identifications pulled back to Δ_F . Consequently every column, after restriction to the diagonal and this common one-dimensional twist, is evaluated from a global section on Δ_F of

$$\mathcal{L}_p = \mathcal{O}_{\Delta_F}(B_{p,1} + B_{p,2} + pqS_{0,1}).$$

The rank on the diagonal is therefore at most $h^0(\Delta_F, \mathcal{L}_p)$.

The first projection $\Delta_F \rightarrow W$ is finite flat of degree s . Filter its direct image by powers of the ideal of the reduced diagonal. Étale-locally in smooth coordinates its algebra is

$$\mathcal{O}_W[\delta_1, \dots, \delta_n]/(\delta_1^p, \dots, \delta_n^p).$$

Its graded pieces, after the line twist, are thus the vector bundles

$$\mathcal{G}_j = \mathcal{O}_W(2B_p + pqS_0) \otimes A_j(\Omega_W^1), \quad 0 \leq j \leq u = n(p-1), \quad (224)$$

where $A_j(V)$ is the degree- j piece of the symmetric algebra of V modulo p -th powers. This coordinate calculation is also the canonical Frobenius filtration [71, Theorem 3.7]. If $e_j = \text{rk } A_j(\Omega_W^1)$, then

$$\sum_{j=0}^u e_j = p^n = s. \quad (225)$$

The sum is s , rather than s^2 ; the second factor of s will come from the bound for sections.

Lemma H.2 (Uniform slope bound). *With all characteristic-zero choices fixed, uniformly for $0 \leq j \leq n(p-1)$,*

$$\mu_{\max}(\mathcal{G}_j|_C) \leq 7pl^{n-1}H^n + O(1).$$

The constant implicit in $O(1)$ is independent of p and j .

Proof. Put $V = \Omega_W^1|_C$. For a vector bundle on the smooth curve C , write

$$L_{\min}(V) = \lim_{e \rightarrow \infty} \frac{\mu_{\min}((F_C^e)^*V)}{p^e}.$$

Langer's Frobenius-instability estimate [58, Corollary 2.5] states that

$$\mu_{\min}(V) - L_{\min}(V) \leq \frac{(n-1) \deg A_C}{p-1}$$

if A_C is nef and $T_C(A_C)$ is globally generated. The genus is fixed, so A_C can be chosen with degree bounded independently of p . Equation (220) yields

$$L_{\min}(V) \geq -\frac{c}{p-1} \quad (226)$$

with one fixed c .

We need an estimate for tensor powers whose exponent grows with p . For every $i \geq 0$,

$$\mu_{\min}(V^{\otimes i}) \geq iL_{\min}(V). \quad (227)$$

To prove this directly, twist $(F_C^e)^*V$ by a line bundle of degree at most $-\mu_{\min}((F_C^e)^*V) + 2g(C) + 2$ so that it is globally generated. This follows from Serre duality and the slope criterion for the vanishing of H^1 after subtracting any point. The i -th tensor power is then globally generated with the corresponding i -fold twist. Pull back any quotient of $V^{\otimes i}$, take degrees, divide by p^e , and let e tend to infinity. The bounded genus term disappears, proving Equation (227). Thus every quotient of a tensor power with $i \leq n(p-1)$ has minimum slope at least $-nc$.

Multiplication to the socle of the truncated polynomial algebra is a perfect pairing in complementary degrees. The top monomial transforms by $(\det V)^{p-1}$, so, equivariantly,

$$A_j(V)^* \otimes (\det V)^{p-1} \simeq A_{u-j}(V).$$

The right side is a quotient of $V^{\otimes(u-j)}$. Therefore

$$\mu_{\max}(A_j(V)) \leq (p-1) \deg K_W|_C + nc.$$

By Equations (222) and (219),

$$\begin{aligned} (2B_p + pqS_0) \cdot C &\leq 4pl^{n-1}H^n + O(1), \\ (p-1)K_W \cdot C &\leq \frac{2p}{r} l^{n-1}H^n. \end{aligned}$$

Since $r > 1$, their sum is bounded by the claimed expression, with slack in its coefficient 7. $\square$

Lemma H.3 (Uniform section bound). *For the bundles in Equation (224),*

$$h^0(W, \mathcal{G}_j) \leq e_j p^n (7^n H^n + o(1)),$$

uniformly in j . Consequently, for all sufficiently large p ,

$$h^0(\Delta_F, \mathcal{L}_p) \leq \frac{s^2}{4}. \quad (228)$$

Proof. Choose $M_p = 7pl^{n-1}H^n + c_0$, where c_0 bounds the error in Lemma H.2, and put $d_C = h \cdot C = l^n H^n$. A bundle of rank e_j and maximal slope at most M_p has at most $e_j(M_p + 1)$ sections: evaluation at $\lfloor M_p \rfloor + 1$ distinct points is injective. Its twist by $-kh$ has no sections if $kd_C > M_p$.

The divisor sequences on the fixed flag give

$$h^0(W, \mathcal{G}_j) \leq \sum_{k_1, \dots, k_{n-1} \geq 0} h^0(C, \mathcal{G}_j|_C(-(k_1 + \dots + k_{n-1})h)).$$

One obtains this by successively summing the restriction inequalities; the remainders vanish for sufficiently negative ample twists at each stage. At most $(1 + \lfloor M_p/d_C \rfloor)^{n-1}$ tuples contribute. Hence

$$h^0(W, \mathcal{G}_j) \leq e_j(M_p + 1)(1 + M_p/d_C)^{n-1} = e_j p^n (7^n H^n + o(1)).$$

The powers of l cancel in the leading coefficient. All error constants are fixed before p , and the same bound applies to every j . Summing over the filtration and using Equation (225) gives

$$h^0(\Delta_F, \mathcal{L}_p) \leq s^2(7^n H^n + o(1)).$$

Now $7^n H^n \leq (14\epsilon)^n < 1/4$ by Equations (217) and (219). This proves Equation (228). $\square$

H.5 The determinant contradiction

Put $R = s^2$. Choose R columns of Equation (223) with nonzero determinant. Their determinant is a nonzero section σ of an external product $\mathcal{Q}_1 \boxtimes \mathcal{Q}_2$ of line bundles on $W' \times W'$. Indeed $\det(\mathcal{E} \boxtimes \mathcal{E}) = (\det \mathcal{E})^s \boxtimes (\det \mathcal{E})^s$, and the column twists add the sums of their respective weights. Consequently, under numerical identification with the base twists,

$$\frac{c_1(\mathcal{Q}_i)}{R} = \frac{c_1(\mathcal{E})}{s} + \lambda_i S_0 = \frac{B_p}{p} + \frac{p-1}{2p} K_W + \lambda_i m K, \quad 0 \leq \lambda_i \leq q. \quad (229)$$

The last equality is degree-one Grothendieck–Riemann–Roch for Frobenius [71, Lemma 4.2]:

$$c_1(F_* \mathcal{O}_W(B_p)) = p^{n-1} B'_p + \frac{p^n - p^{n-1}}{2} K_{W'}.$$

Here numerical identification through the base-field twist must not be confused with Frobenius pullback, which multiplies divisor classes by p .

For $0 \leq \lambda \leq q$, write $Q_\lambda = L + K_W/2 + \lambda m K$. Direct calculation gives

$$D^+ - Q_\lambda = (r - \frac{1}{2} - \lambda m) K + 3rtA + \frac{3}{2} E. \quad (230)$$

This is pseudo-effective in characteristic zero because $r > qm + 1$, K is pseudo-effective, and A, E are effective up to the indicated equivalence. Pairing with the fixed class γ therefore bounds $Q_\lambda \cdot \gamma$ by $(\pi_x^* D^+) \cdot \gamma$. This numerical inequality survives the spread: it concerns only intersections of fixed divisors with the fixed family. The difference between Equation (229) and Q_{λ_i} is $O(1/p)$ in a fixed finite-dimensional space, uniformly in the chosen columns.

Let $x' \in W'$ be the twist of x . For a nonzero section of $\mathcal{Q}_i$, its effective divisor, strictly transformed on the blowup at x' , has nonnegative intersection with the spread moving curve. Dividing this inequality by $RJ \cdot \gamma$ and using Equations (221)–(230) yields

$$\frac{\text{ord}_{x'}(\tau)}{R} \leq 40\epsilon + o(1) \quad \text{for every } 0 \neq \tau \in H^0(W', \mathcal{Q}_i). \quad (231)$$

This bounds all sections on the reduction, not merely the sections spread from characteristic zero.

An external-product section can have order at (x', x') at most the sum of the two maximal slot orders. To verify this despite possible cancellation, choose bases in each section space adapted to its filtration by order at x' . Their leading homogeneous terms are linearly independent within each degree. Tensor products of these terms are independent in each bidegree in the two disjoint sets of local parameters. Thus the first nonzero homogeneous part of a nonzero tensor cannot be cancelled beyond the sum of the slot maxima. Künneth’s formula identifies the external-product section space with this tensor product. In particular,

$$\text{ord}_{(x', x')}(\sigma) \leq R(80\epsilon + o(1)). \quad (232)$$

On the other hand, Equation (228) says that the matrix of the chosen columns has rank at most $R/4$ at (x', x') . Trivialize its line bundles locally and perform invertible row and column operations on the constant matrix. At least $3R/4$ of the resulting rows have every entry in the maximal ideal. Every term of the determinant consequently has order at least $3R/4$, so

$$\text{ord}_{(x', x')}(\sigma) \geq \frac{3R}{4}. \quad (233)$$

This is a pointwise maximal-ideal estimate; no stronger assertion about an order along the whole diagonal is required. Equations (232) and (233) contradict $80\epsilon < 3/4$ for all sufficiently large p .

We have made the choices in the order

$$n, \epsilon, q, (t, r), (W, H, h, \text{flag}, x, \gamma), (k_0, T_0, \text{fixed sections}), p.$$

In particular every constant suppressed in the estimates is fixed before the last prime tends to infinity. This contradiction excludes the assumed counterexample and proves Theorem H.1.

I Completion and change of ground field

Proof of Theorem A.2 over $\mathbb{C}$. We prove existence of good log minimal models by dimension induction, in the form used in Proposition B.6. Dimension zero is immediate. Assume that good models exist in all smaller dimensions. The signed-representative argument (Theorem C.1), the exclusions, and the jet estimates are then available with exactly that lower-dimensional hypothesis. Theorem H.1 establishes smooth nonvanishing in the present dimension. Proposition B.6 therefore supplies the good-model statement in this dimension and closes the induction.

Apply this to the rational lc pair (X, B) of Theorem A.2. On a common resolution of its dlt model and a good log minimal model, the pullbacks of the adjoints agree because the original adjoint is nef; this is the nef comparison in Lemma B.2. The pullback of $K_X + B$ is thus semiample. If $p : W \rightarrow X$ is this resolution, normality gives $p_* \mathcal{O}_W = \mathcal{O}_X$. Choose a sufficiently divisible Cartier multiple $m(K_X + B)$ whose pullback is globally generated. The projection formula identifies its sections with those of its pullback. Surjectivity of evaluation upstairs implies surjectivity downstairs: otherwise a base point downstairs would make every pulled-back section vanish on its nonempty fiber. Hence $K_X + B$ is semiample. $\square$

Proposition I.1 (Change of algebraically closed field). *The conclusion of Theorem A.2 over $\mathbb{C}$ implies its conclusion over every algebraically closed field of characteristic zero.*

Proof. Let (X, B) be defined over such a field k . Choose a finitely generated subfield $k_1 \subset k$ over which the projective variety, the rational boundary, a Cartier multiple of its adjoint, and a log resolution are all defined. After enlarging k_1 if necessary, these data base change to the given data. Let k_0 be its algebraic closure inside k . It embeds into $\mathbb{C}$.

The discrepancies on the chosen resolution are unchanged by algebraically closed field extension. Since the resolution has simple normal crossing boundary, it tests log canonicity over both k_0 and $\mathbb{C}$.

Nefness is also invariant under such extensions. Indeed, a curve over an extension is represented by a point of a relative Hilbert scheme after its finite defining data have been spread over a finite-type parameter scheme. The degree of a fixed line bundle is constant in the resulting flat family. Specializing to a closed point over the algebraically closed smaller field preserves a negative degree, if one existed; some irreducible component of the specialized curve would then have negative degree. This contradicts nefness over that field. Conversely, curves over the smaller field remain available after extension. Thus the k_0 -model and its complex base change are nef.

The complex case supplies an integer $m > 0$, enlarged to a multiple of the Cartier index fixed over k_0 , for which the Cartier line bundle $M = \mathcal{O}_{X_{k_0}}(m(K_{X_{k_0}} + B_{k_0}))$ becomes globally generated over $\mathbb{C}$. Proper flat base change for sections identifies

$$H^0(X_{k_0}, M) \otimes_{k_0} \mathbb{C} = H^0(X_{\mathbb{C}}, M_{\mathbb{C}}).$$

The cokernel of the evaluation map for M therefore becomes zero after the faithfully flat extension $k_0 \subset \mathbb{C}$, and is already zero over k_0 . Base change from k_0 to k preserves this surjectivity. The same integer m proves the required semiampleness over k . $\square$

Together with Proposition I.1, the complex proof establishes Theorem A.2 in its stated generality.

References

- [1] Florin Ambro. The moduli b-divisor of an lc-trivial fibration. *Compositio Mathematica*, 141(2):385–403, 2005. Theorem 0.2.
- [2] Thierry Aubin. Équations du type Monge–Ampère sur les variétés kählériennes compactes. *Bulletin des Sciences Mathématiques*, 102(1):63–95, 1978.
- [3] Thierry Aubin. *Some Nonlinear Problems in Riemannian Geometry*. Springer Monographs in Mathematics. Springer-Verlag, Berlin, Heidelberg, 1998.
- [4] Wolf P. Barth, Klaus Hulek, Chris A. M. Peters, and Antonius van de Ven. *Compact Complex Surfaces*, volume 4 of *Ergebnisse der Mathematik und ihrer Grenzgebiete, 3. Folge*. Springer-Verlag, Berlin, Heidelberg, second enlarged edition, 2004.
- [5] Thomas Bauer, Frédéric Campana, Thomas Eckl, Stefan Kebekus, Thomas Peternell, Sławomir Rams, Tomasz Szemberg, and Lorenz Wotzlaw. A reduction map for nef line bundles. In *Complex Geometry: Collection of Papers Dedicated to Hans Grauert*, pages 27–36. Springer, Berlin, 2002. Theorem 2.1 in arXiv version 1. Version cited: <https://arxiv.org/abs/math/0106147v1>.
- [6] Caucher Birkar. On existence of log minimal models. *Compositio Mathematica*, 146(4):919–928, 2010. Lemma 3.6 in the cited arXiv version. Version cited: <https://arxiv.org/abs/0706.1792v3>.
- [7] Caucher Birkar. On existence of log minimal models II. *Journal für die reine und angewandte Mathematik*, 658:99–113, 2011. Theorem 1.5 and Corollary 1.7 in the cited arXiv version. Version cited: <https://arxiv.org/abs/0907.4170v1>.
- [8] Caucher Birkar. Existence of log canonical flips and a special LMMP. *Publications Mathématiques de l’IHÉS*, 115:325–368, 2012. Theorem 1.9, restated as Theorem 4.1. Version cited: <https://arxiv.org/abs/1104.4981v3>.
- [9] Caucher Birkar, Paolo Cascini, Christopher D. Hacon, and James McKernan. Existence of minimal models for varieties of log general type. *Journal of the American Mathematical Society*, 23(2):405–468, 2010.
- [10] Jérémy Blanc. Algebraic structures of groups of birational transformations. In *Algebraic Groups: Structure and Actions*, volume 94 of *Proceedings of Symposia in Pure Mathematics*, pages 17–30. American Mathematical Society, 2017. Theorems 3.8–3.9.
- [11] Sébastien Boucksom. Divisorial Zariski decompositions on compact complex manifolds. *Annales scientifiques de l’École Normale Supérieure*, 37(1):45–76, 2004.
- [12] Sébastien Boucksom, Jean-Pierre Demailly, Mihai Păun, and Thomas Peternell. The pseudo-effective cone of a compact Kähler manifold and varieties of negative Kodaira dimension. *Journal of Algebraic Geometry*, 22(2):201–248, 2013. The uniruledness equivalence in this source is stated for projective manifolds. Theorem locators refer to arXiv version 1: <https://arxiv.org/abs/math/0405285v1>.
- [13] Frédéric Campana. Coréduction algébrique d’un espace analytique faiblement kählérien compact. *Inventiones Mathematicae*, 63:187–223, 1981.

- [14] Frédéric Campana. Orbifolds, special varieties and classification theory: an appendix. *Annales de l'Institut Fourier*, 54(3):631–665, 2004.
- [15] Frédéric Campana, Benoît Claudon, and Philippe Eyssidieux. Représentations linéaires des groupes kählériens: factorisations et conjecture de Shafarevich linéaire. *Compositio Mathematica*, 151(2):351–376, 2015.
- [16] Frédéric Campana, Andreas Höring, and Thomas Peternell. Abundance for Kähler threefolds. *Annales scientifiques de l'École Normale Supérieure*, 49(4):971–1025, 2016.
- [17] Frédéric Campana, Andreas Höring, and Thomas Peternell. Erratum and addendum to the paper: Abundance for Kähler threefolds. <https://arxiv.org/abs/2304.10161v1>, 2023. Version 1, 20 April 2023.
- [18] Frédéric Campana and Mihai Păun. Orbifold generic semi-positivity: an application to families of canonically polarized manifolds. *Annales de l'Institut Fourier*, 65(2):835–861, 2015.
- [19] Omprokash Das, Christopher Hacon, and Mihai Păun. On the 4-dimensional minimal model program for Kähler varieties. *Advances in Mathematics*, 443:109615, 2024.
- [20] Omprokash Das and Wenhao Ou. On the log abundance for compact Kähler threefolds. *Manuscripta Mathematica*, 173:341–404, 2024.
- [21] Omprokash Das and Wenhao Ou. On the log abundance for compact Kähler threefolds II. <https://arxiv.org/abs/2306.00671v4>, 2025. Version 4, 12 June 2025; first submitted 2023.
- [22] Jean-Pierre Demailly. Regularization of closed positive currents and intersection theory. *Journal of Algebraic Geometry*, 1(3):361–409, 1992.
- [23] Jean-Pierre Demailly. L^2 vanishing theorems for positive line bundles and adjunction theory. In Fabrizio Catanese and Ciro Ciliberto, editors, *Transcendental Methods in Algebraic Geometry*, volume 1646 of *Lecture Notes in Mathematics*, pages 1–97. Springer, Berlin, Heidelberg, 1996.
- [24] Jean-Pierre Demailly. *Analytic Methods in Algebraic Geometry*, volume 1 of *Surveys of Modern Mathematics*. International Press and Higher Education Press, Somerville, MA, and Beijing, 2012. Theorem numbering refers to the author manuscript dated July 2011. Version cited: https://www-fourier.univ-grenoble-alpes.fr/~demailly/manuscripts/analmeth_book.pdf.
- [25] Jean-Pierre Demailly. Complex analytic and differential geometry. <https://www-fourier.univ-grenoble-alpes.fr/~demailly/manuscripts/agbook.pdf>, 2012. Online book, version of 21 June 2012.
- [26] Jean-Pierre Demailly and János Kollár. Semi-continuity of complex singularity exponents and Kähler–Einstein metrics on Fano orbifolds. *Annales scientifiques de l'École Normale Supérieure*, 34(4):525–556, 2001.
- [27] Jean-Pierre Demailly and Thomas Peternell. A Kawamata–Viehweg vanishing theorem on compact Kähler manifolds. *Journal of Differential Geometry*, 63(2):231–277, 2003.
- [28] Jean-Pierre Demailly, Thomas Peternell, and Michael Schneider. Pseudo-effective line bundles on compact Kähler manifolds. *International Journal of Mathematics*, 12(6):689–741, 2001.

- [29] Osamu Fujino. Introduction to the log minimal model program for log canonical pairs. Unpublished manuscript, <https://www.math.kyoto-u.ac.jp/~fujino/MMP21.pdf>, 2009. Version 6.01, January 8, 2009; Theorem 3.74(3), p. 109.
- [30] Osamu Fujino. Corrigendum to “On subadditivity of the logarithmic Kodaira dimension”. *Journal of the Mathematical Society of Japan*, 72(4):1181–1187, 2020. Accepted author version: <https://www.math.kyoto-u.ac.jp/~fujino/corrigendum-fujino.pdf>.
- [31] Osamu Fujino. Minimal model program for projective morphisms between complex analytic spaces. <https://arxiv.org/abs/2201.11315v1>, 2022. Version 1, 27 January 2022; theorem numbering follows this version.
- [32] Osamu Fujino. Cone and contraction theorem for projective morphisms between complex analytic spaces. <https://www.math.kyoto-u.ac.jp/~fujino/cone-and-contraction.pdf>, 2023. Author manuscript, version 0.00, 9 October 2023; Theorem 4.3.1 and Remark 4.3.4.
- [33] Osamu Fujino and Yoshinori Gongyo. Log pluricanonical representations and the abundance conjecture. *Compositio Mathematica*, 150(4):593–620, 2014. Theorem 1.1 (finiteness of log pluricanonical representations); Theorem 1.5, restated as Theorem 4.3 (projective slc gluing). Final author version: <https://www.math.kyoto-u.ac.jp/~fujino/fg-comp-final.pdf>.
- [34] Osamu Fujino and Yoshinori Gongyo. On the moduli b-divisors of lc-trivial fibrations. *Annales de l’Institut Fourier*, 64(4):1721–1735, 2014.
- [35] Osamu Fujino and Yoshinori Gongyo. On log canonical rings. In *Higher Dimensional Algebraic Geometry in honour of Professor Yujiro Kawamata’s sixtieth birthday*, volume 74 of *Advanced Studies in Pure Mathematics*, pages 159–169. Mathematical Society of Japan, Tokyo, 2017. Lemma 3.5. Final author version: <https://www.math.kyoto-u.ac.jp/~fujino/log-ring-aspmp.pdf>.
- [36] Osamu Fujino and Kenta Hashizume. Existence of log canonical modifications and its applications. *European Journal of Mathematics*, 9(1), 2023. Paper No. 13, 32 pp.; Theorem 2.10 in the accepted author version. Version cited: <https://www.math.kyoto-u.ac.jp/~fujino/lc-model-fujino-hashizume-draft-7-EJM-rev-normalstylefile.pdf>.
- [37] Yoshinori Gongyo. Abundance theorem for numerically trivial log canonical divisors of semi-log canonical pairs. *Journal of Algebraic Geometry*, 22(3):549–564, 2013.
- [38] Yoshinori Gongyo and Brian Lehmann. Reduction maps and minimal model theory. *Compositio Mathematica*, 149(2):295–308, 2013. Read with the numerical-dimension erratum on Lehmann’s research page. Erratum: <https://sites.bc.edu/lehmann/research/>. Final author version: <https://www.ms.u-tokyo.ac.jp/~gongyo/papers/redmapsmmp-final.pdf>.
- [39] Tom Graber, Joe Harris, and Jason Starr. Families of rationally connected varieties. *Journal of the American Mathematical Society*, 16(1):57–67, 2003.
- [40] Vincent Guedj and Ahmed Zeriahi. Intrinsic capacities on compact Kähler manifolds. *The Journal of Geometric Analysis*, 15(4):607–639, 2005.
- [41] Henri Guenancia and Mihai Păun. Bogomolov–Gieseker inequality for log terminal Kähler threefolds. *Communications on Pure and Applied Mathematics*, 78(11):2206–2244, 2025. With an appendix by Frédéric Campana, Andreas Höring, and Thomas Peternell.

- [42] Christopher D. Hacon, James McKernan, and Chenyang Xu. Boundedness of varieties of log general type. In *Algebraic Geometry: Salt Lake City 2015, Part 1*, volume 97.1 of *Proceedings of Symposia in Pure Mathematics*, pages 309–348. American Mathematical Society, 2018. Published Theorem 4.0.1; Theorem 3.0.1 in arXiv version 2. Published offprint: <https://www.claymath.org/wp-content/uploads/2022/03/Hacon-AG2015.pdf>.
- [43] Masaki Hanamura. Structure of birational automorphism groups, I: non-uniruled varieties. *Inventiones Mathematicae*, 93:383–403, 1988. Theorems 2.1–2.2; statements also reproduced in Blanc, Theorems 3.8–3.9.
- [44] Kenta Hashizume. On the non-vanishing conjecture and existence of log minimal models. *Publications of the Research Institute for Mathematical Sciences*, 54(1):89–104, 2018. Theorem 1.4; arXiv version 4. Version cited: <https://arxiv.org/abs/1609.00121v4>.
- [45] Heisuke Hironaka. Bimeromorphic smoothing of a complex-analytic space. *Acta Mathematica Vietnamica*, 2(2):103–168, 1977.
- [46] Andreas Höring, Vladimir Lazić, and Christian Lehn. Nonvanishing results for Kähler varieties. <https://arxiv.org/abs/2508.14634v2>, 2025. Version 2, 21 October 2025.
- [47] Andreas Höring and Thomas Peternell. Minimal models for Kähler threefolds. *Inventiones Mathematicae*, 203(1):217–264, 2016.
- [48] Yujiro Kawamata. A generalization of Kodaira–Ramanujam’s vanishing theorem. *Mathematische Annalen*, 261(1):43–46, 1982. For the relative formulation used here, see also Kollár–Mori, *Birational Geometry of Algebraic Varieties*, Theorem 2.64.
- [49] Yujiro Kawamata. The minimal discrepancy of a 3-fold terminal singularity. *Russian Academy of Sciences. Izvestiya Mathematics*, 40(1):193–195, 1993. Appendix to V. V. Shokurov, Three-fold log flips.
- [50] Stefan Kebekus and Christian Schnell. Extending holomorphic forms from the regular locus of a complex space to a resolution of singularities. *Journal of the American Mathematical Society*, 34(2):315–368, 2021.
- [51] Sean Keel, Kenji Matsuki, and James McKernan. Log abundance theorem for threefolds. *Duke Mathematical Journal*, 75(1):99–119, 1994. Read with the 2004 corrections.
- [52] Sean Keel, Kenji Matsuki, and James McKernan. Corrections to “Log abundance theorem for threefolds”. *Duke Mathematical Journal*, 122(3):625–630, 2004.
- [53] János Kollár, editor. *Flips and abundance for algebraic threefolds*. Number 211 in *Astérisque*. Société Mathématique de France, 1992.
- [54] János Kollár. Sources of log canonical centers. In *Minimal Models and Extremal Rays*, volume 70 of *Advanced Studies in Pure Mathematics*, pages 29–48. Mathematical Society of Japan, Tokyo, 2016.
- [55] János Kollár and Sándor J. Kovács. Log canonical singularities are Du Bois. *Journal of the American Mathematical Society*, 23(3):791–813, 2010.
- [56] János Kollár and Shigefumi Mori. *Birational Geometry of Algebraic Varieties*, volume 134 of *Cambridge Tracts in Mathematics*. Cambridge University Press, Cambridge, 1998.

- [57] Sándor J. Kovács and Zolt Patakfalvi. Projectivity of the moduli space of stable log-varieties and subadditivity of log-Kodaira dimension. *Journal of the American Mathematical Society*, 30(4):959–1021, 2017. Theorem 9.9 in the author version. Version cited: https://sites.math.washington.edu/~kovacs/current/papers/Kovacs_Patakfalvi__Projectivity.pdf.
- [58] Adrian Langer. Semistable sheaves in positive characteristic. *Annals of Mathematics*, 159(1):251–276, 2004.
- [59] Yoichi Miyaoka and Shigefumi Mori. A numerical criterion for uniruledness. *Annals of Mathematics*, 124(1):65–69, 1986.
- [60] Yoshinori Namikawa. Projectivity criterion of Moishezon spaces and density of projective symplectic varieties. *International Journal of Mathematics*, 13(2):125–135, 2002.
- [61] OpenAI. Abundance after nonvanishing for compact Kähler fourfolds. OpenAI Math Release preprint [OAI:Abundance-after-nonvanishing-for-compact-Kahler-fourfolds-September-27-2026](#), 2026. Theorem 1.1; semiampleness of the actual nef adjoint under nonnegative Kodaira dimension.
- [62] OpenAI. Finite ordinary minimal model programs on compact Kähler fourfolds. OpenAI Math Release preprint [OAI:Finite-ordinary-minimal-model-programs-on-compact-Kahler-fourfolds-October-6-2026](#), 2026. Theorem 6.24, Lemma 3.2, and Propositions 3.4 and 6.23.
- [63] OpenAI. Orbifold and logarithmic Iitaka subadditivity. OpenAI Math Release preprint [OAI:Orbifold-and-logarithmic-Iitaka-subadditivity-September-26-2026](#), 2026. Theorem 1.1.
- [64] Wenhao Ou. A characterization of uniruled compact Kähler manifolds. <https://arxiv.org/abs/2501.18088v1>, 2025. Version 1, 30 January 2025.
- [65] Mihai Păun. Siu’s invariance of plurigena: a one-tower proof. *Journal of Differential Geometry*, 76(3):485–493, 2007. Theorem 1.
- [66] Morihiko Saito. Mixed Hodge modules. *Publications of the Research Institute for Mathematical Sciences*, 26(2):221–333, 1990.
- [67] Morihiko Saito. Mixed Hodge complexes on algebraic varieties. *Mathematische Annalen*, 316(2):283–331, 2000. Theorem 0.2 and Corollary 0.3 in arXiv version 1. Version cited: <https://arxiv.org/abs/math/9906088v1>.
- [68] Jean-Pierre Serre. *A Course in Arithmetic*, volume 7 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1973.
- [69] Yum-Tong Siu. Analyticity of sets associated to Lelong numbers and the extension of closed positive currents. *Inventiones Mathematicae*, 27:53–156, 1974.
- [70] Henri Skoda. Sous-ensembles analytiques d’ordre fini ou infini dans C^n . *Bulletin de la Société Mathématique de France*, 100:353–408, 1972.
- [71] Xiaotao Sun. Direct images of bundles under Frobenius morphism. *Inventiones Mathematicae*, 173(2):427–447, 2008. Theorem 3.7 and Lemma 4.2 cited from arXiv version 2. Version cited: <https://arxiv.org/abs/math/0611360v2>.

- [72] The Stacks Project Authors. The Stacks Project: Grothendieck’s existence theorem, I. <https://stacks.math.columbia.edu/tag/087V>, 2026. Tag 087V, Lemmas 30.24.1 and 30.24.3; accessed September 18, 2026.
- [73] Frédéric Touzet. Feuilletages holomorphes admettant une mesure transverse invariante. <https://arxiv.org/abs/1409.3445v1>, 2014.
- [74] Jean Varouchas. Kähler spaces and proper open morphisms. *Mathematische Annalen*, 283:13–52, 1989.
- [75] Pinya Vikash. Classification of smooth minimal Kähler fourfolds without effective divisors and surfaces. <https://arxiv.org/abs/2607.04536v1>, 2026. Version 1, 5 July 2026.
- [76] Shing-Tung Yau. On the Ricci curvature of a compact Kähler manifold and the complex Monge–Ampère equation, I. *Communications on Pure and Applied Mathematics*, 31(3):339–411, 1978.